

Computational Finance

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Preface

Computational finance lies at the intersection of stochastic modelling, numerical analysis, and financial economics. These notes develop that intersection as a coherent workflow: we begin with the empirical properties of financial data, build continuous-time models for asset prices, derive arbitrage-free pricing relations, and finally study the numerical methods required when closed-form solutions are unavailable.

Throughout the course, the emphasis is on the interaction between theory and computation. Financial models are introduced not only as mathematical objects but also as tools that must ultimately be implemented, tested, and interpreted using numerical methods.

Scope of the Notes

The material progresses from conceptual foundations to practical implementation.

The first part introduces financial market data, stochastic processes, and the principle of no-arbitrage as the central organizing idea of modern financial modelling. The middle part develops static option relationships, the Black–Scholes framework, the associated pricing PDE, and the concept of dynamic hedging. The final part turns to computational methods, including Monte Carlo simulation, discretization of stochastic differential equations, stochastic-volatility models, calibration techniques, and variance-reduction methods.

The goal is not only to derive formulas, but also to understand when numerical methods are reliable, where model risk enters, and how computational choices influence financial conclusions.

How to Use These Notes

The notes are designed for use in lectures, computer labs, and independent study. Mathematical derivations are paired with Python implementations so that theoretical ideas can be explored computationally.

For most chapters, a useful workflow is to first read the mathematical discussion and understand the modelling framework. The accompanying code can then be used to experiment with simulations, explore approximation errors, examine calibration choices, and study model behaviour in practice.

Learning Objectives

By the end of the course, students should be able to:

- understand the key stochastic-process concepts and no-arbitrage principles used in modern financial modelling;
- implement core numerical methods in Python, including Monte Carlo simulation and PDE-based pricing techniques;
- model and calibrate derivative products, with particular attention to volatility modelling and implied-volatility surfaces;
- critically assess numerical output, including statistical error, discretization error, and model limitations;
- communicate technical findings clearly and develop practical computational skills through collaborative lab work.

Prerequisites

Recommended prior knowledge includes:

- basic programming skills, preferably in Python;
- foundational mathematics, particularly calculus and probability theory.

Prior exposure to basic finance is helpful, but the essential financial concepts required for the course are introduced and developed within the notes.

Materials

The main course material consists of the lecture notes collected in this document. The lectures are accompanied by code examples, notebook-based demonstrations, and additional computational material used during lectures and lab sessions.

Teaching Format

The course combines several complementary formats:

- lectures introducing the theoretical foundations;
- computer lab sessions focused on practical implementation;
- practitioner sessions with industry experts discussing real-world applications.

Part I

Lectures

1 Statistical Foundations

Objectives

By the end of this lecture, you should be able to:

- Work with financial time series data and compute log-returns at different time scales
- Understand the relationship between:
 - observed discrete-time market data
 - continuous-time stochastic process models
- Empirically verify the main **stylized facts of financial returns** as documented by Cont (2001).

1.1 Overview

Modern computational finance rests on the interaction between **data**, **probabilistic models**, and **numerical methods**. This lecture begins that journey at the empirical foundation: what do real financial markets actually look like?

Before introducing stochastic models for asset prices and volatility, it is instructive to understand the *empirical properties* that real financial markets consistently exhibit. These properties are known as **stylized facts**, and they appear across asset classes, time periods, and sampling frequencies. Understanding them is the first step toward building models that are both mathematically tractable and financially realistic.

The empirical findings here will directly motivate the modeling choices made in the following lectures. The non-Gaussianity of returns motivates the use of stochastic volatility; volatility clustering motivates mean-reverting variance processes such as the Heston model; and the leverage effect shapes the sign of the spot-variance correlation. These connections will be revisited explicitly as the course progresses.

At the same time, we emphasize an important distinction that runs throughout the course: stylized facts characterize historical returns under the **physical measure** $\mathbb{P}$, whereas derivative pricing is primarily concerned with dynamics under the **risk-neutral measure** $\mathbb{Q}$. As we will see in the next lecture, matching option-implied information is often more important for pricing than reproducing every feature of historical time-series data. The two measures are linked via Girsanov's theorem, and keeping them conceptually separate is essential for the rest of the course.

1.2 Financial Markets

In practice, we begin with observed financial data representing the prices (or quotes) of assets such as equities, bonds, foreign exchange rates, commodities, cryptocurrencies, and other traded instruments. These prices are typically obtained from market data providers or brokerage platforms.

Observations are available only at discrete points in time, yielding a time series of asset prices:

$$S_{t_0}, S_{t_1}, \dots, S_{t_N}, \quad (1.1)$$

where the sampling frequency may range from nanoseconds (high-frequency trading) to minutes, days, or even years.

When modeling asset prices or volatilities, we postulate an underlying **stochastic process**

$$(S_t)_{t \in [0, T]}, \quad (1.2)$$

defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

What we see in the market is usually only one **single realized path** of this process observed at discrete time points $t_0, t_1, \dots, t_N$:

$$S_{t_0}(\omega), S_{t_1}(\omega), \dots, S_{t_N}(\omega), \quad \text{for some } \omega \in \Omega. \quad (1.3)$$

More precisely, in financial modeling we consider a continuous-time financial market over a fixed time horizon $[0, T]$ with $d \in \mathbb{N}$ traded assets.

A **financial market** is described by a filtered probability space together with an asset price process,

$$(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P}, \mathbf{S}), \quad (1.4)$$

where:

- Ω represents the set of all **outcomes** $\omega \in \Omega$,
- $\mathcal{F}$ is a σ -algebra on Ω containing all **events** $A \in \mathcal{F}$,
- $\mathbb{F} = (\mathcal{F}_t)_{t \in [0, T]}$ is a filtration, i.e. $\mathcal{F}_t \subseteq \mathcal{F}$ for $t \in [0, T]$ being a sub- σ -algebra such that $\mathcal{F}_s \subseteq \mathcal{F}_t$ for all $0 \leq s \leq t \leq T$, and describes the flow of information in the market,
- $\mathbb{P}$ denotes the **physical probability measure** defined on the measurable space $(\Omega, \mathcal{F})$,
- $\mathbf{S} = (\mathbf{S}_t)_{t \in [0, T]}$ with $\mathbf{S}_t = (S_t^{(1)}, \dots, S_t^{(d)})$ denotes an $\mathbb{R}^d$ -valued stochastic process on $[0, T]$, describing the time-evolution of some $d \in \mathbb{N}$ asset prices.

1.2.1 The filtered probability space $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$

Below, we provide explanations and intuitive interpretations of the abstract definition of a financial market.

Outcomes and Random Variables

- The **set of all outcomes**, denoted by Ω , represents all possible scenarios from which random outcomes can be sampled.
- Drawing samples $\omega_1, \dots, \omega_N$ from Ω according to the probability measure $\mathbb{P}$ allows us to simulate real-world randomness.
- A **random variable** defined on the measurable space $(\Omega, \mathcal{F})$ is a mapping $X: \Omega \rightarrow \mathbb{R}$ such that $X^{-1}(A) \in \mathcal{F}$ for all Borel sets $A \subseteq \mathbb{R}$, i.e. a measurable function on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$.
- The distribution (law) of X is given by $\mathbb{P} \circ X^{-1}$.
- In practice, to simulate the outcomes of the random variable X , we evaluate $X(\omega_1), \dots, X(\omega_N)$. These realizations are typically denoted as samples $x^{(1)}, \dots, x^{(N)}$.

Filtration and the Flow of Information

When modeling time-dependent random phenomena, we must respect the natural chronological ordering implied by the flow of time. Information obtained at an earlier time remains available at later times.

- This structure is captured by the concept of a **filtration**:

$$\mathbb{F} = (\mathcal{F}_t)_{t \in [0, T]}. \quad (1.5)$$

Typically, we assume:

- **All information of today is known to us:** The initial sigma-algebra $\mathcal{F}_0$ is $\mathbb{P}$ -trivial, meaning:
 - $\mathbb{P}(A) \in \{0, 1\}$ for all $A \in \mathcal{F}_0$.
 - All $\mathcal{F}_0$ -measurable random variables are constant almost surely.
- **Terminal information:** The final sigma-algebra contains all events within the time horizon: $\mathcal{F} = \mathcal{F}_T$.

Moreover, a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ is said to satisfy the **usual conditions** if:

- The initial sigma-algebra $\mathcal{F}_0$ contains all the $\mathbb{P}$ -null sets from $\mathcal{F}$.
- The filtration is **right-continuous**:

$$\mathcal{F}_t = \bigcap_{u > t} \mathcal{F}_u \quad \text{for all } t \in [0, T]. \quad (1.6)$$

Remark: In practice, a filtration can be thought of as the information stream of the market: trading signals, news, quotes, and any other data available up to the current time. It is the formal object that prevents a model or strategy from using future information when making decisions.

Stochastic Processes and Adaptedness

The object that we will mostly work with in any financial market model is the **asset price process**.

- An $\mathbb{R}^d$ -valued **stochastic process** $(S_t)_{t \in [0, T]}$ is a family of random variables defined on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$:

$$\mathbf{S} = (\mathbf{S}_t)_{t \in [0, T]} \quad \text{with } \mathbf{S}_t : \Omega \rightarrow \mathbb{R}^d \text{ measurable for each } t \in [0, T]. \quad (1.7)$$

To align the stochastic process with the notion of information flow, we introduce the concept of **adaptedness**:

- A stochastic process $\mathbf{S}$ is said to be **adapted** to the filtration $\mathbb{F}$ if each random vector $\mathbf{S}_t$ is $\mathcal{F}_t$ -measurable for all $t \in [0, T]$. Intuitively, this means the value of the process at time t depends only on information known up to that time.
- We call $\omega \mapsto S_t(\omega)$ a **sample at time t** and $t \mapsto S_t(\omega)$ a **sample path** of $(S_t)_{t \in [0, T]}$ (see below for a visualisation of these important concepts).

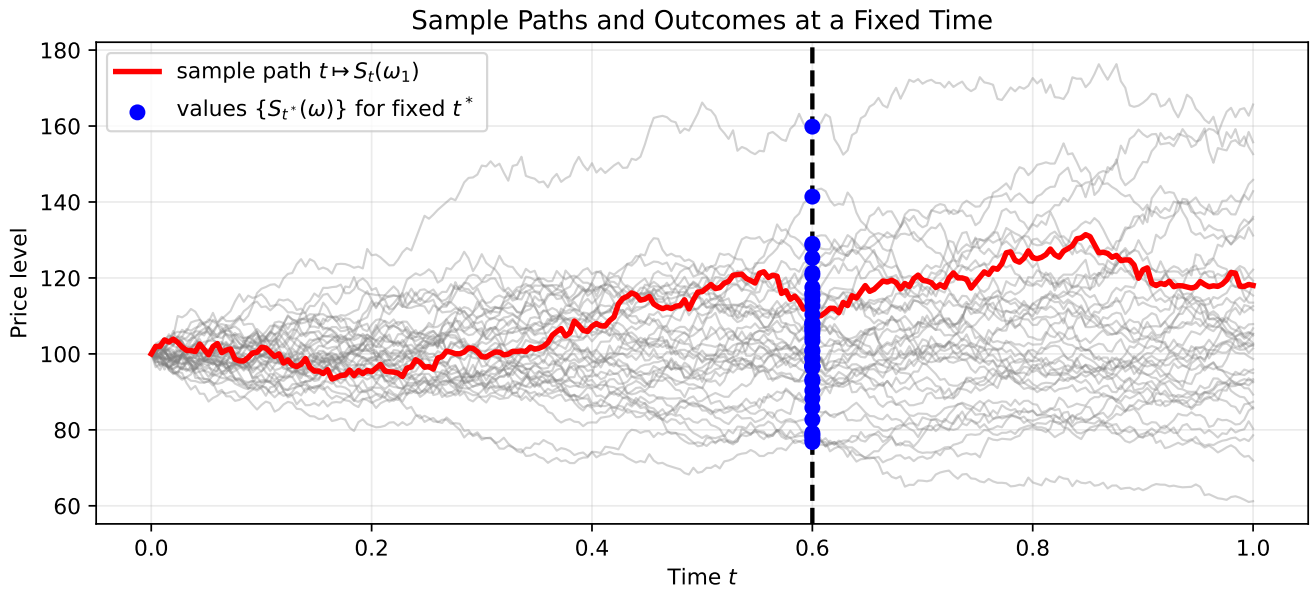


Figure 1.1: Sample paths and cross-sectional outcomes of a stochastic process

1.2.2 Stochastic Models of Financial Markets

Modeling financial markets amounts to specifying the probabilistic structure

$$(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P}, \mathbf{S}), \quad (1.8)$$

which in practice reduces to formulating a stochastic law governing the evolution of asset prices.

Such models may be constructed either in **discrete time**, or in **continuous time**. We will study both approaches in the next lecture on stochastic modelling. Continuous-time models are most commonly formulated in terms of **stochastic differential equations (SDEs)**. A fundamental class is given by *Itô diffusions*, for example:

$$dX_t = b_t dt + \sigma_t dW_t, \quad (1.9)$$

where b_t denotes the drift process, σ_t the (possibly time-varying) volatility process, and $(W_t)_{t \geq 0}$ a standard Brownian motion adapted to the filtration $\mathbb{F}$. In the following, we give some intuition for such stochastic differential equation models and introduce the main empirical regularities they are meant to capture.

1.3 Empirical Workflow

We will proceed as follows:

1. Load historical market data for a liquid equity-index proxy
2. Compute log-returns and volatility proxies
3. Visualize statistical properties
4. Identify stylized facts
5. Interpret consequences for financial modeling

i Note

To keep the lecture notes readable, the rendered chapter suppresses most code by default. The full implementation remains in the source file and can still be used in labs or for independent experimentation.

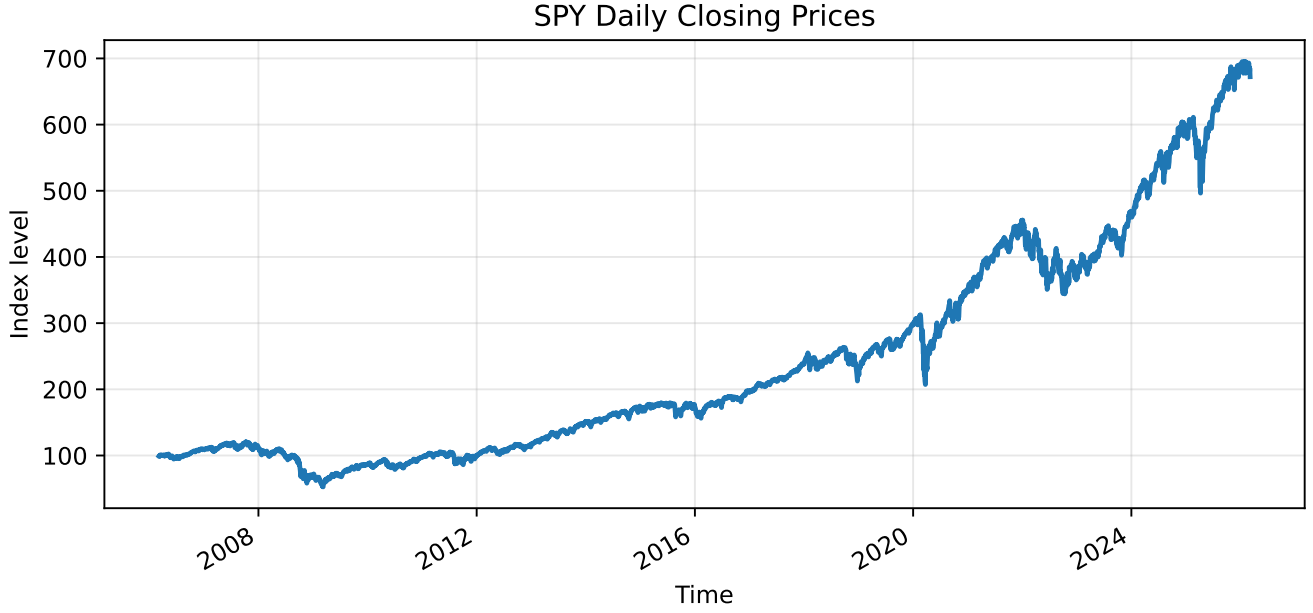


Figure 1.2: SPY daily closing prices over the sample period

1.3.1 Log-Prices and Returns

We introduce the **log-prices**:

$$X_{t_k} = \log S_{t_k}, \quad k \geq 0 \quad (1.10)$$

and define the (discrete-time) **log-returns**

$$r_k = X_{t_k} - X_{t_{k-1}} = \log S_{t_k} - \log S_{t_{k-1}}. \quad (1.11)$$

Log-returns are preferred because they are:

- additive over time, i.e. for any time indices $t_0 < t_1 < t_2$,

$$\log \frac{S_{t_2}}{S_{t_0}} = \log \frac{S_{t_2}}{S_{t_1}} + \log \frac{S_{t_1}}{S_{t_0}}. \quad (1.12)$$

- scale-invariant: If prices are expressed in a different currency or units $\tilde{S}_t = cS_t$ (e.g. dollars vs euros), prices are multiplied by a constant factor $c > 0$, then: $\log \tilde{S}_{t_k} - \log \tilde{S}_{t_{k-1}} = \log S_{t_k} - \log S_{t_{k-1}}$ for $k \geq 1$.

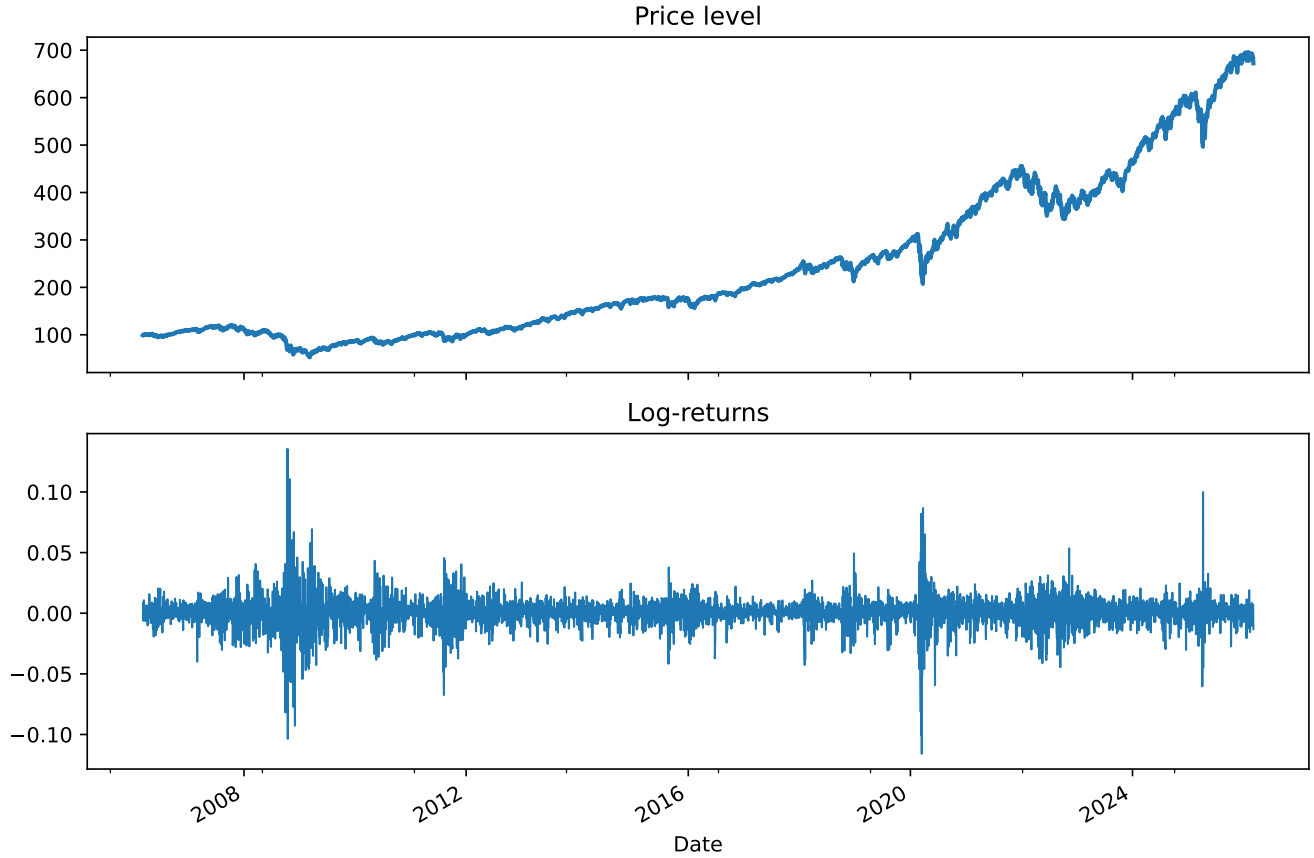


Figure 1.3: Price levels and daily log-returns of SPY

1.4 Stylized Facts of Financial Returns

The following sections document the key empirical regularities that appear consistently across asset classes and time periods. Each stylized fact carries direct modeling implications, which are noted alongside the empirical evidence.

1.4.1 Absence of Linear Autocorrelation in Returns

Empirical studies across asset classes consistently show that **linear autocorrelations of asset returns are close to zero** at most time scales.

Formally, for daily (and lower-frequency) returns:

$$\text{Corr}(r_t, r_{t+k}) \approx 0, \quad k \geq 1. \quad (1.13)$$

This indicates that past returns contain little to no linear predictive power for future returns.

Remark: At very small intraday horizons (on the order of minutes), weak autocorrelations may appear due to:

- bid–ask bounce
- order processing delays
- market microstructure frictions

These effects vanish at daily and longer time scales.

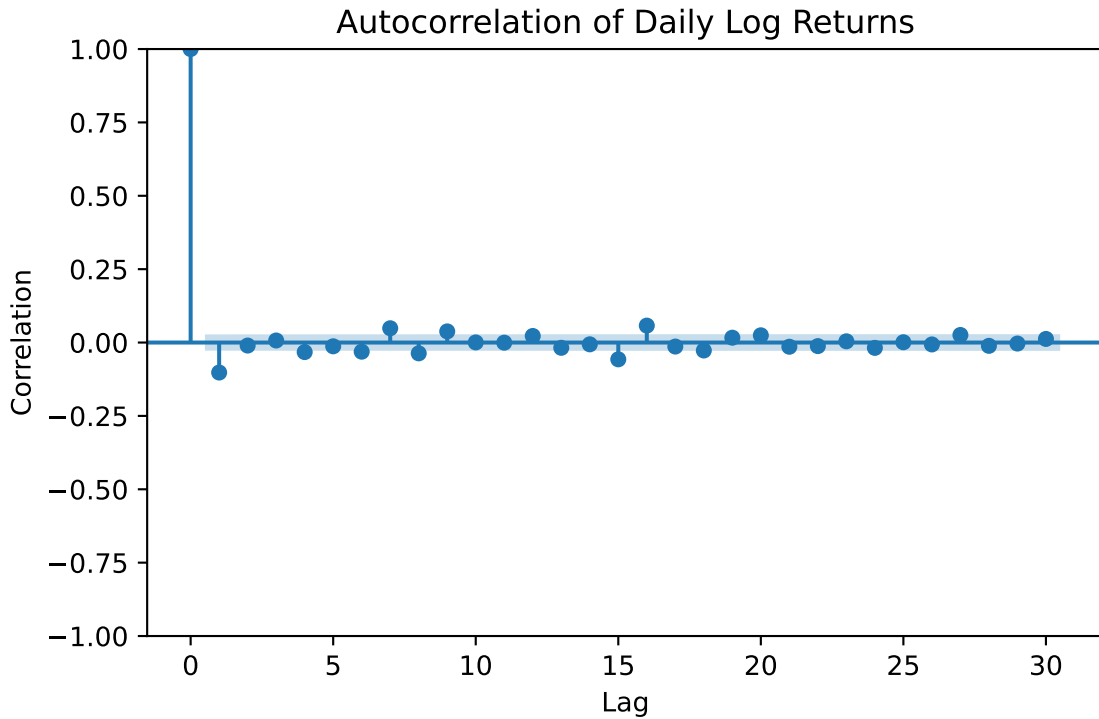


Figure 1.4: Autocorrelation of daily log-returns

Modeling Implication

The absence of linear autocorrelation motivates modeling price dynamics with **martingale-difference-type innovations after removing the predictable drift**.

Consider a continuous-time log-price process following an Itô diffusion:

$$dX_t = \mu_t dt + \sigma_t dW_t. \quad (1.14)$$

The discrete-time return over the interval $[t_{i-1}, t_i]$ can be decomposed as:

$$r_i = X_{t_i} - X_{t_{i-1}} = \int_{t_{i-1}}^{t_i} \mu_s ds + \underbrace{\int_{t_{i-1}}^{t_i} \sigma_s dW_s}_{\text{innovation } \epsilon_i}. \quad (1.15)$$

By the martingale property of the Itô integral (see Section 3.2 of the lecture on stochastic calculus), the innovation term ϵ_i satisfies:

$$\mathbb{E}[\epsilon_i \mid \mathcal{F}_{t_{i-1}}] = 0, \quad (1.16)$$

implying that deviations from the drift term behave as unpredictable shocks rather than systematic signals. This is consistent with the empirical absence of linear return autocorrelation.

Remark: *Uncorrelated does not imply independent.* While raw returns exhibit little linear dependence, strong nonlinear dependence remains present in quantities such as absolute returns $|r_i|$ and squared returns r_i^2 (as we will see in the following stylized facts).

1.4.2 Heavy-Tailed Return Distributions

Empirical studies across financial markets show that the unconditional distribution of asset returns exhibits **heavy tails**, often approximated in the tails by Pareto-type decay.

Specifically, tail probabilities behave approximately as

$$\mathbb{P}(|r_t| > x) \sim x^{-\alpha}, \quad (1.17)$$

with a tail index typically satisfying

$$2 < \alpha < 5. \quad (1.18)$$

This implies far more extreme events than predicted by Gaussian models.

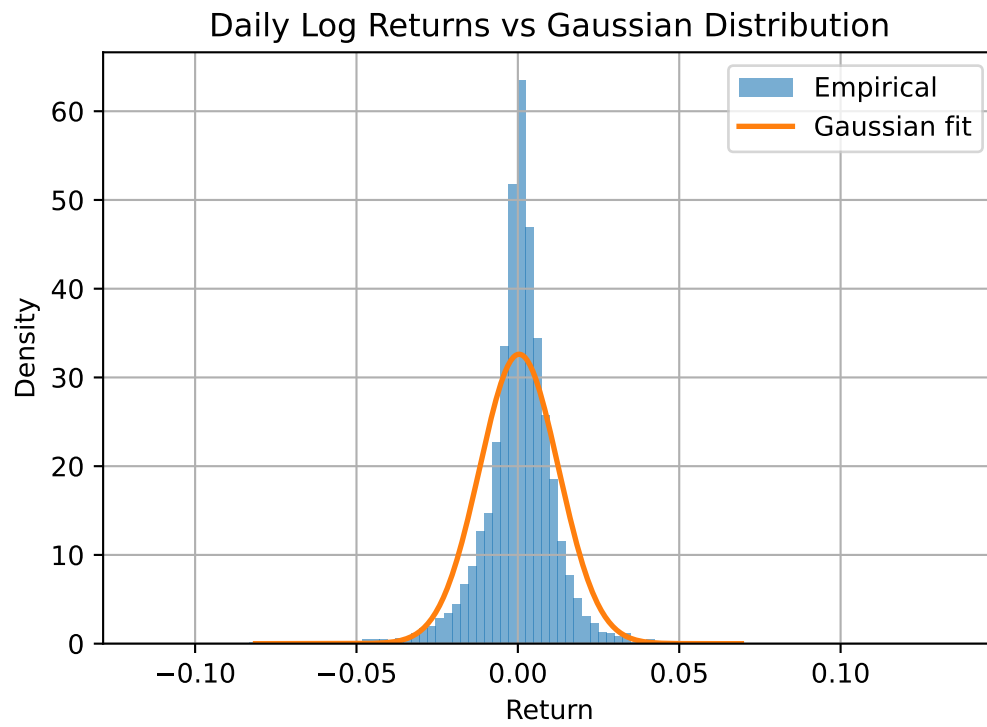


Figure 1.5: Daily log-returns with a Gaussian benchmark

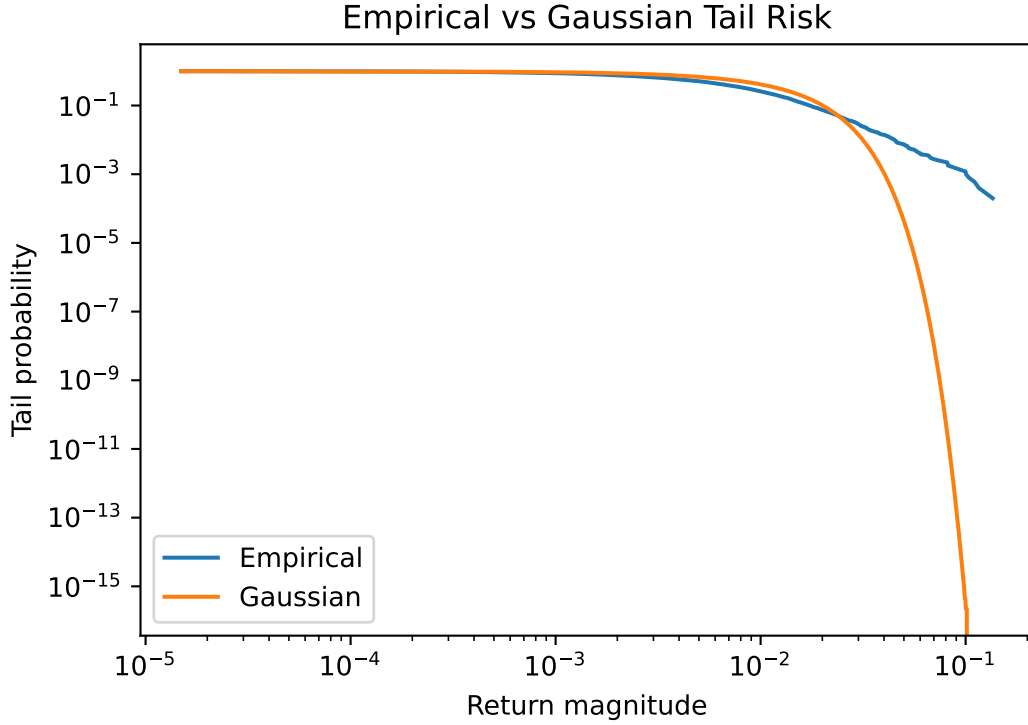


Figure 1.6: Empirical and Gaussian tail decay of daily returns

Modeling Implication

If asset returns were Gaussian, that is,

$$r_t \sim \mathcal{N}(\mu, \sigma^2), \quad (1.19)$$

then extreme events would be exceedingly rare. In particular, the tail probabilities would decay exponentially fast:

$$\mathbb{P}(|r_t| > x) \sim \exp\left(-\frac{x^2}{2\sigma^2}\right), \quad \text{as } x \rightarrow \infty. \quad (1.20)$$

Empirical return distributions, however, exhibit **much slower tail decay**, implying a far higher frequency of extreme price movements than predicted by Gaussian models. This provides strong motivation for employing **non-Gaussian return dynamics**, such as heavy-tailed innovations, jump processes, or stochastic volatility models.

1.4.3 Remark on Probability Measures and Observed Data

Throughout, the probability symbol $\mathbb{P}$ refers to the (physical) probability measure of the financial market model. In practice, however, this measure itself is never directly observed.

What we observe are realizations of returns, that is, samples drawn from the induced distribution

$$\mathbb{P} \circ r_t^{-1}, \tag{1.21}$$

the *law of the return process*. Empirical analysis therefore concerns estimating distributional features, such as tail probabilities $\mathbb{P}(|r_t| > x)$, from observed data rather than accessing the underlying probability measure explicitly.

1.4.4 Gain/Loss Asymmetry

Empirical financial markets exhibit a pronounced **asymmetry** between large negative and positive price movements.

In particular, equity prices and stock indices tend to experience:

- sudden and large drawdowns (crashes)
- but comparatively fewer extreme upward movements of similar magnitude

This implies that the distribution of returns is not symmetric in the tails, with downside extremes occurring more frequently than upside extremes. In practice this typically shows up as negative sample skewness and more pronounced downside quantiles than upside quantiles of the same nominal probability level.

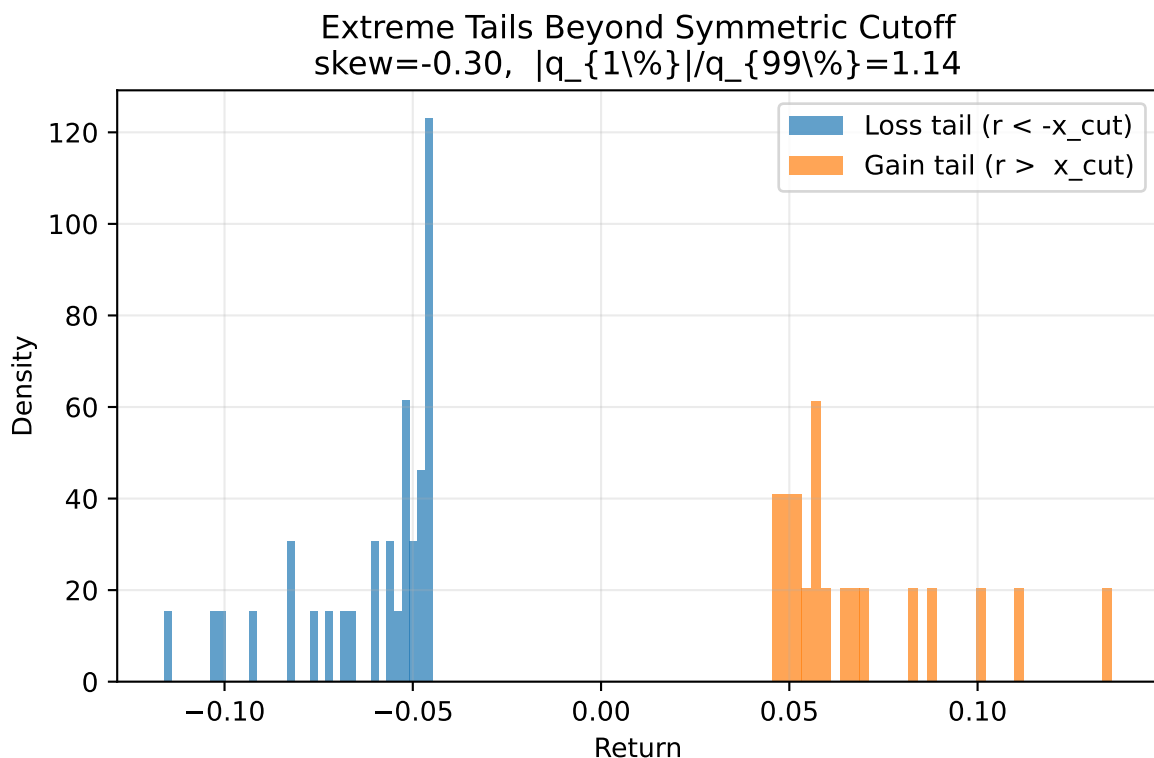


Figure 1.7: Downside versus upside tail asymmetry

1.4.5 Aggregational Gaussianity and Time Scaling

Empirical return distributions depend strongly on the time scale of observation.

At short horizons (daily or intraday):

- returns are strongly non-Gaussian
- heavy tails dominate

As the time horizon increases:

- distributions become progressively more bell-shaped
- tails become less pronounced

This phenomenon is known as **aggregational Gaussianity**.

Modeling Consequences of Aggregational Gaussianity

Let $X_t = \log S_t$ denote the log-price process and define returns over horizon Δt :

$$r_t(\Delta t) = X_{t+\Delta t} - X_t. \quad (1.22)$$

In the simplest benchmark with constant drift and volatility, classical diffusion models imply:

$$r_t(\Delta t) \sim \mathcal{N}(\mu\Delta t, \sigma^2\Delta t), \quad (1.23)$$

and hence Gaussianity at all scales. After centering the returns, or over short horizons where the drift contribution is negligible, this reduces to the familiar variance-scaling relation.

Empirical markets contradict this at short horizons but partially recover Gaussian behavior at longer scales.

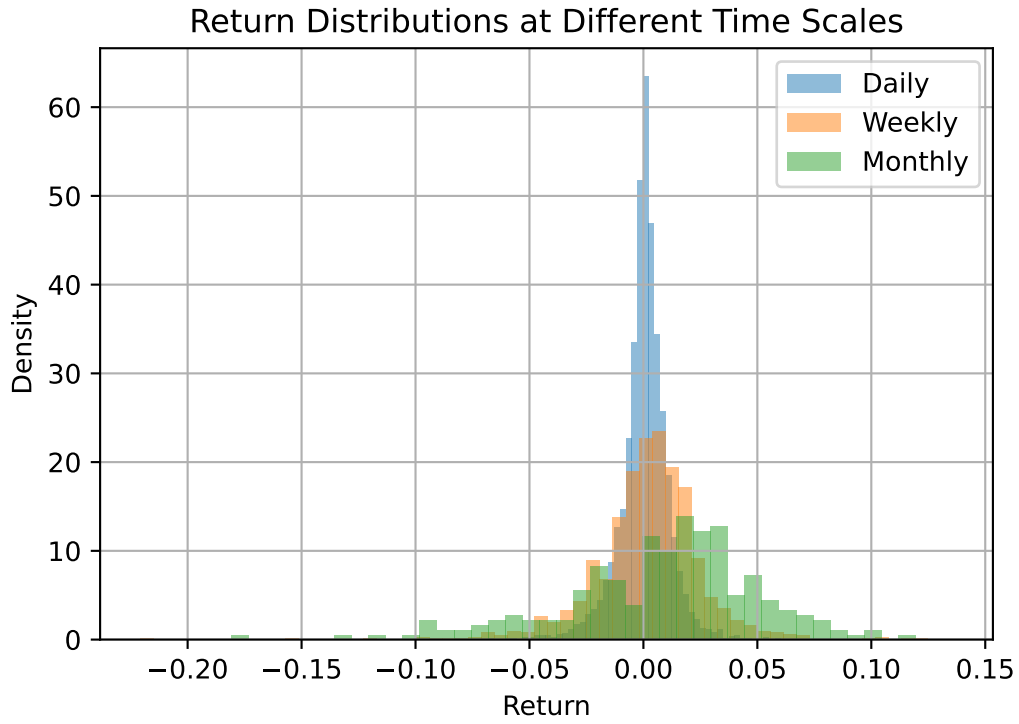


Figure 1.8: Return distributions across daily, weekly, and monthly horizons

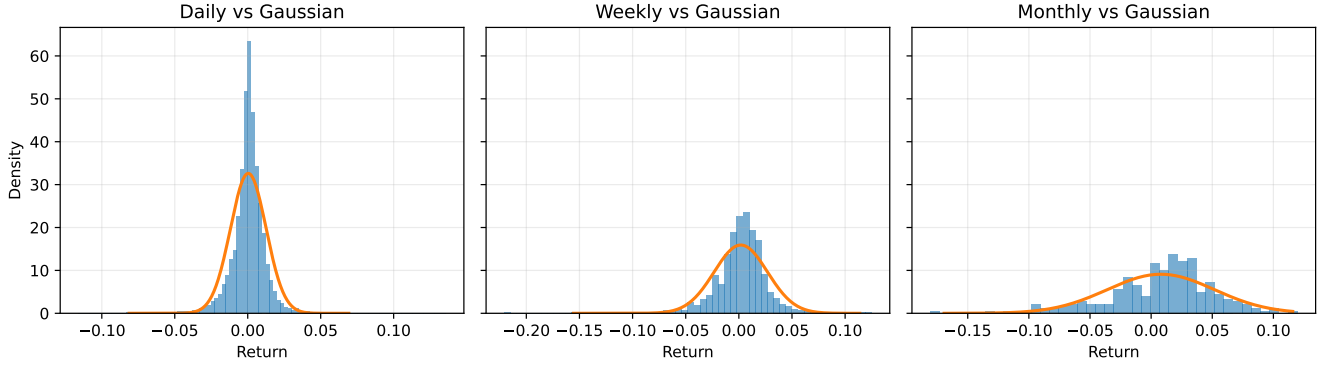


Figure 1.9: Gaussian benchmarks across sampling frequencies

1.4.6 Intermittency

Intermittency refers to the empirical observation that market fluctuations are not “smoothly random” over time. Instead, financial returns exhibit **irregular bursts of activity**: periods of calm are interrupted by short episodes of intense volatility.

A practical way to quantify intermittency is to examine the time series of volatility estimators, for example:

- rolling standard deviation of returns
- exponentially weighted volatility
- realized volatility proxies computed from aggregated returns

Quadratic Variation and Integrated Volatility

A fundamental quantity describing the cumulative variability of the log-price process over time is its **quadratic variation**, defined as

$$[X]_T = \lim_{|\Pi| \rightarrow 0} \sum_i (X_{t_i} - X_{t_{i-1}})^2, \quad (1.24)$$

where $\Pi = \{0 = t_0 < t_1 < \dots < t_n = T\}$ denotes a partition of the interval $[0, T]$, and $|\Pi|$ its mesh size.

For diffusion-type processes, this limit exists and takes the explicit form

$$[X]_T = \int_0^T \sigma_t^2 dt. \quad (1.25)$$

Quadratic variation therefore measures the **integrated volatility** accumulated over time. The process σ_t itself represents the **instantaneous (spot) volatility**, governing the local intensity of price fluctuations at each point in time.

Thus:

- historical or realized volatility captures **time-averaged volatility** over an interval,
- while σ_t describes the **latent, continuously evolving volatility state** of the market.

Intermittency and volatility clustering observed in empirical data correspond to strong temporal variation in this spot volatility process.

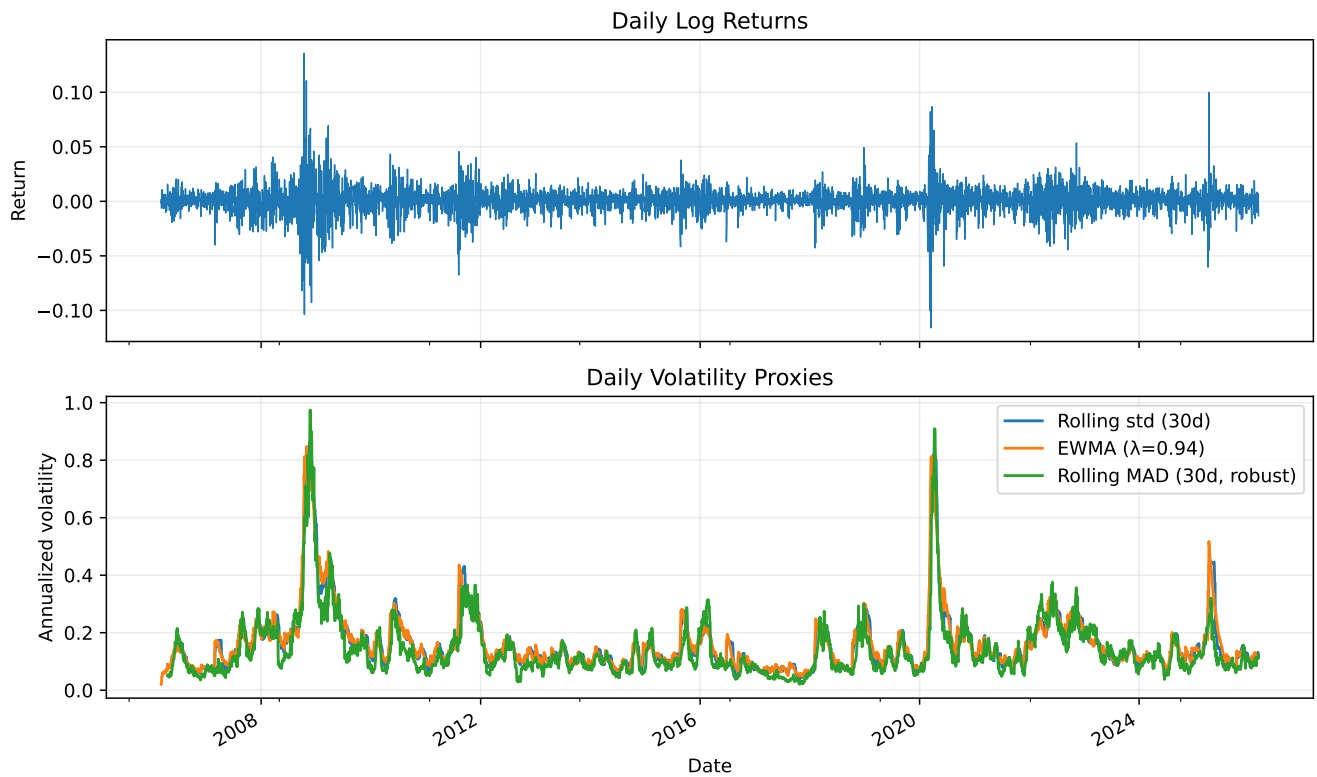


Figure 1.10: Intermittency in daily returns and volatility proxies

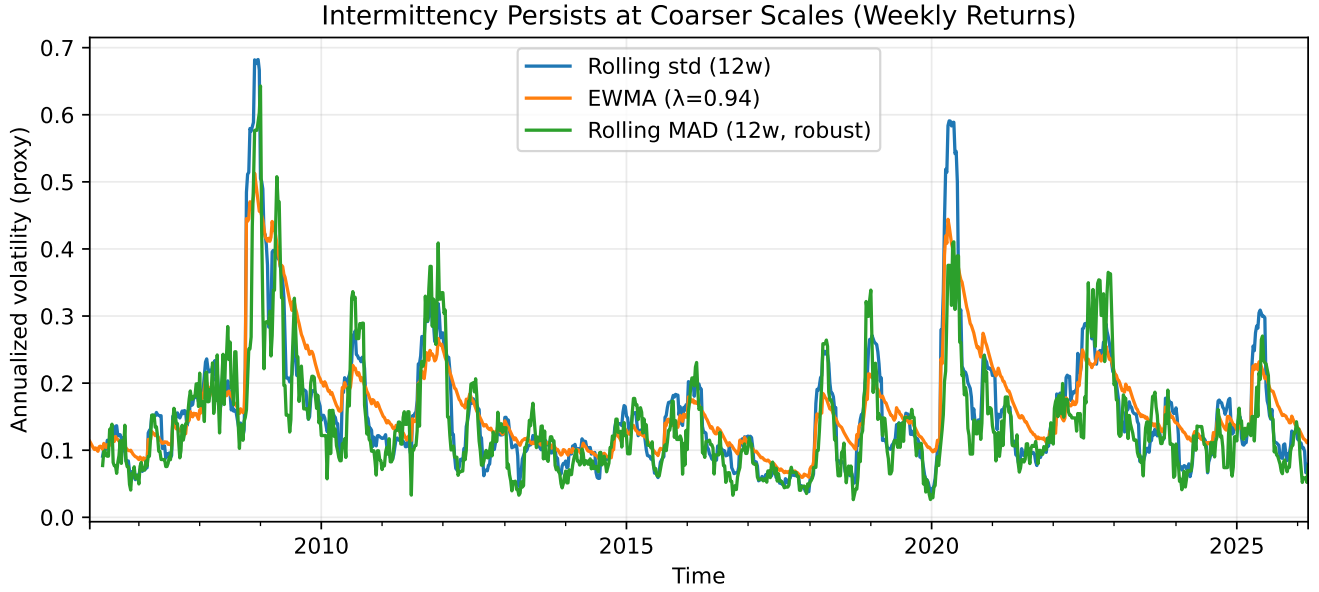


Figure 1.11: Intermittency persists at the weekly scale

1.4.7 Volatility Clustering and Long Memory

Empirical studies show that various measures of volatility exhibit strong positive autocorrelation over time, often persisting across many days or even longer horizons.

While raw returns display little linear autocorrelation, their magnitudes reveal pronounced temporal dependence.

Formally, one typically observes:

$$\text{Corr}(|r_t|, |r_{t+k}|) > 0 \quad \text{for many lags } k, \quad (1.26)$$

and similarly for squared returns r_t^2 .

Moreover, this dependence decays slowly with the lag k , often approximately as a power law. This phenomenon is sometimes interpreted as **long-range dependence**.

Volatility Clustering

This empirical behavior is commonly summarized as:

periods of high volatility tend to be followed by high volatility,
periods of low volatility tend to persist.

Volatility therefore appears in **clusters** rather than being randomly scattered through time.

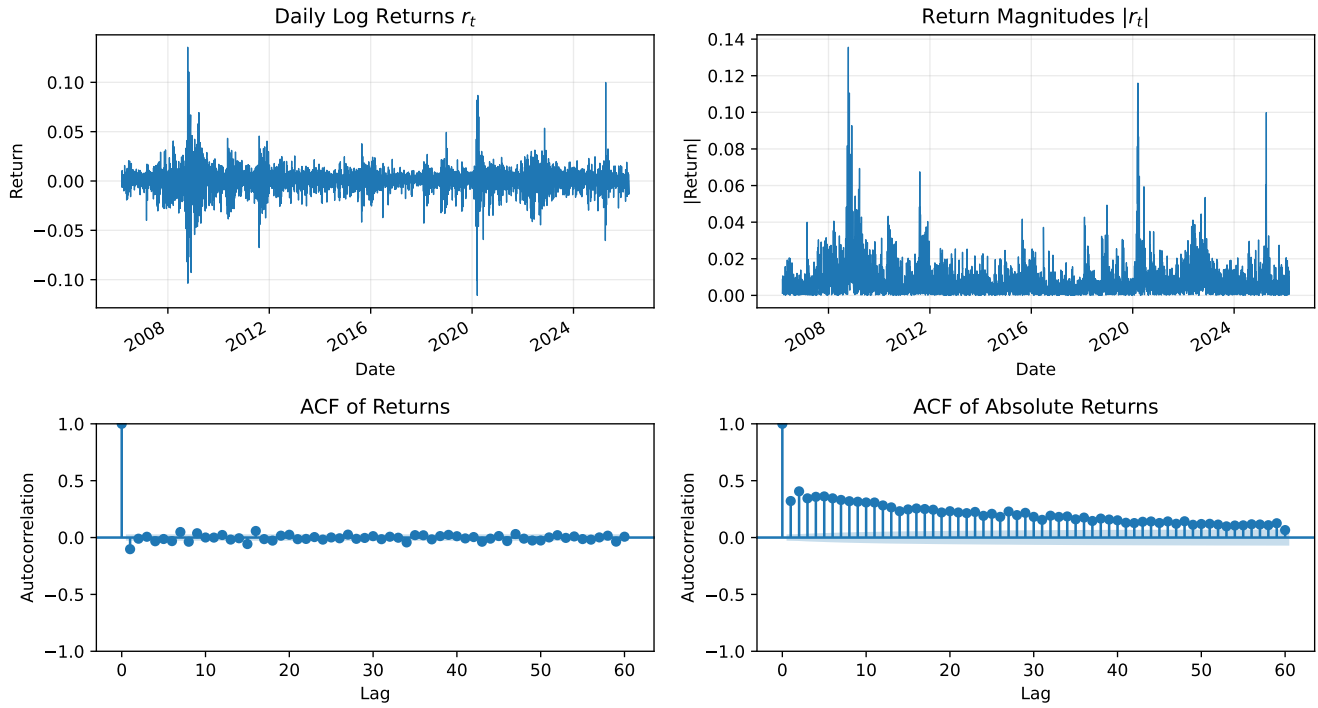


Figure 1.12: Volatility clustering in returns and autocorrelations

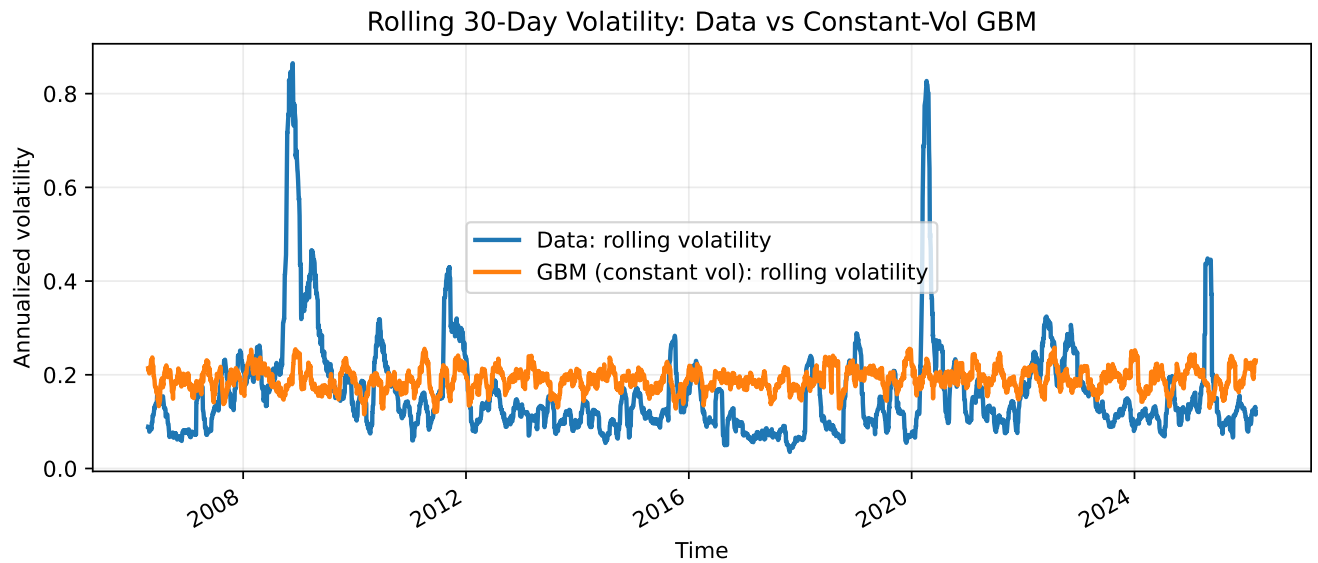


Figure 1.13: Rolling volatility in data versus constant-volatility GBM

Interpretation in Stochastic Process Terms

In continuous-time financial models, the log-price process X_t is typically represented as a semimartingale, for example,

$$dX_t = b_t dt + \sigma_t dW_t, \quad (1.27)$$

where σ_t denotes the (possibly stochastic) spot volatility process.

For discrete-time returns $r_i = X_{t_i} - X_{t_{i-1}}$, the conditional variance satisfies

$$\text{Var}(r_i \mid \mathcal{F}_{t_{i-1}}) = \mathbb{E} \left[\int_{t_{i-1}}^{t_i} \sigma_s^2 ds \mid \mathcal{F}_{t_{i-1}} \right]. \quad (1.28)$$

Thus, fluctuations in return magnitudes are driven by the evolution of the integrated volatility over each sampling interval. Persistent dynamics in σ_t naturally generate volatility clustering and long-memory effects observed in empirical data.

1.4.8 Conditional Heavy Tails

Volatility clustering explains a large fraction of the time-variation in return magnitudes. A standard correction is to model returns as

$$r_t = \mu + \sigma_t \varepsilon_t, \quad (1.29)$$

where σ_t is a time-varying volatility process (for example estimated by EWMA or GARCH) and ε_t are standardized shocks.

If volatility clustering were the only reason for heavy tails, then the standardized residuals

$$z_t = \frac{r_t - \mu}{\hat{\sigma}_t} \quad (1.30)$$

should look approximately Gaussian.

Empirically, however, even after volatility normalization, residuals still exhibit heavy tails, that is, **conditional heavy tails**. Typically, these tails are less heavy than those of the unconditional return distribution.

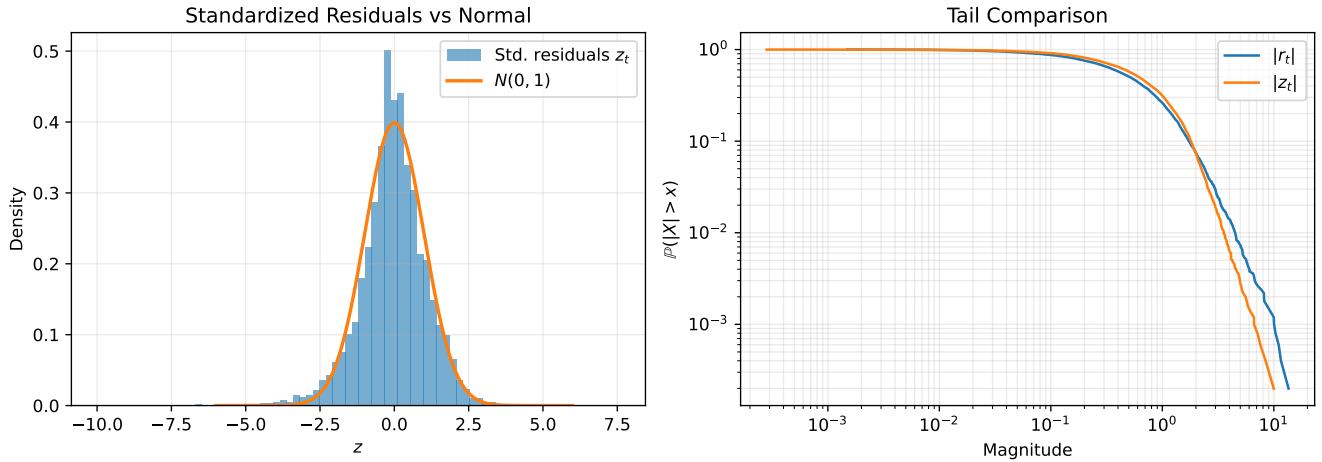


Figure 1.14: Conditional heavy tails before and after volatility standardization

1.5 Further Empirical Regularities

The previous sections contain the core stylized facts needed for the rest of the course. The following diagnostics refine the volatility story and introduce the leverage effect, which will reappear in the Heston model (Lecture 8) where the parameter ρ directly captures the negative correlation between asset returns and variance shocks.

1.5.1 Slow Decay of Autocorrelation in Absolute Returns

A key empirical feature of financial markets is that dependence in volatility proxies persists for long horizons.

While returns themselves have little linear autocorrelation, the autocorrelation of absolute returns decays slowly with lag:

$$\rho_{|r|}(k) := \text{Corr}(|r_t|, |r_{t+k}|) \quad (1.31)$$

Empirically, many studies find an approximate power-law decay:

$$\rho_{|r|}(k) \approx Ck^{-\beta}, \quad \beta \in [0.2, 0.4]. \quad (1.32)$$

Such a slow (hyperbolic) decay is often interpreted as evidence of **long-range dependence** in volatility.

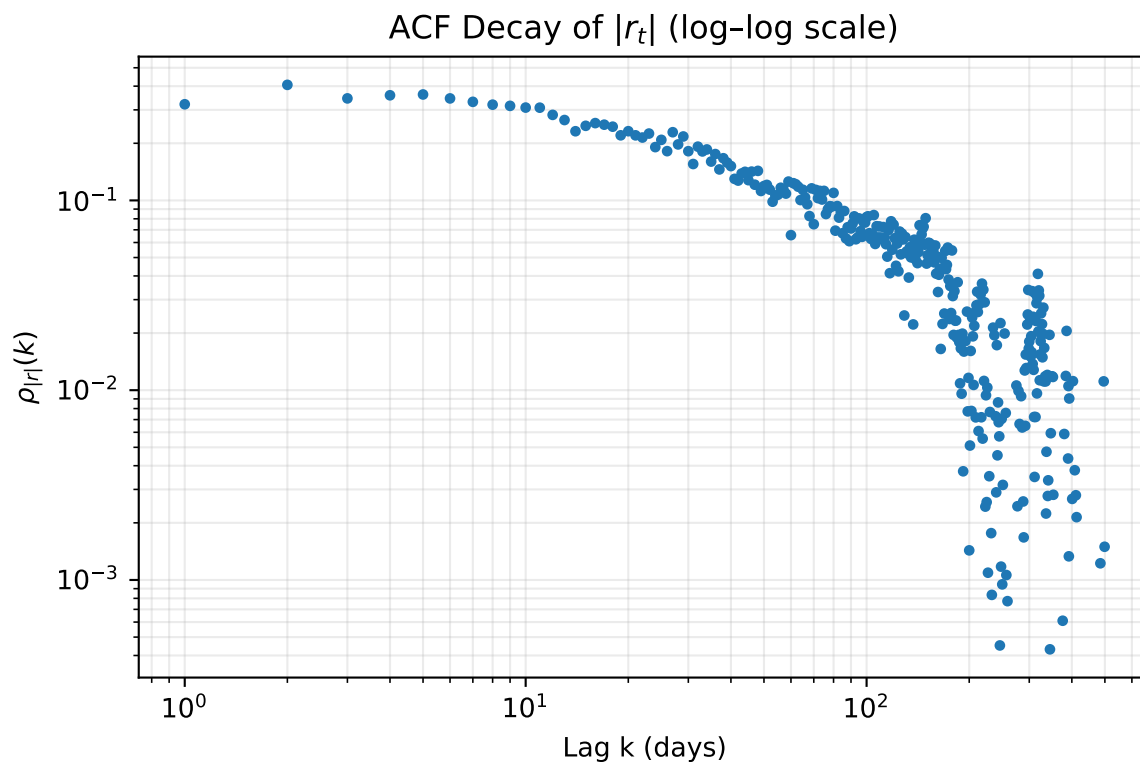


Figure 1.15: Slow decay of autocorrelation in absolute returns

1.5.2 Leverage Effect: Returns and Future Volatility

Empirical financial markets exhibit a systematic asymmetry:

negative returns tend to be followed by higher volatility than positive returns of the same magnitude.

This phenomenon is known as the **leverage effect**.

Formally, one observes:

$$\text{Corr}(r_t, |r_{t+k}|) < 0 \quad \text{for } k > 0. \quad (1.33)$$

That is, downward price movements are associated with increased future volatility.

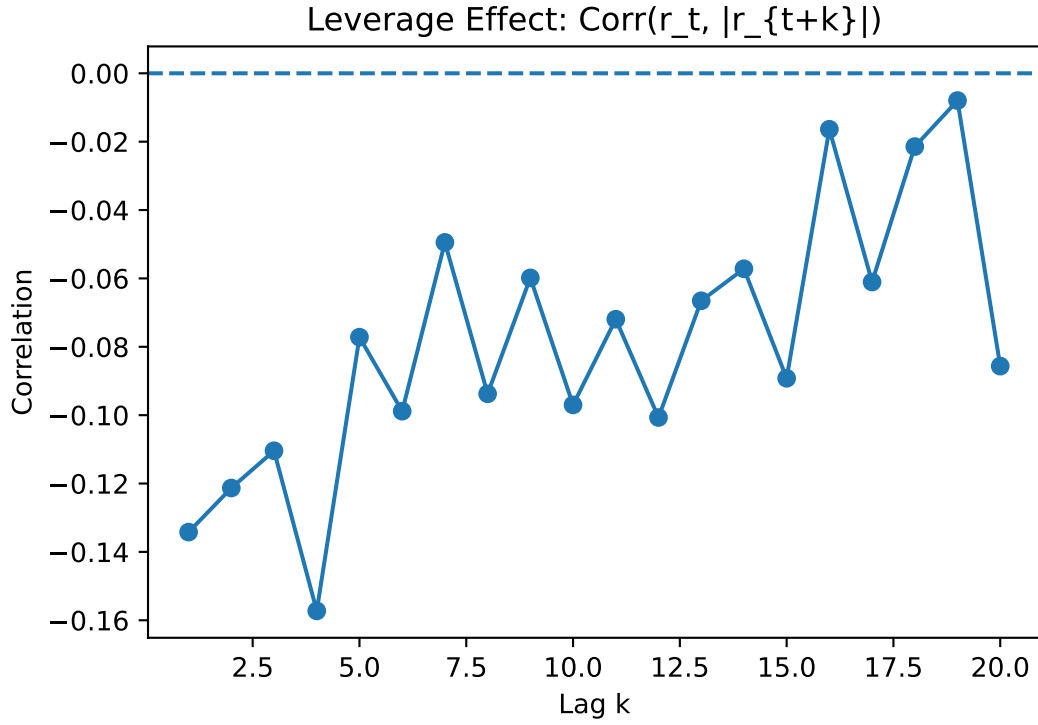


Figure 1.16: Lead correlations between returns and future absolute returns

1.5.3 Volume/Volatility Correlation

Empirically, trading volume is positively correlated with volatility: periods of intense trading activity tend to coincide with periods of large price fluctuations.

A simple way to express this is:

$$\text{Corr}(V_t, \text{VolProxy}_t) > 0, \quad (1.34)$$

where V_t denotes trading volume and VolProxy_t can be a proxy such as $|r_t|$, r_t^2 , or rolling volatility.

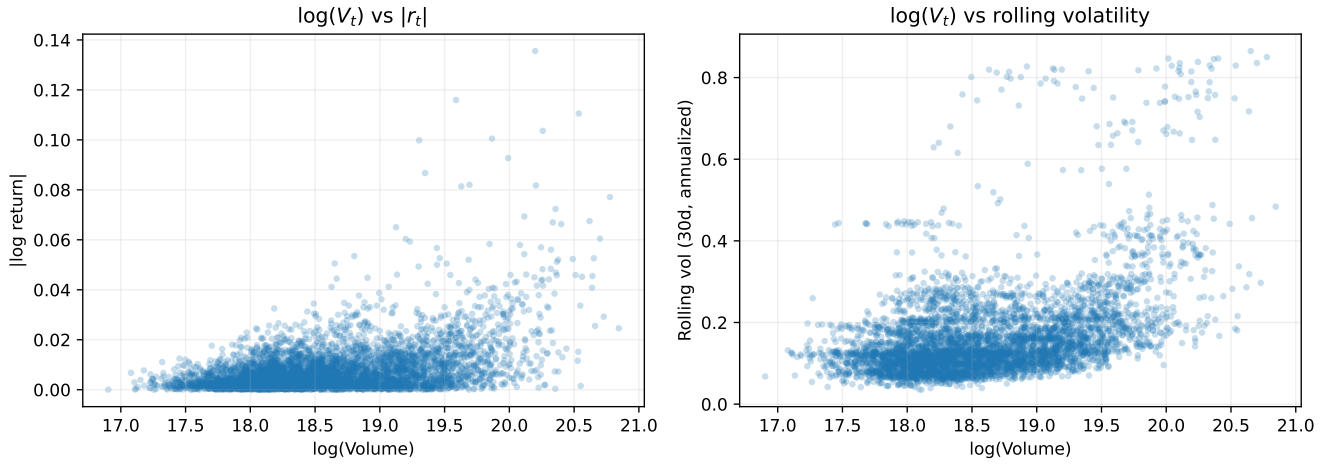


Figure 1.17: Volume-volatility relation

1.5.4 Asymmetry in Time Scales

Empirically, volatility exhibits an asymmetry across time scales:

coarse-grained (smoothed / long-horizon) volatility measures tend to predict fine-scale (short-horizon) volatility better than fine-scale measures predict coarse-grained volatility.

Intuition: - Volatility has persistent components (slow-moving regimes). - A coarse-grained estimator averages out noise and captures the latent regime. - A fine-scale estimator (single-day squared return) is extremely noisy, and therefore less informative about the long-horizon volatility level.

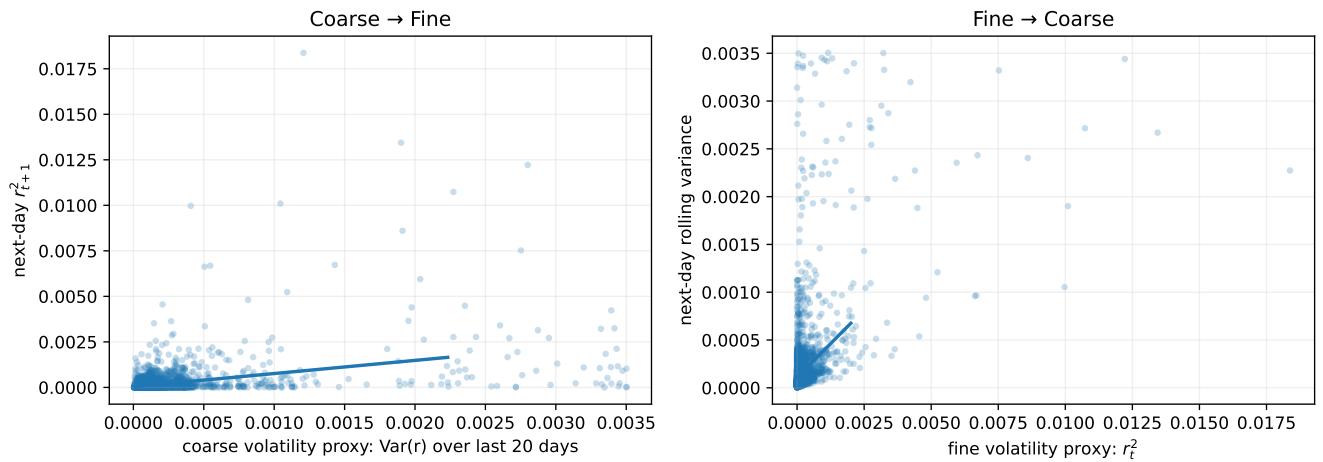


Figure 1.18: Coarse-to-fine versus fine-to-coarse volatility predictability

1.6 Summary and Connection to the Next Lecture

This lecture has introduced the empirical landscape that any realistic financial model must address:

- Returns are approximately uncorrelated but not independent, with nonlinear dependence concentrated in their magnitudes.
- Return distributions have heavy tails and are negatively skewed.
- Volatility is time-varying, clusters in bursts, and tends to rise after negative returns (the leverage effect).

These observations are made under the **physical measure** $\mathbb{P}$, the probability measure that governs observable market data. In the next lecture we introduce the mathematical tools — Itô calculus, the change of measure via Girsanov's theorem, and the risk-neutral measure $\mathbb{Q}$ — that transform these empirical insights into tractable pricing models. The distinction between $\mathbb{P}$ and $\mathbb{Q}$ will remain a central theme throughout the course.

2 Stochastic Modelling of Financial Markets

Objectives

By the end of this lecture, you should be able to:

- Understand financial market models formulated in both **discrete time** and **continuous time**
- Develop intuition for modeling asset prices using **stochastic differential equations (SDEs)**
- Understand **multinomial (tree-based) models** as discrete-time approximations of continuous dynamics
- Derive and interpret the **Black–Scholes model** as a fundamental continuous-time price process
- Apply **Itô’s formula** to functions of stochastic processes and interpret its financial meaning

The previous lecture identified the empirical regularities that any useful asset-price model should capture: fat-tailed returns, volatility clustering, and the leverage effect. We now introduce the stochastic models and continuous-time tools that translate these observations into tractable mathematical structures. The models developed here — geometric Brownian motion, the bank account, and self-financing strategies — are the building blocks for derivative pricing, which begins in Lecture 3.

From this point onward, we use $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ for the filtered probability space, $\mathbb{F} = (\mathcal{F}_t)_{0 \leq t \leq T}$ for the filtration, B_t for the bank account, bold symbols such as $\mathbf{S}_t$ for multi-asset state vectors, and plain symbols such as S_t for one-dimensional processes.

2.1 Market Models in Discrete and Continuous Time

2.1.1 Discrete and Continuous-time Models

Asset prices are modeled as a stochastic process taking values in $\mathbb{R}^d$ over a time index set $\mathbf{T}$, which may be either

- a continuous-time horizon $\mathbf{T} = [0, T]$, or
- a discrete-time grid $\mathbf{T} = \{t_0 < t_1 < \dots < t_N\}$.

We write the risky-asset price process as

$$\mathbf{S} = (\mathbf{S}_t)_{t \in \mathbf{T}}, \quad \mathbf{S}_t = (S_t^{(1)}, \dots, S_t^{(d)}).$$

This stochastic process describes the temporal evolution of $d \in \mathbb{N}$ distinct **risky assets**, commonly referred to as the **underlyings**.

2.1.2 The (locally) Risk-Free Asset

In addition to the risky assets, we introduce a $(d + 1)$ -st asset price process $(S_t^{(0)})_{t \geq 0}$, called the **bank account process** and denoted by $B_t := S_t^{(0)}$. It represents a risk-free (or locally risk-free) cash investment and serves as the numéraire for pricing, discounting, and arbitrage arguments.

2.1.3 Example: Multinomial and Cox–Ross–Rubinstein (CRR) Models

A simple yet flexible class of discrete-time financial market models is given by **multinomial models**, in which asset prices evolve through multiplicative returns.

Let

- $r_1, r_2, \dots, r_N$ denote (possibly random) interest rates over the periods $(0, 1], (1, 2], \dots, (N - 1, N]$,
- $R_1^{(i)}, R_2^{(i)}, \dots, R_N^{(i)}$ denote the simple returns of risky asset i , for $i = 1, \dots, d$.

The bank account evolves according to

$$B_t = B_0 \prod_{j=1}^t (1 + r_j), \quad t = 0, 1, \dots, N,$$

with initial capital $B_0 = 1$ \$.

Each risky asset follows

$$S_t^{(i)} = S_0^{(i)} \prod_{j=1}^t (1 + R_j^{(i)}), \quad t = 0, 1, \dots, N, \quad i = 1, \dots, d,$$

where $S_0^{(i)} > 0$ denotes the initial price and where we assume $r_j > -1$, $R_j^{(i)} > -1$ so that prices remain strictly positive.

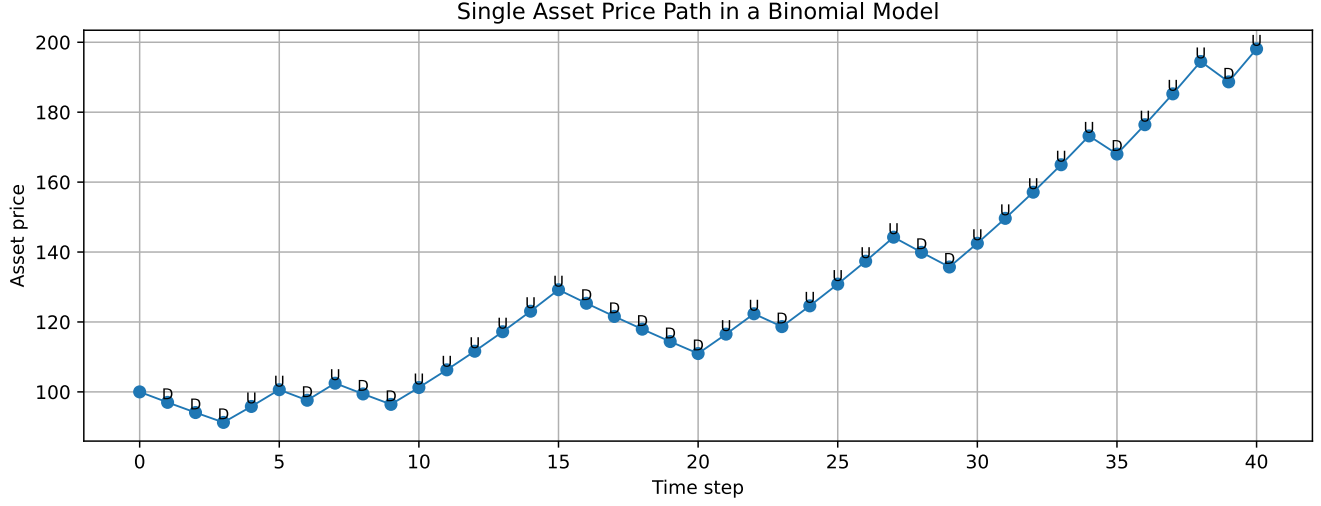
If the returns $R_j^{(i)}$ take only finitely many values (possibly jointly), the model is called a **multinomial model**.

A particularly important special case is:

- constant interest rate $r_j = r$,
- independent returns across time,
- single risky asset $d = 1$,
- each return takes values

$$R_j \in \{U, D\}, \quad \mathbb{P}(R_j = U) = p, \quad \mathbb{P}(R_j = D) = 1 - p.$$

This is the **binomial model**, also known as the **Cox–Ross–Rubinstein (CRR) model**.



2.1.4 Geometric (Log-Return) Representation

A closely related and mathematically convenient formulation uses **log-returns**.

Let $\Delta R_j^{(i)}$ denote the log-return increments and define

$$R_t^{(i)} = \sum_{j=1}^t \Delta R_j^{(i)}.$$

Then asset prices evolve as

$$S_t^{(i)} = S_0^{(i)} \exp(R_t^{(i)}).$$

The corresponding simple return implied by a log-return is

$$\widehat{R}_j^{(i)} = \exp(\Delta R_j^{(i)}) - 1.$$

Hence,

$$1 + \widehat{R}_j^{(i)} = \exp(\Delta R_j^{(i)}), \quad \Delta R_j^{(i)} = \log(1 + \widehat{R}_j^{(i)}).$$

2.2 Fundamentals of Stochastic Calculus in Finance

Multinomial models are useful for intuition, but continuous-time pricing and hedging require stochastic calculus. The key new object is the **Itô integral**, which extends integration to the irregular paths of Brownian motion.

2.2.1 Itô Processes

Financial asset prices are commonly modeled by stochastic processes whose future evolution depends on their current state and time. A general scalar-valued **Itô process** $(S_t)_{t \geq 0}$ satisfies the stochastic differential equation (SDE)

$$dS_t = \mu(t, S_t) dt + \sigma(t, S_t) dW_t, \quad S_0 \in \mathbb{R},$$

where:

- $\mu(t, S_t)$ is the **drift coefficient**, representing the local deterministic trend or expected growth rate.
- $\sigma(t, S_t)$ is the **diffusion (or volatility) coefficient**, controlling the magnitude of random market fluctuations.
- $(W_t)_{t \geq 0}$ is a standard **Brownian motion** (Wiener process) adapted to the market filtration $\mathbb{F}$.

Because Brownian paths are nowhere differentiable, the differential notation dS_t is shorthand. Formally, the SDE means

$$S_t = S_0 + \int_0^t \mu(u, S_u) du + \int_0^t \sigma(u, S_u) dW_u.$$

The first integral is an ordinary Lebesgue integral, accumulating the deterministic drift. The second is an **Itô integral**, capturing the cumulative random shocks.

2.2.2 The Itô Integral and Non-Anticipation

The Itô integral can be understood as the mean-square limit of discrete stochastic sums. Let $0 = t_0 < t_1 < \dots < t_n = T$ be a partition of $[0, T]$ with mesh size $\|\pi\| = \max_i (t_{i+1} - t_i)$. For an adapted integrand φ_t , we define

$$\int_0^T \varphi_t dW_t = \lim_{\|\pi\| \rightarrow 0} \sum_{i=0}^{n-1} \varphi_{t_i} (W_{t_{i+1}} - W_{t_i}).$$

Crucial property: the integrand φ_{t_i} is evaluated at the *left endpoint* of the interval $[t_i, t_{i+1}]$. In financial terms, this is **non-anticipativity**: a trading strategy can only use information available before the next shock $(W_{t_{i+1}} - W_{t_i})$ occurs.

2.3 Infinitesimal Drift and Variance

To gain deeper insight into the coefficients, consider a small time increment Δt . Integrating the SDE over $[t, t + \Delta t]$ yields:

$$S_{t+\Delta t} - S_t \approx \mu(t, S_t)\Delta t + \sigma(t, S_t)\Delta W_t,$$

where the Brownian increment $\Delta W_t = W_{t+\Delta t} - W_t \sim \mathcal{N}(0, \Delta t)$.

2.3.1 The Drift

Taking the conditional expectation given the current information $\mathcal{F}_t$, the Brownian increment vanishes ($\mathbb{E}[\Delta W_t | \mathcal{F}_t] = 0$). The drift $\mu(t, S_t)$ is formally defined as the instantaneous expected rate of change:

$$\mu(t, S_t) = \lim_{\Delta t \rightarrow 0} \frac{\mathbb{E}[S_{t+\Delta t} - S_t | \mathcal{F}_t]}{\Delta t}.$$

2.3.2 The Variance and Itô's Isometry

Similarly, we can determine the local variance of the process. A foundational result in stochastic calculus is **Itô's Isometry**, which states that for any square-integrable adapted process φ_u :

$$\mathbb{E} \left[\left(\int_t^{t+\Delta t} \varphi_u dW_u \right)^2 \middle| \mathcal{F}_t \right] = \mathbb{E} \left[\int_t^{t+\Delta t} \varphi_u^2 du \middle| \mathcal{F}_t \right].$$

Applying this to our process, the volatility coefficient squared, $\sigma^2(t, S_t)$, captures the instantaneous accumulation of variance:

$$\sigma^2(t, S_t) = \lim_{\Delta t \rightarrow 0} \frac{\text{Var}(S_{t+\Delta t} - S_t | \mathcal{F}_t)}{\Delta t}.$$

The process $(\sigma_t)_{t \geq 0} = (\sigma(t, S_t))_{t \geq 0}$ is referred to as the **instantaneous (or spot) volatility**.

2.4 Returns, Innovations, and Market Efficiency

In empirical finance, we rarely model the raw price S_t directly. Instead, we model the log-price $X_t = \ln S_t$. For a strictly positive asset price following proportional dynamics

$$dS_t = \mu_t S_t dt + \sigma_t S_t dW_t,$$

Itô's lemma yields

$$dX_t = d \ln S_t = \left(\mu_t - \frac{1}{2} \sigma_t^2 \right) dt + \sigma_t dW_t.$$

In this notation, the log-price drift is $\tilde{\mu}_t := \mu_t - \frac{1}{2} \sigma_t^2$.

The discrete-time log-return over the interval $[t_{i-1}, t_i]$ can then be decomposed into a predictable component and a random shock:

$$\Delta X_i = X_{t_i} - X_{t_{i-1}} = \int_{t_{i-1}}^{t_i} \tilde{\mu}_s ds + \underbrace{\int_{t_{i-1}}^{t_i} \sigma_s dW_s}_{\text{innovation } \epsilon_i}.$$

Because the Itô integral is a martingale, the conditional expectation of the innovation is zero:

$$\mathbb{E}[\epsilon_i \mid \mathcal{F}_{t_{i-1}}] = 0.$$

This means that deviations from the expected drift behave like unpredictable shocks. It matches the empirical absence of linear autocorrelation in returns and connects naturally with the Efficient Market Hypothesis.

Remark: Uncorrelated $\neq$ Independent. While raw returns r_i exhibit virtually zero linear correlation, strong *nonlinear* dependence remains highly prevalent in financial markets. For example, absolute returns $|r_i|$ and squared returns r_i^2 exhibit significant positive autocorrelation, a phenomenon widely known as **volatility clustering**.

2.5 Quadratic Variation and Realized Volatility

For any Itô process $dS_t = \mu_t dt + \sigma_t dW_t$, the total accumulation of variance over a time interval $[0, t]$ is measured by its **Quadratic Variation**, denoted as $\langle S \rangle_t$:

$$\langle S \rangle_t = \int_0^t \sigma_u^2 du.$$

Notice that the quadratic variation is entirely independent of the drift μ_t ; it relies solely on the diffusion path.

2.5.1 From Continuous Theory to High-Frequency Data

In modern empirical finance, we bridge this theoretical concept to observable market data using high-frequency observations of the log-price process $X_t = \ln S_t$.

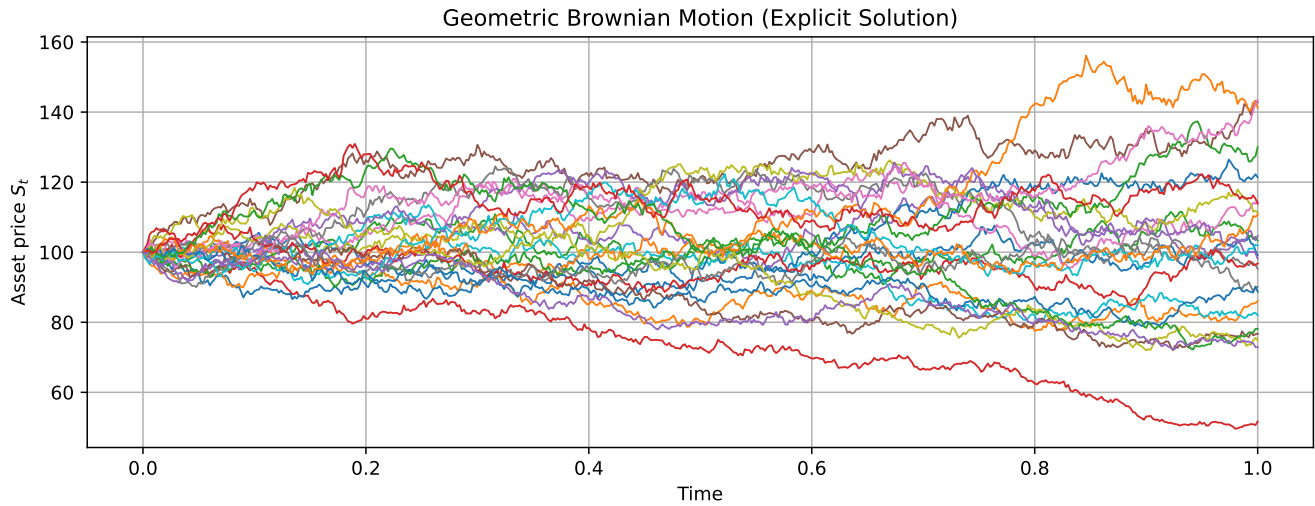
If we sample returns at a high frequency over a trading day $[0, T]$, we can compute the sum of squared discrete returns, known as **Realized Volatility (RV)**:

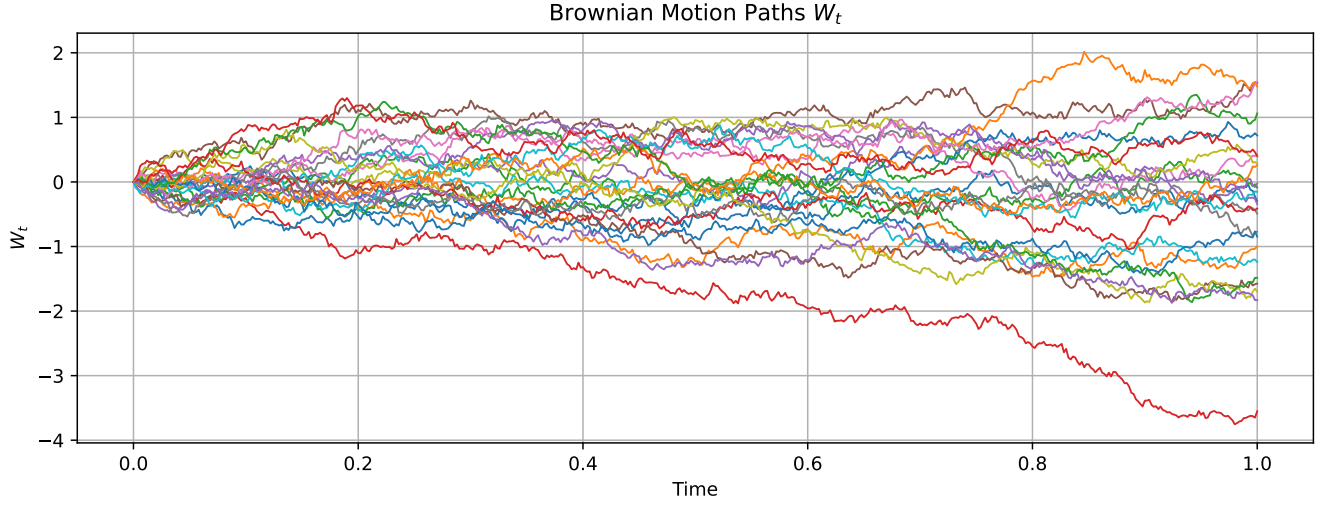
$$\text{RV}_T = \sum_{i=1}^n (X_{t_i} - X_{t_{i-1}})^2.$$

A central theorem in stochastic calculus shows that as the sampling frequency increases (that is, as the mesh size goes to zero), the empirical realized volatility converges in probability to the true integrated quadratic variation:

$$\text{plim}_{\|\pi\| \rightarrow 0} \text{RV}_T = \langle X \rangle_T = \int_0^T \sigma_t^2 dt.$$

This result explains why high-frequency data can be used to estimate integrated volatility, even though the continuous-time volatility path itself is not observed directly.





2.6 Multivariate Stochastic Calculus

When pricing complex derivatives like basket options or multi-asset structures, we must extend Itô's calculus to higher dimensions.

Let $\mathbf{W}_t$ be an m -dimensional standard Brownian motion. Consider an n -dimensional Itô process $\mathbf{X}_t$ governed by the stochastic differential equation:

$$d\mathbf{X}_t = \mu(t, \mathbf{X}_t) dt + \sigma(t, \mathbf{X}_t) d\mathbf{W}_t$$

where μ is an n -dimensional drift vector and σ is an $n \times m$ volatility matrix.

2.6.1 Multivariate Itô's Formula

Let $f \in C^{1,2}([0, T] \times \mathbb{R}^n; \mathbb{R})$ be a smooth scalar-valued function. Then the transformed process $f(t, \mathbf{X}_t)$ is also an Itô process, and its dynamics are given by:

$$\begin{aligned} f(t, \mathbf{X}_t) = f(0, \mathbf{X}_0) &+ \int_0^t \left(\frac{\partial f}{\partial s} + \langle \mu_s, \nabla f_s \rangle + \frac{1}{2} \text{Tr}(\sigma_s^\top (\nabla^2 f_s) \sigma_s) \right) ds \\ &+ \int_0^t (\nabla f_s)^\top \sigma_s d\mathbf{W}_s, \end{aligned}$$

where all derivatives are evaluated at $(s, \mathbf{X}_s)$, ∇f is the **gradient vector** of f with respect to the spatial variables, $\nabla^2 f$ is the **Hessian matrix** of f , $\text{Tr}(\cdot)$ denotes the trace, and $\langle \cdot, \cdot \rangle$ denotes the standard Euclidean inner product.

2.6.2 The Product Rule and Integration by Parts

Corollary (Product Rule for Itô Processes) Let X_t and Y_t be two continuous scalar Itô processes. The differential of their product is:

$$d(X_t Y_t) = X_t dY_t + Y_t dX_t + d\langle X, Y \rangle_t$$

Written in integral form, this yields the **Stochastic Integration by Parts** formula:

$$\int_0^t X_s dY_s = X_t Y_t - X_0 Y_0 - \int_0^t Y_s dX_s - \langle X, Y \rangle_t$$

where $\langle X, Y \rangle_t$ represents the **quadratic covariation process** between X and Y over the interval $[0, t]$. (Informally, this is often written as the cross-variation term $\int_0^t dX_s dY_s$).

2.7 Girsanov's Theorem: Changing the Measure

Girsanov's Theorem is the mathematical engine behind risk-neutral pricing. It provides the exact mechanism to shift the probability measure such that the physical drift of an asset is replaced by the risk-free rate.

Theorem (Girsanov Theorem) Let $(W_t)_{t \geq 0}$ be a standard Brownian motion defined on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$. Consider an adapted process λ_t (the **market price of risk**), satisfying **Novikov's condition**:

$$\mathbb{E}^{\mathbb{P}} \left[\exp \left(\frac{1}{2} \int_0^T \lambda_s^2 ds \right) \right] < \infty.$$

Define the **Radon-Nikodym density process** Z_t as the solution to the SDE $dZ_t = -Z_t \lambda_t dW_t$, which explicitly is:

$$Z_t = \exp \left(- \int_0^t \lambda_s dW_s - \frac{1}{2} \int_0^t \lambda_s^2 ds \right), \quad 0 \leq t \leq T.$$

Novikov's condition guarantees that Z_t is a strictly positive, true martingale with expectation $\mathbb{E}^{\mathbb{P}}[Z_T] = 1$. Using Z_T , we define a new probability measure $\mathbb{Q}$ that is equivalent to $\mathbb{P}$ ($\mathbb{Q} \sim \mathbb{P}$) via the density:

$$\left. \frac{d\mathbb{Q}}{d\mathbb{P}} \right|_{\mathcal{F}_T} = Z_T.$$

Then, under this new measure $\mathbb{Q}$, the shifted process $(\tilde{W}_t)_{0 \leq t \leq T}$ defined by:

$$\tilde{W}_t = W_t + \int_0^t \lambda_s ds,$$

is a standard Brownian motion.

2.8 Discounting and the Bank Account (The Numéraire)

In financial mathematics, future cash flows must be continuously adjusted (or **discounted**) to reflect their present value. This accounts for the **time value of money** and establishes a baseline for risk-free growth.

To formalize this, we designate one particular asset as special and refer to it as the **bank account** (or **numéraire**). It is typically modeled as the 0-th asset in the market and is denoted by B_t .

Definition (The Bank Account Process) The **bank account process** $(B_t)_{t \in [0, T]}$ models the value of a continuously compounded risk-free investment growing at a deterministic short rate r_t :

$$B_t = \exp \left(\int_0^t r_u du \right), \quad t \in [0, T], \quad B_0 = 1.$$

where: - r_t is the **instantaneous risk-free interest rate** at time t . - B_t acts as a **numéraire**, meaning we express the prices of all other assets in units of the bank account rather than in an arbitrary fiat currency (like USD or EUR).

The Discount Factor We define the **discount factor** between times s and t as:

$$D_{s,t} := \frac{B_s}{B_t} = \exp \left(- \int_s^t r_u du \right).$$

For discounting back to time zero, we use $D_{0,t} = 1/B_t$. Since r_t is deterministic (and thus has finite variation), the continuous-time dynamics of the discount factor are simply:

$$dD_{0,t} = -r_t D_{0,t} dt.$$

Discounted Asset Prices Given the bank account, the **discounted asset price process** for any risky asset vector $\mathbf{S}_t$ is defined by:

$$\tilde{\mathbf{S}}_t = D_{0,t} \mathbf{S}_t = \frac{\mathbf{S}_t}{B_t}.$$

2.9 Continuous-Time Trading Strategies

To price derivatives via replication, we must formalize the mathematical definition of a dynamic portfolio.

In a continuous-time financial market comprising d risky assets and one risk-free bank account, a **trading strategy** is described by a predictable stochastic vector process:

$$\theta = (\theta_t)_{t \in [0, T]} = (\theta_t^{(1)}, \dots, \theta_t^{(d)})_{t \in [0, T]},$$

where: - $\theta_t^{(i)}$ denotes the number of units (shares) of asset i held in the portfolio at time t . - The process θ must be **predictable** with respect to the market filtration $(\mathcal{F}_t)_{t \in [0, T]}$.

Mathematical Note: Predictability is a slightly stronger condition than adaptedness. It ensures that the trading decision at time t is formulated using strictly prior information (e.g., $t - \epsilon$), mathematically guaranteeing that the trader cannot “peek” at the instantaneous price jump $d\mathbf{S}_t$ before executing the trade.

2.9.1 The Gains Process

Definition (Gains Process) The **gains process** of a trading strategy θ measures the cumulative financial profit or loss generated strictly from the asset price fluctuations over time:

$$G_t(\theta) = \int_0^t \theta_u \cdot d\mathbf{S}_u, \quad t \in [0, T],$$

where: - $\mathbf{S}_u$ is the vector of asset prices at time u . - The stochastic integral dynamically accumulates the infinitesimal changes in the asset prices ($d\mathbf{S}_u$), weighted exactly by the portfolio holdings (θ_u) held precisely at that instant.

A trading strategy is said to be **self-financing** if the entire value of the portfolio at any time t depends *only* on the initial capital V_0 and the gains process $G_t(\theta)$, with no exogenous cash injected or withdrawn.

Consider a continuous-time financial market containing the risk-free bank account B_t and d risky assets $\mathbf{S}_t = (S_t^{(1)}, \dots, S_t^{(d)})$.

A dynamic trading strategy is defined by a predictable stochastic process $(\theta_t^{(0)}, \theta_t)$, where $\theta_t^{(0)}$ is the number of units held in the bank account and θ_t is the vector of shares held in the risky assets.

The **total wealth process** V_t of this strategy at any time t is the mark-to-market value of the portfolio:

$$V_t = \theta_t^{(0)} B_t + \theta_t \cdot \mathbf{S}_t.$$

2.9.2 The Self-Financing Condition

Definition (Self-Financing Strategy) A trading strategy is called **self-financing** if all changes in the portfolio's total wealth V_t are due *exclusively* to the price movements of the underlying assets and the accumulation of interest. No exogenous capital is injected, and no wealth is withdrawn during the trading horizon $[0, T]$.

In continuous-time mathematics, this requires the differential of the wealth process to satisfy:

$$dV_t = \theta_t^{(0)} dB_t + \theta_t \cdot d\mathbf{S}_t, \quad t \in [0, T].$$

Integrating this equation over time yields the wealth process formulation:

$$V_t = V_0 + \int_0^t \theta_u^{(0)} dB_u + \int_0^t \theta_u \cdot d\mathbf{S}_u.$$

This explicitly states that total wealth at time t equals the initial investment (V_0) plus the accumulated risk-free interest on cash balances, plus the accumulated trading gains from the risky assets.

2.10 The Discounted Wealth Process

Analyzing nominal wealth V_t is mathematically cumbersome due to the continuous accumulation of interest dB_t . A profound simplification occurs when we transition to the discounted framework.

We define the **discounted wealth process** as:

$$\tilde{V}_t = D_{0,t} V_t = \frac{V_t}{B_t}.$$

Theorem (Invariance of the Self-Financing Condition) A trading strategy is self-financing in nominal terms if and only if its discounted wealth process is entirely driven by the discounted risky assets:

$$d\tilde{V}_t = \theta_t \cdot d\tilde{\mathbf{S}}_t.$$

Proof: Using the product rule for semimartingales on $\tilde{V}_t = D_{0,t} V_t$, and noting that $D_{0,t}$ has finite variation so the quadratic covariation term vanishes:

$$d\tilde{V}_t = D_{0,t} dV_t + V_t dD_{0,t}$$

Substitute the nominal self-financing condition (dV_t) and the wealth definition (V_t):

$$d\tilde{V}_t = D_{0,t}(\theta_t^{(0)} dB_t + \theta_t \cdot d\mathbf{S}_t) + (\theta_t^{(0)} B_t + \theta_t \cdot \mathbf{S}_t) dD_{0,t}$$

Rearranging the terms to group the cash position $\theta_t^{(0)}$ and the risky asset position θ_t :

$$d\tilde{V}_t = \theta_t^{(0)} \underbrace{(D_{0,t}dB_t + B_t dD_{0,t})}_{=d(D_{0,t}B_t)=d(1)=0} + \theta_t \cdot \underbrace{(D_{0,t}d\mathbf{S}_t + \mathbf{S}_t dD_{0,t})}_{=d(D_{0,t}\mathbf{S}_t)=d\tilde{\mathbf{S}}_t}$$

Because $D_{0,t}$ and B_t are exact inverses, the cash term perfectly cancels out. We are left with the elegant result:

$$\boxed{d\tilde{V}_t = \theta_t \cdot d\tilde{\mathbf{S}}_t}$$

Integrating this gives:

$$\tilde{V}_t = \tilde{V}_0 + \int_0^t \theta_u \cdot d\tilde{\mathbf{S}}_u.$$

This proves that in discounted units, the evolution of wealth depends strictly on the portfolio allocation to the risky assets.

2.11 The Martingale Representation Theorem

Why do we care so deeply about expressing the discounted wealth process as a stochastic integral $\int \theta_u \cdot d\tilde{\mathbf{S}}_u$? Because it bridges the gap between probability theory and practical derivative hedging.

Theorem (Martingale Representation under Brownian Filtrations)

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T]}, \mathbb{Q})$ be a filtered probability space where $(\mathcal{F}_t)$ is generated by a d -dimensional Brownian motion $\mathbf{W}^{\mathbb{Q}}$. If $M = (M_t)_{t \in [0, T]}$ is a square-integrable $\mathbb{Q}$ -martingale, then there exists a unique predictable process φ with

$$\mathbb{E}^{\mathbb{Q}} \left[\int_0^T \|\varphi_u\|^2 du \right] < \infty$$

such that

$$M_t = M_0 + \int_0^t \varphi_u d\mathbf{W}_u^{\mathbb{Q}}, \quad t \in [0, T].$$

2.11.1 Application to Pricing and Replication

In a frictionless market, we assume the existence of an Equivalent Martingale Measure (EMM) $\mathbb{Q}$. In a complete Brownian market, the *discounted* asset price process $\tilde{\mathbf{S}}_t := D_{0,t}\mathbf{S}_t$ is a $\mathbb{Q}$ -martingale, governed by the SDE:

$$d\tilde{\mathbf{S}}_t = \sigma_t d\mathbf{W}_t^{\mathbb{Q}}$$

where σ_t is the volatility matrix and $\mathbf{W}_t^{\mathbb{Q}}$ is a d -dimensional Brownian motion.

2.11.2 The Derivative Pricing Engine

If we wish to value a derivative with discounted terminal payoff $\tilde{H}$, its discounted price process is given by the conditional expectation

$$M_t = \mathbb{E}^{\mathbb{Q}}[\tilde{H} \mid \mathcal{F}_t].$$

By the **Martingale Representation Theorem**, since M_t is a $\mathbb{Q}$ -martingale, there exists a predictable process φ_t such that

$$M_t = M_0 + \int_0^t \varphi_u d\mathbf{W}_u^{\mathbb{Q}}.$$

2.11.3 The Perfect Hedge: Strategy Identification

A self-financing discounted wealth process $\tilde{V}_t$ with trading strategy θ_t evolves as:

$$\tilde{V}_t = \tilde{V}_0 + \int_0^t \theta_u \cdot d\tilde{\mathbf{S}}_u = \tilde{V}_0 + \int_0^t \theta_u^\top \sigma_u d\mathbf{W}_u^{\mathbb{Q}}.$$

By equating the wealth process to the derivative price process ($\tilde{V}_t = M_t$), we see that a replicating strategy must satisfy

$$\theta_t^\top \sigma_t = \varphi_t^\top.$$

If σ_t has full rank, this relation uniquely identifies the hedge ratio. This is the link between martingale representation and dynamic replication in complete diffusion models.

2.12 Summary and Connection to the Next Lecture

This lecture has built the mathematical infrastructure for the rest of the course. The key objects are:

- **The filtered probability space** and the distinction between $\mathbb{P}$ (physical) and $\mathbb{Q}$ (risk-neutral) measures.
- **Itô's lemma** and stochastic differential equations as the language of continuous-time finance.
- **Girsanov's theorem**, which shows how changing the drift of a Brownian motion corresponds to a change of probability measure. This is the mechanism by which the physical drift μ is replaced by the risk-free rate r under $\mathbb{Q}$.
- **Self-financing strategies** and the discounted wealth process, which reduces hedging to a stochastic integral with respect to discounted asset prices.
- **The Martingale Representation Theorem**, which guarantees that every sufficiently integrable payoff can be replicated in a complete Brownian filtration market.

In the next lecture we introduce the main objects to be priced — European contingent claims, options, and exotic derivatives — and connect the theoretical framework developed here to the Black–Scholes formula and the principle of risk-neutral valuation.

3 Derivatives Pricing

Objectives

After this lecture, you should be able to:

1. Understand the basic structure of financial derivatives.
2. Distinguish between different types of options.
3. Interpret option payoffs and their economic meaning.
4. Apply the **risk-neutral valuation principle**.
5. Compute prices of European options in the **Black–Scholes model**.
6. Understand how option prices depend on
 - the underlying price,
 - volatility,
 - time to maturity,
 - and interest rates.

In the previous lecture, we introduced the filtered probability space, self-financing strategies, Girsanov's theorem, and the Martingale Representation Theorem. We can now put these tools to work: this lecture defines the contingent claims to be priced, establishes the no-arbitrage principles that determine their values, and derives the Black–Scholes formula for European call options.

Throughout this lecture, H denotes the terminal payoff random variable, h denotes a deterministic payoff function when such a representation exists, $\pi_t(H)$ denotes the arbitrage-free price process of the claim, and C_{BS} is reserved for the Black-Scholes call formula.

3.1 Derivatives

A **European contingent claim** is an $\mathcal{F}_T$ -measurable random variable H representing the payoff at maturity T . In pricing applications one typically assumes at least integrability under the relevant pricing measure, and many standard option payoffs are non-negative. In other words, it is a cash flow paid at a fixed **expiry date** T and determined by the information available up to time T .

These claims are the basic objects we aim to **price** and **hedge**.

Typical examples include:

- Options

- Insurance-type payoffs
- Structured products
- Contracts written on interest rates, commodities, or indices

3.2 Why Do Derivatives Exist?

Derivatives are financial instruments designed to **transfer, transform, and manage risk** between market participants.

Different agents in the market use derivatives for different purposes:

Banks

- Hedge risks arising from lending and trading activities
- Manage interest rate and credit exposure

Asset Managers and Pension Funds

- Protect portfolios against downside risk
- Implement portfolio insurance strategies

Corporates

- Hedge foreign exchange risk
- Stabilize input costs (e.g. energy, commodities)
- Manage interest rate exposure

Hedge Funds and Trading Firms

- Exploit relative mispricings
- Trade volatility
- Gain leveraged exposure

3.3 Lifecycle of an Option Contract

An option is characterized by two contract dates and one time-varying valuation concept:

- **Inception (start date):**
The contract comes into existence (“goes live”).

- **Expiry date:**

The contract ceases to exist after this date.

(Strictly speaking, options **expire**, whereas bonds **mature**.)

- **Mark-to-Market (MtM) value:**

At any time between inception and expiry, the option has a **fair value**, which is the price at which the contract can be traded in the market.

3.3.1 Derivatives of an Underlying Process

Let $X = (X_t)_{t \in [0, T]}$ be a stochastic process describing an underlying quantity (e.g., a stock price, interest rate, or commodity price).

If the claim H is measurable with respect to

$$\sigma(X_t : 0 \leq t \leq T),$$

then H is called a **(European) derivative written on the underlying process X** .

In words:

A derivative is a payoff whose value depends on the evolution of an underlying asset.

3.3.2 Payoffs as Functions of the Terminal Value

In many important models, the payoff depends only on the terminal value of an underlying asset S . In that case, the contingent claim can be written as

$$H = h(S_T),$$

where

- $S = (S_t)_{t \in [0, T]}$ is the underlying price process,
- $h : \mathbb{R} \rightarrow \mathbb{R}$ is a deterministic, measurable function called the **payoff function**.

Restricting attention to payoffs of the form $H = h(S_T)$ has several important advantages:

- It reduces the **state space** of the problem to the current asset price.
- Pricing formulas become significantly simpler.
- In some models (e.g. Black–Scholes) explicit, closed-form solutions are available.
- This framework already covers many of the most liquid and widely traded market products (e.g. vanilla options).

3.4 Taxonomy of Option Contracts

The terminal-value case is central for vanilla products, but it is not the whole story. If the payoff depends on the **entire price path**, then we obtain a **path-dependent derivative** of the more general form

$$H = h((S_t)_{0 \leq t \leq T}).$$

Typical examples include:

- **Asian options** (depend on the average price),
- **Barrier options** (depend on whether certain levels are crossed),
- **Lookback options** (depend on running minima or maxima).

3.4.1 Linear Derivatives

Linear derivatives are financial instruments whose payoffs depend *linearly* on the price of the underlying asset.

Forward Contract

- A contract to buy or sell an asset at a specified future date T for a predetermined price K .
- **Payoff at maturity:** $h(S_T) = S_T - K$.

Futures Contract

- Similar to a forward contract in payoff structure.
- Standardized and traded on regulated exchanges.
- Settled daily through a mechanism known as **marking-to-market**, which reduces counterparty risk.

The **Forward Price** F is the agreed-upon price today for a transaction that will occur at a future date T . In the simple frictionless setting with no dividends or storage costs, it represents the no-arbitrage future delivery price implied by carrying the asset from today to time T .

The No-Arbitrage Derivation (Cash-and-Carry) Suppose interest rates are $r > 0$. To prevent arbitrage, the cost of buying the stock **today** and holding it until time T must equal the cost of buying it **forward** for payment at T .

1. **Spot Strategy:** Borrow S_0 cash, buy the stock now. At time T , you own the stock, but you owe the bank $S_0 e^{rT}$.
2. **Forward Strategy:** Enter a forward contract to buy the stock at time T for price F .
3. **Equilibrium:** To ensure no risk-free profit exists, the two costs must be equal:

$$F = S_0 e^{rT}$$

Swap Contract

- An agreement between two parties to exchange a series of future cash flows.

- The most common example is an **interest rate swap**, where one party pays a fixed rate and the other a floating rate.
- Payoffs depend linearly on interest rates, making swaps linear derivatives.

3.4.2 Exchange-Traded (non-linear) Options

An **option** gives the holder the right — but not the obligation — to buy or sell an asset at maturity T for a predetermined **strike price** K .

European Call Option

- Grants the right to **buy** the underlying asset at strike price K at time T .
- **Payoff:**

$$h(S_T) = (S_T - K)^+.$$

European Put Option

- Grants the right to **sell** the underlying asset at strike price K at time T .
- **Payoff:**

$$h(S_T) = (K - S_T)^+.$$

These options are actively traded on exchanges with standardized terms across a wide range of underlying assets.

3.4.3 Over-The-Counter (OTC) Derivatives

OTC derivatives are customized, privately negotiated contracts traded directly between two parties, outside of centralized exchanges.

These contracts offer great flexibility and can be tailored to meet specific investment goals or risk exposures. However, they can also lead to highly complex **structured products**.

Examples of Exotic OTC Derivatives:

- **Arithmetic Asian Option**

Payoff depends on the average price of the asset over time:

$$H = \left(\frac{1}{T} \int_0^T S_t dt - K \right)^+.$$

- **Barrier Option (Down-and-Out Call)**

Payoff is conditional on the underlying never breaching a barrier level B :

$$H = (S_T - K)^+ \cdot \mathbf{1}_{\{\min_{0 \leq t \leq T} S_t > B\}}, \quad \text{with } B < S_0.$$

- **Lookback Option**

Payoff depends on the minimum price reached during the option's lifetime:

$$H = S_T - \min_{0 \leq t \leq T} S_t.$$

- **Spread Call Option:** $H = (S_T^{(1)} - S_T^{(2)} - K)^+$ with strike $K > 0$, written on two asset prices $(S_t^{(1)})_{t \in [0, T]}$ and $(S_t^{(2)})_{t \in [0, T]}$.
- **Digital Barrier Option:** $H = \mathbf{1}_{\{S_t \geq B \text{ for some } t \in [0, T]\}}$ with barrier $B > 0$.

3.5 Arbitrage-Free Pricing of European Contingent Claims

Once a payoff has been specified, two questions arise immediately: what is its fair price, and how can the seller hedge the resulting liability?

3.5.1 Pricing Problem

- Given a contingent claim H maturing at time T , determine its **fair value** $\pi_t(H)$ at some earlier time $t < T$.
- What is the **fair value**? We define it as a price that rules out arbitrage opportunities, that is, it is consistent with an **arbitrage-free market**.

3.5.2 Hedging Problem

- When a seller issues a contingent claim H , they commit to paying a potentially uncertain amount at time T .
- The **hedging problem** asks:
 - How can the seller construct a trading strategy to mitigate or eliminate this risk?
 - What is the minimum price the seller should charge today to ensure they can meet their liability at time T with no arbitrage?

3.5.3 Arbitrage Opportunities

A trading strategy

$$(\theta_t^{(0)}, \theta_t^{(1)}, \dots, \theta_t^{(d)})_{t \in [0, T]}$$

is said to be **admissible** over the horizon $[0, T]$ if its associated wealth process V_t is bounded from below $\mathbb{P}$ -almost surely at all times. Formally, there exists a constant $M \in \mathbb{R}$ such that:

$$\mathbb{P} \left(\inf_{0 \leq t \leq T} V_t \geq M \right) = 1.$$

We denote the set of all admissible strategies by $\mathcal{A}$.

A self-financing admissible trading strategy is called an **arbitrage opportunity** if:

- Initial wealth is zero:

$$V_0 = 0,$$

- Terminal wealth is non-negative almost surely:

$$V_T \geq 0 \quad \text{a.s.},$$

- There is a strictly positive probability of a gain:

$$\mathbb{P}(V_T > 0) > 0.$$

Interpretation:

Arbitrage refers to the possibility of making a profit with **no net initial investment**, **no downside risk**, and a strictly positive chance of gain.

3.6 The Arbitrage-Free Market Framework

Before discussing derivative pricing, we must establish the equilibrium condition of our financial market.

Definition: Arbitrage-Free Market Let $\mathcal{A}$ be the set of all admissible, self-financing trading strategies. A financial market is **arbitrage-free** if there exists no strategy $\theta \in \mathcal{A}$ with initial wealth $V_0 = 0$ such that:
1. $\mathbb{P}(V_T \geq 0) = 1$ (No risk of losing money) 2. $\mathbb{P}(V_T > 0) > 0$ (A strictly positive probability of making a profit)

Remark: This condition is the bedrock of modern financial economics. It serves as a foundational equilibrium assumption, ensuring consistency in pricing and guaranteeing the absence of “free lunches.”

3.6.1 The Two-Asset Financial Market

Consider a continuous-time financial market defined on a filtered probability space $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$, where $\mathbb{P}$ denotes the **physical (real-world) measure**, estimated from historical market data.

The market contains two primary continuous-time assets:

1. The Bank Account (Risk-Free Asset) The bank account grows deterministically at the short rate r :

$$dB_t = rB_t dt, \quad B_t = B_0 e^{rt}$$

2. The Stock (Risky Asset) The stock carries a risk premium $\mu - r$ to compensate investors for bearing market risk:

$$dS_t = \mu S_t dt + \sigma S_t dW_t^{\mathbb{P}}, \quad \text{with } \mu > r$$

Under the physical measure, the expected future spot price reflects this premium:

$$\mathbb{E}^{\mathbb{P}}[S_T] = S_0 e^{\mu T}$$

3.7 The Pricing Paradox: Why Not Use the Physical Measure $\mathbb{P}$?

Now, consider adding a **derivative** to this market. Its terminal payoff at maturity T is defined by a deterministic function of the stock price:

$$H = h(S_T)$$

We seek to determine its fair present value, π_0 .

3.7.1 The Naive Approach (and Why It Fails)

A natural first guess is to discount the expected payoff under the real-world measure:

$$\pi_0 \stackrel{?}{=} e^{-rT} \mathbb{E}^{\mathbb{P}}[h(S_T)]$$

This is not correct. One might try to repair the formula by introducing some claim-specific risk-adjusted discount rate ρ and writing

$$\pi_0 = e^{-\rho T} \mathbb{E}^{\mathbb{P}}[h(S_T)]$$

3.7.2 The Core Problem: ρ is Unknowable

We do **not** know ρ . Unlike the underlying stock, whose risk characteristics may be relatively stable: * The risk of an option is highly **state-dependent** (it changes dynamically as S_t moves). * Its leverage changes instantaneously (e.g., a deep Out-of-the-Money call is massively riskier than an In-the-Money call). * Its sensitivities (Delta, Gamma) evolve continuously as time decays.

Consequently, ρ is not a constant; it is a complex, time-varying function of both S_t and t . Direct pricing under $\mathbb{P}$ therefore depends on investor risk preferences and does not lead to a clean, unique no-arbitrage price.

3.8 The Solution: Risk-Neutral Pricing

To obtain a preference-free price, we use the No-Arbitrage Principle via replication. If a **self-financing portfolio** of the stock and the bank account reproduces the payoff $h(S_T)$, then the derivative's price must equal the cost of setting up that replicating portfolio. In complete markets this leads to a unique arbitrage-free price.

This is the key reason the stock's physical expected return μ does not enter derivative pricing. What matters is the cost of hedging, not the real-world expected return.

3.8.1 The First Fundamental Theorem of Asset Pricing (FTAP)

Theorem (FTAP): Under the standard technical no-free-lunch assumptions, a financial market is **arbitrage-free** if and only if there exists an **Equivalent Local Martingale Measure (ELMM)** $\mathbb{Q} \sim \mathbb{P}$ such that the discounted prices of all tradeable assets are local martingales. In the Black-Scholes model considered below, these discounted prices are in fact true martingales.

If this equivalent measure $\mathbb{Q}$ is unique, the market is called **complete**, and every sufficiently integrable contingent claim can be perfectly replicated.

3.8.2 The Risk-Neutral Measure $\mathbb{Q}$

In the Black-Scholes-type market above, two critical transformations occur under the new probability measure $\mathbb{Q}$:

1. **Asset Dynamics Change:** The drift of the stock is shifted from μ to the risk-free rate r .

$$dS_t = rS_t dt + \sigma S_t dW_t^{\mathbb{Q}}$$

$$\mathbb{E}^{\mathbb{Q}}[S_T] = S_0 e^{rT}$$

All risk premia are absorbed into the change of measure (via Girsanov's Theorem).

2. **Universal Discounting:** Because discounted traded asset prices are martingales under $\mathbb{Q}$, we no longer need the unknowable risk-adjusted rate ρ . Any replicable payoff can therefore be discounted using the observable risk-free rate r :

$$\pi_0 = e^{-rT} \mathbb{E}^{\mathbb{Q}}[h(S_T)]$$

3.9 Key Takeaway: $\mathbb{P}$ vs. $\mathbb{Q}$

We do **not** use the measure $\mathbb{Q}$ because we believe the real world is risk-neutral. Real investors are highly risk-averse.

We use $\mathbb{Q}$ as a mathematical tool because:

1. **Risk premia are hidden:** The change of measure from $\mathbb{P}$ to $\mathbb{Q}$ implicitly encodes all risk aversion, meaning we do not have to model subjective utility functions.
2. **Standardized Discounting:** It allows us to apply a single, observable discount rate (r) universally across all derivatives, regardless of their specific payoff structures.

Summary: $\mathbb{P}$ (**Physical Measure**): Used for **Risk Management** and forecasting. It answers, “Where is the market likely to go, and how much money might I lose?” $\mathbb{Q}$ (**Risk-Neutral Measure**): Used exclusively for **Pricing**. It answers, “What is the fair value of this derivative today relative to the cost of hedging it?”

3.10 Probability Measures and the ELMM

Before proving the absence of arbitrage, we must establish the mathematical framework that allows us to transition from the real-world physical probabilities to risk-neutral pricing probabilities. Let $\mathbb{P}$ and $\mathbb{Q}$ be two probability measures defined on the same measurable space $(\Omega, \mathcal{F})$.

3.10.1 Absolute Continuity and Equivalence

Definition: Absolute Continuity The measure $\mathbb{P}$ is said to be **absolutely continuous** with respect to $\mathbb{Q}$, denoted as:

$$\mathbb{P} \ll \mathbb{Q}$$

if for every event $A \in \mathcal{F}$, the following implication holds:

$$\mathbb{Q}(A) = 0 \implies \mathbb{P}(A) = 0$$

(In financial terms: anything that is impossible under $\mathbb{Q}$ must also be impossible under $\mathbb{P}$.)

Definition: Equivalence of Measures The measures $\mathbb{P}$ and $\mathbb{Q}$ are said to be **equivalent**, denoted by:

$$\mathbb{P} \sim \mathbb{Q}$$

if they are mutually absolutely continuous:

$$\mathbb{P} \ll \mathbb{Q} \quad \text{and} \quad \mathbb{Q} \ll \mathbb{P}$$

This means that $\mathbb{P}$ and $\mathbb{Q}$ assign zero probability to exactly the same events (i.e., they share the exact same null sets). They agree on what is *possible*, even if they disagree on how *probable* it is.

3.10.2 The Equivalent Local Martingale Measure (ELMM)

Definition: ELMM A probability measure $\mathbb{Q}$ is called an **Equivalent Local Martingale Measure (ELMM)**—often referred to as a risk-neutral measure—if it satisfies two conditions: 1. $\mathbb{Q} \sim \mathbb{P}$ (It is equivalent to the physical measure). 2. The **discounted asset price process** $\tilde{\mathbf{S}}$ is a $\mathbb{Q}$ -local martingale.

3.11 The No-Arbitrage Lemma

We can now mathematically prove that the existence of an ELMM instantly guarantees a market free of “free lunches.”

Lemma: No Arbitrage via Local Martingales If there exists at least one Equivalent Local Martingale Measure $\mathbb{Q} \sim \mathbb{P}$ under which the discounted asset prices $\tilde{\mathbf{S}}$ are local martingales, then the market admits **no arbitrage**.

Proof: 1. Suppose, for the sake of contradiction, that an arbitrage strategy $\theta = (\theta_t^{(0)}, \dots, \theta_t^{(d)})$ exists. 2. By definition of a self-financing strategy starting with zero initial capital ($\tilde{V}_0 = 0$), the **discounted wealth process** is the stochastic integral:

$$\tilde{V}_t = \int_0^t \theta_u \cdot d\tilde{\mathbf{S}}_u$$

3. Because an arbitrage strategy must be **admissible**, its wealth process is bounded from below by some constant $-M$ (this prevents infinitely leveraged “doubling strategies”). 4. Since $\tilde{\mathbf{S}}$ is a $\mathbb{Q}$ -local martingale and θ is predictable, the integral $\tilde{V}_t$ is also a $\mathbb{Q}$ -local martingale. 5. By Fatou’s Lemma, a local martingale that is bounded from below is a true **supermartingale**. Therefore, its expectation cannot increase:

$$\mathbb{E}^{\mathbb{Q}}[\tilde{V}_T] \leq \tilde{V}_0 = 0$$

6. However, the definition of arbitrage requires that under the physical measure $\mathbb{P}$:

$$\mathbb{P}(\tilde{V}_T \geq 0) = 1 \quad \text{and} \quad \mathbb{P}(\tilde{V}_T > 0) > 0$$

7. Because $\mathbb{P}$ and $\mathbb{Q}$ are equivalent ($\mathbb{P} \sim \mathbb{Q}$), they share the same null sets. Therefore, the same must hold under $\mathbb{Q}$:

$$\mathbb{Q}(\tilde{V}_T \geq 0) = 1 \quad \text{and} \quad \mathbb{Q}(\tilde{V}_T > 0) > 0$$

8. A random variable that is non-negative almost surely and strictly positive with non-zero probability must have a strictly positive expectation:

$$\mathbb{E}^{\mathbb{Q}}[\tilde{V}_T] > 0$$

9. This contradicts step 5 ($\mathbb{E}^{\mathbb{Q}}[\tilde{V}_T] \leq 0$). Thus, no such arbitrage strategy can exist. ■

3.12 Replication and Market Completeness

Having established the conditions for a fair market, we now ask whether we can synthetically construct any desired financial derivative.

Definition: Attainable Payoffs (Replicability) A contingent claim with terminal payoff H is said to be **attainable** (or replicable) if there exists: * A strictly **self-financing, admissible** trading strategy $\theta \in \mathcal{A}$. * A scalar value $v \in \mathbb{R}$ representing the initial setup cost.

such that the discounted terminal wealth perfectly matches the discounted claim $\tilde{H}$:

$$v + \int_0^T \theta_t \cdot d\tilde{\mathbf{S}}_t = \frac{H}{B_T}, \quad \mathbb{P}\text{-almost surely.}$$

Equivalently, in nominal units the terminal wealth satisfies $V_T = H$ almost surely. Under an ELMM, the discounted wealth of an admissible replicating strategy is then a local martingale, and under the usual integrability assumptions a true martingale.

3.12.1 Market Completeness and the Second FTAP

Definition: Complete Market A financial market is said to be **complete** if **every** sufficiently integrable contingent claim H can be attained (replicated) by a self-financing admissible portfolio.

$$\text{Every claim } H \text{ is attainable} \iff \text{Market is complete.}$$

In a complete market, there is no unhedgeable risk. Every derivative is redundant because it can be synthesized by trading the primary underlying assets.

Theorem: The Second Fundamental Theorem of Asset Pricing (FTAP) Assuming an arbitrage-free market, the market is **complete** if and only if there exists a **unique** Equivalent Local Martingale Measure (ELMM) $\mathbb{Q}$.

$$\text{Completeness} \iff \exists! \text{ risk-neutral measure } \mathbb{Q}$$

(If multiple ELMMs exist, the market is incomplete, and derivatives cannot be uniquely priced by replication; they instead require utility-based pricing or super-hedging bounds).

Theorem: The Completeness Criterion (Volatility Rank) In a continuous-time diffusion framework with d risky assets and m underlying Brownian motions, the market is complete if and only if the **volatility matrix** $\sigma_t \in \mathbb{R}^{d \times m}$ admits a **left inverse** $\mathbb{P}$ -almost surely.

$$\text{Completeness} \iff \text{Rank}(\sigma_t) = m \quad \text{a.s. for all } t \in [0, T]$$

Financial Intuition for the Rank Criterion: To hedge against m sources of random risk (the m Brownian motions), a trader needs at least m linearly independent assets. * If $\text{Rank}(\sigma) < m$, there are more sources of risk than there are unique assets to trade. The market is **incomplete** (e.g., stochastic volatility models where volatility is a risk factor but not a directly tradable asset). * If $\text{Rank}(\sigma) = m$, every source of stochastic shock can be perfectly neutralized by adjusting the portfolio weights, making the market **complete** (e.g., the standard Black-Scholes model where 1 stock perfectly hedges 1 Brownian motion)

3.13 Example: European Call Option in the Black–Scholes Model

Consider the Black–Scholes model

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

where $(W_t)_{t \geq 0}$ is a Brownian motion.

The payoff of a European call option with strike $K > 0$ is

$$H(S_T) = (S_T - K)^+.$$

3.13.1 Risk-Neutral Valuation

Assume a **zero interest rate** $r = 0$.

Under the **risk-neutral measure** $\mathbb{Q}$, the drift must equal the short rate, hence

$$dS_t = \sigma S_t dW_t^{\mathbb{Q}}.$$

The solution of this SDE is

$$S_T = S_0 \exp\left(\sigma W_T^{\mathbb{Q}} - \frac{1}{2}\sigma^2 T\right).$$

Since

$$W_T^{\mathbb{Q}} = \sqrt{T}Z, \quad Z \sim \mathcal{N}(0, 1),$$

we obtain

$$S_T = S_0 \exp\left(\sigma\sqrt{T}Z - \frac{1}{2}\sigma^2 T\right).$$

3.13.2 Black–Scholes Call Price: Derivation (zero interest rate)

We assume $r = 0$ and the risk-neutral dynamics

$$dS_t = \sigma S_t dW_t^{\mathbb{Q}}, \quad S_0 > 0.$$

Then

$$S_T = S_0 \exp\left(\sigma W_T^{\mathbb{Q}} - \frac{1}{2}\sigma^2 T\right) = S_0 \exp\left(\sigma\sqrt{T}Z - \frac{1}{2}\sigma^2 T\right), \quad Z \sim \mathcal{N}(0, 1).$$

For a European call $H = (S_T - K)^+$, the time-0 price is

$$C_0 = \pi_0(H; S) = \mathbb{E}_{\mathbb{Q}}[(S_T - K)^+] = \mathbb{E}\left[(S_0 e^{\sigma\sqrt{T}Z - \frac{1}{2}\sigma^2 T} - K)^+\right].$$

Equivalently,

$$C_0 = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \left(S_0 e^{\sigma\sqrt{T}x - \frac{1}{2}\sigma^2 T} - K\right)^+ e^{-x^2/2} dx. \quad (1)$$

Step 1: Identify the exercise region

Define

$$Y := \sigma\sqrt{T}Z - \frac{1}{2}\sigma^2T.$$

Then $S_T = S_0e^Y$ and the call is in-the-money iff

$$S_T > K \iff S_0e^Y > K \iff Y > \log(K/S_0).$$

In terms of Z ,

$$\sigma\sqrt{T}Z - \frac{1}{2}\sigma^2T > \log(K/S_0) \iff Z > a,$$

where

$$a := \frac{\log(K/S_0) + \frac{1}{2}\sigma^2T}{\sigma\sqrt{T}}.$$

Hence

$$(S_T - K)^+ = (S_T - K) \mathbf{1}_{\{Z > a\}}.$$

Step 2: Split the expectation

Using $S_T = S_0e^{\sigma\sqrt{T}Z - \frac{1}{2}\sigma^2T}$,

$$C_0 = \mathbb{E}[(S_T - K)\mathbf{1}_{\{Z > a\}}] = \mathbb{E}[S_T\mathbf{1}_{\{Z > a\}}] - K\mathbb{P}(Z > a). \quad (2)$$

The second term is immediate:

$$\mathbb{P}(Z > a) = 1 - \Phi(a) = \Phi(-a). \quad (3)$$

Step 3: Compute $\mathbb{E}[S_T\mathbf{1}_{\{Z > a\}}]$

We compute

$$\mathbb{E}[S_T\mathbf{1}_{\{Z > a\}}] = S_0e^{-\frac{1}{2}\sigma^2T} \mathbb{E}[e^{\sigma\sqrt{T}Z}\mathbf{1}_{\{Z > a\}}].$$

Write the expectation as an integral with $\varphi(z) = \frac{1}{\sqrt{2\pi}}e^{-z^2/2}$:

$$\mathbb{E}[e^{\sigma\sqrt{T}Z}\mathbf{1}_{\{Z > a\}}] = \int_a^\infty e^{\sigma\sqrt{T}z}\varphi(z) dz.$$

Complete the square:

$$e^{\sigma\sqrt{T}z}\varphi(z) = \frac{1}{\sqrt{2\pi}} \exp\left(\sigma\sqrt{T}z - \frac{z^2}{2}\right) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(z - \sigma\sqrt{T})^2}{2} + \frac{1}{2}\sigma^2T\right) = e^{\frac{1}{2}\sigma^2T} \varphi(z - \sigma\sqrt{T}).$$

Therefore,

$$\int_a^\infty e^{\sigma\sqrt{T}z}\varphi(z) dz = e^{\frac{1}{2}\sigma^2T} \int_a^\infty \varphi(z - \sigma\sqrt{T}) dz.$$

Substitute $u = z - \sigma\sqrt{T}$:

$$\int_a^\infty \varphi(z - \sigma\sqrt{T}) dz = \int_{a - \sigma\sqrt{T}}^\infty \varphi(u) du = 1 - \Phi(a - \sigma\sqrt{T}) = \Phi(\sigma\sqrt{T} - a).$$

Putting it together:

$$\mathbb{E}[S_T \mathbf{1}_{\{Z > a\}}] = S_0 e^{-\frac{1}{2}\sigma^2 T} \cdot e^{\frac{1}{2}\sigma^2 T} \Phi(\sigma\sqrt{T} - a) = S_0 \Phi(\sigma\sqrt{T} - a). \quad (4)$$

Step 4: Express the result in the standard d_1, d_2 form

Define the usual Black–Scholes quantities (with $r = 0$):

$$d_1 := \frac{\log(S_0/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \quad d_2 := d_1 - \sigma\sqrt{T}.$$

Note that

$$a = \frac{\log(K/S_0) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}} = -d_2, \quad \sigma\sqrt{T} - a = d_1.$$

Therefore,

$$\Phi(-a) = \Phi(d_2), \quad \Phi(\sigma\sqrt{T} - a) = \Phi(d_1).$$

Using (2), (3), (4) and $1 - \Phi(a) = \Phi(-a)$, we obtain

$$C_0 = S_0 \Phi(d_1) - K \Phi(d_2).$$

3.13.3 Final Black–Scholes Call Price (with $r = 0$)

$$C_0 = S_0 \Phi(d_1) - K \Phi(d_2), \quad d_1 = \frac{\log(S_0/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

(If $r \neq 0$, the only change is the discount factor and drift under $\mathbb{Q}$, giving

$C_0 = S_0 \Phi(d_1) - K e^{-rT} \Phi(d_2)$ with $d_1 = \frac{\log(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$ and $d_2 = d_1 - \sigma\sqrt{T}$.)

3.13.4 Interpretation

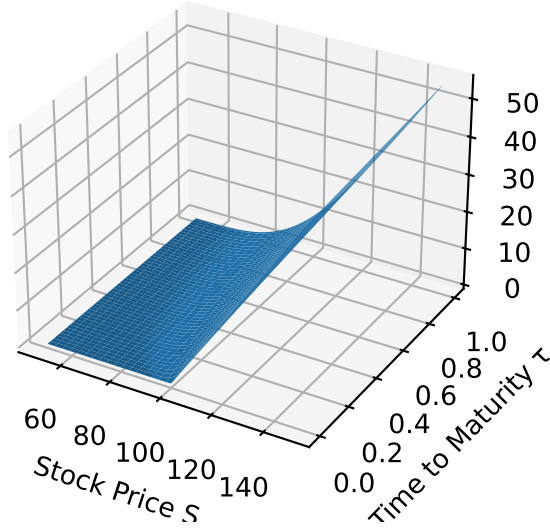
Key observation:

In the Black–Scholes model, the stock price is **lognormally distributed** under the risk-neutral measure. Therefore, pricing the option reduces to computing a **one-dimensional deterministic integral**.

Computational implication:

- No simulation is required.
- Standard numerical quadrature is extremely fast and accurate.
- This is one of the main reasons why the Black–Scholes model is computationally efficient.

European Call Option Surface



3.13.5 Pricing and Hedging in the Black–Scholes Model: The Complete Workflow

Step 1: Setup and Dynamics under the Physical Measure ($\mathbb{P}$)

Consider a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$, equipped with a standard Brownian motion $W_t^{\mathbb{P}}$. Under the physical (real-world) measure $\mathbb{P}$, we model the underlying asset price S_t via the following stochastic differential equation (SDE):

$$dS_t = \mu S_t dt + \sigma S_t dW_t^{\mathbb{P}}, \quad S_0 > 0,$$

where: * μ is the expected return (drift) of the asset under the real-world measure. * $\sigma > 0$ is the constant volatility parameter.

By applying Itô's Lemma to $\ln(S_t)$, we find that this SDE has a unique strong solution given explicitly by the exponential formula (Geometric Brownian Motion):

$$S_t = S_0 \exp \left(\left(\mu - \frac{\sigma^2}{2} \right) t + \sigma W_t^{\mathbb{P}} \right), \quad t \geq 0.$$

Step 2: Introduction of the Bank Account and Discounting

Consider a risk-free **bank account** B_t , which evolves deterministically at a constant continuously compounded interest rate r :

$$B_t = e^{rt}, \quad t \geq 0.$$

We define the **discount factor** between times 0 and t as the inverse of the bank account:

$$D_{0,t} = \frac{1}{B_t} = e^{-rt}.$$

The discounted asset price is thus given by $\tilde{S}_t = D_{0,t}S_t = e^{-rt}S_t$. Applying Itô's lemma to this discounted price, we find its dynamics under the physical measure $\mathbb{P}$:

$$d\tilde{S}_t = \tilde{S}_t [(\mu - r) dt + \sigma dW_t^{\mathbb{P}}].$$

Step 3: Risk-Neutral Measure and Girsanov's Theorem

To obtain an arbitrage-free valuation framework, we must eliminate the subjective drift μ . We introduce a change of measure via **Girsanov's Theorem**.

First, we define the **market price of risk** (the Sharpe ratio) as:

$$\lambda := \frac{\mu - r}{\sigma}.$$

We then define a new equivalent probability measure $\mathbb{Q} \sim \mathbb{P}$ via the Radon–Nikodym derivative (the density process):

$$\left. \frac{d\mathbb{Q}}{d\mathbb{P}} \right|_{\mathcal{F}_T} = \exp \left(-\frac{1}{2} \lambda^2 T - \lambda W_T^{\mathbb{P}} \right).$$

Under this new measure $\mathbb{Q}$, Girsanov's theorem states that the shifted process:

$$W_t^{\mathbb{Q}} := W_t^{\mathbb{P}} + \lambda t$$

is a standard Brownian motion. Substituting this back into our discounted asset dynamics completely eliminates the μ term, turning $\tilde{S}_t$ into a **$\mathbb{Q}$ -local martingale**:

$$d\tilde{S}_t = \tilde{S}_t \sigma dW_t^{\mathbb{Q}}, \quad \text{or equivalently,} \quad dS_t = S_t(r dt + \sigma dW_t^{\mathbb{Q}}).$$

This measure $\mathbb{Q}$ is the **Equivalent Martingale Measure (EMM)**, commonly called the **risk-neutral measure**.

Step 4: Martingale Representation and Replication of Payoffs

Consider a European contingent claim with terminal payoff H_T , which is $\mathcal{F}_T$ -measurable and square-integrable under $\mathbb{Q}$. Let

$$M_t := \mathbb{E}^{\mathbb{Q}}[D_{0,T}H_T \mid \mathcal{F}_t], \quad 0 \leq t \leq T.$$

Then M_t is a $\mathbb{Q}$ -martingale, and the **Martingale Representation Theorem** guarantees the existence of a predictable, square-integrable process φ_t such that:

$$M_t = M_0 + \int_0^t \varphi_u dW_u^{\mathbb{Q}}.$$

Since the discounted stock satisfies $d\tilde{S}_t = \sigma\tilde{S}_t dW_t^{\mathbb{Q}}$, a self-financing replicating portfolio with discounted value $\tilde{V}_t = M_t$ must satisfy

$$d\tilde{V}_t = \Delta_t d\tilde{S}_t = \Delta_t \sigma \tilde{S}_t dW_t^{\mathbb{Q}}.$$

Matching diffusion coefficients gives

$$\Delta_t = \frac{\varphi_t}{\sigma \tilde{S}_t} = \frac{\varphi_t}{\sigma D_{0,t} S_t}.$$

To replicate the claim in nominal units, we hold: * Δ_t shares of the underlying asset S_t . * $\theta_t^{(0)}$ units invested in the risk-free bank account B_t .

The bank-account holding is determined by the portfolio identity

$$V_t = \Delta_t S_t + \theta_t^{(0)} B_t,$$

that is,

$$\theta_t^{(0)} = \frac{V_t - \Delta_t S_t}{B_t} = D_{0,t}(V_t - \Delta_t S_t).$$

Because this portfolio perfectly replicates the payoff H_T without introducing arbitrage, the initial capital required to form it must equal the fair price of the derivative:

$$V_0 = \mathbb{E}^{\mathbb{Q}}[D_{0,T} H_T].$$

Step 5: Explicit Black–Scholes Formula for Call Options

For a European Call option with maturity T and strike K , the terminal payoff is $H_T = (S_T - K)^+$. Evaluating the risk-neutral expectation explicitly (or equivalently, solving the Black-Scholes PDE) yields the fundamental **Black–Scholes formula**:

$$C_{\text{BS}}(S_t, K, t, T, r, \sigma) = S_t \Phi(d_+) - K e^{-r(T-t)} \Phi(d_-),$$

with

$$d_{\pm} = \frac{\ln\left(\frac{S_t}{K}\right) + \left(r \pm \frac{\sigma^2}{2}\right)(T-t)}{\sigma\sqrt{T-t}},$$

where $\Phi(\cdot)$ denotes the cumulative distribution function (CDF) of the standard normal distribution $\mathcal{N}(0, 1)$.

Step 6: Simplified Parametrization in Terms of Log-Forward Moneyness

In quantitative development and volatility surface modeling, we often compress the formula by transforming the state variables into dimensionless quantities.

Let $\tau = T - t$ be the time to maturity, and $F_t = S_t e^{r\tau}$ be the Forward price. We define: * **Log-forward moneyness**: $k = \ln\left(\frac{F_t}{K}\right)$ * **Total variance**: $v = \sigma^2 \tau$

By factoring S_t out of the Black-Scholes equation, the Call price becomes strictly proportional to the spot price:

$$C_{\text{BS}}(S_t, K, t, T, r, \sigma) = S_t \cdot \text{BS}(k, v),$$

where the normalized, dimensionless pricing function $\text{BS}(k, v)$ is:

$$\text{BS}(k, v) = \Phi(d_+(k, v)) - e^{-k} \Phi(d_-(k, v)),$$

and the auxiliary variables simplify beautifully to:

$$d_{\pm}(k, v) = \frac{k}{\sqrt{v}} \pm \frac{\sqrt{v}}{2}.$$

This parametrization is incredibly powerful. It highlights that, once normalized for the forward rate, the percentage premium of the option depends *exclusively* on the log-moneyness k and the total accumulated variance v .

Below, we plot the European Call Option Surface, computing the value of call option prices in the Black-Scholes model across a continuous grid of underlying stock prices and times-to-maturity to visualize these non-linear dependencies.

3.14 Summary and Connection to the Next Lecture

This lecture has established three essential pillars of derivative pricing:

1. **Payoff structures.** European, path-dependent, and exotic payoffs are all special cases of the general contingent claim H , an $\mathcal{F}_{T-}$ -measurable random variable.
2. **No-arbitrage and the risk-neutral measure.** The First Fundamental Theorem of Asset Pricing connects absence of arbitrage to the existence of an equivalent martingale measure $\mathbb{Q}$. Under $\mathbb{Q}$, discounted prices are local martingales, and every sufficiently integrable claim is priced as a discounted expectation.
3. **The Black-Scholes formula.** In the complete Black-Scholes model, risk-neutral pricing reduces to a single Gaussian integral, yielding the closed-form formula C_{BS} .

The next two lectures study the structural properties of this formula in more depth. Lecture 4 analyzes how the call price depends on moneyness, volatility, and time, and introduces implied volatility as the market's quoting convention. Lecture 5 derives the Greeks and connects them to the PDE obtained from the delta-hedging argument.

4 European Vanilla Options: Static Properties and Functional Dependencies

The previous lecture derived the Black–Scholes pricing formula from risk-neutral valuation and showed how a complete market with a single Brownian driver admits a unique equivalent martingale measure. Before turning to dynamic replication and the Black–Scholes PDE in Lecture 5, it is useful to isolate the static structure already encoded in vanilla option prices.

This lecture studies how moneyness, time to maturity, and volatility enter the Black–Scholes formulas, and why implied volatility becomes the market’s preferred calibration language. The structural properties developed here — linear homogeneity, put-call parity, and the symmetry identity $F\phi(d_1) = K\phi(d_2)$ — will reappear when computing Greeks and calibrating more complex models such as Heston (Lecture 8). The concept of the implied volatility surface connects directly to calibration, which is the subject of Lecture 8.

4.1 The Black–Scholes Pricing Formulas

Let us establish the exact formulas for European vanilla options. We assume we are pricing at time $t = 0$, meaning T represents the time to maturity. $\Phi(\cdot)$ denotes the cumulative distribution function (CDF) of the standard normal distribution $\mathcal{N}(0, 1)$.

The European Call Option

$$C_{\text{BS}}(S_0, K, T, r, \sigma) = S_0 \Phi(d_1) - K e^{-rT} \Phi(d_2)$$

The European Put Option

$$P_{\text{BS}}(S_0, K, T, r, \sigma) = K e^{-rT} \Phi(-d_2) - S_0 \Phi(-d_1)$$

where the standardized auxiliary variables d_1 and d_2 are defined as:

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad \text{and} \quad d_2 = d_1 - \sigma\sqrt{T} = \frac{\ln(S_0/K) + (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}.$$

4.1.1 Financial Interpretation of the Formula

The Black-Scholes formula is not just an algebraic identity; it reflects the structure of the replicating portfolio:

- $S_0 \Phi(d_1)$ is the stock component. Here $\Phi(d_1)$ is the call Delta, so it tells us how many shares appear in the hedge.
- $Ke^{-rT} \Phi(d_2)$ is the discounted strike component, weighted by the exercise event. In the zero-dividend Black-Scholes model, $\Phi(d_2) = \mathbb{Q}(S_T > K)$.

4.2 Functional Dependencies and Market Data

To implement pricing models and risk systems cleanly, it helps to separate market observables, contract terms, and model inputs. We write the price as

$$C = C(S_0, K, T, r, \sigma),$$

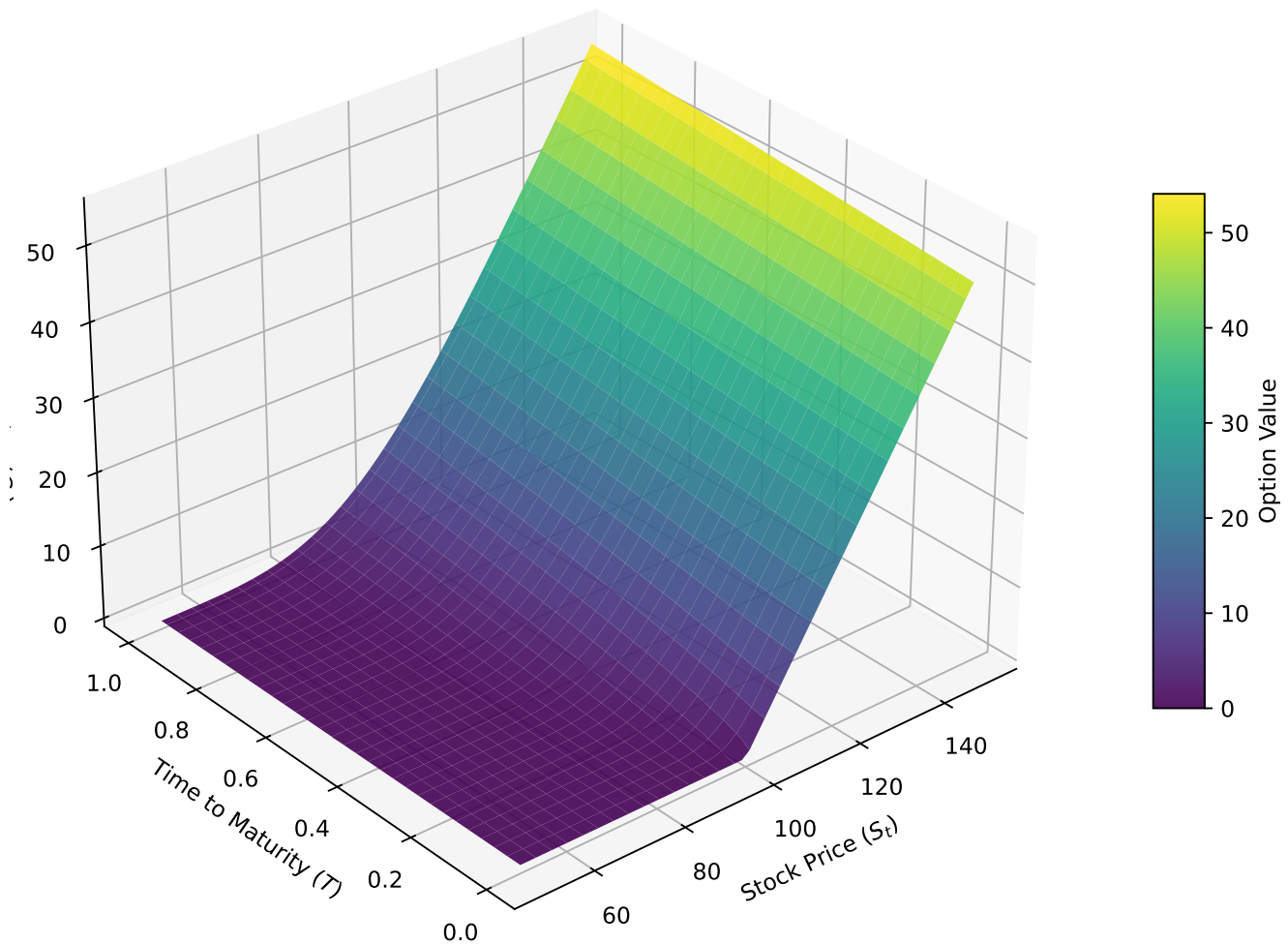
and, when the current market state (S_0, r) is fixed, often as $C(K, T; \sigma)$.

The inputs decompose into three distinct categories:

- **State variables (market observables):** S_0 is the current spot price and r is the relevant continuously compounded risk-free rate.
- **Contract specifications:** K is the strike and T is the time to maturity.
- **Model input:** σ is the annualized volatility parameter.

Unlike the other inputs, σ is not directly observable. In practice it must be estimated from data or inferred from option prices via implied-volatility inversion.

European Call Option Surface
 $(K = 100, \sigma = 0.2, r = 0.05)$



4.2.1 The Concept of Moneyness

Before simplifying the Black-Scholes formulas, it is useful to introduce **moneyness**. Moneyness compares the current asset price S_0 with the strike K and indicates whether immediate exercise would generate a positive cash flow.

Moneyness Regime	Call Option	Put Option	Financial Definition
In-the-Money (ITM)	$S_0 > K$	$S_0 < K$	The option currently possesses intrinsic value .

Moneyiness Regime	Call Option	Put Option	Financial Definition
At-the-Money (ATM)	$S_0 \approx K$	$S_0 \approx K$	The underlying asset price is close to the strike.
Out-of-the-Money (OTM)	$S_0 < K$	$S_0 > K$	The option has zero intrinsic value (only time value).

We formalize this by the dimensionless **moneyiness ratio** ξ :

$$\xi = \frac{S_0}{K}$$

(Later, especially for volatility surfaces, we will often use the forward log-moneyiness $k = \ln(F_0/K)$ instead. The present section keeps $\xi = S_0/K$ because it makes the scale invariance most transparent.)

4.2.2 Linear Homogeneity and Dimensionality Reduction

A central structural property of the Black-Scholes formula is **linear homogeneity of degree 1** in the spot price S_0 and the strike K .

Mathematically, for any positive scaling constant $\lambda > 0$, the pricing function satisfies:

$$C_{\text{BS}}(\lambda S_0, \lambda K, T, r, \sigma) = \lambda C_{\text{BS}}(S_0, K, T, r, \sigma)$$

Financial intuition: the model is scale-invariant. If a company executes a 2-for-1 stock split, both the spot and the strike halve, and the option value halves as well.

4.2.2.1 Formulation in Normalized Moneyiness

We can exploit this homogeneity to reduce the number of variables in the pricing problem. Choosing the scaling factor $\lambda = 1/K$ normalizes the strike to one. This leads to the **normalized price** $c(\xi, T, r, \sigma)$:

$$c(\xi, T, r, \sigma) := \frac{C_{\text{BS}}(S_0, K, T, r, \sigma)}{K} = C_{\text{BS}}\left(\frac{S_0}{K}, 1, T, r, \sigma\right) = C_{\text{BS}}(\xi, 1, T, r, \sigma)$$

4.2.2.2 The Moneyness Pricing Formulas

Dividing the Black-Scholes call price by K and substituting $S_0/K = \xi$ gives the normalized formula

$$c(\xi, T, r, \sigma) = \xi \Phi(d_1) - e^{-rT} \Phi(d_2)$$

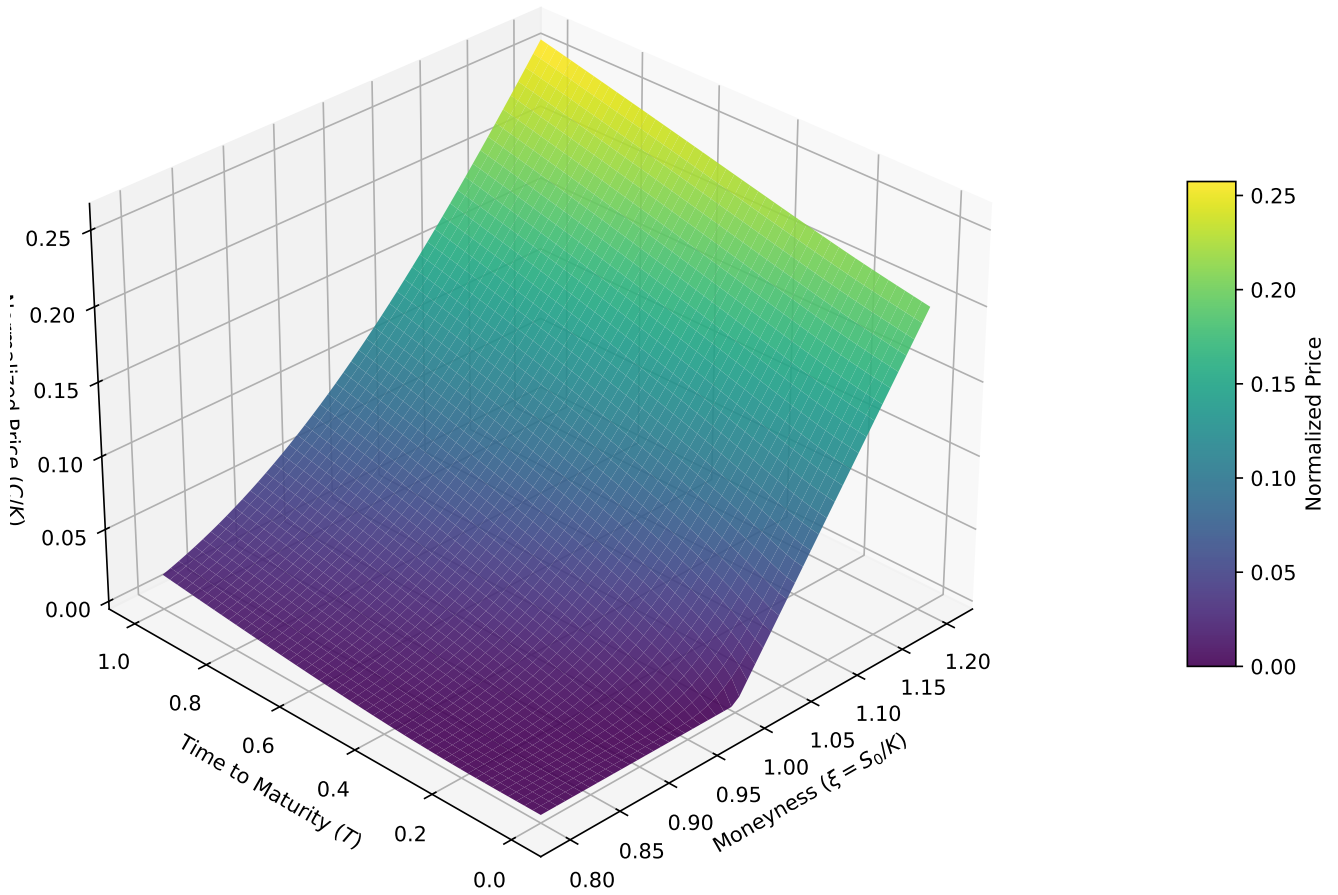
Because K has been factored out, the auxiliary variables d_1 and d_2 now depend on the moneyness ξ , time to maturity T , interest rate r , and volatility σ :

$$d_1 = \frac{\ln(\xi) + (r + \frac{1}{2}\sigma^2) T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

4.2.2.3 Computational Advantage

This normalization is not just algebraic. It reduces the state space from (S_0, K, T, r, σ) to (ξ, T, r, σ) , which is useful in computation. In practice this matters when: 1. Constructing finite difference grids for PDE solvers (reducing memory overhead). 2. Building interpolation grids for fast pricing in trading systems. 3. Training Machine Learning models (e.g., Deep Neural Networks) to learn the implied volatility surface, as normalized inputs prevent the network from memorizing arbitrary price scales.

Black-Scholes Surface: Moneyness Coordinates



4.3 Implied Volatility: The Market's Calibration Metric

So far, volatility σ has been treated as a given model input. In practice, however, the instantaneous volatility is not observed directly. What we do observe are market option prices, denoted by C_{mkt} .

4.3.1 The Inverse Problem and Implied Volatility

Using our established notation, the theoretical Black-Scholes pricing function is parameterized by σ :

$$C_{\text{BS}}(S_0, K, T, r, \sigma) = S_0 \Phi(d_1) - Ke^{-rT} \Phi(d_2)$$

Given the market price $C_{\text{mkt}}(K, T)$, we invert the pricing formula to infer the volatility parameter implied by the market.

Definition: The **Implied Volatility**, denoted $\sigma_{\text{imp}}(K, T)$, is defined as the unique value of the volatility parameter that, when plugged into the theoretical Black-Scholes formula, exactly equates the model price to the observed market price.

Mathematically, it is the root of the following equation:

$$C_{\text{BS}}(S_0, K, T, r, \sigma_{\text{imp}}) - C_{\text{mkt}}(K, T) = 0$$

Existence and Uniqueness: Does a unique solution σ_{imp} always exist? A direct differentiation of the Black-Scholes formula shows that the Black-Scholes vega,

$$\nu = \frac{\partial C_{\text{BS}}}{\partial \sigma},$$

is strictly positive for all $S_0, K, T, \sigma > 0$. Hence the option price is strictly increasing in volatility, so any arbitrage-free market price can correspond to at most one implied volatility. Existence for a strictly positive implied volatility requires the market price to lie strictly between the Black-Scholes no-arbitrage bounds; a price on the lower bound corresponds to the degenerate case $\sigma_{\text{imp}} = 0$. In practice this root is found numerically, for example by Newton-Raphson or Brent's method.

4.3.2 Financial Interpretation

- **The Universal Quoting Convention:** In professional trading, options are rarely discussed in cash terms (e.g., "\$2.50"). Instead, traders quote prices directly in terms of implied volatility (e.g., "Bid 18.5 vols, Ask 19.0 vols"). This translates the price into a standardized, dimensionless metric that can be immediately compared across different underlying asset levels and varying time horizons.
- **The "Wrong" Number:** Implied volatility is famously described by quantitative finance pioneer Riccardo Rebonato as: *> "The wrong number to put in the wrong formula to get the right price."* It acts as a universal calibration factor. It forces the idealized, simplified Black-Scholes model to mechanically agree with the complex, messy realities of market supply and demand.

4.4 The Implied Volatility Surface

If the Black-Scholes assumptions were exact, then the calibrated σ_{imp} would be constant across all strikes K and maturities T .

Empirically, this does not happen. Instead, liquid option quotes produce a non-flat **implied-volatility surface** $\Sigma(K, T)$. The surface records, in a compact way, the market's pricing of fat tails, skewness, and maturity effects.

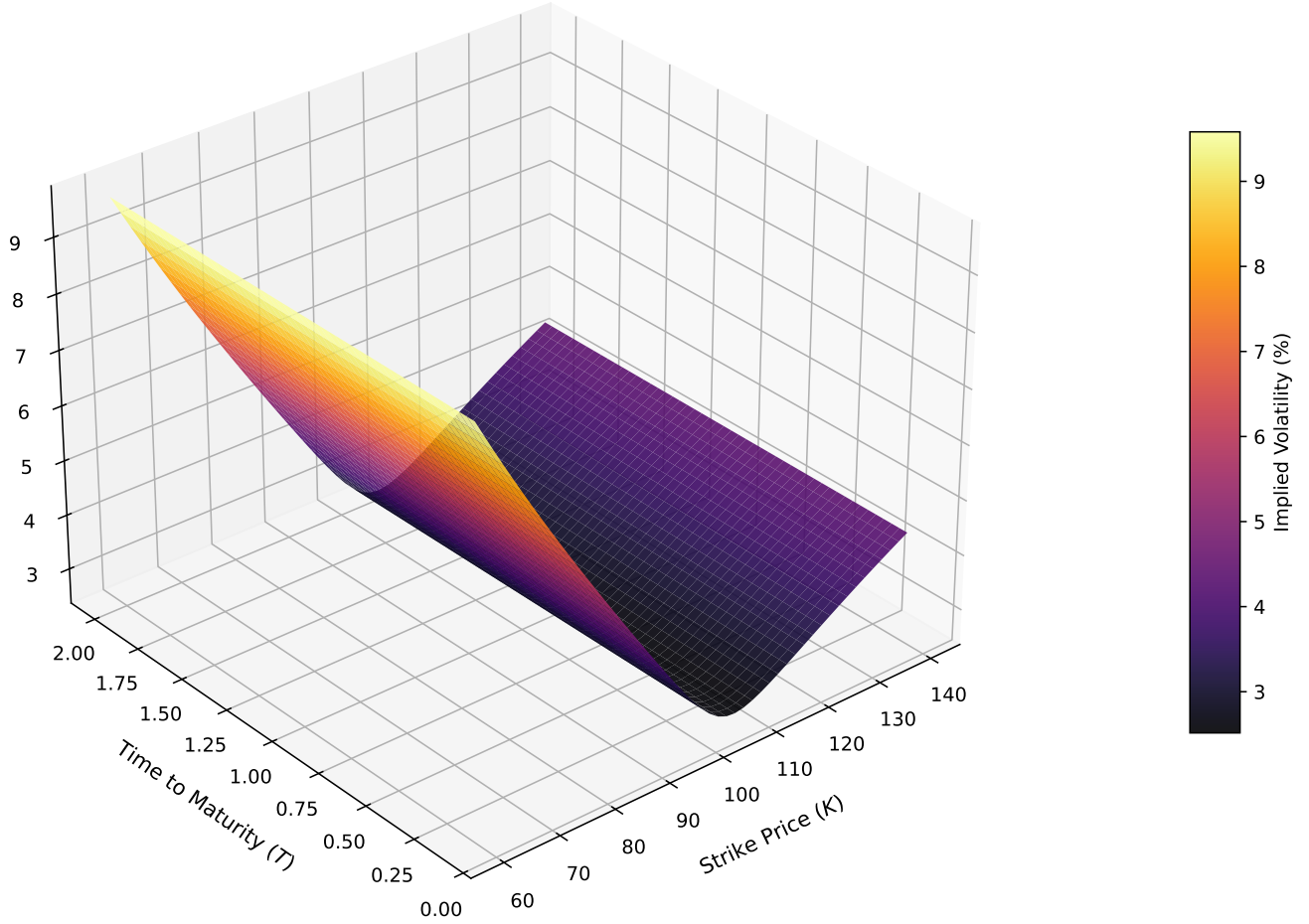
4.4.1 Stylized Facts of the Volatility Surface

The surface exhibits two primary structural deformations:

1. The Volatility Smile and Skew (Dependence on Strike K): For a fixed time to maturity T , implied volatility varies significantly with the strike price. - **Equity Skew (The “Smirk”):** In equity indices (like the S&P 500), σ_{imp} typically decreases strictly monotonically as the strike increases. Out-of-the-money (OTM) puts trade at substantially higher implied volatilities than OTM calls. This represents the market’s “crash premium”—investors aggressively overpay for downside protection due to the fear of sudden market collapses. - **FX Smile:** In foreign exchange markets, the curve is typically U-shaped (a true “smile”). Both OTM puts and OTM calls trade at a premium to At-the-Money (ATM) options. This reflects the symmetric probability of large jumps (fat tails) in either direction due to macroeconomic shocks.

2. The Term Structure of Volatility (Dependence on Maturity T): For a fixed strike K , implied volatility varies with the time to maturity. - The term structure can be in *contango* (upward sloping, typical in calm markets as uncertainty increases over time) or *backwardation* (downward sloping, typical during market panics where short-term fear is highly elevated). - Short-term options possess much steeper smiles/skews because short-term distributions are dominated by jump risk (e.g., earnings reports, macro data releases), whereas long-term options benefit from the Central Limit Theorem, causing their smiles to flatten out.

SABR Volatility Surface
 $\alpha = 0.3, \beta = 0.5, \rho = -0.6, \nu = 0.4$



4.4.2 Alternative Integral Representation via Mills' Ratio

We define the **Mills' ratio** $R(x)$ as the ratio of the survival function (or CDF of $-x$) to the density:

$$R(x) := \frac{\Phi(-x)}{\phi(x)} = \frac{1 - \Phi(x)}{\phi(x)}.$$

Using the integral identity for the normal cumulative distribution:

$$\Phi(-x) = \phi(x) \int_0^\infty e^{-xt} e^{-t^2/2} dt,$$

we can substitute this into the pricing equations to obtain the **deterministic integral representations**:

Put Option Integral:

$$P_{\text{BS}}(F, K, \sigma_{\text{tot}}) = K\phi(d_2) \int_0^\infty e^{-d_2 t} (1 - e^{-\sigma_{\text{tot}} t}) e^{-t^2/2} dt$$

Call Option Integral:

$$C_{\text{BS}}(F, K, \sigma_{\text{tot}}) = F\phi(d_1) \int_0^\infty e^{d_1 t} (1 - e^{-\sigma_{\text{tot}} t}) e^{-t^2/2} dt$$

4.5 Black–Scholes Analytics: The Forward Measure and Core Identities

To expose the core algebraic symmetries of the Black-Scholes framework, it is convenient to work with **zero interest rates** ($r = 0$), or equivalently with the **forward price**. This separates the deterministic carry from the stochastic volatility input.

4.5.1 Notation and Setup

Throughout this subsection we work in **forward units**. Accordingly, $C_{\text{BS}}(F, K, \sigma_{\text{tot}})$ and $P_{\text{BS}}(F, K, \sigma_{\text{tot}})$ denote the undiscounted Black formulas. When the short rate is nonzero, the corresponding time-0 prices are obtained by multiplying by the discount factor e^{-rT} .

- S_0 : Current Spot price of the underlying asset.
- F : **Forward price** for maturity T .
 - In the general case: $F = S_0 e^{(r-q)T}$.
 - In our simplified zero-rate, zero-dividend case ($r = 0, q = 0$): $F = S_0$.
- K : Strike price.
- $\sigma_{\text{tot}} := \sigma\sqrt{T}$: **Total Volatility** (the standard deviation of the asset's return over the entire life of the option).
- $\Phi(\cdot)$ and $\phi(\cdot)$: The standard normal Cumulative Distribution Function (CDF) and Probability Density Function (PDF), where:

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

4.5.2 The Forward Pricing Formulas

By expressing the Black-Scholes equation using the Forward price F and total volatility σ_{tot} , the pricing formulas take on a highly symmetric, elegant form (often referred to as Black's Model of 1976):

$$\begin{aligned} C_{\text{BS}}(F, K, \sigma_{\text{tot}}) &= F\Phi(d_1) - K\Phi(d_2) \\ P_{\text{BS}}(F, K, \sigma_{\text{tot}}) &= K\Phi(-d_2) - F\Phi(-d_1) \end{aligned}$$

where the standardized coordinates d_1 and d_2 are defined strictly by **log-moneyness** and **total volatility**:

$$d_1 = \frac{\ln(F/K) + \frac{1}{2}\sigma_{\text{tot}}^2}{\sigma_{\text{tot}}}, \quad d_2 = d_1 - \sigma_{\text{tot}}$$

4.5.3 The Fundamental Symmetry Identity

A crucial algebraic relationship exists between the standard normal density evaluated at d_1 and d_2 . This lemma is the engine behind many advanced proofs in option pricing (including the derivation of the Greeks).

Lemma (The Symmetry Identity):

$$F\phi(d_1) = K\phi(d_2)$$

Proof: Recall the definition $d_1 = d_2 + \sigma_{\text{tot}}$. We begin by expanding the ratio of the normal densities:

$$\frac{\phi(d_1)}{\phi(d_2)} = \frac{e^{-d_1^2/2}}{e^{-d_2^2/2}} = \exp\left(-\frac{1}{2}(d_1^2 - d_2^2)\right)$$

Using the difference of squares algebraic identity, $d_1^2 - d_2^2 = (d_1 - d_2)(d_1 + d_2)$. From our definitions, we have the system:

$$\begin{aligned} d_1 - d_2 &= \sigma_{\text{tot}} \\ d_1 + d_2 &= \frac{2\ln(F/K)}{\sigma_{\text{tot}}} \end{aligned}$$

Multiplying these two terms yields the exact log-moneyness:

$$d_1^2 - d_2^2 = \sigma_{\text{tot}} \times \frac{2\ln(F/K)}{\sigma_{\text{tot}}} = 2\ln\left(\frac{F}{K}\right)$$

Substituting this result back into the exponent of our density ratio:

$$\frac{\phi(d_1)}{\phi(d_2)} = \exp\left(-\frac{1}{2}\left[2\ln\left(\frac{F}{K}\right)\right]\right) = \exp\left(-\ln\left(\frac{F}{K}\right)\right) = \frac{K}{F}$$

Cross-multiplying yields the final identity:

$$\boxed{F\phi(d_1) = K\phi(d_2)}$$

(Note: In advanced martingale theory, this identity perfectly mirrors the Radon-Nikodym derivative used when executing a change of numeraire from the Risk-Neutral Measure to the Asset/Share Measure).

4.5.3.1 Application: The “Cancellation Trick”

This identity acts as a mathematical “cancellation trick” that vastly simplifies the calculation of the Greeks. For example, consider deriving Vega ($\nu = \frac{\partial C}{\partial \sigma_{\text{tot}}}$) using the product and chain rules on $C_{\text{BS}} = F\Phi(d_1) - K\Phi(d_2)$:

$$\frac{\partial C_{\text{BS}}}{\partial \sigma_{\text{tot}}} = F\phi(d_1)\frac{\partial d_1}{\partial \sigma_{\text{tot}}} - K\phi(d_2)\frac{\partial d_2}{\partial \sigma_{\text{tot}}}$$

Because $F\phi(d_1) = K\phi(d_2)$, we can factor it out:

$$\frac{\partial C_{\text{BS}}}{\partial \sigma_{\text{tot}}} = F\phi(d_1) \left(\frac{\partial d_1}{\partial \sigma_{\text{tot}}} - \frac{\partial d_2}{\partial \sigma_{\text{tot}}} \right)$$

Since $d_1 - d_2 = \sigma_{\text{tot}}$, their derivatives with respect to σ_{tot} differ by exactly 1 (i.e., $\frac{\partial d_1}{\partial \sigma_{\text{tot}}} - \frac{\partial d_2}{\partial \sigma_{\text{tot}}} = 1$). The messy differential terms instantly collapse, leaving the elegant result: $\nu = F\phi(d_1)$.

4.5.4 Put-Call Parity and Computational Optimization

Before proceeding to integral representations, we must verify that our Forward Black-Scholes formulas satisfy the fundamental, model-independent **Put-Call Parity** relationship.

The Parity Relationship (for $r = 0$)

$$C_{\text{BS}}(F, K) - P_{\text{BS}}(F, K) = F - K$$

Proof: We substitute the respective pricing formulas into the left-hand side:

$$\begin{aligned} C_{\text{BS}} - P_{\text{BS}} &= [F\Phi(d_1) - K\Phi(d_2)] - [K\Phi(-d_2) - F\Phi(-d_1)] \\ &= F[\Phi(d_1) + \Phi(-d_1)] - K[\Phi(d_2) + \Phi(-d_2)] \end{aligned}$$

Using the fundamental symmetry property of the Standard Normal CDF, $\Phi(x) + \Phi(-x) = 1$ (which simply states that the total probability mass over the entire domain is 1), we obtain:

$$\begin{aligned} C_{\text{BS}} - P_{\text{BS}} &= F(1) - K(1) \\ &= F - K \quad \blacksquare \end{aligned}$$

Industry Computational Note: In a high-frequency trading engine or a large-scale risk management system (where millions of options are priced per second), the `norm.cdf` (or `erf`) function is computationally expensive to evaluate.

Therefore, you should *never* call the CDF function four times to price both a Call and a Put for the same strike. The standard algorithmic practice is: 1. Compute the Call price directly: $C = F\Phi(d_1) - K\Phi(d_2)$ (Requires 2 CDF calls). 2. Compute the Put price strictly via Parity: $P = C - (F - K)$ (Requires 0 CDF calls, only basic arithmetic).

This simple structural optimization immediately reduces the CPU overhead of the pricing module by ~50%.

4.5.5 Compact Representation via Mills' Ratio

We can achieve a highly compact and analytically powerful representation of the Black-Scholes formulas by introducing **Mills' Ratio**, a well-known function in probability theory. For a standard normal distribution, Mills' ratio $R(x)$ is defined as the ratio of the survival function to the probability density function:

$$R(x) := \frac{1 - \Phi(x)}{\phi(x)} = \frac{\Phi(-x)}{\phi(x)}$$

Using the fundamental symmetry identity from the previous section ($F\phi(d_1) = K\phi(d_2)$), we can factor out the density. Let us define an auxiliary function $\Omega(x, y)$:

$$\Omega(x, y) := \phi(x) [R(x) - R(x + y)]$$

This allows us to express the Black-Scholes Forward option prices purely in terms of Ω , the standardized coordinates, and total volatility (σ_{tot}):

$$\begin{aligned} P_{\text{BS}}(F, K) &= K\Omega(d_2, \sigma_{\text{tot}}) \\ C_{\text{BS}}(F, K) &= F\Omega(-d_1, \sigma_{\text{tot}}) \end{aligned}$$

Applications in Computational Finance: This representation is not merely an algebraic curiosity; it is heavily utilized in quantitative development because it facilitates: 1. **Asymptotic Analysis:** Studying pricing behavior in extreme limits (e.g., $T \rightarrow 0$ or $K \rightarrow \infty$). 2. **Tail Bounds:** Deriving tight upper and lower bounds using established algebraic inequalities for Mills' ratio. 3. **Numerical Stability:** Creating fast approximation schemes that avoid calling the computationally expensive and underflow-prone `norm.cdf` function in deep Out-of-the-Money (OTM) regions.

4.6 Static Properties of Black-Scholes Prices

By analyzing the structure of these pricing equations, we can mathematically formalize several fundamental properties of options that are mandated by no-arbitrage arguments.

4.6.1 Strict Positivity

Since the Forward price $F > 0$ and the normal density $\phi(x) > 0$, the sign of the option price is determined entirely by the bracketed term in our Ω function: $[R(x) - R(x + y)]$.

1. **Strict Monotonicity:** Mills' ratio $R(z)$ is a strictly monotonically decreasing function for all $z \in \mathbb{R}$.
2. **Ordering:** Since total volatility is strictly positive ($\sigma_{\text{tot}} > 0$), we have $x < x + \sigma_{\text{tot}}$.
3. **Conclusion:** By the strict decreasing nature of $R(\cdot)$:

$$R(x) > R(x + \sigma_{\text{tot}}) \implies \Omega(x, \sigma_{\text{tot}}) > 0$$

Thus, the Black-Scholes formula guarantees strictly positive prices **mathematically**, which is perfectly consistent with the **financial** requirement of limited liability (an option is a right, not an obligation, and therefore cannot have a negative value).

4.6.2 Monotonicity in Strikes

Basic no-arbitrage logic dictates the direction of prices relative to the strike price K .

- **Calls:** The right to buy an asset at a *higher* price is strictly *less* valuable.

$$K_1 < K_2 \implies C_{\text{BS}}(K_1) \geq C_{\text{BS}}(K_2)$$

- **Puts:** The right to sell an asset at a *higher* price is strictly *more* valuable.

$$K_1 < K_2 \implies P_{\text{BS}}(K_1) \leq P_{\text{BS}}(K_2)$$

4.6.3 Monotonicity in Volatility (Vega)

Differentiating the pricing formula with respect to the volatility parameter σ yields the **Vega** (ν):

$$\frac{\partial C_{\text{BS}}}{\partial \sigma} = \nu = F\phi(d_1)\sqrt{T} > 0$$

Computational Consequence: Because the derivative is strictly positive for all valid inputs, the option price is strictly monotonically increasing in σ . This guarantees: 1. **Uniqueness:** For any valid market price, there exists exactly one unique Implied Volatility σ_{imp} . 2. **Convergence:** The pricing function is well-behaved, guaranteeing that gradient-based root-finding algorithms (like Newton-Raphson) will converge rapidly to the implied volatility without getting trapped in local minima.

4.6.4 Tail Bounds via Mills' Ratio

For deep Out-of-the-Money (OTM) options, evaluating the cumulative distribution $\Phi(\cdot)$ is numerically expensive and highly prone to floating-point underflow (evaluating to exactly 0 in standard 64-bit precision).

We can bypass this by substituting rigorous algebraic bounds for Mills' ratio $R(x)$. One such bound is the **Birnbaum Bound**:

$$R(x) \geq \frac{\sqrt{x^2 + 4} - x}{2} \quad \text{for } x > 0$$

Birnbaum-Type Upper Bound (Deep OTM Call) For tail regimes (for a deep OTM call, this corresponds to $d_1 \ll 0$), the price is bounded algebraically by:

$$C_{\text{BS}} \lesssim F\phi(d_1) \left[\frac{\sqrt{d_1^2 + 4} - d_1}{2} - \frac{\sqrt{(d_1 + \sigma_{\text{tot}})^2 + 4} - (d_1 + \sigma_{\text{tot}})}{2} \right]$$

Applications in Practice: * **Numerical Stability:** Replacing CDF calls with these algebraic bounds prevents crashes in risk engines when calculating deep-tail risks. * **Screening:** Quickly rejecting candidate calibrations during model fitting if the market price falls outside these absolute mathematical bounds.

4.6.5 The ATM Expansion: A Trader's Approximation

While computers handle exact pricing, traders on the floor rely on “mental math” heuristics. The most famous is derived by taking the Taylor expansion of the Black-Scholes formula around the **At-The-Money (ATM)** point where $F = K$.

The Derivation: At exactly $F = K$, we have $\ln(F/K) = 0$, which simplifies our standardized coordinates to:

$$d_1 = \frac{\sigma_{\text{tot}}}{2}, \quad d_2 = -\frac{\sigma_{\text{tot}}}{2}$$

Substituting this into the Forward Call formula:

$$C_{\text{BS}}(F, F) = F \left[\Phi\left(\frac{\sigma_{\text{tot}}}{2}\right) - \Phi\left(-\frac{\sigma_{\text{tot}}}{2}\right) \right] = F \left[2\Phi\left(\frac{\sigma_{\text{tot}}}{2}\right) - 1 \right]$$

Using the first-order Taylor expansion for the standard normal CDF around zero, $2\Phi(x) - 1 \approx \frac{2}{\sqrt{2\pi}}x$:

$$C_{\text{BS}}(F, F) \approx F \cdot \frac{2}{\sqrt{2\pi}} \cdot \frac{\sigma_{\text{tot}}}{2} = \frac{F\sigma_{\text{tot}}}{\sqrt{2\pi}}$$

Since $1/\sqrt{2\pi} \approx 0.3989$, we arrive at the famous **Brenner-Subrahmanyam approximation**:

$$\boxed{C_{\text{BS}}(F, F) \approx 0.4 F \sigma \sqrt{T}}$$

4.6.6 The Trader's Rule of Thumb (The ATM Straddle)

An **ATM Straddle** involves simultaneously buying a Call and a Put at the strike $K = F$. Because $C_{\text{ATM}} = P_{\text{ATM}}$ (when $r = 0$), the price of the straddle is simply $2 \times C_{\text{ATM}}$:

$$\text{Straddle}_{\text{ATM}} \approx 2 \times (0.4 F \sigma \sqrt{T}) = \mathbf{0.8 F \sqrt{T}}$$

Practical Trading Usage: * Sanity Checks (The Implied Volatility Backout): If a trader sees a 1-year ATM straddle trading at \$16 on a \$100 stock, they can instantly estimate the implied volatility without a computer: $\sigma \approx 16 / (0.8 \times 100) = 20\%$. *** The “Rule of 16” for Daily Break-evens:** A trading year contains roughly 256 trading days. Therefore, the conversion from annualized to daily time is $\sqrt{T_{\text{day}}} = \sqrt{1/256} = 1/16$. The expected daily break-even move for an ATM straddle is:

$$\text{Daily Straddle} \approx \frac{0.8}{16} \sigma F = \mathbf{0.05 F}$$

If a stock is at \$100 with an implied volatility of 20%, the straddle costs $0.05 \times 0.20 \times 100 = \1.00 . The trader knows the stock must move more than \$1.00 today just to break even on theta decay!

4.7 Calibration to Vanilla Options and the Volatility Surface

In a functioning, liquid financial market, we do not observe just a single option price in isolation. We observe a vast **matrix** of quoted option prices spanning various strike prices (K_i) and expiration dates (T_j).

4.7.1 The Global Calibration Problem

Calibration, in the context of the Black-Scholes framework, is the process of solving the **inverse problem** for every quoted option in the market simultaneously.

Given a discrete set of market prices $C_{\text{mkt}}(K_i, T_j)$, we seek to construct a corresponding matrix of implied volatilities $\Sigma_{i,j}$ such that the theoretical model perfectly matches the market observations:

$$C_{\text{BS}}(S_0, K_i, T_j, r, \Sigma_{i,j}) = C_{\text{mkt}}(K_i, T_j) \quad \forall i, j$$

4.7.2 Empirical Reality: The Smile and Term Structure

If the core assumptions of the Black-Scholes model (specifically, that the underlying asset follows a Geometric Brownian Motion with constant variance) were perfectly true, the calibrated volatility matrix $\Sigma_{i,j}$ would be a flat, horizontal plane.

When we calibrate to real market data, we instead observe a dynamic, deformed surface characterized by two main structural features:

1. **The Smile/Skew (Strike Dependence):** For a fixed maturity T_j , implied volatility varies with the strike K_i .
 - *Equity Markets:* Typically downward-sloping (Volatility Skew). Out-of-the-money (OTM) Puts trade at significantly higher implied volatilities than OTM Calls because investors demand (and market makers charge a premium for) protection against sudden market crashes.
 - *FX Markets:* Typically U-shaped (Volatility Smile). Extreme currency moves in either direction are more likely than the log-normal distribution predicts, reflecting “fat-tailed” distributions.
2. **The Term Structure (Maturity Dependence):** Implied volatility varies across time to maturity T_j . Short-term expirations often exhibit much steeper smiles and higher volatility-of-volatility than long-term expirations, largely due to the eventual mean-reversion of volatility and the Central Limit Theorem smoothing out long-horizon returns.

4.7.3 Computational Algorithm: Point-wise Calibration

To construct this surface computationally, we must apply a numerical root-finding algorithm to each data point in the option grid independently.

Inputs: * Vector of available Strikes: $\mathbf{K}$ * Vector of available Maturities: $\mathbf{T}$ * Matrix of observed Market Prices: $\mathbf{C}_{\text{mkt}}$

The Point-Wise Algorithm: Iterate over every option (i, j) in the market chain:

1. **Check Arbitrage Bounds:** Before attempting to invert the formula, mathematically verify that the market price is arbitrage-free. For a European Call, the absolute bounds are:

$$\max(S_0 - K_i e^{-rT_j}, 0) \leq C_{\text{mkt}} < S_0$$

If the market price violates these bounds (e.g., due to stale quotes or wide bid-ask spreads), no real-valued implied volatility exists. If the price lies exactly on the lower bound, the corresponding implied volatility is the degenerate value $\sigma_{\text{imp}} = 0$.

2. **Root Finding:** Solve the non-linear equation $C_{\text{BS}}(\sigma) - C_{\text{mkt}} = 0$ using a numerical solver (e.g., Newton-Raphson).
3. **Store:** Place the resulting σ_{imp} into the surface matrix $\Sigma_{i,j}$.

The Interpolation Challenge: Market data is fundamentally discrete (existing only at specific standardized strikes and monthly expirations). However, continuous pricing and risk management (calculating Greeks) require a continuous, differentiable surface. After point-wise calibration, quantitative analysts typically fit a parametric, arbitrage-free model (such as the **SABR** or **SVI** models) to interpolate the discrete points into a smooth surface $\Sigma(K, T)$.

4.8 The Newton-Raphson Method for Implied Volatility

Because the standard normal CDF $\Phi(\cdot)$ inside the Black-Scholes formula is non-algebraic, we cannot isolate σ analytically. We must solve for it numerically. Fortunately, as established in the previous section, the strict monotonicity of the pricing function with respect to volatility ($\nu > 0$) guarantees that the inversion is well-posed and possesses a unique root.

Numerically, we frame the calibration as finding the root of the objective function $f(\sigma)$:

$$f(\sigma) = C_{\text{BS}}(\sigma) - C_{\text{mkt}} = 0$$

The **Newton-Raphson method** finds this root rapidly by iteratively approximating the non-linear function $f(\sigma)$ via its tangent line. Given an initial guess σ_0 , we update our estimate by subtracting the ratio of the pricing error to the local slope:

$$\sigma_{n+1} = \sigma_n - \frac{f(\sigma_n)}{f'(\sigma_n)}$$

Substituting our specific financial terms into this purely mathematical framework: 1. **The Numerator (Pricing Error):** $f(\sigma_n) = C_{\text{BS}}(\sigma_n) - C_{\text{mkt}}$ 2. **The Denominator (The Slope):** $f'(\sigma_n) = \frac{\partial C_{\text{BS}}}{\partial \sigma} = \nu(\sigma_n)$ (The Black-Scholes Vega)

The iterative update step thus becomes beautifully grounded in financial sensitivities:

$$\sigma_{n+1} = \sigma_n - \frac{C_{\text{BS}}(\sigma_n) - C_{\text{mkt}}}{\nu(\sigma_n)}$$

4.8.1 Computational Challenges and Robustness

While the Newton-Raphson algorithm boasts quadratic convergence (meaning the number of correct decimal places roughly doubles with each step), it possesses severe, known pitfalls when applied blindly to option pricing:

1. The “Small Vega” Problem (Numerical Explosion): For deep Out-of-the-Money (OTM) or deep In-the-Money (ITM) options, the Vega decays exponentially toward zero ($\nu \rightarrow 0$). - *The Result:* The denominator in the Newton step becomes infinitesimally small, causing the update step $\Delta\sigma$ to explode toward infinity. The algorithm will overshoot wildly and fail to converge. - *The Fix:* Industry-grade pricing libraries rarely use raw Newton-Raphson. Instead, they use hybrid algorithms like **Brent’s Method**, which combines the rapid convergence of secant/inverse quadratic interpolation with the absolute safety of the **Bisection Method**. If a Newton step threatens to jump outside realistic bounds (e.g., $\sigma < 0$ or $\sigma > 500\%$), Brent’s method forces a safe bisection step.

2. The Initial Guess (σ_0): Newton-Raphson is highly sensitive to the initial starting seed. A poor guess can lead to divergence or excessive iterations. A highly effective heuristic utilizes the Brenner-Subrahmanyam At-The-Money approximation ($C_{\text{ATM}} \approx 0.4F\sigma\sqrt{T}$) rearranged to solve for σ . Using the precise constant $1/\sqrt{2\pi} \approx 0.4$, we derive an excellent starting seed:

$$\sigma_0 \approx \frac{C_{\text{mkt}}}{F} \sqrt{\frac{2\pi}{T}}$$

Starting the algorithm with this seed practically guarantees convergence in 3 to 4 iterations for options near the money.

4.8.2 Python Implementation: Implied Volatility Solver

Below is a simple implementation with guards against vanishing vega and non-convergence.

```
import numpy as np
from scipy.stats import norm

def implied_volatility(price_mkt, F, K, T, tol=1e-6, max_iter=100):
    """
    Calculate Implied Volatility using Newton-Raphson.

    Parameters:
    price_mkt : float - Observed market price of the option
    F          : float - Forward Price
    K          : float - Strike Price
    T          : float - Time to maturity
    Returns:
    sigma_imp : float - Implied Volatility
    """
    if T <= 0:
        raise ValueError("T must be positive.")

    intrinsic = max(F - K, 0.0)
    if price_mkt < intrinsic - tol or price_mkt >= F:
        return np.nan
    if abs(price_mkt - intrinsic) <= tol:
        return 0.0

    # 1. Initial Guess (Brenner-Subrahmanyam approximation)
    # This places us close to the solution for ATM options
    sigma = max(np.sqrt(2 * np.pi / T) * (price_mkt / F), 1e-4)

    for i in range(max_iter):
        # Calculate d1, d2
        sigma_sqrt_T = sigma * np.sqrt(T)
        d1 = (np.log(F / K) + 0.5 * sigma**2 * T) / sigma_sqrt_T
        d2 = d1 - sigma_sqrt_T
```



```

# Calculate Price and Vega
# We work in forward units, so the undiscounted Black formula applies.
price_model = F * norm.cdf(d1) - K * norm.cdf(d2)
vega = F * norm.pdf(d1) * np.sqrt(T)

# Calculate Error
diff = price_mkt - price_model

# Check Convergence
if abs(diff) < tol:
    return sigma

# Newton-Raphson Step
# Protection against Zero Vega (Deep OTM/ITM)
if abs(vega) < 1e-8:
    print("Warning: Vega is zero, Newton method fails.")
    return np.nan

sigma = max(sigma + diff / vega, 1e-8)

print("Warning: Max iterations reached without convergence.")
return sigma

# Example Usage
F = 100
K = 100
T = 1.0
market_price = 7.96 # Assume this is the observed price

iv = implied_volatility(market_price, F, K, T)
print(f"Market Price: {market_price}")
print(f"Implied Vol: {iv:.4f} (Expected ~20%)")

```

Market Price: 7.96

Implied Vol: 0.1999 (Expected ~20%)

4.9 Summary and Connection to the Next Lecture

This lecture has analyzed the static properties of Black–Scholes prices, establishing several key results:

- The formula is linearly homogeneous in (S_0, K) , enabling dimensionality reduction via moneyness coordinates.
- The symmetry identity $F\phi(d_1) = K\phi(d_2)$ simplifies derivations of Greeks and underlies the computation of vega and other sensitivities.

- Put-call parity is a model-independent no-arbitrage identity that connects call and put prices.
- Implied volatility is the market's standard quoting convention. Empirically, it is not flat — it exhibits a smile or skew across strikes and a term structure across maturities, both of which reflect departures from log-normality.

In the next lecture we turn to the dynamic perspective: how does the option price change as the underlying moves, and how can an option seller hedge this exposure continuously? This leads to the Black–Scholes PDE and to the Greeks as its coefficients.

5 Black-Scholes PDE and the Greeks

Objectives

After this lecture, you should be able to:

1. Derive the Black–Scholes Partial Differential Equation (PDE) using a dynamic replication (delta-hedging) argument.
2. Explain why the drift μ of the stock disappears from the pricing equation and why pricing depends only on r , σ , and the payoff.
3. Formulate the Black–Scholes PDE together with appropriate terminal and boundary conditions for European options.
4. Understand the connection between the PDE approach and the probabilistic (risk-neutral expectation) approach via the Feynman–Kac theorem.
5. Define and interpret the main Greeks:
 - Delta (Δ),
 - Gamma (Γ),
 - Vega (ν),
 - Theta (Θ),
 - Rho (ρ).
6. Compute Greeks analytically in the Black–Scholes model and explain their economic meaning in terms of hedging and risk management.

In the previous lecture we studied the static structure of Black–Scholes prices as a function of moneyness and implied volatility. This lecture takes the dynamic view: how does the option price change continuously as the underlying moves? The central insight is that maintaining a zero-risk portfolio forces the option price to satisfy a PDE — the Black–Scholes equation — and that the partial derivatives of the price function are the Greeks. The Feynman–Kac theorem then reveals that this PDE and the risk-neutral expectation formula are two representations of the same pricing function. Later, when we move to finite-difference methods in Lectures 9 and 10, it is the PDE representation that forms the numerical basis.

5.1 The Black–Scholes Equation: Replication, Hedging, and Break-Even Volatility

5.1.1 From Risk-Neutral Valuation to Replication

Risk-neutral valuation expresses the arbitrage-free price of a contingent claim as a discounted expectation under an equivalent martingale measure ($\mathbb{Q}$). For hedging, however, it is often more instructive to work from the replication viewpoint.

The core idea is:

If a derivative can be dynamically replicated by continuously trading in the underlying asset and a risk-free bank account, then its price must be such that the replicating portfolio is self-financing and free of arbitrage.

This reasoning leads directly to a parabolic partial differential equation (PDE) for the option price.

5.1.2 Market Model and the Self-Financing Portfolio

Assume the underlying asset follows a geometric Brownian motion under the real-world measure $\mathbb{P}$:

$$dS_t = \mu S_t dt + \sigma S_t dW_t^{\mathbb{P}},$$

where: * $\mu \in \mathbb{R}$ is the real-world expected return (drift), * $\sigma > 0$ is the constant volatility, * $W_t^{\mathbb{P}}$ is a standard Brownian motion under the physical measure.

Let $V(S, t)$ denote the price of a European-style derivative with maturity T written on this asset.

Suppose a market maker has sold one derivative contract. To hedge this exposure, they construct a portfolio Π_t at time t consisting of: * Short one option: $-V(S_t, t)$ * Long Δ_t shares of the underlying asset S_t

The total value of this portfolio is:

$$\Pi_t = -V(S_t, t) + \Delta_t S_t.$$

Any residual cash needed to finance the stock position is implicitly invested in, or borrowed from, the bank account. The self-financing condition ensures that once the hedge is set up, subsequent changes in portfolio value come only from changes in the traded asset prices.

5.1.3 Dynamics via Itô's Lemma

Assuming $V(S, t)$ is sufficiently smooth ($C^{2,1}$), we apply Itô's lemma to find the continuous-time dynamics of the option price:

$$dV = \left(\frac{\partial V}{\partial t} + \mu S \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \sigma S \frac{\partial V}{\partial S} dW_t^{\mathbb{P}}.$$

Because the portfolio is strictly self-financing, any change in its value $d\Pi$ is driven solely by changes in the asset prices, with no external cash inflows or outflows:

$$d\Pi = -dV + \Delta_t dS_t.$$

Substituting the explicit expressions for dV and dS_t into this equation yields:

$$d\Pi = - \left(\frac{\partial V}{\partial t} + \mu S \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt - \sigma S \frac{\partial V}{\partial S} dW_t^{\mathbb{P}} + \Delta_t (\mu S dt + \sigma S dW_t^{\mathbb{P}}).$$

Grouping the terms in the underlying dynamics $dS_t = \mu S dt + \sigma S dW_t$ gives

$$d\Pi = - \left(\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \left(\Delta_t - \frac{\partial V}{\partial S} \right) dS_t.$$

5.1.4 Delta Hedging: Eliminating Risk

The portfolio risk is entirely contained in the dS_t term. To hedge it, the trader chooses

$$\Delta_t = \frac{\partial V}{\partial S}(S_t, t).$$

This quantity is the **Delta** (Δ) of the option. Holding exactly Δ_t shares eliminates the stochastic term, so the portfolio dynamics become independent of both the Brownian shock and the physical drift μ .

The portfolio now evolves deterministically:

$$d\Pi = - \left(\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt.$$

5.1.5 The No-Arbitrage Condition and the Black-Scholes PDE

Because the hedged portfolio is locally risk-free ($d\Pi$ contains no dW_t term), no-arbitrage requires it to earn the risk-free rate r .

Therefore, we must have:

$$d\Pi = r\Pi_t dt.$$

Substitute the portfolio value $\Pi_t = -V + \frac{\partial V}{\partial S} S$ and the deterministic dynamics $d\Pi$ into this no-arbitrage condition:

$$- \left(\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt = r \left(-V + \frac{\partial V}{\partial S} S \right) dt.$$

Dividing by dt and rearranging all terms to one side yields the fundamental **Black-Scholes Partial Differential Equation**:

$$\frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV = 0.$$

Any derivative written on the underlying S_t must satisfy this PDE. The particular contract enters through its terminal condition $V(S, T) = h(S)$.

5.1.6 Discrete-Time Hedging and P&L Decomposition

To connect the continuous-time PDE with actual trading, we now analyze the portfolio over a small but finite time interval δt .

We now generalize the setup to include a continuous dividend yield. Assume the market parameters: * r : Risk-free interest rate. * q : Continuous dividend yield (or repo rate) of the underlying asset. When $q > 0$, the holder of the stock receives a continuous income stream, and the cost-of-carry rate becomes $r - q$ instead of r .

Consider a self-financing, delta-hedged portfolio Π where the trader is **short** one derivative $V(t, S)$. To hedge this, they hold Δ shares of the underlying asset and invest/borrow the remainder in the risk-free bank account. Over the interval δt , the Profit and Loss ($\delta\Pi$) of this portfolio is:

$$\delta\Pi = - \underbrace{[V(t + \delta t, S + \delta S) - V(t, S)]}_{\text{Change in option value}} + \underbrace{rV(t, S) \delta t}_{\text{Interest on short proceeds}} + \underbrace{\Delta(\delta S - rS\delta t + qS\delta t)}_{\text{Return on stock position minus financing}}$$

5.1.6.1 Taylor Expansion and the Hedged Portfolio

Assuming the pricing function V is sufficiently smooth, we expand the future option price $V(t + \delta t, S + \delta S)$ using a Taylor series to first order in time and second order in space:

$$\delta V \approx \frac{\partial V}{\partial t} \delta t + \frac{\partial V}{\partial S} \delta S + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} (\delta S)^2.$$

Recognizing the standard “Greeks” — Theta ($\Theta = \frac{\partial V}{\partial t}$), Delta ($\Delta = \frac{\partial V}{\partial S}$), and Gamma ($\Gamma = \frac{\partial^2 V}{\partial S^2}$) — we can rewrite the expansion as:

$$\delta V \approx \Theta \delta t + \Delta \delta S + \frac{1}{2} \Gamma (\delta S)^2.$$

Here $\Theta = \partial_t V$ follows the mathematical convention used throughout these notes. If we switch to time to maturity $\tau = T - t$, then $\partial_\tau V = -\Theta$. On trading desks, “theta” is often reported with the opposite sign because time to maturity decreases as calendar time increases.

Substituting this back into the $\delta\Pi$ equation, the linear directional risk ($\Delta \delta S$) cancels out perfectly. This is the discrete-time analogue of eliminating the stochastic dW_t term in continuous time. The resulting delta-neutral P&L simplifies to:

$$\delta\Pi = -(\Theta - rV + (r - q)S\Delta) \delta t - \frac{1}{2} \Gamma (\delta S)^2.$$

5.1.6.2 Order-of-Magnitude Argument

Under standard diffusive scaling (Geometric Brownian Motion), price increments scale with the square root of time:

$$\delta S \sim \sigma S \sqrt{\delta t} \implies (\delta S)^2 \sim \sigma^2 S^2 \delta t.$$

Consequently, both the deterministic time-decay term (δt) and the quadratic variation term $((\delta S)^2)$ are of the exact same order of magnitude, $\mathcal{O}(\delta t)$. This rigorously confirms that truncating the Taylor expansion at the first order for space (ignoring Gamma) would completely invalidate the P&L analysis.

We can rewrite the P&L by isolating the normalized squared return:

$$\delta \Pi = - \underbrace{(\Theta - rV + (r - q)S\Delta)}_{\text{Deterministic Carry } (\mathcal{C})} \delta t - \underbrace{\left(\frac{1}{2}\Gamma S^2\right)}_{\text{Convexity Cost}} \left(\frac{\delta S}{S}\right)^2.$$

5.1.7 Break-Even Volatility and Model Consistency

Empirically, over repeated observations, the squared relative returns satisfy the leading-order relation

$$\mathbb{E} \left[\left(\frac{\delta S}{S} \right)^2 \right] \approx \sigma_{real}^2 \delta t,$$

where σ_{real} denotes the actual **realized volatility** observed in the market.

For a dynamically consistent pricing model, the *expected* P&L of a continuously delta-hedged portfolio should vanish at leading order when the model variance matches the realized local variance. Setting $\mathbb{E}[\delta \Pi] \approx 0$ then requires:

$$-(\Theta - rV + (r - q)S\Delta) \delta t - \frac{1}{2}\Gamma S^2 (\sigma_{real}^2 \delta t) = 0.$$

Dividing by δt and substituting the partial derivatives back in, we recover:

$$\frac{\partial V}{\partial t} + (r - q)S \frac{\partial V}{\partial S} + \frac{1}{2}\sigma_{real}^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV = 0.$$

This is exactly the **Black–Scholes PDE**, now obtained from a discrete-time break-even hedging argument rather than directly from Itô calculus.

5.1.7.1 The Gamma-Theta Trade-Off (P&L Factorization)

In practice, the option is priced and hedged using the market's **implied volatility**, σ_{imp} . Because $V(t, S)$ is derived using σ_{imp} , it strictly satisfies the Black-Scholes PDE parameterized by σ_{imp} . Therefore, we can substitute the carry term:

$$(\Theta - rV + (r - q)S\Delta) = -\frac{1}{2}\sigma_{imp}^2 S^2 \Gamma.$$

Inserting this into our discrete P&L equation yields the fundamental equation of quantitative option trading:

$$\delta\Pi = \frac{1}{2}\Gamma S^2 \left[\sigma_{imp}^2 \delta t - \left(\frac{\delta S}{S} \right)^2 \right].$$

5.1.7.2 Interpretation for the Trader

Assume a standard vanilla option, which possesses strictly positive Gamma ($\Gamma > 0$). The bracketed term reveals the exact nature of the risk:

- **Realized > Implied:** If the market moves violently and realized variance exceeds the implied break-even level ($(\frac{\delta S}{S})^2 > \sigma_{imp}^2 \delta t$), the bracket is negative. The **short** option position incurs a net loss, as the cost of re-hedging (buying high, selling low) exceeds the theta collected.
- **Realized < Implied:** If the market remains quiet, the realized variance is smaller than the implied variance. The bracket is positive, and the short position generates a deterministic profit.

Thus, selling a delta-hedged option is equivalent to being short realized variance. The implied volatility parameter σ_{imp} is the corresponding **break-even volatility** for the market maker.

5.1.8 Criteria for Usable Pricing Models

A mathematically robust and market-consistent pricing model must satisfy two fundamental conditions:

1. **Existence of a Break-Even Volatility:** There must exist a specific volatility level at which the expected second-order hedging P&L vanishes to leading order.
2. **Payoff Independence:** The break-even volatility condition must not depend on the specific structural payoff $H(S_T)$ of the derivative.

The second condition is vital for preventing local inconsistencies across products. If a portfolio has zero Gamma (a locally linear exposure profile), its Theta must also vanish after financing adjustments. Otherwise, a trader could combine instruments into a locally riskless portfolio with deterministic carry, contradicting no-arbitrage pricing.

5.1.9 Conclusion

The Black–Scholes equation is a consistency condition imposed by dynamic hedging. In this lecture it appears in three equivalent ways: from risk-neutral valuation, from continuous-time replication, and from discrete-time break-even hedging. This is why it remains central in both theory and practice.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import norm

def exact_bs_call(S, K, tau, r, sigma):
    if tau <= 0:
        return max(S - K, 0.0)
    d1 = (np.log(S / K) + (r + 0.5 * sigma**2) * tau) / (sigma * np.sqrt(tau))
    d2 = d1 - sigma * np.sqrt(tau)
    return S * norm.cdf(d1) - K * np.exp(-r * tau) * norm.cdf(d2)

def simulate_delta_hedging(S0, K, T, r, sigma, mu=0.08, N_steps=252):
    dt = T / N_steps
    t = np.linspace(0, T, N_steps + 1)

    # Simulate one stock path under the physical measure
    rng = np.random.default_rng(123)
    dW = rng.standard_normal(N_steps) * np.sqrt(dt)
    W = np.zeros(N_steps + 1)
    W[1:] = np.cumsum(dW)
    S = S0 * np.exp((mu - 0.5 * sigma**2) * t + sigma * W)

    portfolio_value = np.zeros(N_steps + 1)
    cash = np.zeros(N_steps + 1)

    # Initial Hedge
    d1 = (np.log(S[0]/K) + (r + 0.5*sigma**2)*T) / (sigma*np.sqrt(T))
    delta = norm.cdf(d1)
    option_price = exact_bs_call(S[0], K, T, r, sigma)

    # We sell the option, buy delta shares, borrow the rest
    cash[0] = option_price - delta * S[0]

    for i in range(1, N_steps + 1):
        tau = T - t[i]

        # Grow cash at risk free rate
        cash[i] = cash[i-1] * np.exp(r * dt)

        if tau > 0.001:
```

```

    d1 = (np.log(S[i]/K) + (r + 0.5*sigma**2)*tau) / (sigma*np.sqrt(tau))
    new_delta = norm.cdf(d1)
else:
    new_delta = 1.0 if S[i] > K else 0.0

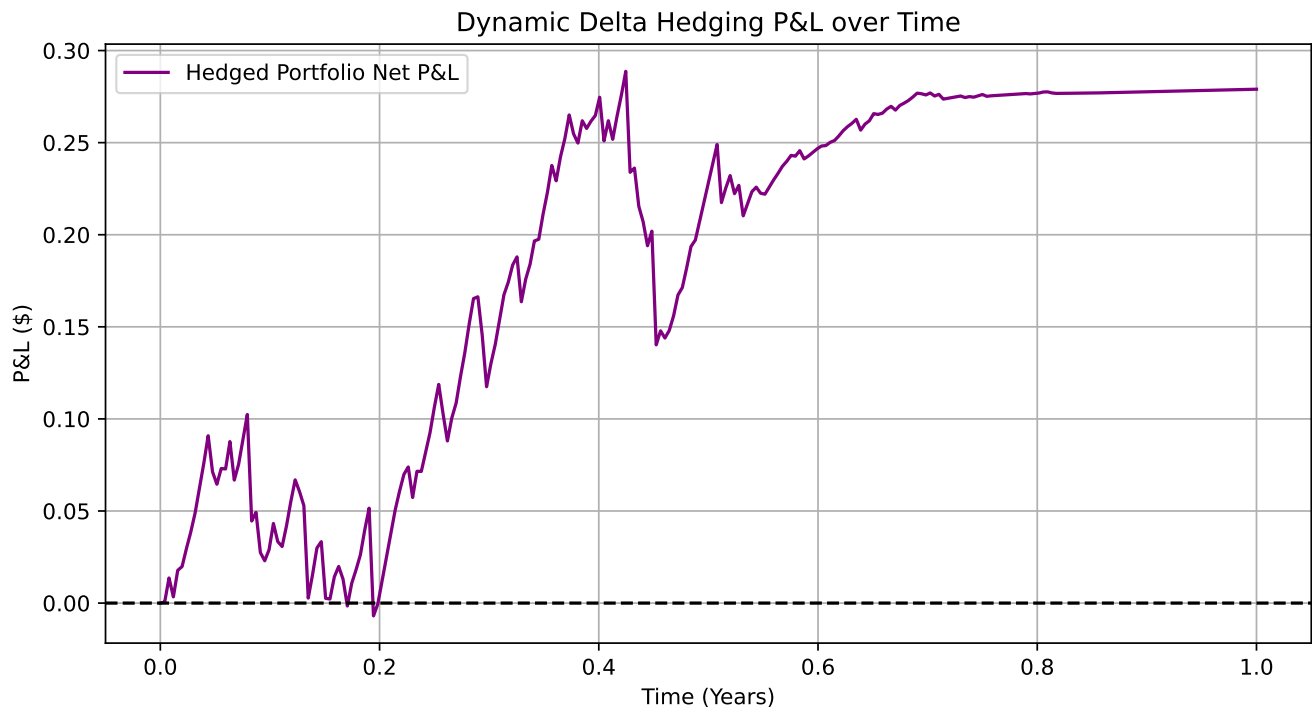
# Rebalancing cost: buying (new_delta - old_delta) shares
cash[i] -= (new_delta - delta) * S[i]
delta = new_delta

# Track portfolio value (Cash + Shares - Option Liability)
opt_val = exact_bs_call(S[i], K, max(tau, 0.0), r, sigma)
portfolio_value[i] = cash[i] + delta * S[i] - opt_val

plt.figure(figsize=(10, 5))
plt.plot(t, portfolio_value, label='Hedged Portfolio Net P&L', color='purple')
plt.axhline(0, color='black', linestyle='--')
plt.title('Dynamic Delta Hedging P&L over Time')
plt.xlabel('Time (Years)')
plt.ylabel('P&L ($)')
plt.legend()
plt.grid(True)
plt.show()

```

```
simulate_delta_hedging(100, 100, 1.0, 0.05, 0.20, mu=0.08)
```



5.2 The Feynman–Kac Theorem: Bridging Probability and PDEs

Throughout this lecture, we have approached the valuation of derivatives from two conceptually distinct perspectives:

1. **The Probabilistic (Risk-Neutral) Approach:** Pricing via the discounted expected payoff under the equivalent martingale measure $\mathbb{Q}$:

$$V(t, S_t) = \mathbb{E}^{\mathbb{Q}} \left[e^{-r(T-t)} h(S_T) \mid \mathcal{F}_t \right].$$

2. **The Deterministic (Replication) Approach:** Pricing via the construction of a risk-free hedged portfolio, which yields a parabolic partial differential equation (PDE):

$$\frac{\partial V}{\partial t} + \mathcal{L}V - rV = 0,$$

subject to the terminal boundary condition $V(T, S) = h(S)$.

The profound **Feynman–Kac Theorem** establishes that these two formulations are not merely analogous, but mathematically identical.

5.2.1 General Statement of the Feynman–Kac Theorem

Let X_s be an Itô diffusion process solving the stochastic differential equation (SDE) under a probability measure $\mathbb{Q}$ for $s \in [t, T]$:

$$dX_s = \mu(X_s, s) ds + \sigma(X_s, s) dW_s,$$

and define the associated infinitesimal generator $\mathcal{L}$ acting on a sufficiently smooth function $V(x, t)$:

$$\mathcal{L}V(x, t) = \mu(x, t) \frac{\partial V}{\partial x} + \frac{1}{2} \sigma^2(x, t) \frac{\partial^2 V}{\partial x^2}.$$

Consider the backward Cauchy problem (PDE) subject to a terminal condition $h(x)$ and a continuous discounting rate $r(x, t)$:

$$\frac{\partial V}{\partial t} + \mathcal{L}V - r(x, t)V = 0, \quad \text{with } V(T, x) = h(x).$$

Theorem 5.1. *Then, provided that h , μ , σ , and r satisfy sufficient regularity and growth conditions, the unique solution to the PDE admits the stochastic representation*

$$V(t, x) = \mathbb{E}^{\mathbb{Q}} \left[\exp \left(- \int_t^T r(X_s, s) ds \right) h(X_T) \mid X_t = x \right].$$

5.2.2 Application to the Black–Scholes Model

Under the risk-neutral measure $\mathbb{Q}$, the underlying asset follows a Geometric Brownian Motion:

$$dS_t = rS_t dt + \sigma S_t dW_t.$$

The infinitesimal generator $\mathcal{L}$ for this process is:

$$\mathcal{L}V = rS \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2}.$$

Substituting this into the Feynman-Kac PDE yields the exact Black–Scholes equation:

$$\frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV = 0, \quad V(T, S) = h(S).$$

By Theorem 5.1, the solution is immediately given by the conditional expectation:

$$V(t, S) = \mathbb{E}^{\mathbb{Q}} [e^{-r(T-t)} h(S_T) \mid S_t = S].$$

Thus, the analytical PDE solution and the probabilistic risk-neutral expectation are perfectly reconciled.

5.2.3 Why This Matters for Computational Finance

The Feynman-Kac theorem provides a fundamental *computational duality*. To price a complex derivative, one can either: 1. Solve a macroscopic **PDE** backward in time. 2. Compute an expectation by simulating microscopic **stochastic paths** forward in time.

Because these are mathematically identical representations of the same price, the choice of numerical methodology is dictated entirely by the characteristics of the specific derivative and the computational constraints:

Derivative Characteristics	Preferred Mathematical Framework	Standard Computational Technique
High Dimensions (e.g., Basket Options, Multi-Asset)	Probabilistic (Avoids the curse of dimensionality)	Monte Carlo Simulation
Strong Path Dependence (e.g., Asian, Lookback Options)	Probabilistic	Monte Carlo Simulation
Early Exercise Features (e.g., American Options)	PDE / Variational Inequality	Finite Difference / Binomial Trees
Complex Trigger Boundaries (e.g., Barrier Options)	PDE	Finite Difference Methods

5.3 The Greeks: Sensitivities in Option Pricing

In quantitative finance, the **Greeks** represent the partial derivatives of an option's theoretical price with respect to various model parameters and market variables.

These sensitivities are the bedrock of modern **risk management** and **dynamic hedging**. They allow traders to linearly decompose the complex, non-linear risk of an option portfolio into isolated, manageable risk factors.

5.3.1 The Standard First- and Second-Order Greeks

Greek	Symbol	Mathematical Definition	Financial Interpretation
Delta	Δ	$\frac{\partial V}{\partial S}$	Directional risk. The expected change in option price for a \$1 change in the underlying asset.
Gamma	Γ	$\frac{\partial^2 V}{\partial S^2} = \frac{\partial \Delta}{\partial S}$	Convexity risk. The rate of change of Delta; dictates how frequently a portfolio must be re-hedged.
Theta	Θ	$\frac{\partial V}{\partial t}$	Time decay. The rate at which the option loses value as maturity approaches, all else being equal.
Vega	ν	$\frac{\partial V}{\partial \sigma}$	Volatility risk. The sensitivity of the option's value to a unit change in volatility; desks often rescale it to a 1-vol-point move.

Greek	Symbol	Mathematical Definition	Financial Interpretation
Rho	ρ	$\frac{\partial V}{\partial r}$	Interest rate risk. The sensitivity to a unit change in the risk-free rate; market convention often reports the effect of a 1 bp or 1% move after rescaling.

(Note: “Vega” is not a letter in the Greek alphabet, but is universally referred to as a Greek in finance. In these notes we denote it by ν ; some texts instead write $\mathcal{V}$.)

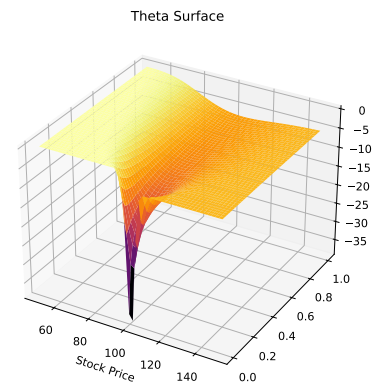
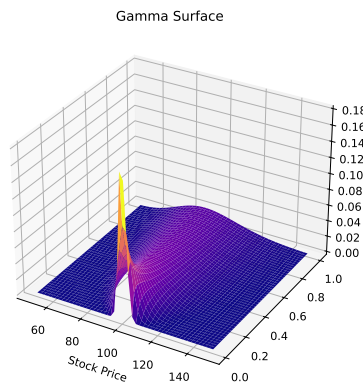
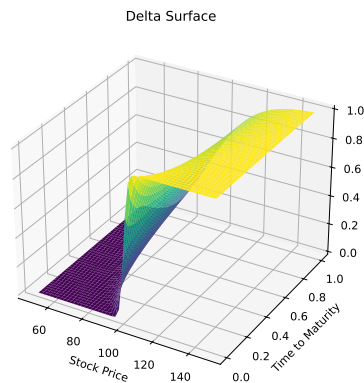
5.3.2 Closed-Form Greeks for a European Call (Black-Scholes)

By differentiating the Black-Scholes closed-form formula, we obtain explicit analytical expressions for the Greeks. Let $\tau = T - t$ denote the time to maturity, $\Phi(\cdot)$ the cumulative distribution function (CDF) of the standard normal distribution, and $\phi(\cdot)$ its probability density function (PDF):

- **Delta:** $\Delta = \Phi(d_1)$
- **Gamma:** $\Gamma = \frac{\phi(d_1)}{S\sigma\sqrt{\tau}}$
- **Theta:** $\Theta = -\frac{S\phi(d_1)\sigma}{2\sqrt{\tau}} - rKe^{-r\tau}\Phi(d_2)$
- **Vega:** $\nu = S\phi(d_1)\sqrt{\tau}$
- **Rho:** $\rho = K\tau e^{-r\tau}\Phi(d_2)$

Recall the standardized variables from the pricing formula:

$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \quad d_2 = d_1 - \sigma\sqrt{\tau}.$$



5.4 Summary and Connection to the Next Lectures

This lecture has developed two complementary views of the same pricing problem:

- **The replication view** produces the Black–Scholes PDE by constructing a locally risk-free hedged portfolio. The drift μ drops out because it appears symmetrically in the option and in the delta-hedge, leaving only the risk-free rate and the volatility.
- **The probabilistic view** (Feynman–Kac) identifies the PDE solution with the risk-neutral conditional expectation, confirming that both approaches are mathematically equivalent.
- **The Greeks** are the partial derivatives of the pricing function and serve as the fundamental risk measures in any options trading book. At leading order, the Gamma-Theta trade-off implies $\mathbb{E}[\delta\Pi] \approx \frac{1}{2}\Gamma S^2(\sigma_{\text{imp}}^2 - \sigma_{\text{real}}^2)\delta t$, capturing the core economics of delta-hedged option positions.

The next lecture moves from analytic methods to simulation. When payoffs depend on many risk factors or on the entire price path, closed-form formulas are not available, and Monte Carlo becomes the primary computational tool. The connection between the PDE and the simulation viewpoint established by Feynman–Kac will be essential for understanding why both approaches target the same quantity.

6 Monte Carlo Methods in Finance

Closed-form formulas and PDE methods are powerful, but they become restrictive once payoffs depend on many sources of randomness or on the full path of the underlying. This lecture shifts the storyline from analytic pricing to simulation-based pricing.

6.1 Monte Carlo Pricing as Expectation Estimation

The central question is how to compute the risk-neutral price $\pi_0(H, \mathbf{S})$ when the distribution of the underlying is unknown in closed form or too complex for direct analysis.

Assume in the following that the payoff $H(\mathbf{S})$ is integrable with respect to the risk-neutral measure $\mathbb{Q}$, i.e., $\mathbb{E}_{\mathbb{Q}}[|H(\mathbf{S})|] < \infty$. By generating a sequence $\mathbf{S}^{(1)}, \mathbf{S}^{(2)}, \dots$ of independent and identically distributed realizations of the asset path $\mathbf{S}$, the Strong Law of Large Numbers (SLLN) assures us that the sample average converges *almost surely* (a.s.) to the true expectation:

$$\frac{1}{N} \sum_{i=1}^N H(\mathbf{S}^{(i)}) \xrightarrow{\text{a.s.}} \mathbb{E}_{\mathbb{Q}}[H(\mathbf{S})], \quad \text{as } N \rightarrow \infty.$$

This convergence is the basis of the **Monte Carlo method** for derivative pricing. Once the expected payoff has been estimated by simulation, the present value is obtained by applying the appropriate discount factor.

6.1.1 Essential Monte Carlo Method for Option Pricing

In practice the method has four steps:

1. **Simulate Price Paths:** Generate sample paths $\{(S_t(\omega_i))_{t \in [0, T]} : i = 1, \dots, N\}$ for the underlying asset over the interval $[0, T]$ according to the chosen stochastic price model under $\mathbb{Q}$.
2. **Calculate Payoff:** At the end of each sampled path, calculate the option's payoff. For example:
 - **Plain Vanilla Call Option:** $H(S_T) = (S_T - K)^+$
 - **Discrete Asian Call Option:** $H((S_t)_{t \in [0, T]}) = \left(\frac{1}{M} \sum_{j=1}^M S_{t_j} - K \right)^+$ for fixed observation dates $t_1, \dots, t_M$.

3. **Average the Payoffs:** Compute the arithmetic mean of all simulated payoffs to estimate the expected future payoff:

$$\bar{H}_N = \frac{1}{N} \sum_{i=1}^N H(\mathbf{S}(\omega_i))$$

4. **Discount to Present Value:** Calculate the option's present value $\hat{\pi}_0$ by discounting the expected payoff using the risk-free rate r :

$$\hat{\pi}_0 = e^{-rT} \frac{1}{N} \sum_{i=1}^N H(\mathbf{S}(\omega_i))$$

6.2 Basics of Monte Carlo Methods for Derivative Pricing

We now analyze the error of the Monte Carlo estimator. *For simplicity, we absorb the discount factor into the payoff function $H(\mathbf{S})$, so the true price is $\pi_0 = \mathbb{E}_{\mathbb{Q}}[H(\mathbf{S})]$.*

6.2.1 Monte Carlo Integration Error

We define the Monte Carlo integration error ε_N for N samples as:

$$\varepsilon_N := \varepsilon_N(H, \mathbf{S}) := \hat{\pi}_0^N(H, \mathbf{S}) - \pi_0(H, \mathbf{S}),$$

where $\hat{\pi}_0^N(H, \mathbf{S})$ denotes the Monte Carlo estimator:

$$\hat{\pi}_0^N(H, \mathbf{S}) := \frac{1}{N} \sum_{i=1}^N H(\mathbf{S}^{(i)}).$$

Because the samples are drawn directly from the target distribution under $\mathbb{Q}$, $\hat{\pi}_0^N(H, \mathbf{S})$ is an **unbiased estimator** for $\pi_0(H, \mathbf{S})$. Therefore:

$$\mathbb{E}_{\mathbb{Q}}[\hat{\pi}_0^N(H, \mathbf{S})] = \pi_0(H, \mathbf{S}), \quad \text{and thus} \quad \mathbb{E}_{\mathbb{Q}}[\varepsilon_N] = 0.$$

To quantify the magnitude of the error, we introduce the **Mean Square Error (MSE)**:

$$\text{MSE}(\hat{\pi}_0^N) := \mathbb{E}_{\mathbb{Q}}[\varepsilon_N(H, \mathbf{S})^2],$$

and its square root, the **Root Mean Square Error (RMSE)**:

$$\text{RMSE}(\hat{\pi}_0^N) := \sqrt{\mathbb{E}_{\mathbb{Q}}[\varepsilon_N(H, \mathbf{S})^2]}.$$

Proposition 6.1. Let $\sigma_H := \sqrt{\text{Var}(H(\mathbf{S}))} < \infty$ denote the standard deviation of the discounted payoff $H(\mathbf{S})$. Then the root mean square error satisfies

$$\text{RMSE}(\hat{\pi}_0^N) = \frac{\sigma_H}{\sqrt{N}}.$$

Moreover, the scaled error $\sqrt{N} \cdot \varepsilon_N$ is asymptotically normally distributed with variance σ_H^2 . Equivalently, for any real numbers $a < b$, the probability that the actual error falls within a specific confidence bound is given by

$$\lim_{N \rightarrow \infty} \mathbb{Q} \left(a \frac{\sigma_H}{\sqrt{N}} \leq \varepsilon_N \leq b \frac{\sigma_H}{\sqrt{N}} \right) = \Phi(b) - \Phi(a),$$

where Φ denotes the cumulative distribution function of the standard normal distribution $\mathcal{N}(0, 1)$.

Proof. Using the independence of the simulated paths $\mathbf{S}^{(i)}$ and the fact that $\hat{\pi}_0^N(H, \mathbf{S})$ is an unbiased estimator, the variance of the sample mean is:

$$\mathbb{E}_{\mathbb{Q}}[\varepsilon_N^2] = \text{Var} \left(\frac{1}{N} \sum_{i=1}^N H(\mathbf{S}^{(i)}) \right) = \frac{1}{N^2} \sum_{i=1}^N \text{Var}(H(\mathbf{S}^{(i)})) = \frac{N\sigma_H^2}{N^2} = \frac{\sigma_H^2}{N}.$$

Taking the square root yields $\text{RMSE} = \sigma_H / \sqrt{N}$.

Furthermore, by the Central Limit Theorem (CLT), the sum of independent, identically distributed random variables with finite variance converges in distribution to a normal random variable:

$$\frac{1}{\sigma_H \sqrt{N}} \sum_{i=1}^N \left(H(\mathbf{S}^{(i)}) - \pi_0(H, \mathbf{S}) \right) \xrightarrow{d} \mathcal{N}(0, 1), \quad \text{as } N \rightarrow \infty.$$

Rewriting the left side in terms of ε_N , we get $\frac{\sqrt{N}}{\sigma_H} \varepsilon_N \xrightarrow{d} \mathcal{N}(0, 1)$. Evaluating the probability limits of this normal distribution yields the stated result. ■

Implications of Proposition 6.1

Proposition 6.1 highlights two basic facts about Monte Carlo error:

1. **Probabilistic Nature of Error:** The simulation error lacks a strict deterministic bound. For any given simulation run, the exact error is random. However, the CLT allows us to construct precise asymptotic confidence intervals (e.g., we can be roughly 95% confident that the true price lies within $\hat{\pi}_0^N \pm 1.96 \frac{\sigma_H}{\sqrt{N}}$).
2. **Convergence Rate:** The typical error (RMSE) diminishes at the rate of $\mathcal{O}(N^{-1/2})$. To improve the error by a factor of 10, the sample size must therefore increase by a factor of 100.

6.2.2 High-Dimensionality and Monte Carlo Integration

To illustrate the advantage of Monte Carlo methods in high-dimensional settings, consider a simplified scenario where the underlying risk factors $\mathbf{S}$ are represented by a d -dimensional uniform random variable. In this case, the fair price of a European-style derivative is given by the volume under the payoff function H :

$$\pi_0(H, \mathbf{S}) = \int_{[0,1]^d} H(x) dx.$$

Traditional numerical integration (quadrature) methods typically rely on a deterministic grid defined by $0 \leq x_1 < x_2 < \dots < x_N \leq 1$ along each dimension. For a d -dimensional space, this grid expands by the Cartesian product to $\{x_1, \dots, x_N\}^d$, encompassing $n = N^d$ evaluation points. The integral approximation is derived by evaluating the function H at these grid points and interpolating between them (e.g., via the Trapezoidal rule or Simpson's rule).

For a numerical integration method of order k , the approximation error scales as $\mathcal{O}((1/N)^k)$. However, the computational effort is driven by the total number of points n . Expressing the error in terms of computational complexity n , the accuracy is proportional to $n^{-k/d}$. Consequently, the effective rate of convergence rapidly degrades as the dimensionality d increases. This exponential explosion in computational cost required to maintain a given level of accuracy is famously known as the **curse of dimensionality**.

In contrast, the Monte Carlo method's convergence rate of $\mathcal{O}(n^{-1/2})$ is strictly *independent* of the dimension d , making it one of the few robust general-purpose techniques for pricing complex, multi-asset, or path-dependent derivatives.

6.2.3 Confidence Intervals of the Monte Carlo Method

Because the error is random, we also need a probabilistic way to control it. The practical question is: how large must N be to ensure that the probability of exceeding a tolerance level $\varepsilon > 0$ is at most some threshold $\delta > 0$?

Mathematically, we seek an N such that:

$$\mathbb{Q}(|\varepsilon_N(H, \mathbf{S})| > \varepsilon) \leq \delta.$$

This inequality forms the foundation for rigorous error management in Monte Carlo frameworks.

The solution is derived from the Central Limit Theorem (CLT) discussed in Proposition 6.1. Because the scaled error converges in distribution to a standard normal, we can approximate the finite-sample error probability. For large N , we have:

$$\mathbb{Q}(|\varepsilon_N| > \varepsilon) = 1 - \mathbb{Q}\left(-\frac{\varepsilon}{\sigma_H/\sqrt{N}} \leq \frac{\varepsilon_N}{\sigma_H/\sqrt{N}} \leq \frac{\varepsilon}{\sigma_H/\sqrt{N}}\right) \approx 2 - 2\Phi\left(\frac{\varepsilon\sqrt{N}}{\sigma_H}\right),$$

where $\Phi(\cdot)$ is the cumulative distribution function of the standard normal distribution.

Note: Because the Monte Carlo error is only asymptotically normal, this equation is an approximation (indicated by “ $\approx$ ”) that becomes exact as $N \rightarrow \infty$.

To align the error probability with our desired threshold δ , we set the approximated probability equal to δ and solve for N :

$$2 - 2\Phi\left(\frac{\varepsilon\sqrt{N}}{\sigma_H}\right) \leq \delta \quad \Rightarrow \quad \Phi\left(\frac{\varepsilon\sqrt{N}}{\sigma_H}\right) \geq 1 - \frac{\delta}{2}.$$

Applying the inverse standard normal CDF, $\Phi^{-1}(\cdot)$, yields the required sample size:

$$N \geq \left(\Phi^{-1}\left(1 - \frac{\delta}{2}\right)\right)^2 \frac{\sigma_H^2}{\varepsilon^2}.$$

This reveals the required computational budget N scales inversely with the square of the tolerance (ε^2) and linearly with the variance of the payoff (σ_H^2).

6.2.4 Remark: Challenges in Variance Estimation

The preceding asymptotic analysis assumes that the true variance $\sigma_H^2 = \text{Var}(H(\mathbf{S}))$ is known. In practice it is not, so we replace σ_H by the unbiased **sample standard deviation** $\hat{\sigma}_N$:

$$\hat{\sigma}_N = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (H(\mathbf{S}^{(i)}) - \hat{\pi}_0^N(H, \mathbf{S}))^2}.$$

By Slutsky's theorem, replacing σ_H by the consistent estimator $\hat{\sigma}_N$ preserves asymptotic normality. Still, the resulting confidence intervals are only asymptotic and may be less reliable for small N or very skewed payoffs.

For that reason, one should not report only the point estimate. Standard practice is to report the estimator together with an asymptotic **confidence interval**:

$$\text{CI}_{1-\delta} = \left[\hat{\pi}_0^N - z_{1-\delta/2} \frac{\hat{\sigma}_N}{\sqrt{N}}, \hat{\pi}_0^N + z_{1-\delta/2} \frac{\hat{\sigma}_N}{\sqrt{N}} \right],$$

where $z_{1-\delta/2} = \Phi^{-1}(1 - \delta/2)$ is the critical value (e.g., 1.96 for a 95% confidence level). This provides a comprehensive, rigorous measure of the estimator's accuracy and the computational confidence in the derived derivative price.

6.2.5 Asymptotic Confidence Intervals

Definition 6.1. Let $Z \sim \mathcal{N}(0, 1)$ be a standard normal random variable, and consider a significance level $\alpha \in (0, 1)$. The critical value $z_{1-\frac{\alpha}{2}}$ is defined such that the probability of Z falling within the symmetric interval $(-z_{1-\frac{\alpha}{2}}, z_{1-\frac{\alpha}{2}})$ is exactly $1 - \alpha$:

$$\mathbb{P}(|Z| \leq z_{1-\frac{\alpha}{2}}) = 1 - \alpha.$$

The critical number $z_{1-\frac{\alpha}{2}}$ for a given confidence level $(1 - \alpha)$ is computed directly using the inverse cumulative distribution function (quantile function) of the standard normal distribution:

$$z_{1-\frac{\alpha}{2}} = \Phi^{-1}\left(1 - \frac{\alpha}{2}\right).$$

Commonly used critical values in quantitative finance include: * For $\alpha = 0.05$ (95% confidence level): $z_{0.975} \approx 1.96$ * For $\alpha = 0.01$ (99% confidence level): $z_{0.995} \approx 2.58$

We can leverage the asymptotic normality of the Monte Carlo error established in Proposition 6.1 to derive practical confidence intervals for our estimator $\hat{\pi}_0^N(H, \mathbf{S})$. Denoting the true standard deviation of the discounted payoff as σ_H , the Central Limit Theorem tells us that for large N :

$$\begin{aligned} 1 - \alpha &\approx \mathbb{Q}\left(-z_{1-\frac{\alpha}{2}} \leq \frac{\hat{\pi}_0^N - \pi_0}{\sigma_H/\sqrt{N}} \leq z_{1-\frac{\alpha}{2}}\right) \\ &= \mathbb{Q}\left(-\frac{\sigma_H z_{1-\frac{\alpha}{2}}}{\sqrt{N}} \leq \hat{\pi}_0^N - \pi_0 \leq \frac{\sigma_H z_{1-\frac{\alpha}{2}}}{\sqrt{N}}\right) \\ &= \mathbb{Q}\left(\hat{\pi}_0^N - \frac{\sigma_H z_{1-\frac{\alpha}{2}}}{\sqrt{N}} \leq \pi_0 \leq \hat{\pi}_0^N + \frac{\sigma_H z_{1-\frac{\alpha}{2}}}{\sqrt{N}}\right). \end{aligned}$$

Thus, the $(1 - \alpha)$ -level **confidence interval** for the true option price π_0 is given by:

$$\text{CI}_{1-\alpha}(\hat{\pi}_0^N) = \left[\hat{\pi}_0^N - \frac{\sigma_H z_{1-\frac{\alpha}{2}}}{\sqrt{N}}, \hat{\pi}_0^N + \frac{\sigma_H z_{1-\frac{\alpha}{2}}}{\sqrt{N}} \right].$$

6.2.6 Example: Monte Carlo Simulation in the Black–Scholes Model

We now apply this framework to the classical Black–Scholes model. Consider the underlying asset dynamics

$$\begin{cases} dS_t &= rS_t dt + \sigma S_t dW_t, \\ S_0 &= s_0 \in \mathbb{R}^+. \end{cases}$$

where W_t is a standard Brownian motion under the risk-neutral martingale measure $\mathbb{Q}$, r is the risk-free rate, and σ is the asset's return volatility.

We aim to compute the time-0 price π_0 of a European call option with payoff $H(S_T) = (S_T - K)^+$, together with its 95% and 99% confidence intervals.

Algorithm: 1. Simulate Exact Terminal Values: Because Geometric Brownian Motion has a known exact terminal distribution, we avoid time-discretization errors (like Euler-Maruyama) and directly simulate N independent terminal asset prices:

$$S_T^{(i)} = s_0 \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) T + \sigma \sqrt{T} Z^{(i)} \right), \quad \text{for } i = 1, \dots, N$$

where $Z^{(i)} \sim \mathcal{N}(0, 1)$ are independent standard normal draws.

2. Compute Discounted Payoffs: For each simulated path, calculate the present value of the payoff:

$$Y^{(i)} = e^{-rT} (S_T^{(i)} - K)^+$$

3. Estimate Price and Variance:

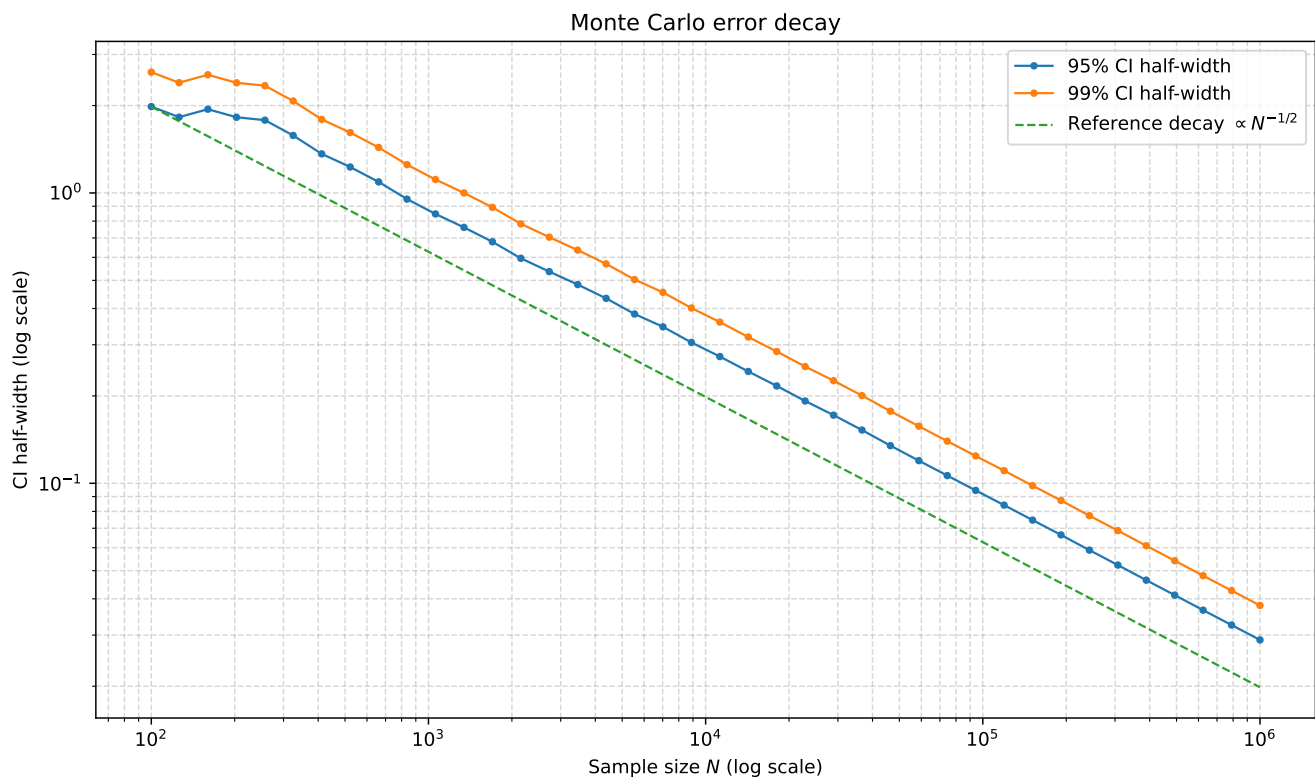
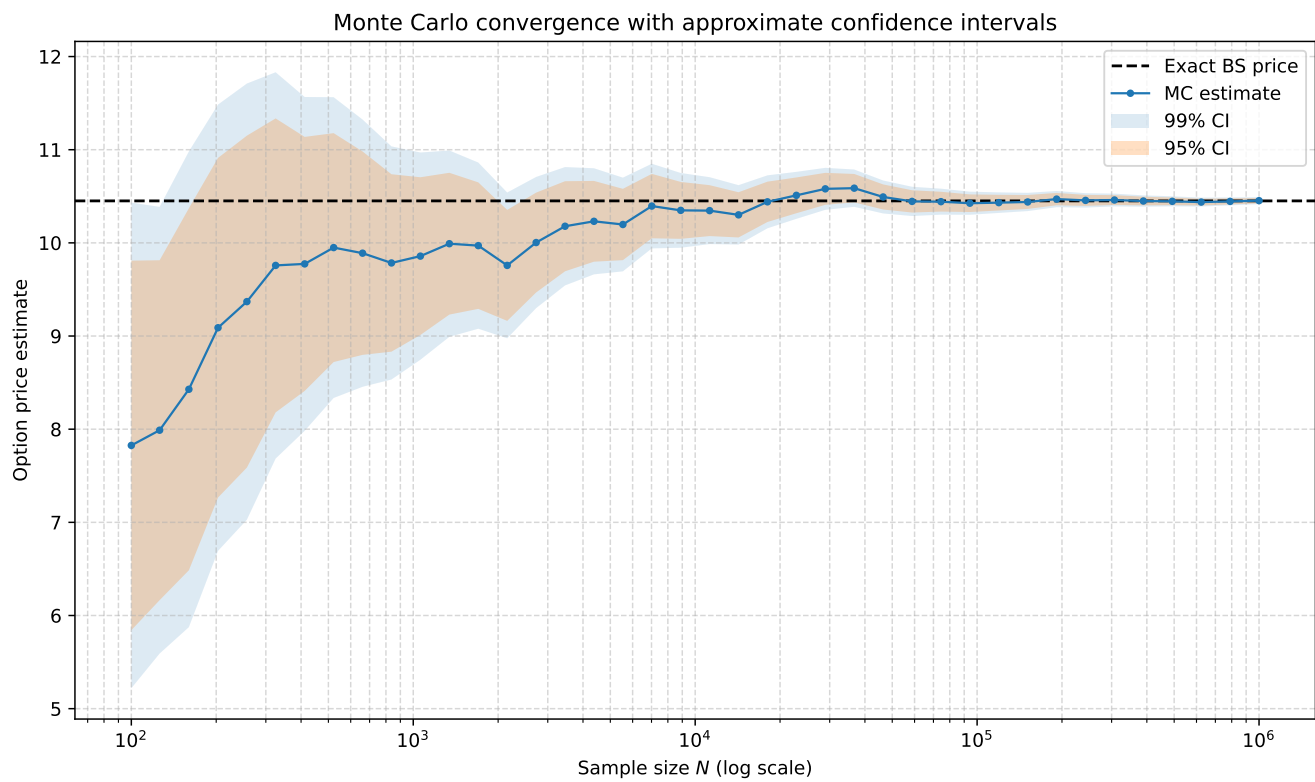
- Compute the Monte Carlo price estimate: $\hat{\pi}_0^N = \frac{1}{N} \sum_{i=1}^N Y^{(i)}$
- Compute the sample standard deviation: $\hat{\sigma}_H = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (Y^{(i)} - \hat{\pi}_0^N)^2}$

4. Construct Confidence Intervals:

- **95% CI:** $\left[\hat{\pi}_0^N - 1.96 \frac{\hat{\sigma}_H}{\sqrt{N}}, \hat{\pi}_0^N + 1.96 \frac{\hat{\sigma}_H}{\sqrt{N}} \right]$
- **99% CI:** $\left[\hat{\pi}_0^N - 2.58 \frac{\hat{\sigma}_H}{\sqrt{N}}, \hat{\pi}_0^N + 2.58 \frac{\hat{\sigma}_H}{\sqrt{N}} \right]$

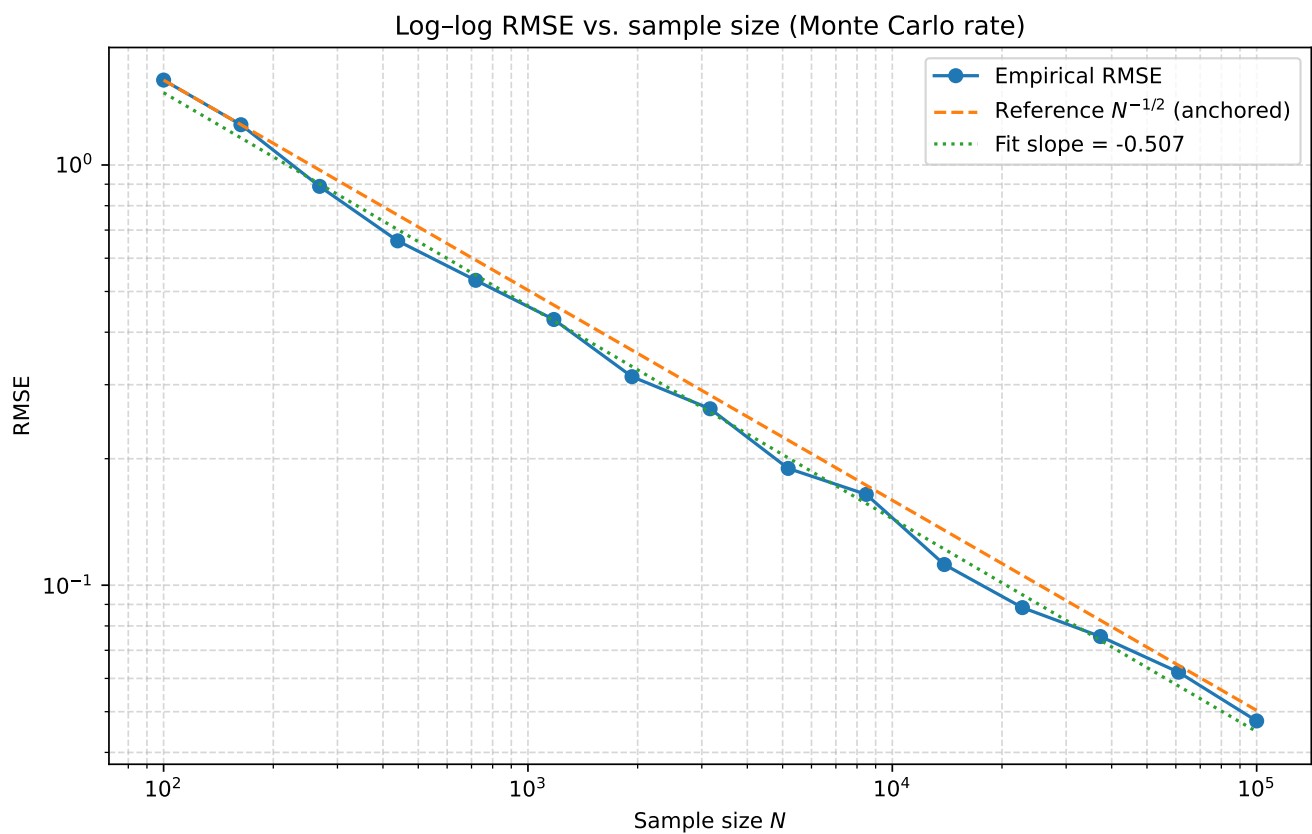
Because the exact Black-Scholes price can be calculated via the closed-form analytical formula, this example serves as a standard benchmark to empirically verify that, across repeated large-sample simulations, the true price is covered by the nominal $(1 - \alpha)\%$ confidence intervals at approximately the advertised frequency.

Exact Black-Scholes price: 10.450584



Final sample size: $N = 1000000$
 Monte Carlo estimate: 10.453195
 Estimated standard error: 0.014731
 Approx. 95% CI: [10.424322, 10.482068]
 Approx. 99% CI: [10.415250, 10.491141]
 Exact Black-Scholes price: 10.450584

Running Experiment 2: Empirical RMSE Convergence Rate...
 Empirical convergence slope: -0.5069 (theory: -0.5000)



```

(array([ 100,   163,   268,   439,   719,  1178,  1930,  3162,
        5179,  8483, 13894, 22758, 37275, 61054, 100000]),
 array([1.59248892, 1.24684983, 0.88942298, 0.65982195, 0.53155246,
        0.42887372, 0.31349766, 0.26294124, 0.18962492, 0.16446242,
        0.11197823, 0.08843807, 0.07547747, 0.06208433, 0.04752514])),
 np.float64(-0.5068637857029131))

```


6.3 Summary and Connection to the Next Lecture

Monte Carlo pricing rests on three pillars established in this lecture:

1. **The SLLN** guarantees almost-sure convergence of the sample average to the true expectation.
2. **The CLT** provides the $\mathcal{O}(N^{-1/2})$ RMSE rate and enables asymptotic confidence intervals.
3. **Dimension-independence** makes Monte Carlo the method of choice for high-dimensional and path-dependent payoffs where PDE methods suffer from the curse of dimensionality.

The critical remaining question is: how do we generate the paths $(S_t)_{t \in [0, T]}$ when no exact simulation formula is available? In the Black–Scholes model we can simulate terminal values exactly via the log-normal formula. In more complex models — such as Heston with stochastic volatility — the continuous SDE must be approximated by a discrete scheme. The next lecture develops exactly this: Euler-Maruyama and Milstein discretization schemes, their strong and weak convergence orders, and their application to the Heston model.

7 Discretization Schemes for SDEs

The previous lecture showed how Monte Carlo handles high-dimensional expectations once we can simulate the relevant risk factors. This lecture addresses the missing step in that workflow: how to approximate continuous-time dynamics when exact simulation is unavailable.

We begin with a one-dimensional Itô SDE for an asset price process S :

$$dS_t = \mu(t, S_t) dt + \sigma(t, S_t) dW_t, \quad S_0 > 0,$$

where:

- $\mu(t, S_t)$ is the drift coefficient, representing deterministic trends.
- $\sigma(t, S_t)$ is the volatility coefficient, representing the randomness.
- $(W_t)_{t \geq 0}$ is a standard Brownian motion.

Unless a measure superscript is written explicitly, the Brownian motion in this lecture is understood under the measure relevant for the application at hand; for pricing examples this is typically the risk-neutral measure $\mathbb{Q}$.

We simulate $(S_t)_{t \in [0, T]}$ over a discrete grid

$$0 = t_0 < t_1 < \dots < t_N = T,$$

and, for most formulas below, we assume equal step size $\Delta t = t_i - t_{i-1}$.

Remark: Equally-spaced time increments are primarily used for notational convenience, because it allows us to write $t_i - t_{i-1}$ as simply Δt . All the results derived with equally-spaced increments are easily generalized to unequal spacing.

7.1 When Do We Need to Discretize?

In derivative pricing, we typically want to evaluate the expected discounted payoff under the risk-neutral measure:

$$\pi_0 = e^{-rT} \mathbb{E}^{\mathbb{Q}} \left[h \left((S_t)_{t \in [0, T]} \right) \right].$$

The need to **discretize** the stochastic differential equation (SDE) for $(S_t)_{t \in [0, T]}$ arises whenever we need a path simulation but no exact simulation method is available. A first important case is:

Exotic and Path-Dependent Payoffs

If the payoff h depends on the **entire path** of the asset price — not just the terminal value S_T — then we must approximate/sample the full trajectory of S_t . Examples include:

- **Asian Options** (depend on the average of S_t),
- **Barrier Options** (depend on whether S_t crosses a certain level),
- **Lookback Options**, and many other exotic derivatives.

In many practically relevant models, the law of the full path $(S_t)_{t \in [0, T]}$ is not available in a form that can be sampled exactly, making simulation via discretization (e.g. Euler–Maruyama) necessary.

7.2 Generation of Sample Paths for Brownian Motions

Simulating Brownian paths is the starting point for numerically solving SDEs and pricing path-dependent derivatives. Because a continuous-time Brownian path $(W_t)_{t \in [0, T]}$ is infinite-dimensional, a computer can only generate a finite-dimensional **skeleton** $(W_{t_1}, \dots, W_{t_N})$ on a discrete time grid $0 = t_0 < t_1 < \dots < t_N = T$.

We begin with the one-dimensional case. The same ideas extend directly to several dimensions.

Example 7.1. Once we have generated a Brownian skeleton, it can be used directly. For instance, in the Black-Scholes framework, the stock price process S follows a Geometric Brownian Motion. Because its SDE has an exact analytical solution, we can perfectly simulate the asset price at any time t using the sampled Brownian motion W_t :

$$S_t = S_0 \exp \left(\left(\mu - \frac{\sigma^2}{2} \right) t + \sigma W_t \right).$$

7.2.1 Method 1: Discretization via Random Walk (Sequential Construction)

The most intuitive and computationally efficient construction is sequential, using the fact that Brownian increments are independent and normally distributed.

Objective: Generate sampled values $W_{t_1}, W_{t_2}, \dots, W_{t_N}$ of a Brownian motion at specified time points $0 = t_0 < t_1 < \dots < t_N = T$.

1. Initialization:

- Set $W_{t_0} = 0$, as a standard Brownian motion always starts at zero.
- Calculate the time increments $\Delta t_i = t_i - t_{i-1}$ for $i = 1, 2, \dots, N$.

2. Generate Independent Shocks:

- Draw an N -dimensional vector of independent standard normal random variables $X = (X_1, X_2, \dots, X_N)$ where $X_i \sim \mathcal{N}(0, 1)$.

3. Construct the Path Sequentially:

- For each step $i = 1$ to N , compute the Brownian increment: $\Delta W_i = \sqrt{\Delta t_i} X_i$.
 - Update the path: $W_{t_i} = W_{t_{i-1}} + \Delta W_i$.
4. **Output:** Return the generated skeleton $(W_{t_1}, \dots, W_{t_N})$.

7.2.2 Method 2: Sampling via Covariance and Cholesky Factorization

An alternative, global construction uses the joint distribution of the full skeleton. By definition,

$$(W_{t_1}, \dots, W_{t_N}) \sim \mathcal{N}(\mathbf{0}, \Sigma)$$

where the covariance matrix Σ is defined by $\Sigma_{i,j} = \min(t_i, t_j)$ for $1 \leq i, j \leq N$.

To simulate a vector with this specific covariance structure, we can generate a vector of independent standard normal variables $X = (X_1, \dots, X_N)^T$ and apply a linear transformation A , such that our simulated path is $W = AX$. For this to hold, the matrix A must satisfy $AA^T = \Sigma$.

We find A using the **Cholesky decomposition**, which yields a lower triangular matrix. Because a Brownian increment at time t_j permanently affects all future values W_{t_i} for $i \geq j$, the Cholesky factor A takes a highly structured, cumulative form:

$$A = \begin{pmatrix} \sqrt{t_1} & 0 & 0 & \dots & 0 \\ \sqrt{t_1} & \sqrt{t_2 - t_1} & 0 & \dots & 0 \\ \sqrt{t_1} & \sqrt{t_2 - t_1} & \sqrt{t_3 - t_2} & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sqrt{t_1} & \sqrt{t_2 - t_1} & \sqrt{t_3 - t_2} & \dots & \sqrt{t_N - t_{N-1}} \end{pmatrix}$$

Multiplying this lower triangular matrix A by the independent normal vector X reproduces the sequential random-walk construction and embeds the correct correlation structure across the whole time horizon.

Note: while mathematically elegant, the Cholesky method requires $\mathcal{O}(N^2)$ operations for the matrix-vector multiplication. The sequential random-walk method only requires $\mathcal{O}(N)$ operations and is therefore preferred in practice unless the covariance structure is modified.

7.2.3 Python Implementation: Brownian Motion via Cholesky Factorization

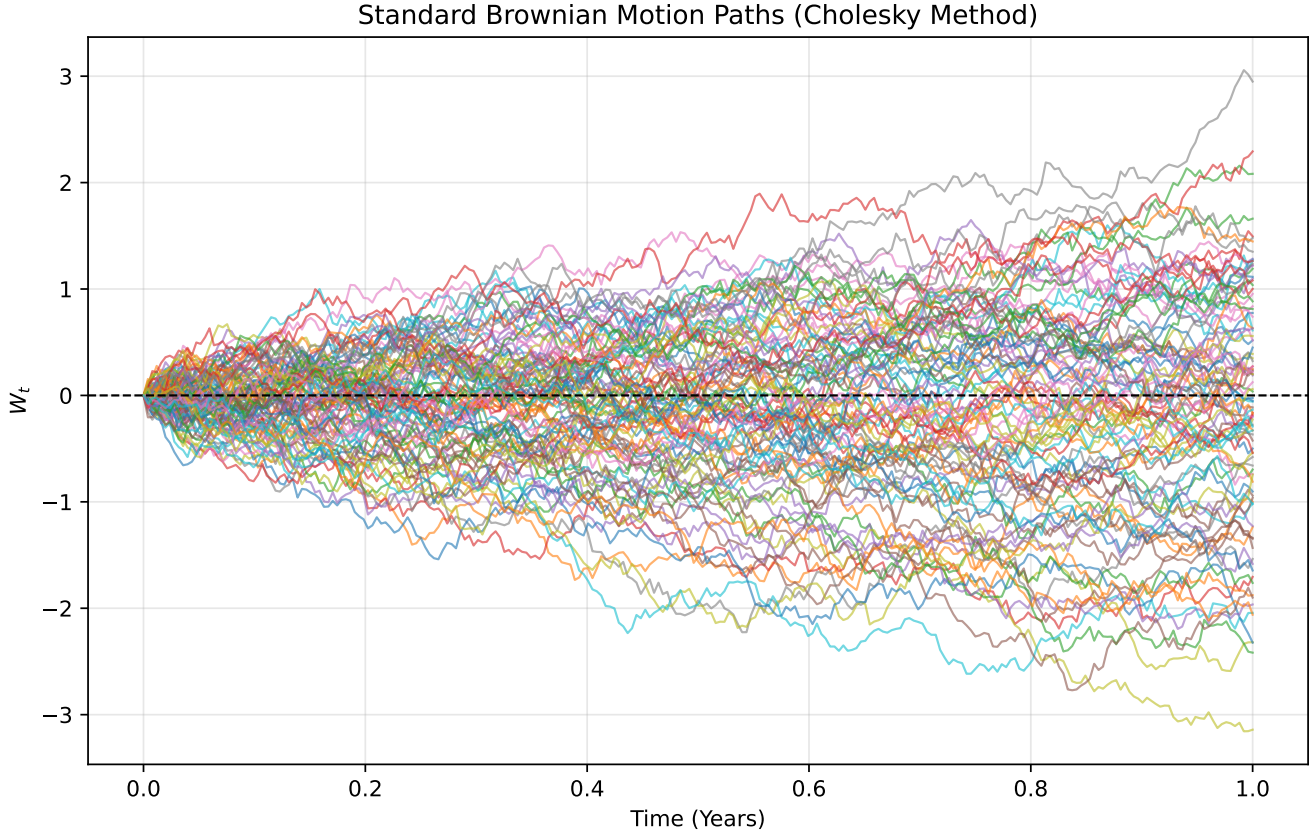
While the sequential random-walk method builds the path step by step, the **Cholesky factorization method** generates the full finite-dimensional skeleton in one global step. It transforms independent standard normal variables into a correlated vector with the exact Brownian covariance structure.

Recall that for a standard Brownian motion evaluated at time points $t_1, t_2, \dots, t_N$, the covariance between any two points is strictly determined by the earlier time:

$$\Sigma_{i,j} = \text{Cov}(W_{t_i}, W_{t_j}) = \min(t_i, t_j)$$

The algorithm proceeds by constructing this $N \times N$ covariance matrix Σ , finding its lower-triangular Cholesky factor A (where $AA^T = \Sigma$), and multiplying it by an N -dimensional vector of independent standard normal shocks X .

The following Python code vectorizes this process to generate M independent Monte Carlo paths simultaneously:



7.3 Discretization Schemes and Convergence

In computational finance, exact analytical solutions to stochastic differential equations are rare. Therefore, our primary goal in this section is to construct a numerical approximation $\hat{S}$ to the true solution S on a fixed time horizon $[0, T]$.

These approximations are constructed on a discrete time grid Δ , defined as:

$$\Delta := \{0 = t_0 < t_1 < \dots < t_N = T\}$$

where N is the number of time steps.

We denote the **mesh-size** (or maximum step size) of the grid by:

$$|\Delta| := \max_{1 \leq i \leq N} |t_i - t_{i-1}|,$$

and we define the time increments and process increments along the grid by

$$\Delta t_i := t_i - t_{i-1}, \quad \Delta S_i := S_{t_i} - S_{t_{i-1}},$$

for $1 \leq i \leq N$.

To compare our discrete approximation with the continuous process at any given time $t \in [0, T]$, we introduce the step-function time index:

$$\bar{t} := \sup\{t_i \mid 0 \leq i \leq N, t_i \leq t\}$$

This allows us to define the continuous-time extension of our discrete approximation $\hat{S}^\Delta$ as a piecewise-constant process:

$$\hat{S}_t^\Delta := \hat{S}_{\bar{t}} \quad \text{for } t \in [0, T].$$

A fundamental question: In what sense should $\hat{S}^\Delta$ approximate S ?

Because SDEs involve randomness, “closeness” can be measured in different ways. We rely on two primary notions of convergence, each serving a distinct purpose in financial engineering:

7.3.1 Strong Convergence (Pathwise Approximation)

Strong convergence measures how closely the simulated path tracks the true solution. This is essential when the trajectory itself matters, for example in dynamic hedging or for path-dependent derivatives such as Asian or barrier options.

Definition: We say that the scheme $\hat{S}^\Delta$ converges **strongly** to S if

$$\lim_{|\Delta| \rightarrow 0} \mathbb{E}_{\mathbb{Q}} [|\hat{S}_T^\Delta - S_T|] = 0.$$

Furthermore, the scheme has **strong order** $\gamma > 0$ if, for sufficiently small $|\Delta|$, there exists a constant $C > 0$ independent of $|\Delta|$ such that

$$\mathbb{E}_{\mathbb{Q}} [|\hat{S}_T^\Delta - S_T|] \leq C|\Delta|^\gamma.$$

7.3.2 Weak Convergence (Distributional Approximation)

Weak convergence is less restrictive. It does not require individual paths to be close; it only requires the terminal distribution of the approximation to match the true one. This is often sufficient for pricing European options, where only the expected terminal payoff matters.

Definition: Given a suitable class $\mathcal{G}$ of test functions $f : \mathbb{R} \rightarrow \mathbb{C}$, we say that the scheme $\hat{S}^\Delta$ converges **weakly** with respect to $\mathcal{G}$ if, for all $f \in \mathcal{G}$,

$$\lim_{|\Delta| \rightarrow 0} \mathbb{E}_{\mathbb{Q}}[f(\hat{S}_T^\Delta)] = \mathbb{E}_{\mathbb{Q}}[f(S_T)].$$

Similarly, $\hat{S}^\Delta$ has **weak order** $\gamma > 0$ if, for every $f \in \mathcal{G}$, there is a constant $C > 0$ independent of $|\Delta|$ such that

$$|\mathbb{E}_{\mathbb{Q}}[f(\hat{S}_T^\Delta)] - \mathbb{E}_{\mathbb{Q}}[f(S_T)]| \leq C|\Delta|^\gamma$$

provided that $|\Delta|$ is small enough.

Remarks on Convergence and Pricing:

1. If our primary objective is the pricing of standard European derivatives, weak convergence is usually sufficient. For a payoff $H \in \mathcal{G}$, weak convergence implies that

$$\mathbb{E}_{\mathbb{Q}}[H(\hat{S}_T^\Delta)] \approx \frac{1}{M} \sum_{j=1}^M H(\hat{S}_T^{\Delta, (j)}),$$

where $\hat{S}_T^{\Delta, (1)}, \dots, \hat{S}_T^{\Delta, (M)}$ are independent simulated samples of the discretized terminal value.

2. Even if the payoff function H does not strictly belong to the initially chosen class $\mathcal{G}$, weak convergence often remains sufficient for pricing. For example, if $\mathcal{G}$ consists of the Fourier basis elements, i.e., $\mathcal{G} = \{x \mapsto e^{iux} : u \in \mathbb{R}\}$, then weak convergence with respect to $\mathcal{G}$ implies pointwise convergence of the characteristic functions. By Lévy's Continuity Theorem, this is equivalent to convergence in distribution. For standard European payoffs, additional continuity and integrability conditions then allow one to pass from convergence in distribution to convergence of option prices.
3. Note that for path-dependent derivatives—such as Asian, Lookback, and Barrier options—the payoff depends not just on the final asset price, but on the specific trajectory taken by the asset over the life of the option. In these cases, weak convergence on the path space is required, and strong convergence is often preferred (or strictly necessary) to accurately capture the dynamics of the path-dependent features and to compute stable hedge ratios (Greeks).

7.3.3 The Euler-Maruyama Scheme

Let us consider a scalar SDE. Integrating the differential form $dS_t = \mu(t, S_t)dt + \sigma(t, S_t)dW_t$ from an arbitrary time t to $t + \Delta t$ yields the following integral equation:

$$S_{t+\Delta t} = S_t + \int_t^{t+\Delta t} \mu(u, S_u)du + \int_t^{t+\Delta t} \sigma(u, S_u)dW_u.$$

This integral representation is the starting point for developing discretization schemes. We assume that the value of the process at time t , denoted as S_t , is known, and we wish to compute the subsequent value $S_{t+\Delta t}$.

The simplest way to discretize this process is to use the **Euler-Maruyama discretization**. This approach is equivalent to approximating the integrals using the left-point rule (also known as the explicit Euler method).

Because the process is adapted, the values $\mu(t, S_t)$ and $\sigma(t, S_t)$ are known exactly at time t . Therefore, the first integral (the drift term) is approximated by holding the integrand constant at its initial value over the interval Δt :

$$\int_t^{t+\Delta t} \mu(u, S_u)du \approx \mu(t, S_t) \int_t^{t+\Delta t} du = \mu(t, S_t)\Delta t.$$

Similarly, the second integral (the diffusion term) is approximated as:

$$\int_t^{t+\Delta t} \sigma(u, S_u)dW_u \approx \sigma(t, S_t) \int_t^{t+\Delta t} dW_u = \sigma(t, S_t)(W_{t+\Delta t} - W_t).$$

By the properties of standard Brownian motion, the increment $W_{t+\Delta t} - W_t$ is normally distributed with mean 0 and variance Δt . Therefore, it is identical in distribution to $\sqrt{\Delta t}Z$, where $Z \sim \mathcal{N}(0, 1)$ is a standard normal random variable.

Combining these approximations, we define the **Euler-Maruyama scheme** for advancing the process from t to $t + \Delta t$ as:

$$\hat{S}_{t+\Delta t} = \hat{S}_t + \mu(t, \hat{S}_t)\Delta t + \sigma(t, \hat{S}_t)\sqrt{\Delta t}Z. \quad (7.1)$$

7.3.3.1 Euler Scheme for the Black-Scholes Model

The Black-Scholes stock price dynamics under the risk-neutral measure are given by:

$$dS_t = rS_t dt + \sigma S_t dW_t.$$

Applying Equation 7.1 to the Black-Scholes model yields the direct Euler discretization:

$$\hat{S}_{t+\Delta t} = \hat{S}_t + r\hat{S}_t\Delta t + \sigma\hat{S}_t\sqrt{\Delta t}Z.$$

Remark on Convergence:

It can be shown that for general SDEs, the Euler-Maruyama scheme has a **strong convergence order of 0.5**, meaning the strong error is bounded by $\mathcal{O}(\Delta t^{1/2})$. This implies that the error of the numerical solution decreases only as the square root of the time step size Δt , which can be computationally slow. In the following sections, we will explore higher-order schemes that perform better.

An Exact Alternative: Simulation via Log-Prices

Instead of applying the Euler scheme directly to the asset price S_t , we can simulate the log-stock price and exponentiate the result.

Using Itô's lemma, the dynamics of $\ln S_t$ are given by:

$$d(\ln S_t) = \left(r - \frac{1}{2}\sigma^2\right) dt + \sigma dW_t.$$

Notice that the drift and diffusion coefficients for this log-process are strictly constant. Applying the Euler discretization Equation 7.1 to this transformed SDE yields:

$$\ln \hat{S}_{t+\Delta t} = \ln \hat{S}_t + \left(r - \frac{1}{2}\sigma^2\right) \Delta t + \sigma\sqrt{\Delta t}Z.$$

Exponentiating both sides gives the simulated asset price:

$$\hat{S}_{t+\Delta t} = \hat{S}_t \exp \left(\left(r - \frac{1}{2}\sigma^2\right) \Delta t + \sigma\sqrt{\Delta t}Z \right).$$

Note: Because the drift and diffusion coefficients of the $\ln S_t$ process do not depend on the state variable, the left-point rule approximation of the integral is perfectly exact. Consequently, this approach yields the **exact solution** for the geometric Brownian motion at the grid points, completely eliminating discretization error!

```

import numpy as np
import matplotlib.pyplot as plt

def simulate_euler_black_scholes(S0, mu, sigma, T, N, seed=None):
    """
    Simulate one path of Black-Scholes using Euler-Maruyama scheme.

    Parameters
    -----
    S0 : float
        Initial asset price
    mu : float
        Drift
    sigma : float
        Volatility
    T : float
        Time horizon
    N : int
        Number of time steps
    seed : int or None
        Random seed for reproducibility

    Returns
    -----
    time : ndarray
        Time grid
    S : ndarray
        Simulated price path
    """

    rng = np.random.default_rng(seed)

    dt = T / N
    time = np.linspace(0, T, N + 1)

    S = np.zeros(N + 1)
    S[0] = S0

    for t in range(1, N + 1):
        dW = rng.normal(0.0, np.sqrt(dt)) # Brownian increment
        S[t] = S[t-1] + mu * S[t-1] * dt + sigma * S[t-1] * dW

    return time, S

```

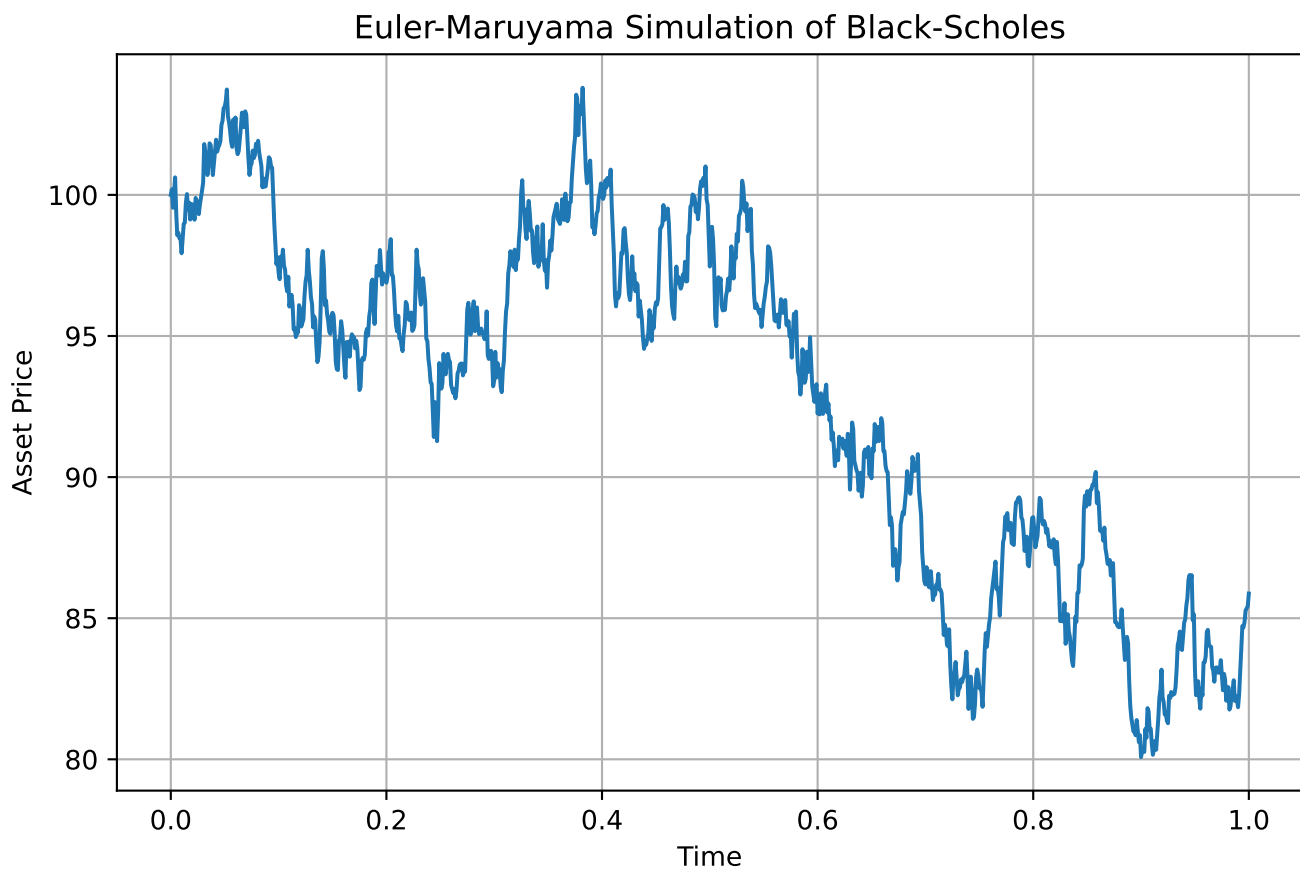
```

# Parameters
S0 = 100      # Initial asset price
mu = 0.05     # Drift
sigma = 0.2   # Volatility
T = 1.0       # Time horizon
N = 1000      # Number of steps

time, S = simulate_euler_black_scholes(S0, mu, sigma, T, N, seed=42)

# Plot
plt.figure(figsize=(8, 5))
plt.plot(time, S)
plt.xlabel("Time")
plt.ylabel("Asset Price")
plt.title("Euler-Maruyama Simulation of Black-Scholes")
plt.grid(True)
plt.show()

```



7.3.3.2 Euler Scheme for the Heston Model

Recall that the Heston model is described by a bivariate system of Stochastic Differential Equations (SDEs) for the asset price S_t and its variance V_t . Under the risk-neutral measure, the continuous-time dynamics are given by:

$$\begin{aligned} dS_t &= rS_t dt + \sqrt{V_t} S_t dW_t^{(1)} \\ dV_t &= \kappa(\theta - V_t) dt + \xi \sqrt{V_t} dW_t^{(2)} \end{aligned}$$

where the two Brownian motions satisfy

$$d\langle W^{(1)}, W^{(2)} \rangle_t = \rho dt,$$

for some correlation parameter $\rho \in [-1, 1]$.

In this section, V_t denotes the **variance process** of the Heston model. This local choice is standard in the stochastic-volatility literature and should not be confused with the option-value notation $V(t, s)$ used in the PDE lectures.

Discretization of the Variance Process V

In contrast to the Black-Scholes model, the Heston model requires us to numerically approximate the unobserved variance process. The SDE for $(V_t)_{t \in [0, T]}$ reads in integral form as:

$$V_{t+\Delta t} = V_t + \int_t^{t+\Delta t} \kappa(\theta - V_u) du + \int_t^{t+\Delta t} \xi \sqrt{V_u} dW_u^{(2)}.$$

The explicit Euler discretization approximates these integrals using the left-point rule:

$$\begin{aligned} \int_t^{t+\Delta t} \kappa(\theta - V_u) du &\approx \kappa(\theta - V_t) \Delta t \\ \int_t^{t+\Delta t} \xi \sqrt{V_u} dW_u^{(2)} &\approx \xi \sqrt{V_t} (W_{t+\Delta t}^{(2)} - W_t^{(2)}) = \xi \sqrt{V_t} \Delta t Z_V \end{aligned}$$

where $Z_V \sim \mathcal{N}(0, 1)$ is a standard normal random variable.

Therefore, the direct Euler discretization of the variance process is:

$$\hat{V}_{t+\Delta t} = \hat{V}_t + \kappa(\theta - \hat{V}_t) \Delta t + \xi \sqrt{\hat{V}_t} \Delta t Z_V.$$

This formula is the formal Euler step. In practical implementations, whenever the square root is evaluated, one typically replaces $\hat{V}_t$ by its positive part $\hat{V}_t^+ := \max(0, \hat{V}_t)$.

Handling Negative Variances

Because the discretization introduces discrete jumps, we cannot guarantee that $\hat{V}_{t+\Delta t}$ remains strictly positive, even if the continuous-time Feller condition ($2\kappa\theta \geq \xi^2$) is satisfied. A negative variance invalidates the square-root computation in the diffusion term. This necessitates the use of truncation or reflection schemes: * **Full Truncation Scheme:** We replace the $\hat{V}_t$ appearing in the drift and diffusion terms with $\hat{V}_t^+ := \max(0, \hat{V}_t)$. * **Reflection Scheme:** We replace the $\hat{V}_t$ appearing in the drift and diffusion terms with its absolute value $|\hat{V}_t|$.

Discretization of the Asset's Price Process S

The integral representation for the asset price is:

$$S_{t+\Delta t} = S_t + \int_t^{t+\Delta t} r S_u du + \int_t^{t+\Delta t} \sqrt{V_u} S_u dW_u^{(1)}.$$

Applying the left-point rule yields the explicit Euler approximation:

$$\begin{aligned} \int_t^{t+\Delta t} r S_u du &\approx r S_t \Delta t \\ \int_t^{t+\Delta t} \sqrt{V_u} S_u dW_u^{(1)} &\approx \sqrt{V_t} S_t (W_{t+\Delta t}^{(1)} - W_t^{(1)}) = \sqrt{V_t \Delta t} S_t Z_S \end{aligned}$$

where $Z_S \sim \mathcal{N}(0, 1)$ is a standard normal random variable that has correlation ρ with Z_V . This leads to the discrete arithmetic scheme:

$$\hat{S}_{t+\Delta t} = \hat{S}_t + r \hat{S}_t \Delta t + \sqrt{\hat{V}_t \Delta t} \hat{S}_t Z_S.$$

Discretization of the Asset's Log-Price Process $\ln S$

To avoid negative asset prices and improve numerical stability, it is often vastly preferable to simulate the log-price. By Itô's lemma, the diffusion for $\ln S_t$ is:

$$d(\ln S_t) = \left(r - \frac{1}{2} V_t \right) dt + \sqrt{V_t} dW_t^{(1)}$$

In integral form over our discrete time step, this is:

$$\ln S_{t+\Delta t} = \ln S_t + \int_t^{t+\Delta t} \left(r - \frac{1}{2} V_u \right) du + \int_t^{t+\Delta t} \sqrt{V_u} dW_u^{(1)}.$$

The Euler discretization for $\ln S_t$ is therefore:

$$\ln \hat{S}_{t+\Delta t} = \ln \hat{S}_t + \left(r - \frac{1}{2} \hat{V}_t \right) \Delta t + \sqrt{\hat{V}_t \Delta t} Z_S.$$

Exponentiating this result provides the geometric Euler discretization for S_t :

$$\hat{S}_{t+\Delta t} = \hat{S}_t \exp \left(\left(r - \frac{1}{2} \hat{V}_t \right) \Delta t + \sqrt{\hat{V}_t \Delta t} Z_S \right).$$

Algorithm: Euler Scheme for (S, V) with Full Truncation

1. **Initialization:** Start with the initial values S_0 for the stock price and V_0 for the variance. Specify all model parameters $(r, \kappa, \theta, \xi, \rho)$ and the time step Δt .
2. **Iteration:** For each time step from t to $t + \Delta t$, perform the following operations sequentially:
 - a. **Generate Correlated Random Variables (Cholesky Decomposition):**
 - Generate two independent standard normal random variables, Z_1 and Z_2 .
 - Set the variance shock: $Z_V = Z_1$.
 - Set the asset shock: $Z_S = \rho Z_1 + \sqrt{1 - \rho^2} Z_2$.

b. **Update Variance:** Compute the next variance state $\hat{V}_{t+\Delta t}$ applying the full truncation scheme ($\hat{V}_t^+ = \max(0, \hat{V}_t)$):

$$\hat{V}_{t+\Delta t} = \hat{V}_t + \kappa(\theta - \hat{V}_t^+) \Delta t + \xi \sqrt{\hat{V}_t^+ \Delta t} Z_V$$

c. **Update Asset Price:** Compute the next asset price $S_{t+\Delta t}$ using the geometric model (preferred):

$$\hat{S}_{t+\Delta t} = \hat{S}_t \exp \left(\left(r - \frac{1}{2} \hat{V}_t^+ \right) \Delta t + \sqrt{\hat{V}_t^+ \Delta t} Z_S \right)$$

(Alternatively, use the arithmetic update: $\hat{S}_{t+\Delta t} = \hat{S}_t + r \hat{S}_t \Delta t + \sqrt{\hat{V}_t^+ \Delta t} \hat{S}_t Z_S$)

7.3.4 The Milstein Scheme

The Milstein scheme is designed to enhance the strong convergence of our numerical approximation. In this section, we focus on time-homogeneous SDEs, meaning the drift and diffusion coefficients depend only on the state variable S_t and not explicitly on time t .

Let us assume the stock price S_t is driven by the following SDE:

$$dS_t = \mu(S_t)dt + \sigma(S_t)dW_t$$

which in integral form over the time step $[t, t + \Delta t]$ is equivalent to:

$$S_{t+\Delta t} = S_t + \int_t^{t+\Delta t} \mu(S_s)ds + \int_t^{t+\Delta t} \sigma(S_s)dW_s$$

The key idea behind the Milstein scheme is to increase the accuracy of the discretization by performing an Itô-Taylor expansion on the coefficients $\mu(S_s)$ and $\sigma(S_s)$ rather than just holding them constant (as we did in the Euler-Maruyama scheme).

Assuming that μ and σ are sufficiently smooth, we apply Itô's lemma to both functions. Since they do not depend explicitly on time, the partial derivatives with respect to t are zero:

$$\begin{aligned} d\mu(S_t) &= \left(\mu'(S_t)\mu(S_t) + \frac{1}{2}\mu''(S_t)\sigma(S_t)^2 \right) dt + (\mu'(S_t)\sigma(S_t)) dW_t \\ d\sigma(S_t) &= \left(\sigma'(S_t)\mu(S_t) + \frac{1}{2}\sigma''(S_t)\sigma(S_t)^2 \right) dt + (\sigma'(S_t)\sigma(S_t)) dW_t \end{aligned}$$

Writing these in integral form for a time $s \in (t, t + \Delta t]$, we get:

$$\begin{aligned} \mu(S_s) &= \mu(S_t) + \int_t^s \left(\mu'(S_u)\mu(S_u) + \frac{1}{2}\mu''(S_u)\sigma(S_u)^2 \right) du + \int_t^s \mu'(S_u)\sigma(S_u) dW_u \\ \sigma(S_s) &= \sigma(S_t) + \int_t^s \left(\sigma'(S_u)\mu(S_u) + \frac{1}{2}\sigma''(S_u)\sigma(S_u)^2 \right) du + \int_t^s \sigma'(S_u)\sigma(S_u) dW_u \end{aligned}$$

Now, we substitute these expansions back into our original integral equation for $S_{t+\Delta t}$. This creates a sum of single and double integrals. To keep the scheme computationally tractable, we truncate the expansion by ignoring higher-order terms.

Specifically, integrals involving $dsdu = \mathcal{O}(\Delta t^2)$, $dsdW_u = \mathcal{O}(\Delta t^{3/2})$, and $dW_s du = \mathcal{O}(\Delta t^{3/2})$ vanish too quickly as $\Delta t \rightarrow 0$ to affect our desired convergence order.

However, the double stochastic integral involving $dW_u dW_s$ scales as $\mathcal{O}(\Delta t)$ and must be retained! Truncating the higher-order terms leaves us with:

$$S_{t+\Delta t} \approx S_t + \int_t^{t+\Delta t} \mu(S_t) ds + \int_t^{t+\Delta t} \sigma(S_t) dW_s + \int_t^{t+\Delta t} \int_t^s \sigma'(S_u)\sigma(S_u) dW_u dW_s$$

Notice that the first three terms on the right-hand side exactly match the Euler-Maruyama scheme. The fourth term is the **Milstein correction**. We approximate this double integral by evaluating the integrand at the known time t :

$$\int_t^{t+\Delta t} \int_t^s \sigma'(S_u)\sigma(S_u) dW_u dW_s \approx \sigma'(S_t)\sigma(S_t) \int_t^{t+\Delta t} \int_t^s dW_u dW_s$$

We can evaluate the inner stochastic integral as $\int_t^s dW_u = W_s - W_t$. Substituting this into the outer integral gives:

$$\int_t^{t+\Delta t} (W_s - W_t) dW_s = \int_t^{t+\Delta t} W_s dW_s - W_t \int_t^{t+\Delta t} dW_s$$

Using the standard Itô identity $\int_t^{t+\Delta t} W_s dW_s = \frac{1}{2}(W_{t+\Delta t}^2 - W_t^2 - \Delta t)$, we obtain:

$$\begin{aligned}
\int_t^{t+\Delta t} (W_s - W_t) dW_s &= \frac{1}{2} (W_{t+\Delta t}^2 - W_t^2 - \Delta t) - W_t (W_{t+\Delta t} - W_t) \\
&= \frac{1}{2} (W_{t+\Delta t}^2 - 2W_t W_{t+\Delta t} + W_t^2 - \Delta t) \\
&= \frac{1}{2} ((W_{t+\Delta t} - W_t)^2 - \Delta t) \\
&= \frac{1}{2} ((\Delta W_t)^2 - \Delta t)
\end{aligned}$$

Recall that the Brownian increment ΔW_t is equal in distribution to $\sqrt{\Delta t}Z$, where $Z \sim \mathcal{N}(0, 1)$. Substituting this back, the Milstein correction term becomes:

$$\frac{1}{2} \sigma'(S_t) \sigma(S_t) (\Delta t Z^2 - \Delta t) = \frac{1}{2} \sigma'(S_t) \sigma(S_t) \Delta t (Z^2 - 1)$$

Combining all the components, the general form of the **Milstein discretization** is:

$$\hat{S}_{t+\Delta t} = \hat{S}_t + \mu(\hat{S}_t) \Delta t + \sigma(\hat{S}_t) \sqrt{\Delta t} Z + \frac{1}{2} \sigma'(\hat{S}_t) \sigma(\hat{S}_t) \Delta t (Z^2 - 1)$$

Remark on Convergence:

The Milstein scheme introduces a crucial correction for the convexity of the diffusion term. This makes it particularly suitable for equations where the volatility coefficient depends heavily on the state variable (like the CEV model or stochastic volatility models).

Because it correctly captures the $\mathcal{O}(\Delta t)$ variation of the diffusion term, the Milstein scheme achieves a **strong convergence order of 1.0** (meaning the strong error is bounded by $\mathcal{O}(\Delta t)$), provided the drift, diffusion, and the derivative of the diffusion coefficient are Lipschitz continuous. This represents a significant mathematical and computational improvement over the standard Euler-Maruyama scheme, which is stuck at a strong order of 0.5.

7.3.4.1 Milstein Scheme for the Black-Scholes Model

In the Black-Scholes model we have $\mu(S_t) = rS_t$ and $\sigma(S_t) = \sigma S_t$, so the Milstein scheme becomes:

$$\hat{S}_{t+\Delta t} = \hat{S}_t + r\hat{S}_t \Delta t + \sigma \hat{S}_t \sqrt{\Delta t} Z + \frac{1}{2} \sigma^2 \hat{S}_t \Delta t (Z^2 - 1)$$

which adds the correction term $\frac{1}{2} \sigma^2 \hat{S}_t \Delta t (Z^2 - 1)$ to the Euler scheme from before.

In the Black-Scholes model for the log-stock price $X_t := \ln S_t$, we have

$$dX_t = \left(r - \frac{1}{2} \sigma^2 \right) dt + \sigma dW_t,$$

so the drift and diffusion coefficients of X_t are constants. Their derivatives with respect to the state variable X_t are therefore zero, and the Milstein correction vanishes. The scheme is

$$\ln \hat{S}_{t+\Delta t} = \ln \hat{S}_t + \left(r - \frac{1}{2}\sigma^2\right) \Delta t + \sigma\sqrt{\Delta t}Z$$

which is identical to the Euler scheme from before. Hence, while the Milstein scheme improves the discretization of S_t in the Black-Scholes model, it does not improve the discretization of $\ln S_t$!

7.3.4.2 Milstein Scheme for the Heston Model

Recall the continuous-time Heston model from before, given by the system of SDEs under the risk-neutral measure:

$$\begin{aligned} dS_t &= rS_t dt + \sqrt{V_t}S_t dW_t^{(1)} \\ dV_t &= \kappa(\theta - V_t)dt + \xi\sqrt{V_t}dW_t^{(2)} \end{aligned}$$

Warning

For the full two-factor Heston system, a genuine strong-order-1 Milstein discretization is more subtle than in the scalar case because mixed iterated stochastic integrals appear. The formulas below should therefore be read as **Milstein-style componentwise corrections**, not as a complete strong-order-1 scheme for the correlated system. In practice, full-truncation Euler is often the baseline method used in pricing applications.

Process for the Variance V

For the variance process, the drift and diffusion coefficients are $\mu(V_t) = \kappa(\theta - V_t)$ and $\sigma(V_t) = \xi\sqrt{V_t}$. The derivative of the diffusion coefficient with respect to V is $\sigma'(V_t) = \frac{\xi}{2\sqrt{V_t}}$.

Applying the scalar Milstein correction term $\frac{1}{2}\sigma'(V_t)\sigma(V_t)\Delta t(Z_V^2 - 1)$ yields the variance update

$$\hat{V}_{t+\Delta t} = \hat{V}_t + \kappa(\theta - \hat{V}_t)\Delta t + \xi\sqrt{\hat{V}_t}\Delta t Z_V + \frac{1}{4}\xi^2\Delta t(Z_V^2 - 1)$$

By completing the square, this can be elegantly rewritten as:

$$\hat{V}_{t+\Delta t} = \left(\sqrt{\hat{V}_t} + \frac{1}{2}\xi\sqrt{\Delta t}Z_V\right)^2 + \kappa(\theta - \hat{V}_t)\Delta t - \frac{1}{4}\xi^2\Delta t$$

Writing it this way reveals a structural advantage: the first term is strictly non-negative. Because of this, the Milstein discretization of the variance process naturally produces far fewer negative values than the explicit Euler discretization. Nevertheless, because the drift adjustment $-\frac{1}{4}\xi^2\Delta t$ can still pull the value below zero, a full truncation or reflection scheme must still be applied!

Process for the Asset Price S and Log-Price $\ln S$

For the arithmetic stock-price process, if we freeze V_t over one time step, the coefficients are $\mu(S_t) = rS_t$ and $\sigma(S_t) = \sqrt{V_t}S_t$. The derivative of the diffusion term with respect to S is then $\sqrt{V_t}$, which gives the componentwise correction

$$\hat{S}_{t+\Delta t} = \hat{S}_t + r\hat{S}_t\Delta t + \sqrt{\hat{V}_t\Delta t}\hat{S}_tZ_S + \frac{1}{2}\hat{V}_t\hat{S}_t\Delta t(Z_S^2 - 1)$$

As before, we can also discretize the log-stock process. By Itô's lemma, the continuous dynamics of $\ln S_t$ are:

$$d(\ln S_t) = \left(r - \frac{1}{2}V_t\right) dt + \sqrt{V_t}dW_t^{(1)}$$

Notice that the diffusion coefficient for the log-price, $\sqrt{V_t}$, does not depend on the state variable $\ln S_t$. Its derivative with respect to $\ln S_t$ is identically zero. Therefore, the Milstein correction term vanishes completely!

An application of the Milstein scheme yields:

$$\ln \hat{S}_{t+\Delta t} = \ln \hat{S}_t + \left(r - \frac{1}{2}\hat{V}_t\right) \Delta t + \sqrt{\hat{V}_t\Delta t}Z_S$$

which is identical to the Euler-Maruyama discretization for the log-price. Hence, just as in the Black-Scholes model, simulating $\ln S_t$ removes the scalar Milstein correction in the price equation once V_t is frozen over the step. Exponentiating this gives the geometric update:

$$\hat{S}_{t+\Delta t} = \hat{S}_t \exp \left(\left(r - \frac{1}{2}\hat{V}_t\right) \Delta t + \sqrt{\hat{V}_t\Delta t}Z_S \right).$$

Again, it is strictly necessary to apply the full truncation or reflection scheme to $\hat{V}_t$ before evaluating these equations!

Algorithm: Milstein Scheme for (S, V) or $(\ln S, V)$ with Full Truncation

Given a known state $(\hat{S}_t, \hat{V}_t)$ at time t , the procedure to update to the next time step $t + \Delta t$ is as follows:

1. Generate Correlated Random Variables:

- To generate the standard normal variables Z_V and Z_S with correlation ρ , first draw two independent standard normal variables Z_1 and Z_2 .
- Set $Z_V = Z_1$ and $Z_S = \rho Z_1 + \sqrt{1 - \rho^2}Z_2$.

2. **Update Variance:** Compute the next variance state $\hat{V}_{t+\Delta t}$ using the truncated Milstein formula:

$$\hat{V}_{t+\Delta t} = \hat{V}_t + \kappa(\theta - \hat{V}_t^+)\Delta t + \xi\sqrt{\hat{V}_t^+\Delta t}Z_V + \frac{1}{4}\xi^2\Delta t(Z_V^2 - 1)$$

where $\hat{V}_t^+ = \max(0, \hat{V}_t)$, κ is the rate of mean reversion, θ is the long-term variance, and ξ is the volatility of volatility.

3. **Update Stock Price:** Compute $\hat{S}_{t+\Delta t}$ using either of the following formulas, ensuring you use the truncated variance $\hat{V}_t^+$:

- **For the geometric model (Preferred):**

$$\hat{S}_{t+\Delta t} = \hat{S}_t \exp \left(\left(r - \frac{1}{2}\hat{V}_t^+ \right) \Delta t + \sqrt{\hat{V}_t^+\Delta t}Z_S \right)$$

- **For the arithmetic model:**

$$\hat{S}_{t+\Delta t} = \hat{S}_t + r\hat{S}_t\Delta t + \sqrt{\hat{V}_t^+\Delta t}\hat{S}_tZ_S + \frac{1}{2}\hat{V}_t^+\hat{S}_t\Delta t(Z_S^2 - 1)$$

```
import numpy as np
import matplotlib.pyplot as plt

def simulate_heston_paths(S0, V0, r, kappa, theta, xi, rho, T, N, M, scheme='milstein', seed=None):
    """
    Simulates Heston model paths using either Euler-Maruyama or Milstein scheme.

    Parameters:
    S0      : Initial stock price
    V0      : Initial variance
    r       : Risk-free rate
    kappa   : Rate of mean reversion
    theta   : Long-term variance
    xi      : Volatility of volatility
    rho     : Correlation between asset and variance Brownian motions
    T       : Time horizon
    N       : Number of time steps
    M       : Number of Monte Carlo paths
    scheme  : 'euler' or 'milstein'
    seed    : Random seed for reproducibility

    Returns:
    S, V    : Arrays of shape (N+1, M) containing simulated paths
    """
    dt = T / N
    rng = np.random.default_rng(seed)

    # Initialize arrays to store paths
```

```

S = np.zeros((N + 1, M))
V = np.zeros((N + 1, M))

S[0] = S0
V[0] = V0

for t in range(N):
    # 1. Generate Correlated Random Variables
    Z1 = rng.standard_normal(M)
    Z2 = rng.standard_normal(M)

    Z_V = Z1
    Z_S = rho * Z1 + np.sqrt(1 - rho**2) * Z2

    # Apply full truncation to the variance
    V_plus = np.maximum(0, V[t])

    # 2. Update Variance
    drift_V = kappa * (theta - V_plus) * dt
    diff_V = xi * np.sqrt(V_plus * dt) * Z_V

    if scheme == 'euler':
        V[t+1] = V[t] + drift_V + diff_V
    elif scheme == 'milstein':
        milstein_correction = 0.25 * xi**2 * dt * (Z_V**2 - 1)
        V[t+1] = V[t] + drift_V + diff_V + milstein_correction
    else:
        raise ValueError("Scheme must be 'euler' or 'milstein'")

    # 3. Update Stock Price (Geometric Model)
    drift_S = (r - 0.5 * V_plus) * dt
    diff_S = np.sqrt(V_plus * dt) * Z_S

    S[t+1] = S[t] * np.exp(drift_S + diff_S)

return S, V

# Example Parameters (Feller condition  $2\kappa\theta > \xi^2$  is NOT strictly met here,
# making truncation crucial)
S0 = 100.0
V0 = 0.04
r = 0.05
kappa = 2.0
theta = 0.04
xi = 0.5

```

```

rho = -0.7
T = 1.0
N = 252 # Daily steps for 1 year
M = 10000 # Number of paths

# Simulate using both schemes
S_euler, V_euler = simulate_heston_paths(
    S0, V0, r, kappa, theta, xi, rho, T, N, M, scheme='euler', seed=42
)
S_milstein, V_milstein = simulate_heston_paths(
    S0, V0, r, kappa, theta, xi, rho, T, N, M, scheme='milstein', seed=42
)

```

7.3.5 Comparison Between the Euler and Milstein Schemes

The Euler-Maruyama and Milstein schemes are foundational tools for numerically solving Stochastic Differential Equations (SDEs). As we have seen, the Milstein scheme can significantly improve the approximation in certain cases, such as the arithmetic simulation of the Geometric Brownian Motion model, by accounting for the convexity of the diffusion term.

In other settings, however, the Milstein scheme structurally reduces to the simpler Euler scheme and offers no additional benefit. For example, in an Arithmetic Brownian Motion model ($dS_t = \mu dt + \sigma dW_t$), the diffusion coefficient σ is strictly constant. Consequently, its derivative $\sigma'(S_t)$ is zero, causing the entire Milstein correction term to vanish.

In general, we can summarize the theoretical performance of both schemes as follows:

Convergence Rates Comparison

Assuming the drift and diffusion coefficients satisfy the necessary regularity and Lipschitz conditions, the convergence properties are:

Discretization Scheme	Strong Convergence Order	Weak Convergence Order	Error Bound (Strong)	Error Bound (Weak)
Euler-Maruyama	0.5	1.0	$\mathcal{O}(\Delta t^{1/2})$	$\mathcal{O}(\Delta t)$
Milstein	1.0	1.0	$\mathcal{O}(\Delta t)$	$\mathcal{O}(\Delta t)$

Conclusion and Practical Application

Notice that both the Euler and Milstein schemes demonstrate a **weak convergence rate of $\mathcal{O}(\Delta t)$** . This means that if your goal is solely to estimate the terminal distribution of an asset—for instance, when pricing standard European options—the Milstein scheme generally provides no computational advantage over the simpler Euler scheme.

However, the Milstein scheme’s superiority becomes evident in its **strong convergence rate**. Because it accurately tracks the true stochastic path with an error bounded by $\mathcal{O}(\Delta t)$ rather than $\mathcal{O}(\Delta t^{1/2})$, the Milstein scheme is highly beneficial, and often necessary, when pricing **path-dependent derivatives** (such as Asian, Barrier, or Lookback options) where the precise sequence of the asset’s trajectory dictates the payoff.

7.4 Summary and Connection to the Next Lecture

This lecture has developed the numerical infrastructure for simulating continuous-time stochastic processes on a computer. Three themes deserve emphasis as we move forward.

Two notions of accuracy. Strong convergence measures path-by-path proximity to the true trajectory — essential whenever the payoff depends on the sample path itself, as in Asian, barrier, or lookback options, and whenever stable hedge ratios are needed. Weak convergence measures agreement in distribution at the terminal date — sufficient for European option pricing. Both the Euler-Maruyama scheme (strong order 0.5, weak order 1.0) and the Milstein scheme (strong order 1.0, weak order 1.0) are weakly equivalent for European pricing; Milstein is preferred for path-dependent contracts.

Handling the Heston variance. The CIR variance process can become negative under discretization even when the Feller condition $2\kappa\theta \geq \xi^2$ holds, because discrete steps cannot preserve the continuous boundary behaviour. The full-truncation Euler scheme — replacing $\hat{V}_t$ by $\hat{V}_t^+ = \max(0, \hat{V}_t)$ in both drift and diffusion before the update — is the standard practical remedy. The `simulate_heston_paths` function implements this for both the Euler and Milstein schemes.

The log-price trick eliminates the scalar Milstein correction. Whenever the diffusion coefficient of the log-price does not depend on the log-price itself (as in Black-Scholes and in the Heston price equation with V_t frozen over a step), the scalar Milstein correction vanishes. In Heston this means the geometric log-price update is exact conditional on the frozen variance over that step; the remaining discretization error comes from approximating the variance path itself. This is why practitioners consistently simulate $\ln S_t$ rather than S_t .

The Heston path simulator developed here — `simulate_heston_paths` — is the direct input to the next lecture. Lecture 8 takes the simulated terminal distributions produced by this function and addresses the inverse problem: given a surface of observed market option prices, how do we recover the Heston parameters $(\kappa, \theta, \xi, \rho, V_0)$ that best reproduce that surface? This is the **calibration** problem, and it requires efficient path generation as an inner loop inside a numerical optimisation.

8 Heston Model and Implied-Volatility Calibration

Objectives

After this lecture, you should be able to:

1. Explain why constant-volatility models are insufficient for equity index options.
2. Formulate the Heston stochastic-volatility model under the risk-neutral measure.
3. Simulate Heston paths with a discretization consistent with the previous SDE lecture.
4. Map model prices to implied volatilities and interpret smile/skew effects.
5. Set up and interpret a practical calibration problem for option surfaces.

This lecture extends the discretization tools from Lecture 7 — Euler-Maruyama and Milstein schemes, full-truncation of the CIR variance process, correlated Brownian increments via Cholesky — to a stochastic-volatility setting. It also reconnects with the implied-volatility surface first studied in Lecture 4: the SABR model introduced there was a descriptive interpolation device, whereas here we ask whether a structural model can reproduce that surface from a small set of economically interpretable parameters. The stylized facts motivating this effort — volatility clustering, the leverage effect, and the negative skew — were documented in Lecture 1 under the physical measure $\mathbb{P}$; Lecture 2 supplied the Girsanov change-of-measure technology that carries those dynamics into the risk-neutral world.

8.1 From Black-Scholes to Stochastic Volatility

In the Black-Scholes model, volatility is constant:

$$dS_t = rS_t dt + \sigma S_t dW_t^{\mathbb{Q}}.$$

This yields a flat implied-volatility surface, which is inconsistent with market data.

Empirically, index options exhibit:

- a pronounced negative skew (OTM puts have higher implied vol),
- maturity-dependent smiles/skews,
- time variation in the level and shape of the surface.

These facts motivate models where instantaneous variance is itself random.

8.2 Implied Volatility as the Calibration Interface

For a quoted call price $C_{\text{mkt}}(K, T)$, implied volatility is defined by inversion of Black-Scholes:

$$C_{\text{BS}}(S_0, K, T, r, \sigma_{\text{imp}}(K, T)) = C_{\text{mkt}}(K, T).$$

This inversion is meaningful only when the quoted price satisfies the usual no-arbitrage bounds for a European call.

In practice, calibration is often done either:

- directly on prices C_{mkt} ,
- or on implied volatilities σ_{imp} , after inversion.

Using implied volatilities makes calibration errors more comparable across strikes and maturities, but it requires a robust inversion routine.

8.3 The Heston Model Under $\mathbb{Q}$

We use the same notation as in the SDE-discretization lecture:

$$\begin{aligned} dS_t &= rS_t dt + \sqrt{V_t} S_t dW_t^{(1)}, \\ dV_t &= \kappa(\theta - V_t) dt + \xi \sqrt{V_t} dW_t^{(2)}, \\ d\langle W^{(1)}, W^{(2)} \rangle_t &= \rho dt. \end{aligned}$$

Here the Brownian motions are understood under $\mathbb{Q}$; we suppress the superscript $\mathbb{Q}$ on $(W^{(1)}, W^{(2)})$ to keep the notation readable.

Throughout this lecture, V_t denotes the Heston variance process, while option prices are denoted by C_{mkt} , C_{BS} , and C_{Hes} to keep the distinction from the pricing-function notation of earlier lectures explicit.

Parameters:

- $\kappa > 0$: mean-reversion speed of variance,
- $\theta > 0$: long-run variance level,
- $\xi > 0$: volatility of variance (“vol-of-vol”),
- $\rho \in [-1, 1]$: spot/variance shock correlation,
- $V_0 > 0$: initial variance.

The Feller condition

$$2\kappa\theta \geq \xi^2$$

is sufficient (not necessary) for strict positivity of the CIR variance process in continuous time.

8.4 Correlated Gaussian Shocks (Simulation Block)

For a time step Δt , if $Z_1, Z_2 \sim \mathcal{N}(0, 1)$ i.i.d., define

$$\Delta W^{(1)} = \sqrt{\Delta t} Z_1, \quad \Delta W^{(2)} = \sqrt{\Delta t} (\rho Z_1 + \sqrt{1 - \rho^2} Z_2).$$

Then $\text{Corr}(\Delta W^{(1)}, \Delta W^{(2)}) = \rho$.

Target correlation : -0.7

Empirical correlation: -0.7025

8.5 Full-Truncation Euler Scheme for Heston

To remain consistent with the discretization lecture, we use the full-truncation Euler scheme:

$$V_t^+ := \max(V_t, 0).$$

The one-step updates are

$$\begin{aligned} V_{t+\Delta t} &= V_t + \kappa(\theta - V_t^+) \Delta t + \xi \sqrt{V_t^+} \Delta W_t^{(2)}, \\ S_{t+\Delta t} &= S_t \exp \left[\left(r - \frac{1}{2} V_t^+ \right) \Delta t + \sqrt{V_t^+} \Delta W_t^{(1)} \right]. \end{aligned}$$

```
import numpy as np
import matplotlib.pyplot as plt

def heston_paths_full_truncation(
    S0=100.0,
    V0=0.04,
    r=0.02,
    kappa=2.0,
    theta=0.04,
    xi=0.5,
    rho=-0.7,
    T=1.0,
    n_steps=252,
    n_paths=8,
    seed=123,
):
    rng = np.random.default_rng(seed)
    dt = T / n_steps
    t = np.linspace(0.0, T, n_steps + 1)

    S = np.empty((n_paths, n_steps + 1))
```

```

V = np.empty((n_paths, n_steps + 1))
S[:, 0] = S0
V[:, 0] = V0

for k in range(n_steps):
    z1 = rng.standard_normal(n_paths)
    z2 = rng.standard_normal(n_paths)
    dW1 = np.sqrt(dt) * z1
    dW2 = np.sqrt(dt) * (rho * z1 + np.sqrt(1 - rho**2) * z2)

    V_plus = np.maximum(V[:, k], 0.0)
    V[:, k + 1] = V[:, k] + kappa * (theta - V_plus) * dt + xi * np.sqrt(V_plus) * dW2

    S[:, k + 1] = S[:, k] * np.exp((r - 0.5 * V_plus) * dt + np.sqrt(V_plus) * dW1)

return t, S, V

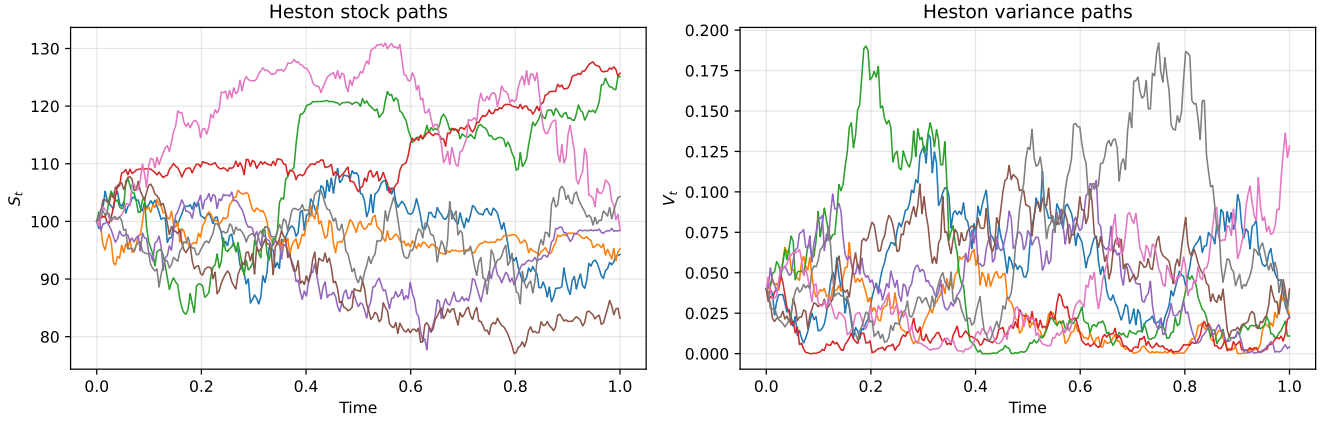
t, S, V = heston_paths_full_truncation()

fig, ax = plt.subplots(1, 2, figsize=(12, 4))
ax[0].plot(t, S.T, lw=1.0)
ax[0].set_title("Heston stock paths")
ax[0].set_xlabel("Time")
ax[0].set_ylabel("$S_t$")
ax[0].grid(True, alpha=0.3)

ax[1].plot(t, np.maximum(V, 0.0).T, lw=1.0)
ax[1].set_title("Heston variance paths")
ax[1].set_xlabel("Time")
ax[1].set_ylabel("$V_t$")
ax[1].grid(True, alpha=0.3)

plt.tight_layout()
plt.show()

```



8.6 From Heston Prices to a Model-Implied Vol Surface

For a parameter vector

$$\eta := (\kappa, \theta, \xi, \rho, V_0),$$

define the model call price

$$C_{\text{Hes}}(K, T; \eta) = e^{-rT} \mathbb{E}^{\mathbb{Q}}[(S_T - K)^+].$$

Then define model-implied volatility by Black-Scholes inversion:

$$\sigma_{\text{imp}}^{\text{Hes}}(K, T; \eta) := \sigma \text{ such that } C_{\text{BS}}(S_0, K, T, r, \sigma) = C_{\text{Hes}}(K, T; \eta).$$

This definition is meaningful only when the Heston model price lies within the Black-Scholes no-arbitrage bounds for the same option.

Negative ρ typically induces left skew, while larger ξ increases the curvature of the smile.

8.7 Calibration Problem

Let market quotes be $\{(K_i, T_i, C_i^{\text{mkt}})\}_{i=1}^M$. A standard price-based weighted least-squares calibration is:

$$\eta^* = \arg \min_{\eta \in \mathcal{A}} \sum_{i=1}^M w_i (C_{\text{Hes}}(K_i, T_i; \eta) - C_i^{\text{mkt}})^2,$$

where

$$\mathcal{A} = \{(\kappa, \theta, \xi, \rho, V_0) \in (0, \infty)^3 \times (-1, 1) \times (0, \infty)\}.$$

An implied-volatility formulation is:

$$\eta^* = \arg \min_{\eta \in \mathcal{A}} \sum_{i=1}^M w_i (\sigma_{\text{imp}}^{\text{Hes}}(K_i, T_i; \eta) - \sigma_i^{\text{mkt}})^2.$$

Typical choices for the weights are:

- $w_i = 1$ (uniform),
- $w_i \propto 1/\text{bid-ask}_i^2$,
- $w_i \propto 1/\text{Vega}_i^2$ if a price-based objective is meant to approximate least squares in implied-volatility errors.

8.7.1 Optional Regularization

To discourage unstable parameter sets, one may add penalties, e.g.

$$\lambda_{\text{Feller}} \max(\xi^2 - 2\kappa\theta, 0)^2.$$

This is a numerical regularization device, not a no-arbitrage condition.

8.8 Minimal Calibration Skeleton

The following is a compact scaffold. It assumes that a reliable Heston pricer `heston_call_price(...)` is already available, for example via the characteristic function.

```
import numpy as np
from scipy.optimize import least_squares
from scipy.stats import norm
from scipy.optimize import brentq

def bs_call_price(S0, K, T, r, sigma):
    if T <= 0:
        return max(S0 - K, 0.0)
    d1 = (np.log(S0 / K) + (r + 0.5 * sigma**2) * T) / (sigma * np.sqrt(T))
    d2 = d1 - sigma * np.sqrt(T)
    return S0 * norm.cdf(d1) - K * np.exp(-r * T) * norm.cdf(d2)

def implied_vol_call(price, S0, K, T, r):
    intrinsic = max(S0 - K * np.exp(-r * T), 0.0)
    upper = S0
    tol = 1e-12

    if price < intrinsic - tol or price >= upper:
        return np.nan
    if abs(price - intrinsic) <= tol:
        return 0.0

    f = lambda sigma: bs_call_price(S0, K, T, r, sigma) - price
    return brentq(f, 1e-8, 5.0)
```

```

# Placeholder signature:
# heston_call_price(S0, K, T, r, kappa, theta, xi, rho, v0) -> float

def residuals(params, S0, r, K_arr, T_arr, market_prices, weights):
    kappa, theta, xi, rho, v0 = params
    model = np.array([
        heston_call_price(S0, K, T, r, kappa, theta, xi, rho, v0)
        for K, T in zip(K_arr, T_arr)
    ])
    return np.sqrt(weights) * (model - market_prices)

# Example call:
# bounds_lower = np.array([1e-6, 1e-6, 1e-6, -0.999, 1e-6])
# bounds_upper = np.array([20.0, 2.0, 5.0, 0.999, 2.0])
# x0 = np.array([2.0, 0.04, 0.5, -0.7, 0.04])
# res = least_squares(residuals, x0, bounds=(bounds_lower, bounds_upper),
#                     args=(S0, r, K_arr, T_arr, market_prices, weights))
# theta_star = res.x

```

8.9 Calibration Diagnostics

After calibration, always check:

1. Price errors and implied-vol errors by maturity and moneyness.
2. Stability across neighboring dates (parameter jumps are a warning sign).
3. Parameter plausibility (especially ρ , ξ , and V_0).
4. Out-of-sample fit (strikes/maturities not used in the objective).

8.10 Summary and Connection to the Next Lectures

This lecture has closed the loop between the market's implied-volatility surface — first encountered in Lecture 4 — and a structurally motivated model. The key steps were:

1. **From constant volatility to stochastic volatility.** The flat implied-volatility surface of Black-Scholes contradicts the empirical skew and smile documented in Lectures 1 and 4. The Heston CIR variance process generates a non-flat, maturity-dependent surface from five interpretable parameters $(\kappa, \theta, \xi, \rho, V_0)$.
2. **Simulation via full-truncation Euler.** The `heston_paths_full_truncation` function directly applies the discretization developed in Lecture 7. The negative correlation $\rho < 0$ between $dW^{(1)}$ and $dW^{(2)}$ is what produces the left-skewed implied-volatility surface, connecting the leverage effect observed under $\mathbb{P}$ (Lecture 1) to the risk-neutral skew under $\mathbb{Q}$ (Lecture 3/Girsanov).

3. **Calibration as a structured least-squares problem.** The calibration skeleton translates “model prices match market prices” into a bounded nonlinear least-squares problem over the admissible parameter set $\mathcal{A}$. Working in implied-volatility space makes residuals comparable across strikes and maturities. Good diagnostics — in-sample fit, parameter stability, out-of-sample performance — are as important as the optimisation itself.

Once calibrated, the Heston parameter vector η^* defines a concrete risk-neutral data-generating mechanism for both Monte Carlo and PDE valuation. The next two lectures — Lectures 9 and 10 on finite-difference methods — temporarily step back to low-dimensional Black-Scholes PDE numerics and develop grid-based solvers that complement the Monte Carlo approach. After that, Lecture 11 returns to simulation and uses the Heston framework to illustrate variance-reduction techniques (antithetic variates, control variates) that can dramatically reduce the Monte Carlo estimator’s standard error.

9 Finite Difference Methods I: From the Black-Scholes PDE to an Explicit Scheme

Monte Carlo became our default tool once the payoff depended on many sources of randomness or on the whole path. In low-dimensional problems, however, the pricing PDE still provides a very efficient numerical route. Finite difference methods exploit that PDE structure directly and therefore complement the simulation methods developed earlier in the course.

The two approaches are in fact two sides of the same mathematical coin. Lecture 5 derived the Black-Scholes PDE from a delta-hedging replication argument and stated the Feynman-Kac theorem, which establishes that the PDE solution equals the risk-neutral expectation: $V(S, t) = e^{-r(T-t)} \mathbb{E}^Q[h(S_T) \mid S_t = S]$. Lecture 6 showed how to evaluate the right-hand side by Monte Carlo simulation. This lecture shows how to solve the left-hand side — the PDE itself — by discretizing it on a grid. The two routes yield the same number; the choice between them is governed by the dimensionality of the problem and the need to compute Greeks.

This lecture develops the basic finite-difference framework for Black-Scholes pricing. The emphasis is on the modelling setup, the transformation to a constant-coefficient heat equation, the construction of a computational grid, and the explicit forward-time central-space (FTCS) scheme.

9.1 Why Finite Differences?

Finite difference methods are particularly attractive when the derivative price depends on only one or two state variables. In that regime they are often faster and more accurate than Monte Carlo for a whole strip of spot values, and they deliver the full price surface together with stable approximations of Greeks.

The trade-off is equally important:

- the method requires a low-dimensional state space,
- the computational domain must be truncated carefully,
- stability and boundary treatment become central numerical issues.

For that reason, finite differences and Monte Carlo should be seen as complementary rather than competing tools. In the present lecture we focus on the low-dimensional Black-Scholes setting where the PDE structure is especially transparent.

9.2 The Black-Scholes PDE as a Backward Evolution Problem

Under the risk-neutral measure and with constant interest rate r and volatility σ , the stock price satisfies

$$dS_t = rS_t dt + \sigma S_t dW_t^{\mathbb{Q}}.$$

Let $V(S, t)$ denote the value of a European-style contingent claim with maturity T and payoff $h(S_T)$. We reserve the symbol Φ for the standard normal cumulative distribution function elsewhere in the course, so the deterministic payoff function is denoted by h . Replication yields the Black-Scholes PDE

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0, \quad 0 < t < T, \quad S > 0,$$

with terminal condition

$$V(S, T) = h(S).$$

For a European call with strike K ,

$$h(S) = (S - K)^+.$$

The PDE evolves **backward in calendar time**: the payoff is known at maturity, and the numerical task is to recover $V(S, 0)$.

Definition 9.1. A Black-Scholes finite-difference problem is specified by:

1. the backward PDE for $V(S, t)$,
2. the terminal payoff at $t = T$,
3. boundary conditions for small and large stock values.

For a European call, the standard asymptotic conditions are

$$V(0, t) = 0, \quad V(S, t) \sim S - Ke^{-r(T-t)} \quad \text{as } S \rightarrow \infty.$$

The boundary conditions matter numerically because a computer cannot work on the infinite domain $S \in (0, \infty)$; we must replace it by a large but finite interval.

9.3 Log-Price and Time-Reversal Transformation

The Black-Scholes PDE has non-constant coefficients in the stock variable S . A standard way to simplify the analysis is to move to log-moneyness and reverse time so that the maturity condition becomes an initial condition.

Set

$$x = \log\left(\frac{S}{K}\right), \quad \tau = \frac{\sigma^2}{2}(T - t), \quad q = \frac{2r}{\sigma^2}.$$

We then define a transformed unknown

$$u(x, \tau) = \frac{V(S, t)}{K} e^{ax+b\tau},$$

where

$$a = \frac{q-1}{2}, \quad b = \frac{(q+1)^2}{4}.$$

Proposition 9.1. *With the transformation above, the Black-Scholes PDE becomes the heat equation*

$$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2}.$$

The gain is substantial: after the transformation the spatial coefficient is constant, so a uniform grid in x becomes natural.

For a European call, the maturity payoff becomes the initial condition

$$u(x, 0) = e^{ax} \max(e^x - 1, 0) = \max\left(e^{\frac{q+1}{2}x} - e^{\frac{q-1}{2}x}, 0\right).$$

At the left boundary, the call value vanishes as $S \downarrow 0$, hence

$$u(x, \tau) \rightarrow 0 \quad \text{as } x \rightarrow -\infty.$$

At the right boundary, the large- S call asymptotics imply

$$u(x, \tau) \approx e^{ax+b\tau} (e^x - e^{-q\tau}) \quad \text{for large } x.$$

Equivalently,

$$u(x, \tau) \approx \exp\left(\frac{q+1}{2}x + \frac{(q+1)^2}{4}\tau\right) - \exp\left(\frac{q-1}{2}x + \frac{(q-1)^2}{4}\tau\right),$$

which is the form typically imposed numerically at the truncated right boundary $x = x_{\max}$.

In practice we replace $\pm\infty$ by finite cutoffs $x_{\min}$ and $x_{\max}$ chosen large enough that the truncation error is negligible relative to the discretization error.

9.4 The Computational Grid

Choose integers $M, N \geq 1$ and define a uniform mesh

$$x_j = x_{\min} + j\Delta x, \quad j = 0, 1, \dots, M,$$

and

$$\tau_n = n\Delta\tau, \quad n = 0, 1, \dots, N,$$

where

$$\Delta x = \frac{x_{\max} - x_{\min}}{M}, \quad \Delta\tau = \frac{\tau_{\max}}{N}, \quad \tau_{\max} = \frac{\sigma^2 T}{2}.$$

We write

$$u_j^n \approx u(x_j, \tau_n).$$

This grid turns the continuous surface (x, τ, u) into a finite table of unknowns. The terminal payoff gives the first row u_j^0 , and the boundary conditions provide the first and last columns u_0^n and u_M^n .

9.5 Taylor Expansions and Finite-Difference Quotients

To discretize the heat equation we approximate derivatives by local Taylor expansions.

For the time derivative, the forward difference gives

$$\frac{\partial u}{\partial \tau}(x_j, \tau_n) = \frac{u_j^{n+1} - u_j^n}{\Delta \tau} + O(\Delta \tau).$$

For the second spatial derivative, the central difference gives

$$\frac{\partial^2 u}{\partial x^2}(x_j, \tau_n) = \frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{\Delta x^2} + O(\Delta x^2).$$

Definition 9.2. The **local truncation error** of a finite-difference approximation is the error made when the exact smooth solution is substituted into the discrete scheme. The formulas above show that the forward difference is first order in time and the central second derivative is second order in space.

These approximations are the building blocks of the explicit scheme.

9.6 The Explicit FTCS Scheme

Substituting the difference quotients into

$$u_\tau = u_{xx}$$

and discarding higher-order terms yields

$$\frac{u_j^{n+1} - u_j^n}{\Delta \tau} = \frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{\Delta x^2}.$$

Solving for the future value u_j^{n+1} gives

$$u_j^{n+1} = \lambda u_{j+1}^n + (1 - 2\lambda)u_j^n + \lambda u_{j-1}^n, \quad \lambda := \frac{\Delta \tau}{\Delta x^2}.$$

This is the **forward-time central-space** (FTCS) scheme.

Proposition 9.2. *For each interior node $j = 1, \dots, M - 1$, the FTCS update is a weighted average of the neighboring values at the previous time level:*

$$u_j^{n+1} = \lambda u_{j+1}^n + (1 - 2\lambda)u_j^n + \lambda u_{j-1}^n.$$

If $0 \leq \lambda \leq \frac{1}{2}$, all weights are nonnegative and add up to one.

The averaging interpretation is useful financially: diffusion smooths the value function, and the explicit step mirrors that smoothing numerically. The restriction $\lambda \leq \frac{1}{2}$ will reappear in the stability analysis of the next lecture.

9.7 Matrix Form and Boundary Vector

Let

$$\mathbf{u}^n = \begin{pmatrix} u_1^n \\ \vdots \\ u_{M-1}^n \end{pmatrix}$$

collect the interior grid values. Then the FTCS scheme can be written as

$$\mathbf{u}^{n+1} = A\mathbf{u}^n + \mathbf{k}^n,$$

where A is the tridiagonal matrix

$$A = \begin{pmatrix} 1-2\lambda & \lambda & & & \\ \lambda & 1-2\lambda & \ddots & & \\ & \ddots & \ddots & \lambda & \\ & & \lambda & 1-2\lambda & \end{pmatrix},$$

and the boundary vector is

$$\mathbf{k}^n = \lambda \begin{pmatrix} u_0^n \\ 0 \\ \vdots \\ 0 \\ u_M^n \end{pmatrix}.$$

This form makes the implementation transparent: at each time step, the new vector is obtained by one sparse matrix-vector product plus a boundary correction.

9.8 Finite-Difference Algorithm

Example 9.1. For a European call in transformed coordinates, the explicit finite-difference algorithm is:

1. Fix $x_{\min}$, $x_{\max}$, M , and N .
2. Set the initial row u_j^0 from the transformed payoff.

3. Set the boundary values u_0^n and u_M^n for all n ; for a call, this means $u_0^n \approx 0$ and u_M^n from the transformed large- x asymptotics above.
4. For $n = 0, \dots, N - 1$, update the interior nodes with the FTCS rule.
5. Recover the option price by undoing the transformation:

$$V(S, t) = K e^{-ax - b\tau} u(x, \tau), \quad x = \log(S/K), \quad \tau = \frac{\sigma^2}{2}(T - t).$$

6. If the desired pair (x, τ) is not exactly on the grid, interpolate between neighboring grid points.

9.9 Practical Remarks

Three implementation choices dominate the quality of the result:

1. **Domain truncation.** If $x_{\max}$ is too small, the right boundary contaminates the interior solution.
2. **Grid ratio λ .** The explicit scheme is easy to implement, but stability forces $\Delta\tau$ to be small relative to Δx^2 .
3. **Boundary modeling.** Dirichlet data are simple when asymptotic prices are known. In other applications one may prefer Neumann conditions that prescribe a Greek, such as delta.

These observations explain why the explicit scheme is the natural starting point but rarely the final production method.

9.10 Summary and Connection to the Next Lecture

Finite differences turn the Black-Scholes pricing problem into a grid-based evolution problem. After transforming the PDE into the heat equation — removing the variable coefficients that arise from the log-normal dynamics — the pricing task reduces to:

- specifying initial data from the terminal payoff and boundary conditions from asymptotic option values,
- replacing derivatives by finite differences (forward difference in τ , central second difference in x),
- iterating the explicit FTCS update $\mathbf{u}^{n+1} = A\mathbf{u}^n + \mathbf{k}^n$ across the grid,
- transforming back to the original option value via $V(S, t) = K e^{-ax - b\tau} u(x, \tau)$.

The advantage is conceptual simplicity, direct access to the entire price surface for all spot values simultaneously, and straightforward computation of Greeks as numerical derivatives of the price grid. The disadvantage is the time-step restriction hidden in the stability parameter $\lambda = \Delta\tau/\Delta x^2 \leq \frac{1}{2}$: halving Δx for better spatial resolution forces $\Delta\tau$ to shrink by a factor of four, making the explicit scheme computationally expensive for high-accuracy calculations.

The next lecture addresses this bottleneck through implicit and Crank-Nicolson discretizations, which are unconditionally stable and eliminate the $\lambda \leq \frac{1}{2}$ restriction. Von Neumann stability analysis provides the theoretical framework for comparing schemes, and the Lax equivalence principle formalizes the connection between consistency, stability, and convergence.

10 Finite Difference Methods II: Implicit Schemes, Stability, and Accuracy

The explicit scheme from the previous lecture gave a clean first finite-difference method, but it also revealed the main weakness of forward stepping: the time step must be kept very small relative to the spatial mesh. This lecture develops the two standard remedies, the implicit scheme and the Crank-Nicolson scheme, and then connects them to the basic convergence theory of finite differences.

We continue to work with the transformed heat equation

$$u_\tau = u_{xx}$$

on a uniform grid (x_j, τ_n) , with

$$\lambda = \frac{\Delta\tau}{\Delta x^2}.$$

10.1 From Explicit to Implicit Time-Stepping

The explicit FTCS method evaluates the spatial second derivative at the old time level n . A natural alternative is to evaluate it at the new time level $n + 1$:

$$\frac{u_j^{n+1} - u_j^n}{\Delta\tau} = \frac{u_{j+1}^{n+1} - 2u_j^{n+1} + u_{j-1}^{n+1}}{\Delta x^2}.$$

Rearranging yields the **backward-time central-space** (BTCS) scheme

$$-\lambda u_{j-1}^{n+1} + (1 + 2\lambda)u_j^{n+1} - \lambda u_{j+1}^{n+1} = u_j^n.$$

Definition 10.1. The BTCS method is called **implicit** because the stencil involves the unknown values at time level $n + 1$. At every time step we must therefore solve a linear system rather than update the grid pointwise.

If we again collect the interior nodes in the vector $\mathbf{u}^n$, then

$$A\mathbf{u}^{n+1} = \mathbf{u}^n + \mathbf{k}^n,$$

with tridiagonal matrix

$$A = \begin{pmatrix} 1 + 2\lambda & -\lambda & & \\ -\lambda & 1 + 2\lambda & \ddots & \\ & \ddots & \ddots & -\lambda \\ & & -\lambda & 1 + 2\lambda \end{pmatrix}.$$

Because A is tridiagonal, each time step can still be solved efficiently by a Thomas algorithm.

10.2 The Crank-Nicolson Scheme

The Crank-Nicolson method averages the explicit and implicit second-derivative terms:

$$\frac{u_j^{n+1} - u_j^n}{\Delta\tau} = \frac{1}{2} \frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{\Delta x^2} + \frac{1}{2} \frac{u_{j+1}^{n+1} - 2u_j^{n+1} + u_{j-1}^{n+1}}{\Delta x^2}.$$

After rearranging,

$$-\frac{\lambda}{2}u_{j-1}^{n+1} + (1 + \lambda)u_j^{n+1} - \frac{\lambda}{2}u_{j+1}^{n+1} = \frac{\lambda}{2}u_{j-1}^n + (1 - \lambda)u_j^n + \frac{\lambda}{2}u_{j+1}^n.$$

In matrix form,

$$A_1 \mathbf{u}^{n+1} = A_2 \mathbf{u}^n + \mathbf{k}^n,$$

with

$$A_1 = \begin{pmatrix} 1 + \lambda & -\lambda/2 & & \\ -\lambda/2 & 1 + \lambda & \ddots & \\ & \ddots & \ddots & -\lambda/2 \\ & & -\lambda/2 & 1 + \lambda \end{pmatrix},$$

and

$$A_2 = \begin{pmatrix} 1 - \lambda & \lambda/2 & & \\ \lambda/2 & 1 - \lambda & \ddots & \\ & \ddots & \ddots & \lambda/2 \\ & & \lambda/2 & 1 - \lambda \end{pmatrix}.$$

Proposition 10.1. *The Crank-Nicolson scheme is second order in time and second order in space:*

$$\text{local truncation error} = O(\Delta\tau^2 + \Delta x^2).$$

In contrast, both FTCS and BTCS are only first order in time and second order in space.

Crank-Nicolson is therefore attractive when one wants higher temporal accuracy without abandoning the tridiagonal linear-system structure.

i Note

Crank-Nicolson is unconditionally stable, but for nonsmooth initial data such as vanilla-option payoffs it may still produce visible grid-scale oscillations near the kink. A common practical remedy is **Rannacher smoothing**: start with a few implicit Euler steps and then switch to Crank-Nicolson.

10.3 Comparing the Main Schemes

Scheme	Discretization	Time accuracy	Space accuracy	Main feature
FTCS	Forward in time, central in space	1	2	Explicit, but conditionally stable
BTCS	Backward in time, central in space	1	2	Implicit and unconditionally stable
Crank-Nicolson	Trapezoidal in time, central in space	2	2	Implicit, unconditionally stable, usually most accurate

The message is not that one scheme dominates in every situation. Rather, the numerical design problem is a balance between simplicity, stability, and accuracy.

10.4 Boundary Treatment Beyond Dirichlet Data

In Lecture 9 we mainly used Dirichlet conditions, meaning that the option value itself is prescribed at the boundary. Another common choice is a **Neumann condition**, where a derivative of the value function is prescribed instead.

For a call option on a sufficiently large stock-price domain, the delta satisfies

$$\frac{\partial V}{\partial S}(S_{\max}, t) \approx 1.$$

On a uniform S -grid with spacing ΔS , a central-difference approximation gives

$$\frac{V_{M+1}^n - V_{M-1}^n}{2\Delta S} \approx 1,$$

so the ghost point can be eliminated as

$$V_{M+1}^n \approx V_{M-1}^n + 2\Delta S.$$

Example 10.1. Ghost-point formulas turn derivative boundary conditions into algebraic corrections of the last matrix row. This is useful when the asymptotic option value is less reliable than the asymptotic Greek.

The same idea applies in transformed coordinates; the algebra becomes different, but the logic is identical.

10.5 Consistency, Stability, and the Lax Principle

Finite-difference methods are not judged only by whether they look plausible. We need a precise criterion for whether the numerical solution approaches the PDE solution as the mesh is refined.

Definition 10.2. Let L be the differential operator of a well-posed linear PDE and $L_{\Delta x, \Delta \tau}$ the finite-difference operator. The scheme is **consistent** if, for every sufficiently smooth test function u ,

$$(L_{\Delta x, \Delta \tau} u)(x_j, \tau_n) - (Lu)(x_j, \tau_n) \rightarrow 0 \quad \text{as } \Delta x, \Delta \tau \rightarrow 0.$$

Consistency is verified by Taylor expansion. For the heat equation:

- FTCS is $O(\Delta \tau + \Delta x^2)$,
- BTCS is $O(\Delta \tau + \Delta x^2)$,
- Crank-Nicolson is $O(\Delta \tau^2 + \Delta x^2)$.

Consistency alone is not enough. Numerical errors must also remain controlled as the scheme is iterated.

Theorem 10.1. *For a well-posed linear initial-value problem, a finite-difference scheme converges if and only if it is both consistent and stable.*

This is the content of the Lax equivalence principle. It is one of the main reasons stability analysis matters so much in practice.

10.6 Von Neumann Stability Analysis

For linear one-step schemes on a uniform grid, stability can often be studied by inserting a Fourier mode

$$u_j^n = \rho^n e^{ij\theta}, \quad \theta \in [-\pi, \pi],$$

into the finite-difference equation. The factor $\rho = \rho(\theta)$ is the **amplification factor**. Stability requires

$$|\rho(\theta)| \leq 1 \quad \text{for all } \theta.$$

10.6.1 FTCS

Substituting the Fourier mode into

$$u_j^{n+1} = \lambda u_{j+1}^n + (1 - 2\lambda)u_j^n + \lambda u_{j-1}^n$$

gives

$$\rho_{\text{FTCS}}(\theta) = 1 - 4\lambda \sin^2\left(\frac{\theta}{2}\right).$$

Hence

$$|\rho_{\text{FTCS}}(\theta)| \leq 1 \quad \text{for all } \theta \iff \lambda \leq \frac{1}{2}.$$

This is the familiar CFL-type restriction for the explicit scheme.

10.6.2 BTCS

For the implicit method,

$$\rho_{\text{BTCS}}(\theta) = \frac{1}{1 + 4\lambda \sin^2(\theta/2)}.$$

Since the denominator is always at least one, we have

$$|\rho_{\text{BTCS}}(\theta)| \leq 1 \quad \text{for every } \lambda > 0.$$

Thus BTCS is **unconditionally stable**.

10.6.3 Crank-Nicolson

For Crank-Nicolson,

$$\rho_{\text{CN}}(\theta) = \frac{1 - 2\lambda \sin^2(\theta/2)}{1 + 2\lambda \sin^2(\theta/2)}.$$

Again,

$$|\rho_{\text{CN}}(\theta)| \leq 1 \quad \text{for every } \lambda > 0,$$

so Crank-Nicolson is also unconditionally stable.

Proposition 10.2. *For the transformed Black-Scholes heat equation on a uniform grid:*

- *FTCS is stable only if $\lambda \leq \frac{1}{2}$,*
- *BTCS is unconditionally stable,*
- *Crank-Nicolson is unconditionally stable.*

Unconditional stability does not mean that arbitrarily coarse time grids are accurate. It only means that the scheme does not blow up numerically as the time-marching proceeds.

10.7 A Practical Solver Template

The production workflow for a one-factor finite-difference pricer is now straightforward:

1. transform the PDE to a convenient coordinate system,
2. choose the truncated spatial domain and the grid,
3. encode the payoff and the boundary conditions,
4. assemble the tridiagonal system for BTCS or Crank-Nicolson,
5. march through the grid in transformed time,
6. interpolate the solution back to the desired spot value.

```

import numpy as np

def crank_nicolson_heat_step(u_old, lam, left_bc, right_bc):
    """
    One Crank-Nicolson step for interior heat-equation values.
    u_old contains the interior nodes only.
    """
    m = len(u_old)
    main_1 = (1.0 + lam) * np.ones(m)
    off_1 = (-0.5 * lam) * np.ones(m - 1)

    main_2 = (1.0 - lam) * np.ones(m)
    off_2 = (0.5 * lam) * np.ones(m - 1)

    A1 = np.diag(main_1) + np.diag(off_1, 1) + np.diag(off_1, -1)
    A2 = np.diag(main_2) + np.diag(off_2, 1) + np.diag(off_2, -1)

    rhs = A2 @ u_old
    rhs[0] += 0.5 * lam * (left_bc["old"] + left_bc["new"])
    rhs[-1] += 0.5 * lam * (right_bc["old"] + right_bc["new"])

    return np.linalg.solve(A1, rhs)

```

The code block is intentionally schematic. In production one would use a dedicated tridiagonal solver rather than dense linear algebra, but the numerical structure is the same.

10.8 Summary and Connection to the Next Lectures

The explicit FTCS method provides a useful entry point, but serious finite-difference pricing relies on implicit or Crank-Nicolson time-stepping:

- BTCS is unconditionally stable and trades first-order time accuracy for the freedom to choose any $\lambda > 0$.
- Crank-Nicolson achieves the best of both worlds: unconditional stability and second-order accuracy in both time and space ($O(\Delta\tau^2 + \Delta x^2)$).
- The Lax equivalence principle gives the rigorous justification: consistency plus stability implies convergence for well-posed linear problems.
- Von Neumann analysis provides a systematic method for computing amplification factors and verifying stability conditions.

This closes the PDE-based portion of the course. Looking back across the full arc: Lecture 5 derived the Black-Scholes PDE and established Feynman-Kac as the theoretical bridge between PDEs and expectations; Lecture 6 evaluated those expectations by Monte Carlo; Lectures 9 and 10 solved the PDE directly on

a grid. All three routes price the same European call; the choice between them depends on dimension, path-dependence, and the need for Greeks.

The next two lectures return to Monte Carlo. Lecture 11 shows how variance-reduction techniques — antithetic variates and control variates — can dramatically reduce the $\mathcal{O}(N^{-1/2})$ standard error without increasing the number of paths. Lecture 12 then addresses the American-option problem from a simulation perspective through the Longstaff-Schwartz least-squares Monte Carlo algorithm, which combines the backward-induction logic of dynamic programming with Monte Carlo path generation.

11 Variance Reduction and Monte Carlo for Greeks

After the two finite-difference lectures, we return to the simulation viewpoint. Lecture 6 established the fundamental limitation of standard Monte Carlo: the RMSE decays as $\mathcal{O}(N^{-1/2})$ and reducing it by a factor of 10 requires 100 times as many paths. This lecture studies how to reduce estimator variance without changing the target expectation — shrinking σ_H rather than increasing N — and then turns to the related question of estimating option sensitivities efficiently within a single simulation run.

The variance-reduction techniques developed here apply broadly, but the Heston model from Lecture 8 provides a natural context. When pricing path-dependent exotics under stochastic volatility, antithetic variates, control variates (using the Black-Scholes price as a control), and importance sampling can all reduce the Monte Carlo standard error substantially relative to crude simulation. The Greeks estimation methods in the second half of the lecture build directly on the Greek definitions and Black-Scholes analytical formulas from Lectures 4 and 5.

As in the Monte Carlo lecture, we work with **discounted payoffs** in the variance-reduction sections below. To keep the formulas compact, $H(\mathbf{S})$ denotes that discounted payoff unless noted otherwise.

11.1 Variance Reduction for Monte Carlo Pricing

Recall that in standard Monte Carlo pricing, the **Root Mean Square Error (RMSE)** of our estimator $\hat{\pi}_0^N$ for the true option price π_0 is given by:

$$\text{RMSE}(\hat{\pi}_0^N) = \sqrt{\mathbb{E}_{\mathbb{Q}}[\varepsilon_N(H, \mathbf{S})^2]} = \frac{\sigma_H}{\sqrt{N}},$$

where: * $H(\mathbf{S})$ is the discounted payoff, which may depend on the entire path of the underlying asset process $\mathbf{S}$. * $\sigma_H := \sqrt{\text{Var}(H(\mathbf{S}))}$ is the standard deviation of the discounted payoff. * N is the number of independent Monte Carlo sample paths.

This equation highlights the main bottleneck of Monte Carlo simulation: the convergence rate is only $\mathcal{O}(N^{-1/2})$. To reduce the pricing error by a factor of 10, the number of simulated paths must therefore increase by a factor of 100.

Since we usually cannot improve the $N^{-1/2}$ rate itself, the practical strategy is to **shrink the numerator**, that is, to reduce $\sigma_H^2 = \text{Var}(H(\mathbf{S}))$.

The essence of **variance reduction** is to construct a new random variable Y and an estimator $g(Y)$ such that two conditions hold:

1. **Unbiasedness:** The new estimator yields the exact same expected value as the original payoff:

$$\mathbb{E}_{\mathbb{Q}}[g(Y)] = \mathbb{E}_{\mathbb{Q}}[H(\mathbf{S})]$$

2. **Reduced Variance:** The variance of the new estimator is strictly smaller than that of the naive estimator:

$$\text{Var}(g(Y)) < \text{Var}(H(\mathbf{S}))$$

If we can successfully construct such an estimator, we simulate $g(Y)$ instead of $H(\mathbf{S})$. This yields a much more statistically efficient estimator for the option price without requiring a larger computational budget N :

$$\text{RMSE}(\hat{\pi}_{\text{reduced}}^N) = \frac{\sqrt{\text{Var}(g(Y))}}{\sqrt{N}} < \frac{\sigma_H}{\sqrt{N}}.$$

11.1.1 Antithetic Variates

Antithetic variates exploit symmetry in the driving random variables to improve the efficiency of Monte Carlo estimators.

Consider a random variable U that is uniformly distributed on $[0, 1]$. Notice that if $U \sim \text{Uniform}(0, 1)$, then its complement $1 - U$ is also distributed as $\text{Uniform}(0, 1)$. Similarly, if Z is a standard normal random variable ($Z \sim \mathcal{N}(0, 1)$), then its negation $-Z$ has the exact same distribution. The variables $(1 - U)$ and $-Z$ are called the **antithetic variates** of U and Z , respectively.

In the context of stochastic processes, if a simulated asset path $\mathbf{S}$ is generated from standard normal shocks $Z_1, Z_2, \dots, Z_n$, then the **antithetic path** $\tilde{\mathbf{S}}$ is obtained by replacing these shocks with $-Z_1, -Z_2, \dots, -Z_n$.

Unbiasedness

Because the driving shocks have the same distribution, the resulting asset paths $\mathbf{S}$ and $\tilde{\mathbf{S}}$ are identically distributed under the risk-neutral measure $\mathbb{Q}$. Therefore, taking the average of their payoffs yields an unbiased estimator of the true option price:

$$\mathbb{E}_{\mathbb{Q}} \left[\frac{H(\mathbf{S}) + H(\tilde{\mathbf{S}})}{2} \right] = \frac{1}{2} \mathbb{E}_{\mathbb{Q}}[H(\mathbf{S})] + \frac{1}{2} \mathbb{E}_{\mathbb{Q}}[H(\tilde{\mathbf{S}})] = \mathbb{E}_{\mathbb{Q}}[H(\mathbf{S})]$$

The Condition for Variance Reduction

To see when this helps, we compare computational cost. Generating the pair $(\mathbf{S}, \tilde{\mathbf{S}})$ requires two payoff evaluations, so the **mean squared error (MSE)** of an estimator based on N antithetic pairs should be compared with standard Monte Carlo using $2N$ **independent samples**.

Let the antithetic estimator be defined as:

$$\hat{\pi}_0^{A,N}(H, \mathbf{S}) := \frac{1}{N} \sum_{i=1}^N \frac{H(\mathbf{S}^{(i)}) + H(\tilde{\mathbf{S}}^{(i)})}{2}$$

For this to be more efficient than standard Monte Carlo with $2N$ paths, we require:

$$\frac{1}{N} \text{Var} \left(\frac{H(\mathbf{S}) + H(\tilde{\mathbf{S}})}{2} \right) < \frac{1}{2N} \text{Var}(H(\mathbf{S}))$$

By expanding the variance of the sum on the left-hand side, we get:

$$\text{Var} \left(\frac{H(\mathbf{S}) + H(\tilde{\mathbf{S}})}{2} \right) = \frac{1}{4} [\text{Var}(H(\mathbf{S})) + \text{Var}(H(\tilde{\mathbf{S}})) + 2\text{Cov}(H(\mathbf{S}), H(\tilde{\mathbf{S}}))]$$

Since $\mathbf{S}$ and $\tilde{\mathbf{S}}$ are identically distributed, their variances are exactly equal ($\text{Var}(H(\mathbf{S})) = \text{Var}(H(\tilde{\mathbf{S}}))$). We can simplify the expression to:

$$\text{Var} \left(\frac{H(\mathbf{S}) + H(\tilde{\mathbf{S}})}{2} \right) = \frac{1}{2} \text{Var}(H(\mathbf{S})) + \frac{1}{2} \text{Cov}(H(\mathbf{S}), H(\tilde{\mathbf{S}}))$$

Comparing this back to our efficiency requirement, the antithetic estimator is superior **if and only if**:

$$\text{Cov}(H(\mathbf{S}), H(\tilde{\mathbf{S}})) < 0$$

This negative covariance condition is the mathematical heart of the technique. It is often satisfied when: * The transformation from $\mathbf{S}$ to $\tilde{\mathbf{S}}$ induces **negative dependence** (which negating the Brownian increments naturally achieves). * The payoff function H is **monotonic** with respect to the underlying asset. For example, standard European call and put options are monotonic, so a large positive shock to one path tends to be offset by a corresponding negative shock to its antithetic counterpart.

For non-monotone or highly discontinuous payoffs, however, the covariance need not be negative, so antithetic variates should be checked empirically rather than assumed to help.

```
import numpy as np
from scipy.stats import norm

# =====
# Black-Scholes Parameters
# =====

S0 = 100.0      # initial stock price
K = 105.0      # strike price
T = 1.0         # maturity (years)
sigma = 0.2     # volatility (20%)
r = 0.05        # risk-free rate (5%)
N = 100_000     # base number of samples

rng = np.random.default_rng(42) # reproducibility
```

```

# =====
# Analytical Black-Scholes Formula
# =====

def black_scholes_call(S0, K, r, sigma, T):
    d1 = (np.log(S0 / K) + (r + 0.5 * sigma**2) * T) / (sigma * np.sqrt(T))
    d2 = d1 - sigma * np.sqrt(T)
    return S0 * norm.cdf(d1) - K * np.exp(-r * T) * norm.cdf(d2)

# =====
# Standard Monte Carlo
# =====

def mc_price_european_call(S0, K, r, sigma, T, N, rng):
    drift = (r - 0.5 * sigma**2) * T
    diffusion = sigma * np.sqrt(T)

    Z = rng.standard_normal(N)
    ST = S0 * np.exp(drift + diffusion * Z)

    payoffs = np.maximum(ST - K, 0.0)
    discounted = np.exp(-r * T) * payoffs

    price = discounted.mean()
    std_error = discounted.std(ddof=1) / np.sqrt(N)

    return price, std_error

# =====
# Antithetic Variates
# =====

def mc_price_european_call_antithetic(S0, K, r, sigma, T, N, rng):
    drift = (r - 0.5 * sigma**2) * T
    diffusion = sigma * np.sqrt(T)

    Z = rng.standard_normal(N)

    ST1 = S0 * np.exp(drift + diffusion * Z)
    ST2 = S0 * np.exp(drift - diffusion * Z)

    payoff1 = np.maximum(ST1 - K, 0.0)

```

```

    payoff2 = np.maximum(ST2 - K, 0.0)

    averaged = 0.5 * (payoff1 + payoff2)
    discounted = np.exp(-r * T) * averaged

    price = discounted.mean()
    std_error = discounted.std(ddof=1) / np.sqrt(N)

    return price, std_error

# =====
# Run Simulations
# =====

price_mc, error_mc = mc_price_european_call(S0, K, r, sigma, T, 2 * N, rng)
price_antithetic, error_antithetic = mc_price_european_call_antithetic(S0, K, r, sigma, T, N, rng)
price_bs = black_scholes_call(S0, K, r, sigma, T)

# =====
# Output
# =====

print("European Call Option Pricing Comparison")
print("=====")
print(f"Analytical Black-Scholes Price      : ${price_bs:.4f}\n")

print(f"Standard Monte Carlo (2N samples)    : ${price_mc:.4f}")
print(f"  Standard Error                      : ±{error_mc:.5f}\n")

print(f"Antithetic Variates (N pairs)          : ${price_antithetic:.4f}")
print(f"  Standard Error                      : ±{error_antithetic:.5f}\n")

efficiency_gain = (error_mc**2) / (error_antithetic**2)
print(f"Efficiency gain (MC Var / Antithetic Var)  {efficiency_gain:.2f}×")

```

```

European Call Option Pricing Comparison
=====
Analytical Black-Scholes Price      : $8.0214

Standard Monte Carlo (2N samples)  : $8.0398
  Standard Error                   : ±0.02971

Antithetic Variates (N pairs)      : $7.9857

```


Standard Error : ± 0.02334

Efficiency gain (MC Var / Antithetic Var) 1.62×

11.1.2 Control Variates

The **control variates** method uses analytical information from a simpler but related problem. The idea is to anchor the Monte Carlo estimator to the known expectation of a correlated random variable.

Assume we want to price a complex derivative with discounted payoff $H(\mathbf{S})$. Suppose we can identify: - A related random variable $\mathbf{Y}$ - A functional $g(\mathbf{Y})$

such that its expected value $\mathbb{E}_{\mathbb{Q}}[g(\mathbf{Y})]$ is known exactly in closed form.

For any real-valued scalar $\lambda \in \mathbb{R}$, we can exploit the linearity of expectations to state the following identity:

$$\mathbb{E}_{\mathbb{Q}}[H(\mathbf{S})] = \mathbb{E}_{\mathbb{Q}}[H(\mathbf{S}) - \lambda(g(\mathbf{Y}) - \mathbb{E}_{\mathbb{Q}}[g(\mathbf{Y})])]$$

For any fixed (non-random) choice of λ , the term $\mathbb{E}_{\mathbb{Q}}[g(\mathbf{Y}) - \mathbb{E}_{\mathbb{Q}}[g(\mathbf{Y})]]$ is zero, so adding it does not change the target expectation. This leads to the **control-variates Monte Carlo estimator**:

$$\hat{\pi}_0^{C,\lambda}(H, \mathbf{S}) := \frac{1}{N} \sum_{i=1}^N (H(\mathbf{S}^{(i)}) - \lambda(g(\mathbf{Y}^{(i)}) - \mathbb{E}_{\mathbb{Q}}[g(\mathbf{Y})]))$$

where $(\mathbf{S}^{(i)}, \mathbf{Y}^{(i)})$ are N independent joint samples.

This exact unbiasedness statement assumes that λ is fixed in advance, or estimated on an independent pilot sample. If λ is fitted from the same Monte Carlo sample, the resulting plug-in estimator remains consistent and is usually much lower-variance, but it carries a small $\mathcal{O}(N^{-1})$ bias.

The Optimal Choice of λ

The effectiveness of this technique hinges entirely on choosing the right penalty parameter, λ . Our goal is to minimize the variance of our new estimator:

$$\text{Var}(H(\mathbf{S}) - \lambda(g(\mathbf{Y}) - \mathbb{E}_{\mathbb{Q}}[g(\mathbf{Y})])) = \text{Var}(H(\mathbf{S})) - 2\lambda \text{Cov}(H(\mathbf{S}), g(\mathbf{Y})) + \lambda^2 \text{Var}(g(\mathbf{Y}))$$

By taking the derivative with respect to λ and setting it to zero, we find that the variance is minimized by the optimal coefficient:

$$\lambda^* = \frac{\text{Cov}(H(\mathbf{S}), g(\mathbf{Y}))}{\text{Var}(g(\mathbf{Y}))}$$

Substituting λ^* back into our variance equation yields the minimized variance of the control variate estimator:

$$\text{Var}(\hat{\pi}_0^{C,\lambda^*}) = \frac{1}{N} \left(\text{Var}(H(\mathbf{S})) - \frac{\text{Cov}(H(\mathbf{S}), g(\mathbf{Y}))^2}{\text{Var}(g(\mathbf{Y}))} \right)$$

We can express this more intuitively using the Pearson correlation coefficient $\rho_{H,g}$ between $H(\mathbf{S})$ and $g(\mathbf{Y})$:

$$\text{Var}(\hat{\pi}_0^{C,\lambda^*}) = \frac{\text{Var}(H(\mathbf{S}))}{N} (1 - \rho_{H,g}^2)$$

Computational Efficiency Gain

This formulation shows that the variance is strictly reduced as long as the payoff and the control variate are correlated ($\rho_{H,g} \neq 0$). The closer $\rho_{H,g}$ is to 1 or -1 , the more dramatic the variance reduction.

The improvement is a multiplicative factor. Let $c \geq 1$ represent the additional computational cost per sample required to simulate the control variate alongside the main payoff. The overall efficiency gain (speed-up factor) of the control variates technique over standard Monte Carlo is:

$$\text{Speed-up} = \frac{1}{c(1 - \rho_{H,g}^2)}$$

For example, if evaluating the control variate doubles the simulation time per path ($c = 2$), but the correlation is high ($\rho_{H,g} = 0.95$), the speed-up factor is $\frac{1}{2(1-0.95^2)} \approx 5.1$. You achieve the same accuracy more than 5 times faster!

11.1.2.1 Finding Good Control Variates

A natural question arises: **how do we find or construct effective control variates?** Unfortunately, there is no universal answer—control variates must be tailored to the specific payoff and model. However, in computational finance, several robust strategies exist:

- **The underlying asset itself** often serves as a powerful, zero-cost control variate. Because we work with discounted payoffs in this lecture, the natural control is the discounted asset $e^{-rT}S_T$, whose expectation is analytically known under the risk-neutral measure: $\mathbb{E}_{\mathbb{Q}}[e^{-rT}S_T] = S_0e^{-qT}$ when the continuous dividend yield is q .
- **Simple options with closed-form solutions** (such as vanilla European calls and puts under Black-Scholes) make excellent control variates when pricing path-dependent exotics like Asian or Barrier options.
- **Simplified dynamics** can be used as a control. For example, when pricing an option under a complex Stochastic Volatility model (like Heston), the price of the same option under the Black-Scholes model with constant volatility can serve as an excellent control variate.

Example: Arithmetic versus Geometric Asian Call Pricing

A particularly instructive control-variate example arises for **Asian options** under the Black-Scholes model. Consider equally spaced monitoring dates

$$t_j = \frac{jT}{m}, \quad j = 1, \dots, m,$$

and define the arithmetic and geometric averages

$$A_m^{\text{arith}} = \frac{1}{m} \sum_{j=1}^m S_{t_j}, \quad A_m^{\text{geo}} = \exp \left(\frac{1}{m} \sum_{j=1}^m \log S_{t_j} \right).$$

Our pricing target is the discounted arithmetic Asian call payoff

$$H = e^{-rT} (A_m^{\text{arith}} - K)^+,$$

which does **not** admit a simple closed form in the Black-Scholes model. As control variate we use the discounted geometric Asian call payoff

$$G = e^{-rT} (A_m^{\text{geo}} - K)^+.$$

This is an excellent choice because it matches the path dependence and averaging structure of the target payoff. Moreover,

$$A_m^{\text{geo}} \leq A_m^{\text{arith}}$$

pathwise, so H and G are typically extremely highly correlated when computed from the same simulated path.

The key advantage is that G can be priced analytically. Indeed, under risk-neutral geometric Brownian motion,

$$\log S_{t_j} = \log S_0 + \left(r - \frac{1}{2} \sigma^2 \right) t_j + \sigma W_{t_j},$$

so

$$Y := \frac{1}{m} \sum_{j=1}^m \log S_{t_j}$$

is Gaussian with

$$\mu_Y = \log S_0 + \left(r - \frac{1}{2} \sigma^2 \right) \frac{T(m+1)}{2m}, \quad v_Y = \sigma^2 T \frac{(m+1)(2m+1)}{6m^2}.$$

Since $A_m^{\text{geo}} = e^Y$ is lognormal, the geometric Asian call price is

$$\pi_0^{\text{geo}} = e^{-rT} \left(e^{\mu_Y + \frac{1}{2} v_Y} \Phi(d_1) - K \Phi(d_2) \right),$$

where

$$d_1 = \frac{\mu_Y - \log K + v_Y}{\sqrt{v_Y}}, \quad d_2 = d_1 - \sqrt{v_Y}.$$

This yields the control-variate estimator

$$\hat{\pi}_0^{\text{cv}} = \frac{1}{N} \sum_{i=1}^N \left(H^{(i)} - \hat{\lambda}(G^{(i)} - \pi_0^{\text{geo}}) \right).$$

To keep the estimator as close as possible to the ideal unbiased setup discussed above, a common practical choice is to estimate $\hat{\lambda}$ on an independent **pilot sample** and then apply it on a fresh main simulation sample.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.stats import norm

# =====
# Parameters
# =====

S0 = 100.0
K = 100.0
T = 1.0
sigma = 0.2
r = 0.05
m = 52          # weekly monitoring over one year
N_pilot = 10_000
N_main = 50_000

rng = np.random.default_rng(42)

# =====
# Closed form for the discretely monitored geometric Asian call
# =====

def geometric_asian_call_closed_form(S0, K, r, sigma, T, m):
    mu_Y = np.log(S0) + (r - 0.5 * sigma**2) * T * (m + 1) / (2 * m)
    v_Y = sigma**2 * T * (m + 1) * (2 * m + 1) / (6 * m**2)
    std_Y = np.sqrt(v_Y)

    d1 = (mu_Y - np.log(K) + v_Y) / std_Y
    d2 = d1 - std_Y

    return np.exp(-r * T) * (
        np.exp(mu_Y + 0.5 * v_Y) * norm.cdf(d1) - K * norm.cdf(d2)
    )
```

```

# =====
# Simulate discounted arithmetic and geometric Asian payoffs
# =====

def simulate_asian_call_payoffs(S0, K, r, sigma, T, m, n_paths, rng):
    dt = T / m
    drift = (r - 0.5 * sigma**2) * dt
    diffusion = sigma * np.sqrt(dt)

    Z = rng.standard_normal((n_paths, m))
    log_paths = np.log(S0) + np.cumsum(drift + diffusion * Z, axis=1)
    S_paths = np.exp(log_paths)

    arithmetic_average = S_paths.mean(axis=1)
    geometric_average = np.exp(log_paths.mean(axis=1))
    discount = np.exp(-r * T)

    arithmetic_payoff = discount * np.maximum(arithmetic_average - K, 0.0)
    geometric_payoff = discount * np.maximum(geometric_average - K, 0.0)

    return arithmetic_payoff, geometric_payoff

# =====
# Pilot sample to estimate lambda
# =====

arith_pilot, geo_pilot = simulate_asian_call_payoffs(
    S0, K, r, sigma, T, m, N_pilot, rng
)

lambda_hat = np.cov(arith_pilot, geo_pilot, ddof=1)[0, 1] / np.var(geo_pilot, ddof=1)
geo_exact = geometric_asian_call_closed_form(S0, K, r, sigma, T, m)

# =====
# Main Monte Carlo sample
# =====

arith_main, geo_main = simulate_asian_call_payoffs(
    S0, K, r, sigma, T, m, N_main, rng
)

cv_samples = arith_main - lambda_hat * (geo_main - geo_exact)

```

```

plain_price = arith_main.mean()
plain_std_error = arith_main.std(ddof=1) / np.sqrt(N_main)

cv_price = cv_samples.mean()
cv_std_error = cv_samples.std(ddof=1) / np.sqrt(N_main)

correlation = np.corrcoef(arith_main, geo_main)[0, 1]
variance_reduction = (plain_std_error / cv_std_error) ** 2

# =====
# Publication-ready summary
# =====

summary = pd.DataFrame(
    {
        "Estimator": [
            "Crude Monte Carlo (Arithmetic Asian)",
            "Control variate (Arithmetic Asian)",
        ],
        "Price": [plain_price, cv_price],
        "Std. Error": [plain_std_error, cv_std_error],
    }
)

print("Arithmetic Asian call with geometric Asian control variate")
print("=====")
print(summary.to_string(index=False, float_format=lambda x: f"{x:0.6f}"))
print()
print(f"Closed-form geometric Asian price : {geo_exact:0.6f}")
print(f"Estimated pilot lambda : {lambda_hat:0.6f}")
print(f"Empirical correlation(H, G) : {correlation:0.6f}")
print(f"Variance reduction factor : {variance_reduction:0.2f}x")

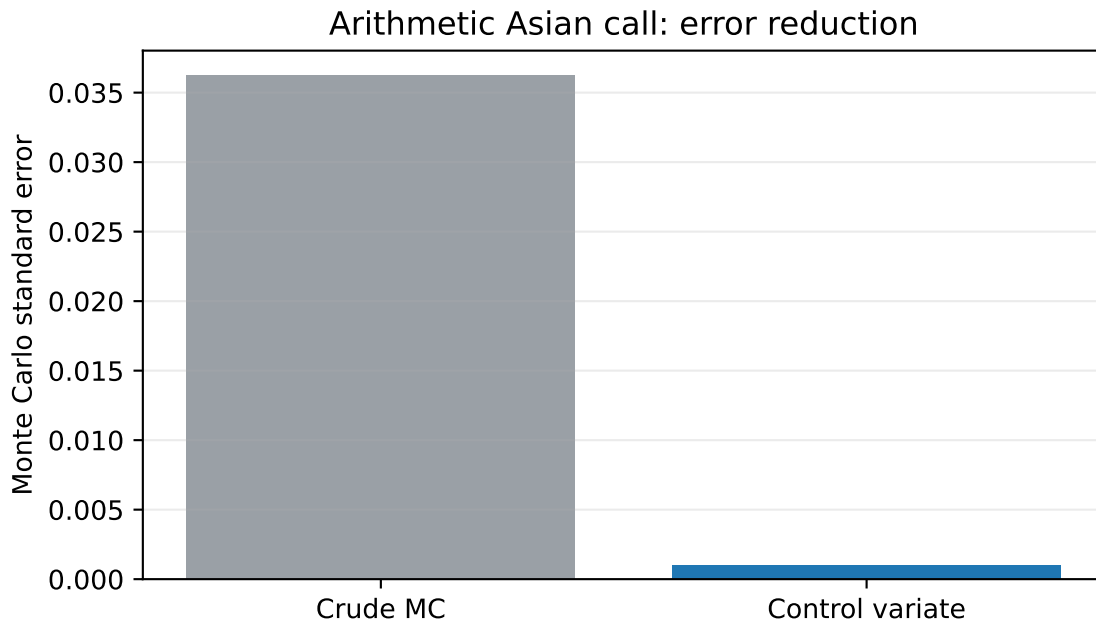
fig, ax = plt.subplots(figsize=(6, 3.5))
ax.bar(
    ["Crude MC", "Control variate"],
    [plain_std_error, cv_std_error],
    color=["#9aa0a6", "#1f77b4"],
)
ax.set_ylabel("Monte Carlo standard error")
ax.set_title("Arithmetic Asian call: error reduction")
ax.grid(axis="y", alpha=0.25)
plt.tight_layout()
plt.show()

```

Arithmetic Asian call with geometric Asian control variate

	Estimator	Price	Std. Error
Crude Monte Carlo (Arithmetic Asian)		5.860920	0.036211
Control variate (Arithmetic Asian)		5.853852	0.000995

Closed-form geometric Asian price : 5.637432
 Estimated pilot lambda : 1.034004
 Empirical correlation(H, G) : 0.999622
 Variance reduction factor : 1323.56x



This example is the canonical control-variate construction for Asian options: the geometric Asian payoff preserves the same path dependence as the arithmetic payoff, yet remains analytically tractable. In practice, that combination of structural similarity and closed-form expectation is exactly what makes a control variate effective.

11.1.3 Importance Sampling

Importance sampling is a variance reduction technique that improves simulation efficiency by allocating more samples to regions of the sample space that contribute most significantly to the expected value. In continuous-time finance, this concept is deeply connected to **Girsanov's Theorem** and the mathematical change of measure.

Assume that the underlying random variable $\mathbf{S} \in \mathbb{R}^d$ has a known probability density function $q(x)$, and let $H(\mathbf{S})$ be the discounted payoff of a derivative. For any alternative density $p(x) > 0$ defined on the same support as q , we can reweight the integral by multiplying and dividing by $p(x)$:

$$\mathbb{E}_q[H(\mathbf{S})] = \int_{\mathbb{R}^d} H(x)q(x)dx = \int_{\mathbb{R}^d} H(x)\frac{q(x)}{p(x)}p(x)dx = \mathbb{E}_p \left[H(\mathbf{Y})\frac{q(\mathbf{Y})}{p(\mathbf{Y})} \right]$$

where $\mathbf{Y} \sim p$ is a random variable sampled under the new **importance density** p . The ratio $L(\mathbf{Y}) = \frac{q(\mathbf{Y})}{p(\mathbf{Y})}$ is the **likelihood ratio** (also known as the Radon–Nikodym derivative).

This gives rise to the **importance sampling Monte Carlo estimator**:

$$\hat{\pi}_0^{\text{Imp},N}(H, \mathbf{S}) = \frac{1}{N} \sum_{i=1}^N H(\mathbf{Y}^{(i)}) \frac{q(\mathbf{Y}^{(i)})}{p(\mathbf{Y}^{(i)})}$$

where $\mathbf{Y}^{(i)} \sim p$ are independent samples drawn from the new distribution.

The efficiency of this method hinges entirely on the variance of the weighted payoff. A robust importance distribution $p(x)$ should: 1. Emphasize regions where the absolute payoff $|H(x)|$ is large. 2. Maintain a manageable likelihood ratio to prevent explosive variance spikes.

Q: How do we choose the optimal importance sampling density p ?

The theoretical goal is to select a density p that minimizes the variance of our new estimator.

The Theoretical Optimum

If the integrand has a fixed sign (in particular, for nonnegative option payoffs), it can be proven that the variance is minimized to exactly zero if we choose:

$$p^*(x) = \frac{|H(x)|q(x)}{\mathbb{E}_q[|H(\mathbf{S})|]} \propto |H(x)| \cdot q(x)$$

However, this is a Catch-22: the denominator requires exact knowledge of $\mathbb{E}_q[|H(\mathbf{S})|]$, which is the very integral we are trying to solve! Therefore, the mathematically optimal density p^* cannot be implemented in practice.

Instead, we aim for a **practical approximation**: we choose $p(x)$ such that the shape of $p(x)$ roughly mimics the shape of $|H(x)|q(x)$. By doing so, the ratio $\frac{H(x)q(x)}{p(x)}$ becomes relatively flat (constant), which dramatically flattens the variance across the sample space.

11.1.3.1 Example: Mean Shift for a Deep OTM European Call

Suppose we want to compute the price of a European call option with payoff $H(S_T) = (S_T - K)^+$. Under the risk-neutral measure q , the terminal asset price is driven by a standard normal random variable $Z \sim \mathcal{N}(0, 1)$:

$$S_T(Z) = S_0 \exp \left(\left(r - \frac{\sigma^2}{2} \right) T + \sigma \sqrt{T} Z \right)$$

If the option is **deep out-of-the-money** ($K \gg S_0$), the vast majority of standard normal draws will result in $S_T < K$, yielding a payoff of zero. We waste immense computational resources simulating paths that contribute nothing to the expectation but still add variance.

To fix this, we shift the mean of the normal distribution upward so that more paths end up in-the-money. Let's sample from a new distribution p where our standard normal is shifted by a constant $\mu_Z > 0$.

Under p , we generate standard normal draws $\tilde{Z}^{(i)} \sim \mathcal{N}(0, 1)$, but we apply the shift to our asset price formula: $Y^{(i)} = \tilde{Z}^{(i)} + \mu_Z$. The likelihood ratio for a normal distribution shifted by μ_Z evaluated at the drawn point $Y^{(i)}$ simplifies elegantly to:

$$\frac{q(Y^{(i)})}{p(Y^{(i)})} = \frac{\exp(-(Y^{(i)})^2/2)}{\exp(-(Y^{(i)} - \mu_Z)^2/2)} = \exp\left(-\mu_Z Y^{(i)} + \frac{1}{2}\mu_Z^2\right)$$

Substituting $Y^{(i)} = \tilde{Z}^{(i)} + \mu_Z$ into the exponent, the likelihood ratio solely in terms of our computer-generated draw $\tilde{Z}^{(i)}$ becomes:

$$\frac{q}{p} = \exp\left(-\mu_Z \tilde{Z}^{(i)} - \frac{1}{2}\mu_Z^2\right)$$

The variance-reduced Monte Carlo estimator is then:

$$\hat{\pi}_0^{\text{Imp}} = e^{-rT} \frac{1}{N} \sum_{i=1}^N \left[(S_T(\tilde{Z}^{(i)} + \mu_Z) - K)^+ \cdot \exp\left(-\mu_Z \tilde{Z}^{(i)} - \frac{1}{2}\mu_Z^2\right) \right]$$

By tuning the shift parameter μ_Z such that the expected asset price lands near the strike K , we force the simulation to focus on the rare but highly impactful events, drastically shrinking the standard error!

11.1.4 Concluding Remarks on Variance Reduction Techniques

Variance reduction techniques are indispensable tools in computational finance. They allow us to enhance the efficiency of our Monte Carlo estimators by lowering the estimator's variance without incurring the heavy computational cost of generating additional sample paths.

While none of the methods discussed alter the fundamental $\mathcal{O}(N^{-1/2})$ convergence rate of standard Monte Carlo, reducing the variance constant can still lead to massive performance gains in practice. To summarize our current arsenal:

- **Antithetic Variates:** Computationally inexpensive and highly straightforward to implement, this method leverages the natural symmetry of Brownian increments. It offers reliable, moderate variance reduction—particularly when the derivative's payoff function is monotonic. However, its ultimate effectiveness is bounded by the degree of negative correlation induced between the original and antithetic samples.

- **Control Variates:** This method can yield dramatic variance reduction by anchoring the simulation to the known, analytical expectation of a correlated variable (such as the underlying asset or a closed-form European option). When a suitable control is available and strongly correlated with the target derivative, this technique is exceptionally powerful. Its primary limitation is the availability and proper identification of a highly correlated, analytically tractable control.
- **Importance Sampling:** By applying a mathematical change of measure, this technique strategically shifts the sampling distribution to focus computationally on the regions of the state space that contribute the most to the expectation. It is the premier choice for pricing deep out-of-the-money options and handling rare-event simulations. However, selecting the optimal importance density is highly problem-specific, making its implementation mathematically and computationally more involved.

11.2 Estimating Option Sensitivities (Greeks)

In this section, we transition from merely pricing derivatives to computing their **sensitivities**, universally known as the **Greeks**. These are the partial derivatives of the option price with respect to various model parameters (e.g., the initial asset price, volatility, interest rate) and non-model parameters.

The Greeks play a fundamental role in financial risk management, dynamic hedging, and portfolio optimization. While they were conceptually introduced in the first part of this course, we now turn our attention to the practical estimation techniques required to compute them, particularly within the context of Monte Carlo simulation.

11.2.1 Problem Formulation

Let us formalize the pricing framework to clearly distinguish between the parameters that drive the underlying asset dynamics and those that define the derivative contract.

Let the price of the derivative at time $t = 0$ be denoted by the function π_0 . Under the risk-neutral pricing framework, this price is the discounted expected value of the payoff:

$$\pi_0(S_0, \sigma, r, \tau, \dots) = \mathbb{E}_{\mathbb{Q}}[e^{-r\tau} H(\mathbf{S}; K, B, \dots)]$$

Notice that we separate the parameters into two distinct categories:

1. **Model Parameters:** These dictate the underlying stochastic process $\mathbf{S}$ and the discounting. * S_0 : Current price of the underlying asset * σ : Volatility parameter of the asset * r : Risk-free interest rate * τ : Time to maturity (where $\tau = T - 0$)
2. **Contractual (Non-Model) Parameters:** These are fixed by the derivative's term sheet and dictate the specific shape of the payoff function H . * K : Strike price of the option * B : Barrier level (for barrier options) * ...: Other contractual terms like rebate levels, averaging windows, etc.

Our primary computational goal is to estimate: * **Model sensitivities:** The partial derivatives of π_0 with respect to the continuous model inputs (S_0, σ, τ, r) . * **Contract sensitivities:** The partial derivatives with respect to structural contract parameters (e.g., K).

11.2.2 Reminder on the Greeks

The Greeks quantify exactly how sensitive the value of our derivative π_0 is to marginal changes in these inputs. Computing them efficiently is arguably more important to a trading desk than computing the price itself, as the Greeks dictate how to hedge the portfolio against market movements.

First-Order Greeks

The first-order derivatives measure the linear sensitivity to a single parameter:

- **Delta (Δ):** The sensitivity of the option price to a change in the initial underlying asset price S_0 .

$$\Delta := \frac{\partial \pi_0}{\partial S_0}$$

- **Vega (ν):** The sensitivity of the option price to a change in the volatility parameter σ .

$$\nu := \frac{\partial \pi_0}{\partial \sigma}$$

- **Theta (Θ):** In the course convention, theta is the calendar-time derivative $\Theta = \partial_t V$. When the price is written as a function of time to maturity τ , the corresponding maturity sensitivity is $\partial_\tau \pi_0 = -\Theta$.

$$\Theta := \frac{\partial V}{\partial t}, \quad \frac{\partial \pi_0}{\partial \tau} = -\Theta$$

- **Rho (ρ):** The sensitivity of the option price to a change in the risk-free interest rate r .

$$\rho := \frac{\partial \pi_0}{\partial r}$$

Second-Order (Higher-Order) Greeks

The second-order derivatives measure the curvature, or how the first-order Greeks themselves change as parameters move. These are crucial for managing large market moves where linear approximations fail:

- **Gamma (Γ):** The rate of change of Delta with respect to the underlying asset price S_0 . It measures the convexity of the option's value relative to the asset.

$$\Gamma := \frac{\partial \Delta}{\partial S_0} = \frac{\partial^2 \pi_0}{\partial S_0^2}$$

- **Vomma (or Volga):** The sensitivity of Vega to changes in volatility. It measures the convexity of the option's value relative to volatility.

$$\text{Vomma} := \frac{\partial \nu}{\partial \sigma} = \frac{\partial^2 \pi_0}{\partial \sigma^2}$$

In the upcoming sections, we will explore **Monte Carlo techniques**—such as the Finite Difference method, the Pathwise Derivative method, and the Likelihood Ratio method—that allow us to compute unbiased and low-variance estimators for these critical derivatives.

11.2.3 Bump-and-Revalue Method for Estimating Greeks

The **bump-and-revalue** method is one of the most intuitive and widely used approaches for estimating Greeks. It approximates partial derivatives via **finite differences**, relying on re-evaluating the option price after a small perturbation (or “bump”) to one of the input parameters.

Estimating Delta (Δ)

To estimate the sensitivity of the option price π_0 with respect to the initial stock price S_0 , proceed as follows:

1. **Bump** the underlying asset price S_0 by a small amount ΔS_0 .
2. **Revalue** the option using both the original and bumped values:
 - Compute $\pi_0(S_0)$ — the original option price.
 - Compute $\pi_0(S_0 + \Delta S_0)$ — the option price with bumped input.
3. **Approximate Delta** using the finite difference formula:

$$\Delta \approx \frac{\pi_0(S_0 + \Delta S_0) - \pi_0(S_0)}{\Delta S_0}, \quad \text{for small } \Delta S_0.$$

Algorithm: Bump-and-Revalue Estimation of Greeks

The **bump-and-revalue** method approximates option sensitivities (Greeks) by perturbing one model input at a time and observing the impact on the derivative price.

Inputs: - Initial parameter vector $\mathbf{p}$ (e.g., $S_0, \sigma, r, \tau, \dots$) - Pricing function or Monte Carlo estimator $\pi_0(\mathbf{p})$
- Small bump size δ - Index or name of the parameter to bump

Output: - Estimated Greek value $\widehat{\text{Greek}}$

Procedure:

1. **Evaluate the base price:**

$$\pi_0^{\text{original}} = \pi_0(\mathbf{p})$$

2. **Apply the bump** to the chosen parameter:

$$\mathbf{p}_{\text{bump}} = \mathbf{p}, \quad \mathbf{p}_{\text{bump}}[i] \leftarrow \mathbf{p}[i] + \delta$$

3. **Re-evaluate the price** with the bumped parameter:

$$\pi_0^{\text{bump}} = \pi_0(\mathbf{p}_{\text{bump}})$$

4. **Estimate the Greek** using the finite difference:

$$\widehat{\text{Greek}} = \frac{\pi_0^{\text{bump}} - \pi_0^{\text{original}}}{\delta}$$

5. **Return** the estimated value $\widehat{\text{Greek}}$

11.2.4 Variance Reduction in Bump-and-Revalue: Common Random Numbers

A critical consideration when applying the **bump-and-revalue** (finite difference) method to estimate Greeks is controlling the **variance of the estimator**. Because Monte Carlo prices naturally contain random noise, taking the difference between two noisy prices can lead to severe numerical instability.

The Finite Difference Estimator Setup

Suppose we wish to estimate a first-order sensitivity—for example, **Delta** (Δ)—using a forward finite difference approximation. The Monte Carlo estimator for this Greek is computed as:

$$\widehat{\Delta} = \frac{\hat{\pi}_0(S_0 + \delta) - \hat{\pi}_0(S_0)}{\delta}$$

where $\hat{\pi}_0(S_0)$ is the base Monte Carlo estimate of the option price, $\hat{\pi}_0(S_0 + \delta)$ is the “bumped” price estimate, and δ is a small perturbation (e.g., 0.01).

The Danger of Variance Blow-Up

By the properties of variance, the variance of our finite difference estimator is:

$$\text{Var}(\widehat{\Delta}) = \frac{1}{\delta^2} \text{Var}(\hat{\pi}_0(S_0 + \delta) - \hat{\pi}_0(S_0))$$

Using the standard identity for the variance of a difference, we expand this to:

$$\text{Var}(\widehat{\Delta}) = \frac{1}{\delta^2} [\text{Var}(\hat{\pi}_0(S_0 + \delta)) + \text{Var}(\hat{\pi}_0(S_0)) - 2\text{Cov}(\hat{\pi}_0(S_0 + \delta), \hat{\pi}_0(S_0))]$$

Imagine if we used **independent** random samples (different seeds) to compute the base price and the bumped price. Because they are independent, the covariance term $\text{Cov}(\hat{\pi}_0(S_0 + \delta), \hat{\pi}_0(S_0))$ would be exactly zero.

Because we require δ to be very small to minimize the finite-difference bias, δ^2 in the denominator becomes infinitesimally small. Dividing the sum of the two variances by a number like 10^{-4} or 10^{-6} causes the

variance of the Greek estimator to explode. The resulting Delta would be utterly submerged in Monte Carlo noise!

The Solution: Common Random Numbers (CRN)

To prevent this variance blow-up, we employ a powerful variance reduction technique known as **Common Random Numbers (CRN)**. The core idea is to use **the exact same sequence of random standard normal draws** (the same seed) for simulating both the base paths and the bumped paths.

By subjecting both the base asset and the slightly bumped asset to the exact same sequence of market shocks, the resulting simulated payoffs move in lockstep. This induces a massive **positive correlation** between the two price estimators.

Because the correlation approaches 1, the covariance term approaches the average of the two variances:

$$2\text{Cov}(\hat{\pi}_0(S_0 + \delta), \hat{\pi}_0(S_0)) \approx \text{Var}(\hat{\pi}_0(S_0 + \delta)) + \text{Var}(\hat{\pi}_0(S_0))$$

This causes the terms inside the brackets of our variance equation to nearly cancel each other out. The numerator becomes exceedingly small, perfectly neutralizing the small δ^2 in the denominator.

Mathematically:

$$\text{Cov}(\hat{\pi}_0^{\text{bump}}, \hat{\pi}_0) \uparrow \Rightarrow \left[\text{Var}(\hat{\pi}_0^{\text{bump}}) + \text{Var}(\hat{\pi}_0) - 2\text{Cov}(\hat{\pi}_0^{\text{bump}}, \hat{\pi}_0) \right] \downarrow \Rightarrow \text{Var}(\widehat{\Delta}) \downarrow$$

In practical computational finance, Common Random Numbers are usually essential for obtaining an efficient bump-and-revalue Greek estimator; without them, the variance is often prohibitively large.

11.3 The Pathwise Derivative Method

While the finite difference (bump-and-revalue) method is universal, it is computationally expensive because it requires multiple pricing runs. The **pathwise method** offers a highly efficient alternative by estimating option sensitivities analytically. It does this by directly differentiating each simulated sample path with respect to the parameter of interest *within a single simulation run*.

Mathematical Setup

Let p be a continuous model parameter (e.g., the initial asset price S_0 for Delta, or volatility σ for Vega). Assume we are given the pathwise representation of the underlying asset's price process:

$$(S_t(\omega, p))_{t \in [0, T]}$$

where we assume that for almost every random realization ω , the path depends smoothly on p .

The price of a derivative with a discounted payoff functional $\tilde{H}$ is given by the risk-neutral expectation:

$$\pi_0(p) = \mathbb{E}_{\mathbb{Q}} \left[\tilde{H}((S_t(p))_{t \in [0, T]}) \right]$$

If the map $p \mapsto \tilde{H}((S_t(p))_{t \in [0, T]})$ is almost surely differentiable and its derivative is dominated by an integrable random variable, then the Dominated Convergence Theorem allows us to interchange the derivative and the expectation operator. A Lipschitz payoff is often helpful in applications, but by itself is not sufficient. This yields the pathwise estimator:

$$\frac{\partial}{\partial p} \pi_0(p) = \mathbb{E}_{\mathbb{Q}} \left[\frac{\partial}{\partial p} \tilde{H}((S_t(p))_{t \in [0, T]}) \right]$$

By applying the chain rule, we can compute this derivative analytically on a path-by-path basis and average the results.

Strengths and Limitations

- The method works exceptionally well when the payoff is **continuously differentiable** (or at least piecewise differentiable and continuous) with respect to p .
- Classic examples where this method shines include standard European call/put options and Asian options with continuous averaging.
- The method **fails completely** if $\tilde{H}$ is discontinuous in p .

11.3.1 The Challenge of Discontinuous Payoffs: Digital Options

To understand this failure, consider a digital (cash-or-nothing) call option, which pays a fixed amount if the underlying asset finishes above a certain strike price K at maturity. Its payoff function can be represented using the indicator function:

$$H(S_T) = \mathbf{1}_{\{S_T > K\}} = \begin{cases} 1 & \text{if } S_T > K \\ 0 & \text{if } S_T \leq K \end{cases}$$

This function has a strict jump discontinuity at $S_T = K$. Its derivative with respect to S_T is zero everywhere, except exactly at K , where it is undefined (formally, it is a Dirac delta function).

In a Monte Carlo simulation, the probability of a continuous random variable landing *exactly* on the strike K is zero. Therefore, the pathwise derivative of this payoff will evaluate to exactly 0 for every single simulated path! Averaging these zeros incorrectly implies that the option's Delta is 0, which is financially absurd.

11.3.2 Smooth Approximation of Discontinuities

To rescue the pathwise method for digital and barrier options, we can replace the discontinuous payoff with a **smooth approximation**.

We define a family of smooth, sigmoid-like functions $H_\epsilon(S_T)$, parameterized by a small bandwidth $\epsilon > 0$, to approximate the rigid step function:

$$H_\epsilon(S_T) = \frac{1}{1 + \exp\left(-\frac{S_T - K}{\epsilon}\right)}$$

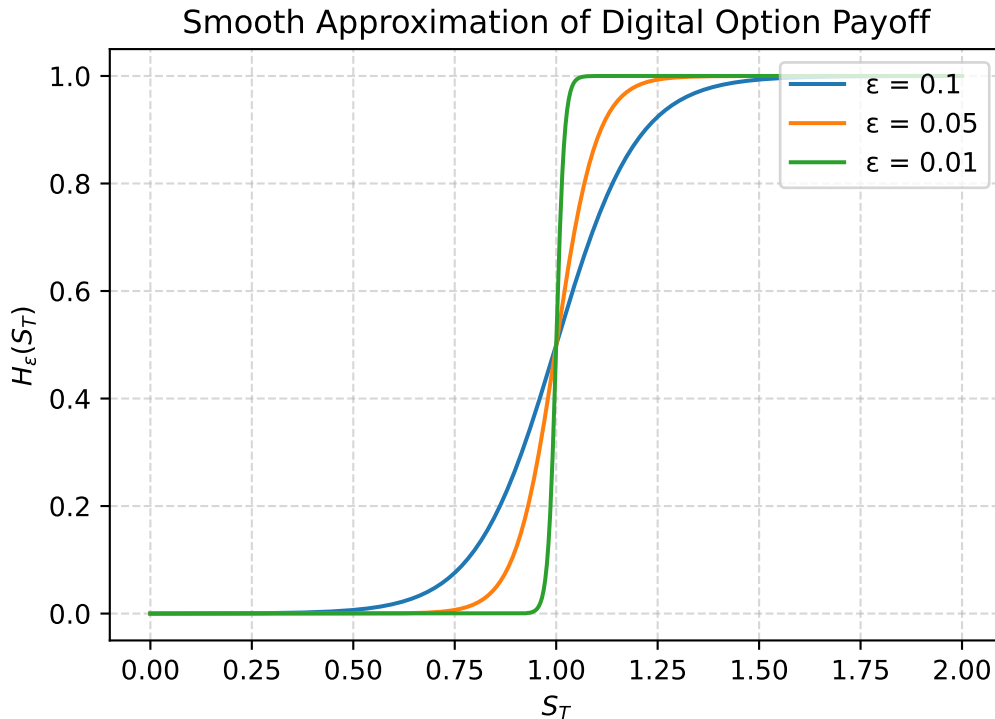
This logistic function possesses several highly desirable mathematical properties: * It is **infinitely differentiable** everywhere for any $\epsilon > 0$. * As the bandwidth shrinks ($\epsilon \rightarrow 0$), the approximation converges to the true digital payoff: $H_\epsilon(S_T) \rightarrow H(S_T)$.

The smoothed function $H_\epsilon(S_T)$ easily satisfies the regularity conditions for interchanging the derivative and the expectation. We can now compute sensitivities like Delta and Vega using the analytic chain rule without triggering the zero-derivative trap.

The Bias-Variance Trade-off

While powerful, this smoothing introduces a structural trade-off: * A **smaller** ϵ makes the approximation financially more accurate (less bias), but the slope of the sigmoid becomes extremely steep. This causes the pathwise derivatives to behave wildly, vastly increasing the variance (numerical instability) of the Greek estimator. * A **larger** ϵ reduces the variance and stabilizes the gradients, but structurally changes the payoff being priced, introducing valuation bias.

In practice, ϵ must be tuned carefully. A common heuristic is to scale it relative to the underlying asset, setting $\epsilon \approx 0.5\%$ to 1% of S_0 , balancing the need for accuracy against the need for stable gradient estimation.



11.3.3 Black-Scholes Delta via the Pathwise Method

Let us apply the pathwise method to the Black-Scholes model. We assume the underlying asset follows a Geometric Brownian Motion, which has the exact terminal solution:

$$S_T = S_0 \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) T + \sigma \sqrt{T} Z \right)$$

where $Z \sim \mathcal{N}(0, 1)$ is a standard normal random variable.

Consider a standard European Call option with strike K . Under the risk-neutral measure $\mathbb{Q}$, its price is the discounted expected payoff:

$$\pi_0 = \mathbb{E}_{\mathbb{Q}} [e^{-rT} (S_T - K)^+]$$

Our goal is to estimate the option's **Delta** (Δ), which is the sensitivity of the price to the initial asset price S_0 :

$$\Delta = \frac{\partial \pi_0}{\partial S_0} = \mathbb{E}_{\mathbb{Q}} \left[e^{-rT} \frac{\partial (S_T - K)^+}{\partial S_0} \right]$$

To evaluate the derivative inside the expectation, we apply the **chain rule**:

$$\frac{\partial (S_T - K)^+}{\partial S_0} = \frac{\partial (S_T - K)^+}{\partial S_T} \cdot \frac{\partial S_T}{\partial S_0}$$

The payoff is differentiable with respect to S_T almost everywhere (except exactly at K , which occurs with probability zero). Its derivative is simply the indicator function:

$$\frac{\partial (S_T - K)^+}{\partial S_T} = \mathbf{1}_{\{S_T > K\}}$$

Next, we differentiate the terminal asset price S_T with respect to the initial price S_0 . Looking at the explicit solution for S_T , we see that S_0 acts as a linear scaling factor, giving us:

$$\frac{\partial S_T}{\partial S_0} = \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) T + \sigma \sqrt{T} Z \right) = \frac{S_T}{S_0}$$

Multiplying these components together, the **pathwise estimator** for Delta is:

$$\widehat{\Delta} = e^{-rT} \mathbf{1}_{\{S_T > K\}} \frac{S_T}{S_0}$$

Remark on Unbiasedness and Generality: * By taking the expectation of this estimator, we recover the exact analytical Black-Scholes Delta, proving the estimator is **unbiased**: $\Delta = \mathbb{E}_{\mathbb{Q}}[\widehat{\Delta}]$. *

Furthermore, the relationship $\frac{\partial S_T}{\partial S_0} = \frac{S_T}{S_0}$ extends beyond Geometric Brownian Motion to models with a spot-multiplicative structure $S_T = S_0 \Xi_T$ where the random factor Ξ_T is independent of S_0 (for example, many stochastic-volatility models). It should not be treated as a generic identity for arbitrary local-volatility dynamics.

11.3.4 Black-Scholes Vega via the Pathwise Method

Using the exact same structural setup, we can compute **Vega** (ν), the sensitivity of the option price to the volatility parameter σ :

$$\nu = \frac{\partial \pi_0}{\partial \sigma} = \mathbb{E}_{\mathbb{Q}} \left[e^{-rT} \frac{\partial (S_T - K)^+}{\partial \sigma} \right]$$

Again, we apply the chain rule:

$$\frac{\partial (S_T - K)^+}{\partial \sigma} = \frac{\partial (S_T - K)^+}{\partial S_T} \cdot \frac{\partial S_T}{\partial \sigma}$$

We already know the first part is $\mathbf{1}_{\{S_T > K\}}$. To find $\frac{\partial S_T}{\partial \sigma}$, we differentiate our explicit formula for S_T using the chain rule on the exponential function:

$$\begin{aligned} \frac{\partial S_T}{\partial \sigma} &= S_0 \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) T + \sigma \sqrt{T} Z \right) \cdot \frac{\partial}{\partial \sigma} \left(\left(r - \frac{1}{2} \sigma^2 \right) T + \sigma \sqrt{T} Z \right) \\ &= S_T \cdot (-\sigma T + \sqrt{T} Z) \end{aligned}$$

Combining these results, the **pathwise estimator** for Vega is given by:

$$\hat{\nu} = e^{-rT} \mathbf{1}_{\{S_T > K\}} S_T (-\sigma T + \sqrt{T} Z)$$

Just like the Delta estimator, the expected value of this expression yields the analytical Black-Scholes vega, confirming it is an unbiased estimator: $\nu = \mathbb{E}_{\mathbb{Q}}[\hat{\nu}]$.

11.3.4.1 Example 1: Asian Option's Delta

As in the previous examples, assume the underlying asset is modeled by a Geometric Brownian Motion, but now let the payoff be path-dependent. Specifically, we examine an **Asian option** written on the discretely monitored arithmetic average:

$$\bar{S}_T = \frac{1}{m} \sum_{i=1}^m S_{t_i}$$

with a discounted payoff given by $H(\mathbf{S}) = e^{-rT} (\bar{S}_T - K)^+$.

Following the pathwise Delta estimation approach, we differentiate the payoff with respect to the initial asset price S_0 :

$$\frac{\partial(\bar{S}_T - K)^+}{\partial S_0} = \mathbf{1}_{\{\bar{S}_T > K\}} \cdot \frac{\partial \bar{S}_T}{\partial S_0}$$

Because the derivative is a linear operator, we can pass it through the summation:

$$\frac{\partial \bar{S}_T}{\partial S_0} = \frac{1}{m} \sum_{i=1}^m \frac{\partial S_{t_i}}{\partial S_0} = \frac{1}{m} \sum_{i=1}^m \frac{S_{t_i}}{S_0} = \frac{\bar{S}_T}{S_0}$$

Substituting this back into the expectation, the pathwise Delta estimator becomes:

$$\widehat{\Delta}_{\text{Asian}} = e^{-rT} \mathbf{1}_{\{\bar{S}_T > K\}} \frac{\bar{S}_T}{S_0}$$

Remark: Because there is no general, exact analytical formula for the price of an arithmetic Asian option, this unbiased estimator has immense practical value. Given that the path average $\bar{S}_T$ must be simulated anyway to estimate the option's price, computing this Greek requires virtually zero additional computational effort!

11.3.4.2 Example 2: Spread Option's Delta

Consider a **Spread option**, which allows an investor to speculate on the price difference between two underlying assets. The discounted payoff is:

$$e^{-rT} \left((S_T^{(1)} - S_T^{(2)}) - K \right)^+$$

We want to find the Delta with respect to each initial asset price. By applying the chain rule, we differentiate the payoff. Note that taking the derivative with respect to $S_0^{(2)}$ introduces a necessary **minus sign** that is often mistakenly omitted by students!

For the first asset, the pathwise Delta estimator is:

$$\widehat{\Delta}^{(1)} = e^{-rT} \mathbf{1}_{\{S_T^{(1)} - S_T^{(2)} > K\}} \frac{S_T^{(1)}}{S_0^{(1)}}$$

For the second asset, applying the chain rule to $-S_T^{(2)}$ yields:

$$\widehat{\Delta}^{(2)} = -e^{-rT} \mathbf{1}_{\{S_T^{(1)} - S_T^{(2)} > K\}} \frac{S_T^{(2)}}{S_0^{(2)}}$$

11.3.4.3 Example 3: Max Option's Delta

Next, we consider a **Rainbow option**, specifically a Max Option, which pays off based on the best-performing asset in a basket of k assets. The discounted payoff is:

$$e^{-rT} \left(\max(S_T^{(1)}, \dots, S_T^{(k)}) - K \right)^+$$

To compute the Delta with respect to the i -th asset, $\Delta^{(i)}$, we recognize that the derivative of the max function is 1 if the i -th asset is the strict maximum, and 0 otherwise.

Thus, the i -th pathwise Delta estimator is:

$$\widehat{\Delta}^{(i)} = e^{-rT} \mathbf{1}_{\{S_T^{(i)} > K\}} \cdot \mathbf{1}_{\{S_T^{(i)} > S_T^{(j)}, \forall j \neq i\}} \cdot \frac{S_T^{(i)}}{S_0^{(i)}}$$

More generally, if we denote the terminal value of a complex multi-asset portfolio by $W(S_T^{(1)}, \dots, S_T^{(k)})$, the i -th Delta of the portfolio follows the exact same chain rule logic:

$$\widehat{\Delta}_W^{(i)} = \frac{\partial W}{\partial S_T^{(i)}} \frac{\partial S_T^{(i)}}{\partial S_0^{(i)}}$$

11.3.5 Option Sensitivities in Stochastic Volatility Models

The elegant scaling property $\frac{\partial S_T}{\partial S_0} = \frac{S_T}{S_0}$ that we utilized in the Geometric Brownian Motion examples is not an isolated mathematical coincidence. It applies much more broadly!

Consider an underlying asset governed by a **Stochastic Volatility (SV) model** (such as the Heston model), where the volatility itself is a random process:

$$\begin{aligned} \frac{dS_t}{S_t} &= r_t dt + \sigma_t dW_t^{(1)} \\ d\sigma_t &= \alpha(\sigma_t) dt + \beta(\sigma_t) dW_t^{(2)} \end{aligned}$$

By applying Itô's lemma to $\ln S_t$, the solution for the asset price is still represented as a geometric exponential process:

$$S_T = S_0 \exp \left(\int_0^T \left(r_t - \frac{1}{2} \sigma_t^2 \right) dt + \int_0^T \sigma_t dW_t^{(1)} \right)$$

Note: In spot-multiplicative stochastic volatility models of the form above, the volatility process σ_t is independent of the initial spot price S_0 . Conditional on a volatility path, we can write $S_T = S_0 \Xi_T$, where the random factor Ξ_T does not depend on S_0 .

Because of this multiplicative structure, the vital relationship remains intact:

$$\frac{\partial S_T}{\partial S_0} = \frac{S_T}{S_0}$$

Consequently, the pathwise Delta estimators we derived previously remain valid for Delta with respect to S_0 in such stochastic volatility models.

- **Standard European Call:**

$$\widehat{\Delta} = e^{-rT} \mathbf{1}_{\{S_T > K\}} \frac{S_T}{S_0}$$

- **Arithmetic Asian Option:**

$$\widehat{\Delta} = e^{-rT} \mathbf{1}_{\{\bar{S}_T > K\}} \frac{\bar{S}_T}{S_0}$$

11.4 The Likelihood Ratio Method

The final method for estimating option sensitivities introduced in this course is the **Likelihood Ratio Method (LRM)** (also known as the Score Function method). Unlike the pathwise method—which relies on differentiating the payoff function itself—the LRM takes a completely different route: it **differentiates the probability density function**.

Rather than requiring the payoff function $H(S_T)$ to be smooth, the LRM only assumes that the probability density function of the asset price at maturity, denoted $g(x, p)$, is known and differentiable with respect to the parameter of interest p .

This makes the LRM exceptionally powerful for pricing derivatives with **discontinuous or non-smooth payoffs** (such as digital or barrier options) where the pathwise method fails, provided the underlying asset's distribution is well understood.

Assume the expected discounted payoff can be written as an integral over the state space:

$$\pi_0(p) = \mathbb{E}_{\mathbb{Q}}[e^{-rT} H(S_T)] = \int_0^{\infty} e^{-rT} H(x) g(x, p) dx$$

where: * $g(x, p)$ is the **probability density function (PDF)** of the terminal asset price S_T . * p is the model parameter with respect to which we want to compute the sensitivity (e.g., S_0, σ).

Notice a crucial shift in perspective: **the parameter p now appears exclusively in the density function g** , completely isolating the payoff function $H(x)$ from the differentiation operator!

11.4.1 Deriving the Likelihood Ratio Estimator

Take the pricing equation and differentiate it with respect to p (assuming regularity conditions allow us to pass the derivative inside the integral):

$$\frac{\partial \pi_0(p)}{\partial p} = \int_0^\infty e^{-rT} H(x) \frac{\partial g(x, p)}{\partial p} dx$$

We can apply a clever mathematical trick by multiplying and dividing the integrand by $g(x, p)$:

$$\frac{\partial \pi_0(p)}{\partial p} = \int_0^\infty e^{-rT} H(x) \left[\frac{1}{g(x, p)} \frac{\partial g(x, p)}{\partial p} \right] g(x, p) dx$$

Recognizing that $\frac{1}{g} \frac{\partial g}{\partial p} = \frac{\partial \log g}{\partial p}$ by the chain rule, we can rewrite this integral back as an expectation:

$$\frac{\partial \pi_0(p)}{\partial p} = \mathbb{E}_{\mathbb{Q}} \left[e^{-rT} H(S_T) \frac{\partial \log(g(S_T, p))}{\partial p} \right]$$

The term $\frac{\partial \log(g(S_T, p))}{\partial p}$ is known as the **Score Function** or the **Likelihood Ratio weight**. Notice that this weight is completely independent of the payoff H . Once derived for a specific model parameter, it can be applied to *any* derivative contract!

11.4.2 Black-Scholes Delta via the Likelihood Ratio Method

Let us apply the LRM to compute the Delta of a European Call option under the Black-Scholes model:

$$\pi_0 = \mathbb{E}_{\mathbb{Q}}[e^{-rT}(S_T - K)^+]$$

Under Geometric Brownian Motion, the terminal asset price S_T follows a log-normal distribution. Its probability density function $g(x, S_0)$ is:

$$g(x, S_0) = \frac{1}{x\sigma\sqrt{2\pi T}} \exp \left[-\frac{1}{2} d(x)^2 \right]$$

where the standardized variable $d(x)$ is defined as:

$$d(x) = \frac{\log(x/S_0) - (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$$

To find the Score Function for Delta, it is mathematically simplest to take the natural logarithm of the density *before* differentiating:

$$\log g(x, S_0) = -\log(x\sigma\sqrt{2\pi T}) - \frac{1}{2} d(x)^2$$

Now, we differentiate this expression with respect to the initial asset price S_0 . The first term is a constant with respect to S_0 , so its derivative is zero. Applying the chain rule to the second term yields:

$$\frac{\partial \log g(x, S_0)}{\partial S_0} = -d(x) \frac{\partial d(x)}{\partial S_0}$$

Next, we differentiate $d(x)$ with respect to S_0 :

$$\frac{\partial d(x)}{\partial S_0} = \frac{\partial}{\partial S_0} \left[\frac{\log(x) - \log(S_0) - (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}} \right] = -\frac{1}{S_0\sigma\sqrt{T}}$$

Substituting this back into our log-derivative equation, we obtain the Score Function Y :

$$Y = \frac{\partial \log g(x, S_0)}{\partial S_0} = -d(x) \left(-\frac{1}{S_0\sigma\sqrt{T}} \right) = \frac{d(x)}{S_0\sigma\sqrt{T}}$$

Expanding $d(x)$ and evaluating it at the simulated terminal price $x = S_T$, the exact mathematical weight is:

$$Y = \frac{\log(S_T/S_0) - (r - \frac{1}{2}\sigma^2)T}{S_0\sigma^2T}$$

The **LRM estimator** for the Black-Scholes Delta is therefore:

$$\widehat{\Delta}_{\text{LRM}} = e^{-rT}(S_T - K)^+ \cdot Y$$

A Crucial Computational Simplification

While the expression for Y derived above is mathematically correct, evaluating logarithms and divisions for every Monte Carlo path is computationally expensive. We can drastically simplify this for our simulation engine!

Recall how we simulate the terminal price S_T using a standard normal draw $Z \sim \mathcal{N}(0, 1)$:

$$\log(S_T/S_0) = \left(r - \frac{1}{2}\sigma^2 \right) T + \sigma\sqrt{T}Z$$

If we substitute this simulated dynamic directly into the numerator of our weight Y , the drift terms cancel out perfectly:

$$Y = \frac{(r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z - (r - \frac{1}{2}\sigma^2)T}{S_0\sigma^2T} = \frac{\sigma\sqrt{T}Z}{S_0\sigma^2T} = \frac{Z}{S_0\sigma\sqrt{T}}$$

This is a profound result. The LRM estimator for Delta simplifies to multiplying the discounted payoff by a strictly linear scaling of the random shock Z :

$$\widehat{\Delta}_{\text{LRM}} = e^{-rT}(S_T - K)^+ \left(\frac{Z}{S_0 \sigma \sqrt{T}} \right)$$

11.4.2.1 Solving the Digital Option Problem

Because the LRM completely isolates the payoff function, we no longer need to construct artificial smooth approximations for discontinuous derivatives.

To compute the exact Delta of a digital call option that pays 1 if $S_T \geq K$, we simply swap the payoff function in our formula while keeping the identical Score Function Y :

$$\widehat{\Delta}_{\text{Digital}} = e^{-rT} \mathbf{1}_{\{S_T \geq K\}} \left(\frac{Z}{S_0 \sigma \sqrt{T}} \right)$$

This provides a mathematically rigorous, unbiased estimator for the digital Greek, showcasing the true power of the Likelihood Ratio Method.

11.5 Summary and Connection to the Next Lecture

This lecture has developed two complementary strategies for making Monte Carlo simulation more useful in practice.

Variance reduction. The fundamental $\mathcal{O}(N^{-1/2})$ convergence rate cannot be improved without changing the problem, but the variance constant σ_H^2 can be reduced dramatically:

- **Antithetic variates** exploit the symmetry of Brownian increments to induce negative covariance between paired payoffs, which lowers variance whenever the antithetic pairing is sufficiently negatively correlated. Monotone payoffs often satisfy this condition, and the method is essentially free.
- **Control variates** anchor the estimator to a correlated random variable whose expectation is known in closed form. The geometric Asian call closed form used here is the canonical example; using the Black-Scholes price as a control for Heston pricing is the analogous construction in stochastic-volatility contexts.
- **Importance sampling** performs a Girsanov-type change of measure to concentrate paths in the payoff-relevant region of the state space. It is most powerful for deep out-of-the-money options and rare-event pricing.

Greeks via simulation. Three methods estimate partial derivatives of the price:

Method	Strengths	Limitations
Bump-and-revalue (FD)	Universal; works for any payoff	Requires separate pricing runs; variance blow-up unless CRN is used
Pathwise (IPA)	Low variance; one simulation run	Requires smooth payoff; fails for digital/barrier payoffs

Method	Strengths	Limitations
Likelihood ratio (score function)	Works for discontinuous payoffs; payoff-agnostic weight	Higher variance than pathwise for smooth payoffs

The pathwise Delta formula $\widehat{\Delta} = e^{-rT} \mathbf{1}_{\{S_T > K\}} \frac{S_T}{S_0}$ extends to stochastic-volatility models by the spot-multiplicative structure argument, connecting this lecture back to the Heston framework of Lecture 8.

The next and final lecture uses simulation in a fundamentally different way: instead of estimating a fixed expectation, it applies **least-squares regression across paths** to approximate the conditional expectation that defines the optimal early-exercise policy for American options. The Longstaff-Schwartz method combines the path generation from Lecture 7, the regression idea from control variates, and the backward-induction logic of dynamic programming into a single unified algorithm.

12 Least-Squares Monte Carlo for American Options

Finite-difference methods handle early exercise very well when the state space is low-dimensional. Lectures 9 and 10 showed how the BTCS and Crank-Nicolson schemes can be adapted to American options by appending a free-boundary condition — at each grid point, the option value is constrained to be at least the intrinsic value — turning the linear tridiagonal system into a complementarity problem. Once the option value depends on several risk factors or on path functionals, however, grid-based PDE methods become expensive very quickly due to the curse of dimensionality established in Lecture 6.

This is precisely the regime where Monte Carlo is attractive, but naive simulation does not by itself solve the early-exercise problem: we cannot simply average discounted terminal payoffs when early exercise may be optimal. This lecture presents the Longstaff-Schwartz insight: use the cross-sectional information already present in the simulated paths to estimate the continuation value by regression, then use that regression to decide whether exercise is optimal at each date along each path. The result is the **least-squares Monte Carlo** (LSM) method, which synthesizes the path generation from Lecture 7, the backward-induction principle that underlies both dynamic programming and PDE time-stepping, and the regression idea seen in the control-variates construction of Lecture 11.

12.1 Motivation: Why American Options Are Hard in Monte Carlo

For a European payoff, Monte Carlo prices the derivative by simulating the terminal cash flow and averaging. For an American-style option, the holder may stop early, so the value depends on an **optimal stopping rule** rather than on a fixed payoff date.

Consider exercise dates

$$0 = t_0 < t_1 < \dots < t_m = T.$$

Let X_n denote the relevant state vector at time t_n . In the simplest Markovian example one can take $X_n = S_{t_n}$, but in path-dependent problems X_n may include additional state variables such as a running average, a short-rate vector, or accumulated cash-flow information.

If the option is exercised at time t_n , the immediate cash flow is

$$h_n(X_n).$$

If the option is not exercised, the holder keeps the option alive and reconsiders the problem at the next exercise date.

Proposition 12.1. *For discrete exercise dates, the American option value satisfies the backward recursion*

$$V_m(X_m) = h_m(X_m),$$

and for $n = m - 1, \dots, 0$,

$$V_n(X_n) = \max(h_n(X_n), C_n(X_n)),$$

where the continuation value is

$$C_n(x) = \mathbb{E}^{\mathbb{Q}} [e^{-r(t_{n+1}-t_n)} V_{n+1}(X_{n+1}) \mid X_n = x].$$

The mathematical problem is therefore not the simulation of the path itself; it is the estimation of the **conditional expectation** $C_n(\cdot)$ at every exercise date. When X_n is chosen to summarize all information relevant for continuation, conditioning on X_n is the Markovian version of conditioning on the full filtration $\mathcal{F}_{t_n}$.

12.2 The Longstaff-Schwartz Insight

The key observation of Longstaff and Schwartz is that the simulated paths already contain cross-sectional information about continuation values. We can turn that information into an approximation of C_n by regression.

Suppose we have simulated N paths. At a given exercise date t_n :

1. identify the paths for which the option is in the money,
2. compute the discounted future cash flow that would be realized on each such path if we continue,
3. regress those discounted cash flows on basis functions of the current state.

If $\psi_0, \dots, \psi_p$ are basis functions, we approximate the continuation value by

$$C_n(x) \approx \widehat{C}_n(x) = \beta_n^\top \psi(x), \quad \psi(x) = \begin{pmatrix} \psi_0(x) \\ \vdots \\ \psi_p(x) \end{pmatrix}.$$

The regression coefficients solve

$$\widehat{\beta}_n = \arg \min_{\beta \in \mathbb{R}^{p+1}} \sum_{i \in \mathcal{J}_n} (Y_i^{(n)} - \beta^\top \psi(X_i^{(n)}))^2,$$

where:

- $\mathcal{J}_n$ is the set of in-the-money paths at time t_n ,
- $X_i^{(n)}$ is the current state on path i ,
- $Y_i^{(n)}$ is the discounted cash flow realized later on path i if exercise is postponed at time t_n .

Definition 12.1. The LSM exercise rule at time t_n is: exercise on path i if and only if

$$h_n(X_i^{(n)}) \geq \widehat{C}_n(X_i^{(n)}),$$

provided the option is in the money.

This turns the optimal stopping problem into a sequence of ordinary least-squares regressions.

12.3 Basis Functions and State Variables

The paper emphasizes that the continuation value can be approximated with relatively few basis functions in many applications. Typical choices are:

- low-order polynomials in the state variables,
- orthogonal polynomial families such as Laguerre or Hermite polynomials,
- cross-products when there are several state variables.

For a one-dimensional stock-price problem, the original paper highlights weighted Laguerre basis functions such as

$$L_0(x) = e^{-x/2}, \quad L_1(x) = e^{-x/2}(1 - x), \quad L_2(x) = e^{-x/2} \left(1 - 2x + \frac{x^2}{2} \right).$$

Remark. The basis does not need to be unique. What matters is that it captures the shape of the continuation value on the relevant region of the state space. In practice, low-order polynomial bases often work very well.

For non-Markovian or path-dependent contracts, the regression should be built on a state vector rich enough to summarize the information relevant for future exercise decisions. This is one of the reasons LSM extends naturally beyond the one-factor American put.

12.4 Backward Induction in Practice

The algorithm works backward from maturity.

12.4.1 Step 1: Simulate Paths Forward

Generate paths under the risk-neutral measure for all state variables needed by the payoff and by the exercise policy. For a Black-Scholes stock,

$$S_{t_{n+1}} = S_{t_n} \exp \left[\left(r - \frac{1}{2} \sigma^2 \right) \Delta t + \sigma \sqrt{\Delta t} Z_{n+1} \right].$$

12.4.2 Step 2: Initialize at Maturity

At the final exercise date, continuation is impossible, so

$$V_m(X_m) = h_m(X_m).$$

Numerically, this means that the current cash-flow vector is just the terminal payoff across all paths.

12.4.3 Step 3: Regress Continuation Values One Date Earlier

At time t_{m-1} , discount the current cash flows back by one step. On the in-the-money paths, regress these discounted realized cash flows on basis functions of X_{m-1} .

The fitted values provide $\hat{C}_{m-1}$.

12.4.4 Step 4: Compare Exercise and Continuation

If the intrinsic value is larger than the fitted continuation value, stop and replace the path's future cash flow by the immediate exercise payoff. Otherwise leave the path alive.

12.4.5 Step 5: Repeat Until Time Zero

Work backward through all exercise dates. Once the stopping rule is fully specified along all paths, discount the realized exercise cash flow on each path back to time zero and average over paths.

Example 12.1. The LSM algorithm for a discrete-exercise American option is:

1. Simulate all state paths $(X_0, \dots, X_m)$ under $\mathbb{Q}$.
2. Set the pathwise cash flow at maturity equal to $h_m(X_m)$.
3. For $n = m - 1, \dots, 1$:
 - select the in-the-money paths,
 - regress discounted future cash flows on basis functions of X_n ,
 - estimate $\hat{C}_n$ from the fitted values,
 - exercise when $h_n(X_n) \geq \hat{C}_n(X_n)$.
4. Discount the resulting exercise cash flow on each path to time zero.
5. Average across paths.

12.5 A Simple American Put Implementation

The following code illustrates the core logic for an American put in the Black-Scholes model. It uses a quadratic polynomial basis in the normalized stock price S/K .

```
import numpy as np

def lsm_american_put(
    S0=100.0,
    K=100.0,
    r=0.05,
    sigma=0.2,
    T=1.0,
```

```

n_steps=50,
n_paths=40_000,
seed=42,
):
    rng = np.random.default_rng(seed)
    dt = T / n_steps
    # Simulate stock paths under Q.
    S = np.empty((n_paths, n_steps + 1))
    S[:, 0] = S0
    drift = (r - 0.5 * sigma**2) * dt
    vol = sigma * np.sqrt(dt)

    for n in range(n_steps):
        Z = rng.standard_normal(n_paths)
        S[:, n + 1] = S[:, n] * np.exp(drift + vol * Z)

    # Start from maturity.
    cashflows = np.maximum(K - S[:, -1], 0.0)
    exercise_index = np.full(n_paths, n_steps, dtype=int)

    for n in range(n_steps - 1, 0, -1):
        intrinsic = np.maximum(K - S[:, n], 0.0)
        itm = intrinsic > 0.0
        if not np.any(itm):
            continue

        x = S[itm, n] / K
        y = cashflows[itm] * np.exp(-r * dt * (exercise_index[itm] - n))

        basis = np.column_stack([np.ones_like(x), x, x**2])
        beta, *_ = np.linalg.lstsq(basis, y, rcond=None)
        continuation = basis @ beta

        exercise_now = intrinsic[itm] >= continuation
        chosen = np.where(itm)[0][exercise_now]

        cashflows[chosen] = intrinsic[chosen]
        exercise_index[chosen] = n

    return np.mean(cashflows * np.exp(-r * dt * exercise_index))

price = lsm_american_put()
print(f"LSM American put estimate: {price:.4f}")

```

LSM American put estimate: 6.0240

The code is intentionally minimal. In a production implementation one would add standard-error estimation, variance reduction, and a more careful treatment of basis selection. For pedagogical brevity, the same simulated paths are used both to fit and to evaluate the stopping rule. In serious implementations, one usually estimates the regression rule on a training sample and then prices that frozen rule on an independent set of paths.

12.6 Why the Regression Uses Realized Future Cash Flows

An important detail is that the regression target at time t_n is the **discounted realized future cash flow** generated by the current stopping policy on that path. We do not regress on a previously fitted continuation function. Using realized future cash flows keeps the backward recursion tied to actual pathwise exercise outcomes.

This is also why the algorithm is naturally recursive: once the stopping decision at time t_{n+1} is updated, the future cash flow seen from time t_n changes as well.

12.7 Practical Implementation Issues

The paper highlights several issues that matter in actual use.

12.7.1 Use Only In-the-Money Paths

Regression is carried out only on paths where exercise is relevant. This concentrates the statistical effort on the region where the exercise decision may actually change.

12.7.2 Normalize the State Variables

When basis functions contain exponentials or high powers, poor scaling can create numerical problems. For simple equity options it is often convenient to work with normalized variables such as S/K rather than with raw prices.

12.7.3 Low R^2 Does Not Mean Failure

The regression tries to estimate a conditional expectation, not to explain all ex post variation in realized cash flows. Realized continuation cash flows are noisy by construction, so a modest R^2 does not by itself indicate a poor exercise policy.

12.7.4 Training and Pricing Should Be Separated in Serious Applications

Once a stopping rule has been estimated, it is good practice to value that rule on an independent set of simulation paths. This reduces look-ahead effects and gives a cleaner lower-bound estimate for the value implied by the fitted policy.

Proposition 12.2. *Any fixed exercise strategy generated by the LSM regression is an admissible stopping rule. Therefore its expected discounted cash flow cannot exceed the true American option value.*

This statement captures the main economic logic: regression can produce a suboptimal exercise rule, but it cannot create a strategy better than the true optimum. The lower-bound interpretation is cleanest when the fitted rule is frozen first and then evaluated out of sample.

12.8 Where LSM Fits in the Course Storyline

The course has now reached a useful division of labor:

- finite differences are strong for low-dimensional PDE problems and give direct access to exercise boundaries,
- Monte Carlo with variance reduction is strong for high-dimensional expectation problems,
- Longstaff-Schwartz extends Monte Carlo to early-exercise problems by approximating continuation values with regression.

This also explains the computational-finance storyline from Lecture 10 to Lecture 12. Lecture 10 closed the PDE route for low-dimensional pricing, Lecture 11 improved the statistical efficiency of simulation, and Lecture 12 uses that simulation machinery to treat American exercise when PDE grids are no longer practical.

12.9 Summary and Course Conclusion

The Longstaff-Schwartz method solves the central Monte Carlo obstacle for American options: the continuation value is unknown because it is a conditional expectation over the future exercise policy, and future policy depends on future continuation values. The method breaks this circular dependence by replacing the true continuation value with a regression fit on simulated paths, working backward in time.

The algorithm is simple in structure:

- simulate paths forward under $\mathbb{Q}$ using the discretization schemes from Lecture 7,
- initialize cash flows at maturity from the terminal payoff,
- work backward through exercise dates, regressing discounted future cash flows on basis functions of the current state,
- at each date, exercise on paths where intrinsic value exceeds the estimated continuation value,
- average discounted exercise cash flows across paths for the final price estimate.

Its power comes from flexibility. The same logic applies to one-factor American puts, path-dependent contracts, multifactor interest-rate derivatives, and stochastic-volatility models like Heston, provided the state variables entering the regression are chosen carefully. The lower-bound interpretation is crucial: any fixed exercise strategy generated by the LSM regression is an admissible stopping rule, so its value cannot exceed the true American option value. Matching an upper bound from dual methods provides a rigorous confirmation of accuracy.

The full course arc. Looking across all twelve lectures, the course has traveled from the empirical facts under $\mathbb{P}$ (Lecture 1) through the mathematical toolkit for the change of measure (Lecture 2), the fundamental theorem of asset pricing and Black-Scholes pricing (Lecture 3), the structure of the implied-volatility surface (Lecture 4), the BS PDE and Feynman-Kac connection (Lecture 5), to Monte Carlo as a direct computation of risk-neutral expectations (Lecture 6). The middle portion built the computational infrastructure: SDE discretization (Lecture 7), the Heston model and calibration (Lecture 8), and finite-difference methods for the PDE route (Lectures 9-10). The final two lectures improved and extended the Monte Carlo route: variance reduction and Greek estimation (Lecture 11) and least-squares Monte Carlo for American options (Lecture 12).

Every method in this course ultimately evaluates the same quantity — the discounted risk-neutral expectation — through different computational strategies. Monte Carlo evaluates it by averaging; finite differences solve the equivalent PDE on a grid; Feynman-Kac provides the mathematical bridge between the two. The choice among them depends on dimensionality, the nature of the payoff, the need for Greeks, and computational constraints. That is the lasting takeaway of the course.

13 Summary

This course developed computational finance as a connected workflow rather than a collection of isolated techniques. We began with financial data and the stylized facts of returns, used those observations to motivate stochastic models for asset prices, and then built the pricing machinery required for derivatives valuation under no-arbitrage principles.

From there, the course moved through the central layers of quantitative finance:

- stochastic modelling in discrete and continuous time;
- derivative pricing and the structure of European option values;
- the Black-Scholes PDE, dynamic hedging, and the Greeks;
- Monte Carlo methods for pricing under uncertainty;
- discretization schemes for stochastic differential equations;
- stochastic-volatility modelling and implied-volatility calibration;
- variance reduction and efficient Monte Carlo estimation of sensitivities.

The practical message is that financial models only become useful once they can be implemented, tested, and interpreted responsibly. A pricing formula, a simulation output, or a calibrated parameter set is never the end of the analysis by itself. Each result must be understood relative to modelling assumptions, numerical approximation error, statistical uncertainty, and the financial question being asked.

Students who have worked through the material should now be able to move between theory and implementation: formulate stochastic models, price and hedge standard derivatives, simulate risk-factor dynamics, calibrate models to market information, and assess whether a numerical result is trustworthy enough to support a financial decision.

In that sense, the course prepares students for more advanced work in derivative modelling, risk management, and computational quantitative finance by establishing the core habit that matters most in practice: combining mathematical structure with critical numerical judgement.

References